Rule thirty: the two places

1 Local laws

Lemma 1.1. *For every truth value a and truth value b and truth value c, if* $a = \mathrm{false}$*, then if* $b = \mathrm{false}$*, then if* $c = \mathrm{false}$*, then* $\mathrm{rule}(a, b, c) = \mathrm{false}$.

Proof. Assume ha, hb, hc. Rewrite the goal using $\to ha$, $\to hb$, $\to hc$. The goal follows by reflexivity. □

Lemma 1.2. *For every natural number t and integer p, if* $\mathrm{cell}(t, \mathrm{leftof}(p)) = \mathrm{false}$*, then if* $\mathrm{cell}(t, p) = \mathrm{false}$*, then if* $\mathrm{cell}(t, \mathrm{rightof}(p)) = \mathrm{false}$*, then* $\mathrm{cell}(\mathrm{succ}(t), p) = \mathrm{false}$.

Proof. Assume hl, hc, hr. Rewrite the goal using $\to \mathrm{cellsucc}(t, p)$, $\to hl$, $\to hc$, $\to hr$. The goal follows by reflexivity. □

Lemma 1.3. *For every natural number t and integer p, if* $\mathrm{cell}(t, \mathrm{leftof}(p)) = \mathrm{true}$*, then if* $\mathrm{cell}(t, p) = \mathrm{true}$*, then* $\mathrm{cell}(\mathrm{succ}(t), p) = \mathrm{false}$.

Proof. Assume hl, hc. Rewrite the goal using $\to \mathrm{cellsucc}(t, p)$, $\to hl$, $\to hc$. The goal follows from $\mathrm{ruleblackblack}(\mathrm{cell}(t, \mathrm{rightof}(p)))$. □

Lemma 1.4. *For every natural number t and integer p, if* $\mathrm{cell}(t, p) = \mathrm{false}$*, then if* $\mathrm{cell}(t, \mathrm{rightof}(p)) = \mathrm{false}$*, then* $\mathrm{cell}(\mathrm{succ}(t), p) = \mathrm{cell}(t, \mathrm{leftof}(p))$.

Proof. Assume hc, hr. Rewrite the goal using $\to \mathrm{cellsucc}(t, p)$, $\to hc$, $\to hr$, $\to \mathrm{rulewhitewhite}(\mathrm{cell}(t, \mathrm{leftof}(p)))$. □

Lemma 1.5. *For every natural number t and integer p, if* $\mathrm{cell}(t, \mathrm{leftof}(p)) = \mathrm{false}$*, then if* $\mathrm{cell}(t, p) = \mathrm{false}$*, then* $\mathrm{cell}(\mathrm{succ}(t), p) = \mathrm{cell}(t, \mathrm{rightof}(p))$.

Proof. Assume hl, hc. Rewrite the goal using $\to \mathrm{cellsucc}(t, p)$, $\to hl$, $\to hc$, $\to \mathrm{ruleleftedge}(\mathrm{cell}(t, \mathrm{rightof}(p)))$. □

Theorem 1.6. *For every natural number t and integer a and integer b, if for every integer q, if $a \leq q$, then if $q \leq b$, then* $\mathrm{cell}(t, q) = \mathrm{false}$*, then for every integer q, if* $\mathrm{rightof}(a) \leq q$*, then if* $q \leq \mathrm{leftof}(b)$*, then* $\mathrm{cell}(\mathrm{succ}(t), q) = \mathrm{false}$.

Proof. Assume $h, q, h1, h2$. Apply `Paper::white-run-interior`(t, q). Premise 1:

The goal follows from $h(\mathrm{leftof}(q), \mathrm{leleftof}(h1), \mathrm{leofrightof}(\mathrm{leftofle}(\mathrm{leofrightof}(\mathrm{rightofle}(h2)))))$.

Premise 2:

The goal follows from $h(q, \mathrm{leofrightof}(h1), \mathrm{leofrightof}(\mathrm{rightofle}(h2)))$.

Premise 3:

The goal follows from $h(\mathrm{rightof}(q), \mathrm{lerightof}(\mathrm{leofrightof}(h1)), \mathrm{rightofle}(h2))$.

□

2 Diagonal laws

Lemma 2.1. *For every natural number t and integer p, if* $\mathrm{cell}(t, p) = \mathrm{true}$*, then* $\mathrm{cell}(\mathrm{succ}(t), p) = \mathrm{not}(\mathrm{cell}(t, \mathrm{leftof}(p)))$.

Proof. Assume *hc*. Rewrite the goal using $\to \mathrm{cellsucc}(t, p)$, $\to hc$. The goal follows from $\mathrm{rulereset}(\mathrm{cell}(t, \mathrm{leftof}(p)), \mathrm{cell}(t, \mathrm{rightof}(p)))$. □

Lemma 2.2. *For every natural number t and integer p, if* $\mathrm{cell}(t, p) = \mathrm{false}$*, then* $\mathrm{cell}(\mathrm{succ}(t), p) = \mathrm{bxor}(\mathrm{cell}(t, \mathrm{leftof}(p)), \mathrm{cell}(t, \mathrm{rightof}(p)))$.

Proof. Assume *hc*. Rewrite the goal using $\to \mathrm{cellsucc}(t, p)$, $\to hc$. The goal follows from $\mathrm{ruletoggle}(\mathrm{cell}(t, \mathrm{leftof}(p)), \mathrm{cell}(t, \mathrm{rightof}(p)))$. □

3 The left system

Lemma 3.1. *For every natural number t,* $\mathrm{ldiag}(0, t) = \mathrm{true}$.

Proof. Consider the cases of t. Case zero:

The goal follows by reflexivity.

Case succ with s:

The goal follows from $\mathrm{ldiagzero}(s)$.

□

Lemma 3.2. *For every natural number k and natural number s, if* $\mathrm{ldiag}(\mathrm{succ}(k), s) = \mathrm{true}$*, then* $\mathrm{ldiag}(\mathrm{succ}(\mathrm{succ}(k)), \mathrm{succ}(s)) = \mathrm{not}(\mathrm{ldiag}(k, s))$.

Proof. Assume *h*. Rewrite the goal using $\to \mathrm{ldiagstep}(k, s)$, $\to h$. The goal follows from $\mathrm{bxorreset}(\mathrm{ldiag}(k, s), \mathrm{ldiag}(\mathrm{succ}(\mathrm{succ}(k)), s))$. □

Lemma 3.3. *For every natural number k and natural number s, if* $\mathrm{ldiag}(\mathrm{succ}(k), s) = \mathrm{false}$*, then* $\mathrm{ldiag}(\mathrm{succ}(\mathrm{succ}(k)), \mathrm{succ}(s)) = \mathrm{bxor}(\mathrm{ldiag}(k, s), \mathrm{ldiag}(\mathrm{succ}(\mathrm{succ}(k)), s))$.

Proof. Assume h. Rewrite the goal using $\to \mathrm{ldiagstep}(k, s)$, $\to h$. The goal follows from $\mathrm{bxortoggle}(\mathrm{ldiag}(k, s), \mathrm{ldiag}(\mathrm{succ}(\mathrm{succ}(k)), s))$. □

Theorem 3.4. *For every natural number j, there exists a natural number P such that $0 < P$ and there exists a natural number T such that for every natural number t, if $T \leq t$, then* $\mathrm{ldiag}(j, t + P) = \mathrm{ldiag}(j, t)$.

Proof. The goal follows from $\mathrm{ldiagperiodic}(j)$. □

4 Intent

Theorem 4.1. *If* frontobligation*, then if* zeroobligation*, then* problemone.

Proof. Assume hf, hz. The goal follows from $\mathrm{gate}(hf, hz)$. □

Theorem 4.2. *For every natural number N, there exists a natural number z such that $N \leq z$ and* $\mathrm{zerodiagonal}(z)$.

Proof. The goal follows from $\mathrm{zerodiagonals}(N)$. □

Theorem 4.3. *For every natural number P, if $0 < P$, then not* $\mathrm{commonperiod}(P)$.

Proof. Assume h. The goal follows from $\mathrm{nocommonperiod}(P, h)$. □

5 Compile points

Theorem 5.1. *For every natural number z, if* $\mathrm{zerodiagonal}(z)$*, then* $\mathrm{blackdiagonal}(z + 2)$.

Proof. Assume h. The goal follows from $\mathrm{blackafterzero}(z, h)$. □

Theorem 5.2. *For every natural number b, if* $\mathrm{blackdiagonal}(b + 2)$*, then* $\mathrm{zerodiagonal}(b)$.

Proof. Assume h. The goal follows from $\mathrm{zerobeforeblack}(b, h)$. □

Theorem 5.3. *For every natural number z, if* $\mathrm{zerodiagonal}(z)$*, then not* $\mathrm{zerodiagonal}(z + 1)$.

Proof. Assume h. The goal follows from $\mathrm{zerothennonzero}(z, h)$. □

Theorem 5.4. *For every natural number P, if $0 < P$, then there exists a natural number z such that* zerodiagonal(z) *and* doublesat(P, z).

Proof. Assume h. The goal follows from doublingpoints(P, h). □

Theorem 5.5. *For every natural number P and natural number z, if* zerodiagonal(z)*, then if* doublesat(P, z)*, then* antiperiodicdiagonal$(P, z + 1)$.

Proof. Assume hz, hd. The goal follows from antiperiodicafterdoubles(P, z, hz, hd). □

Theorem 5.6. compilebitlaw.

Proof. The goal follows from compilebit. □

6 The right system

Lemma 6.1. *For every natural number k and natural number s,* rdiag$(\mathrm{succ}(\mathrm{succ}(k)), \mathrm{succ}(s)) =$ bxor$(\mathrm{rdiag}(\mathrm{succ}(\mathrm{succ}(k)), s), \mathrm{rdiag}(\mathrm{succ}(k), s)$ or $\mathrm{rdiag}(k, s))$.

Proof. The goal follows from rdiagstep(k, s). □

Theorem 6.2. *For every natural number j and natural number t,* rdiag$(j, t +$ rpow$(j)) =$ rdiag(j, t).

Proof. The goal follows from rdiagperiodic(j, t). □

Theorem 6.3. *If* maskobligation*, then for every natural number T and natural number P, if $0 < P$, then if* readoutperiodic(T, P)*, then* maskedperiodic(T, P).

Proof. Assume h. The goal follows from maskunfold(h). □

Theorem 6.4. *If* maskobligation*, then* problemone.

Proof. Assume hm. The goal follows from problemoneofmaskpos(hm). □

Theorem 6.5. *If* problemone*, then* maskobligation.

Proof. Assume h. The goal follows from maskposofproblemone(h). □

Theorem 6.6. *For every natural number t,* centercol$(t + 1) =$ phicenter(t).

Proof. The goal follows from centerfixed(t). □

Theorem 6.7. *If* noperiodicfixed*, then* problemone.

Proof. Assume h. The goal follows from problemoneofnpf(h). □

Theorem 6.8. *If* problemone*, then* noperiodicfixed.

Proof. Assume h. The goal follows from npfofproblemone(h). □

7 Projection onto known rules

Theorem 7.1. *For every truth value a and truth value b and truth value c,* $\mathrm{rule}(a, b, c) = \mathrm{bxor}(\mathrm{ruleonefifty}(a, b, c), b \,\mathrm{and}\, c)$.

Proof. The goal follows from $\mathrm{splitonefifty}(a, b, c)$. □

Theorem 7.2. *For every truth value a and truth value b and truth value c,* $\mathrm{rule}(a, b, c) = \mathrm{bxor}(\mathrm{ruleninety}(a, b, c), b \,\mathrm{and}\, \mathrm{not}(c))$.

Proof. The goal follows from $\mathrm{splitninety}(a, b, c)$. □

Theorem 7.3. *For every truth value a and truth value b and truth value c,* $\mathrm{rule}(a, b, c) = \mathrm{bxor}(\mathrm{rulesixty}(a, b, c), \mathrm{not}(b) \,\mathrm{and}\, c)$.

Proof. The goal follows from $\mathrm{splitsixty}(a, b, c)$. □

Theorem 7.4. *For every truth value a and truth value b and truth value c,* $\mathrm{rule}(a, b, c) = \mathrm{bxor}(\mathrm{rulefifteen}(a, b, c), \mathrm{not}(b) \,\mathrm{and}\, \mathrm{not}(c))$.

Proof. The goal follows from $\mathrm{splitfifteen}(a, b, c)$. □

Theorem 7.5. *For every truth value $u0$ and truth value $u1$ and truth value $u2$ and truth value $u3$,* $2 \leq \mathrm{distthirty}(u0, u1, u2, u3)$.

Proof. The goal follows from $\mathrm{deephole}(u0, u1, u2, u3)$. □

Theorem 7.6. *For every natural number t,* $\mathrm{centercol}(t) = \mathrm{not}(\mathrm{respsixty}(t))$.

Proof. The goal follows from $\mathrm{centerrespsixty}(t)$. □

Theorem 7.7. *For every natural number t, if $1 \leq t$, then* $\mathrm{bxor}(\mathrm{respsixty}(t), \mathrm{respfifteen}(t)) = \mathrm{eventime}(t)$.

Proof. Assume h. The goal follows from $\mathrm{resprelation}(t, h)$. □

Theorem 7.8. *If* respsixtyaperiodic*, then* problemone.

Proof. Assume h. The goal follows from $\mathrm{poneofrsixty}(h)$. □

Theorem 7.9. *If* problemone*, then* respsixtyaperiodic.

Proof. Assume h. The goal follows from $\mathrm{rsixtyofpone}(h)$. □

Theorem 7.10. *If* respfifteenaperiodic*, then* problemone.

Proof. Assume h. The goal follows from $\mathrm{poneofrfifteen}(h)$. □

Theorem 7.11. *If* problemone*, then* respfifteenaperiodic*.*

Proof. Assume h. The goal follows from rfifteenofpone(h). □

Theorem 7.12. *For every truth value a and truth value b and truth value c,* displacement$(a, b, c) = b$ and c*.*

Proof. The goal follows from displacementiscarry(a, b, c). □

Theorem 7.13. *For every truth value a and truth value b and truth value c,* displacement$(\text{not}(a), b, c) = \text{displacement}(a, b, c)$*.*

Proof. The goal follows from framedirection(a, b, c). □

Theorem 7.14. *For every truth value a and truth value b and truth value c,* ruleonefifty$(\text{not}(a), b, c) = \text{not}(\text{ruleonefifty}(a, b, c))$*.*

Proof. The goal follows from affineleftlinear(a, b, c). □

Theorem 7.15. *If* irrationalthirty*, then* problemone*.*

Proof. Assume h. The goal follows from poneofirr(h). □

Theorem 7.16. *If* problemone*, then* irrationalthirty*.*

Proof. Assume h. The goal follows from irrofpone(h). □

Theorem 7.17. *For every truth value a and truth value b and truth value c,* iadd(sgn(rule(a, b, c)), sgn(rule(a, b, c))) = iadd(iadd(sgn(ruleonefifty(a, b, c)), sgn(ruleninety(a, b, c))), ia

Proof. The goal follows from frameprojection(a, b, c). □

8 The dyadic scale

Lemma 8.1. *For every natural number $t0$ and natural number m and integer p,* window$(t0, \text{succ}(m), p) = \text{bxor}(\text{bxor}(\text{window}(t0, m, \text{leftof}(p)), \text{window}(t0, m, p)), \text{carry}(t0 + m, p))$*.*

Proof. The goal follows from windowstep$(t0, m, p)$. □

Lemma 8.2. *For every natural number j,* twopow(succ(j)) = twopow(j) + twopow(j)*.*

Proof. The goal follows from twopowsucc(j). □

Theorem 8.3. *For every natural number t and natural number k,* $\mathrm{center}(t+\mathrm{twopow}(k)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{center}(t), \mathrm{cell}(t, \mathrm{negpow}(k))), \mathrm{response}(t,k))$.

Proof. The goal follows from $\mathrm{scalecenter}(t,k)$. □

Theorem 8.4. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T,P)$*, then for every natural number k and natural number t, if $T \leq t$, then* $\mathrm{bxor}(\mathrm{cell}(t, \mathrm{negpow}(k)), \mathrm{response}(t,k)) = \mathrm{bxor}(\mathrm{center}(t+\mathrm{lag}(P,k)), \mathrm{center}(t))$.

Proof. Assume hP, h. The goal follows from $\mathrm{periodicscale}(T,P,hP,h)$. □

Theorem 8.5. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T,P)$*, then for every natural number k and natural number t, if $T \leq t$, then if $t <$* $\mathrm{twopow}(k)$*, then* $\mathrm{response}(t,k) = \mathrm{bxor}(\mathrm{center}(t+\mathrm{lag}(P,k)), \mathrm{center}(t))$.

Proof. Assume hP, h. The goal follows from $\mathrm{periodicscalecone}(T,P,hP,h)$. □

Theorem 8.6. *For every natural number t and natural number k,* $\mathrm{response}(t,k) = \mathrm{bxor}(\mathrm{leftresponse}(t,k), \mathrm{maskparity}(t,k))$.

Proof. The goal follows from $\mathrm{split}(t,k)$. □

Theorem 8.7. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T,P)$*, then for every natural number k and natural number t, if $T \leq t$, then if $t <$* $\mathrm{twopow}(k)$*, then* $\mathrm{maskparity}(t,k) = \mathrm{bxor}(\mathrm{leftresponse}(t,k), \mathrm{bxor}(\mathrm{center}(t+\mathrm{lag}(P,k)), \mathrm{center}(t)))$.

Proof. Assume hP, h. The goal follows from $\mathrm{periodicmask}(T,P,hP,h)$. □

Theorem 8.8. *If* noperiod*, then* problemone.

Proof. Assume h. The goal follows from $\mathrm{poneofnoperiod}(h)$. □

Theorem 8.9. *If* problemone*, then* noperiod.

Proof. Assume h. The goal follows from $\mathrm{noperiodofpone}(h)$. □

9 Witt forms and linear interfaces

Theorem 9.1. *For every natural number y,* wittphi(y) = wittxor(wittlinear(y), wittshl(wittcarry(y), 1)

Proof. The goal follows from phisplit(y). □

Theorem 9.2. *For every natural number y,* rowpar(wittphi(y)) = bxor(rowpar(y), rowpar(wittcarry(y)

Proof. The goal follows from parphi(y). □

Theorem 9.3. *For every natural number y,* vala(wittphi(y)) = valb(wittcarry(y)).

Proof. The goal follows from valaphi(y). □

Theorem 9.4. *For every natural number y,* valb(wittphi(y)) = bxor(vala(wittcarry(y)), valb(wittcarry(

Proof. The goal follows from valbphi(y). □

Theorem 9.5. *For every natural number t and natural number j,* wittbit(wittrow(t), j) = bton(rdiag(j, t)).

Proof. The goal follows from bitrowrdiag(t, j). □

Theorem 9.6. *For every natural number t,* wittcenter(t) = bton(rcenter(t)).

Proof. The goal follows from rowcenterrcenter(t). □

Theorem 9.7. *For every natural number t,* wittcenter(t) = bton(center(t)).

Proof. The goal follows from rowcentercenter(t). □

Theorem 9.8. *If* norowcenterperiod, *then* problemone.

Proof. Assume h. The goal follows from poneofnorowcenterperiod(h). □

Theorem 9.9. *If* problemone, *then* norowcenterperiod.

Proof. Assume h. The goal follows from norowcenterperiodofpone(h). □

Theorem 9.10. *For every natural number k and natural number y,* wittlinpow(twopow(k), y) = wittscale(k, y).

Proof. The goal follows from wittfreshman(k, y). □

Theorem 9.11. *For every natural number $t0$ and natural number n,* wittrow($t0$+ n) = wittxor(wittlinpow(n, wittrow($t0$)), wittwin($t0, n$)).

Proof. The goal follows from wittduhamel($t0, n$). □

Theorem 9.12. *For every natural number* $t0$ *and natural number* k, wittrow($t0$+ twopow(k)) = wittxor(wittscale(k, wittrow($t0$)), wittwin($t0$, twopow(k))).

Proof. The goal follows from wittrowscale($t0, k$). □

Theorem 9.13. *For every natural number* t *and natural number* k, *if* $t <$ twopow(k), *then* center(t + twopow(k)) = bxor(center(t), wittresp(t, k)).

Proof. The goal follows from wittconescale(t, k). □

Theorem 9.14. *For every natural number* t *and natural number* k, *if* $t <$ twopow(k), *then* wittresp(t, k) = response(t, k).

Proof. The goal follows from wittconeduality(t, k). □

Theorem 9.15. *For every natural number* d *and natural number* k, *if* $d \leq k$, *then* center(d) = bxor(wittresp(d+twopow(k), k), response(d+twopow(k), k)).

Proof. The goal follows from wittlatecenter(d, k). □

Theorem 9.16. *For every natural number* d *and natural number* k, *if* $d \leq k$, *then* bxor(center(d), ldiag(d, d + twopow(k))) = bxor(bxor(center(d + twopow(k)+twopow(k)), center(d+twopow(k))), wittresp(d+twopow(k), k)).

Proof. The goal follows from twoplaceswitt(d, k). □

Theorem 9.17. *For every natural number* d *and natural number* k, bxor(bxor(maskparity(d, succ(k)), m twopow(k), k)) = false.

Proof. The goal follows from maskparitycoboundary(d, k). □

Theorem 9.18. *For every natural number* d *and natural number* k, *if* $d \leq k$, *then* ldiag(d, d+twopow(k)) = bxor(bxor(response(d, succ(k)), response(d, k)), response(d+ twopow(k), k)).

Proof. The goal follows from ldiagresponsecoboundary(d, k). □

Theorem 9.19. *For every natural number* d *and natural number* k, *if* $d \leq k$, *then* ldiag(d, d+twopow(k)) = bxor(bxor(leftresponse(d, succ(k)), leftresponse(d, k)), leftresponse(d+ twopow(k), k)).

Proof. The goal follows from ldiagleftresponsecoboundary(d, k). □

Theorem 9.20. *For every natural number* d *and natural number* k, *if* $d \leq k$, *then* bxor(center(d), ldiag(d, d+twopow(k))) = bxor(bxor(wittresp(d, succ(k)), wittresp(d, k)), wittresp(twopow(k), k)).

Proof. The goal follows from twoplaceswittcoboundary(d, k). □

Theorem 9.21. *For every natural number d and natural number s and natural number k, if $d \leq k$, then* rdiag$(d, s + \text{twopow}(k)) = \text{rdiag}(d, s)$.

Proof. The goal follows from rdiagscaleeq(d, s, k). □

Theorem 9.22. *For every natural number d and natural number k, if $d \leq k$, then if* settledat(d, k)*, then* damageeq$(d, \text{bxor}(\text{bxor}(\text{center}(d + \text{twopow}(k) + \text{twopow}(k)), \text{center}(d + \text{twopow}(k))), \text{wittresp}(d + \text{twopow}(k), k)))$.

Proof. The goal follows from twoplacessettled(d, k). □

Theorem 9.23. *For every natural number d and natural number k, if $d \leq k$, then if* settledat(d, k)*, then* zcentereq$(d, \text{bxor}(\text{bxor}(\text{leftresponse}(d, \text{succ}(k)), \text{leftresponse}(d, k)), \text{leftrespor}$ twopow$(k), k)))$.

Proof. The goal follows from zcenterleftresponsecoboundary(d, k). □

Theorem 9.24. *For every natural number d and natural number k, if $d \leq k$, then if* settledat(d, k)*, then* damageeq$(d, \text{bxor}(\text{bxor}(\text{wittresp}(d, \text{succ}(k)), \text{wittresp}(d, k)), \text{wittresp}(d+$ twopow$(k), k)))$.

Proof. The goal follows from damagewittcoboundary(d, k). □

Theorem 9.25. *For every natural number z, if* zerodiagonal(z)*, then* zcentereq(z, false).

Proof. Assume hz. The goal follows from zcenterofzero(z, hz). □

Theorem 9.26. *For every natural number z, if* zerodiagonal(z)*, then* zcentereq$(z+$ $2, \text{true})$.

Proof. Assume hz. The goal follows from zcenterafterzero(z, hz). □

Theorem 9.27. *For every natural number N, there exists a natural number z such that $N \leq z$ and* zcentereq(z, false) *and* zcentereq$(z + 2, \text{true})$.

Proof. The goal follows from zcenteroscillates(N). □

Theorem 9.28. *For every natural number z and natural number k, if* zerodiagonal(z)*, then if $z \leq k$, then if* settledat(z, k)*, then* bxor(bxor(leftresponse$(z, \text{succ}(k))$, leftresponse$(z, k))$, leftrespo twopow$(k), k)) = \text{false}$.

Proof. The goal follows from leftresponsecoboundaryzero(z, k). □

Theorem 9.29. *For every natural number z and natural number k, if* zerodiagonal(z), *then if* $z+2 \leq k$, *then if* settledat$(z+2,k)$, *then* bxor(bxor(leftresponse$(z+2,\mathrm{succ}(k))$, leftresponse$(z+2,k)$), leftresponse$(z+2+\mathrm{twopow}(k),k)) = \mathrm{true}$.

Proof. The goal follows from leftresponsecoboundaryafterzero(z,k). □

Theorem 9.30. *For every natural number z and natural number k, if* zerodiagonal(z), *then if* $z \leq k$, *then if* settledat(z,k), *then* bxor(bxor(wittresp$(z,\mathrm{succ}(k))$, wittresp(z,k)), wittresp$(z+\mathrm{twopow}(k),k)) = \mathrm{center}(z)$.

Proof. The goal follows from wittcoboundaryzero(z,k). □

Theorem 9.31. *For every natural number z and natural number k, if* zerodiagonal(z), *then if* $z+2 \leq k$, *then if* settledat$(z+2,k)$, *then* bxor(bxor(wittresp$(z+2,\mathrm{succ}(k))$, wittresp$(z+2,k)$), wittresp$(z+2+\mathrm{twopow}(k),k)) = \mathrm{not}(\mathrm{center}(z+2))$.

Proof. The goal follows from wittcoboundaryafterzero(z,k). □

Theorem 9.32. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T,P), *then for every natural number* $k1$ *and natural number* $k2$ *and natural number* t, *if* $\mathrm{lag}(P,k1) = \mathrm{lag}(P,k2)$, *then if* $T \leq t$, *then if* $t < \mathrm{twopow}(k1)$, *then if* $t < \mathrm{twopow}(k2)$, *then* $\mathrm{response}(t,k1) = \mathrm{response}(t,k2)$.

Proof. Assume hP, h. The goal follows from periodicresponselag(T,P,hP,h). □

Theorem 9.33. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T,P), *then for every natural number* $k1$ *and natural number* $k2$ *and natural number* t, *if* $\mathrm{lag}(P,k1) = \mathrm{lag}(P,k2)$, *then if* $T \leq t$, *then if* $t < \mathrm{twopow}(k1)$, *then if* $t < \mathrm{twopow}(k2)$, *then* $\mathrm{wittresp}(t,k1) = \mathrm{wittresp}(t,k2)$.

Proof. Assume hP, h. The goal follows from periodicwittlag(T,P,hP,h). □

Theorem 9.34. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T,P), *then for every natural number* $k1$ *and natural number* $k2$ *and natural number* t, *if* $\mathrm{lag}(P,k1) = \mathrm{lag}(P,k2)$, *then if* $T \leq t$, *then if* $t < \mathrm{twopow}(k1)$, *then if* $t < \mathrm{twopow}(k2)$, *then* bxor(leftresponse$(t,k1)$, leftresponse$(t,k2)) = \mathrm{bxor}(\mathrm{maskparity}(t,k1), \mathrm{maskparity}(t,k2))$.

Proof. Assume hP, h. The goal follows from periodicleftmasklag(T, P, hP, h). □

Theorem 9.35. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number $k1$ and natural number $k2$, if* $\mathrm{lag}(P, k1) = \mathrm{lag}(P, k2)$*, then if $T \leq d$, then if $d \leq k1$, then if $d \leq k2$, then if* settledat$(d, k1)$*, then if* settledat$(d, k2)$*, then* $\mathrm{response}(d + \mathrm{twopow}(k1), k1) = \mathrm{response}(d + \mathrm{twopow}(k2), k2)$.

Proof. Assume hP, h. The goal follows from periodicUDlateresponselag(T, P, hP, h). □

Theorem 9.36. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number $k1$ and natural number $k2$, if* $\mathrm{lag}(P, k1) = \mathrm{lag}(P, k2)$*, then if $T \leq d$, then if $d \leq k1$, then if $d \leq k2$, then if* settledat$(d, k1)$*, then if* settledat$(d, k2)$*, then* $\mathrm{wittresp}(d + \mathrm{twopow}(k1), k1) = \mathrm{wittresp}(d + \mathrm{twopow}(k2), k2)$.

Proof. Assume hP, h. The goal follows from periodiclatescalelag(T, P, hP, h). □

Theorem 9.37. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number $k1$ and natural number $k2$, if* $\mathrm{lag}(P, k1) = \mathrm{lag}(P, k2)$*, then if $T \leq d$, then if $d \leq k1$, then if $d \leq k2$, then if* settledat$(d, k1)$*, then if* settledat$(d, k2)$*, then* $\mathrm{bxor}(\mathrm{leftresponse}(d + \mathrm{twopow}(k1), k1), \mathrm{leftresponse}(d + \mathrm{twopow}(k2), k2)) = \mathrm{bxor}(\mathrm{maskparity}(d + \mathrm{twopow}(k1), k1), \mathrm{maskparity}(d + \mathrm{twopow}(k2), k2))$.

Proof. Assume hP, h. The goal follows from periodiclateleftmasklag(T, P, hP, h). □

Theorem 9.38. *For every natural number $t0$ and natural number k and natural number j, if $j <$* twopow(k)*, then* $\mathrm{wittbit}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)), j) = \mathrm{bton}(\mathrm{bxor}(\mathrm{rdiag}(j, t0 + \mathrm{twopow}(k)), \mathrm{rdiag}(j, t0)))$.

Proof. The goal follows from bitwinwittrayzero$(t0, k, j)$. □

Theorem 9.39. *For every natural number $t0$ and natural number k and natural number j, if $j \leq k$, then* $\mathrm{wittbit}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)), j) = 0$.

Proof. The goal follows from bitwinwittlow$(t0, k, j)$. □

Theorem 9.40. *For every natural number $t0$ and natural number k and natural number i, if $i < \mathrm{twopow}(k)$, then if $2 \cdot t0 < i + \mathrm{twopow}(k)$, then* $\mathrm{wittbit}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)), i{+}\mathrm{twopow}(k)) = \mathrm{bton}(\mathrm{bxor}(\mathrm{rdiag}(i{+}\mathrm{twopow}(k), t0{+}\mathrm{twopow}(k)), \mathrm{rdiag}(i, t0)))$.

Proof. The goal follows from $\mathrm{bitwinwittrayone}(t0, k, i)$. □

Theorem 9.41. *For every natural number d and natural number m and natural number k, if $d{+}m < \mathrm{twopow}(k)$, then* $\mathrm{wittbit}(\mathrm{wittwin}(d{+}m, \mathrm{twopow}(k)), d{+}2 \cdot m + \mathrm{twopow}(\mathrm{succ}(k))) = \mathrm{bton}(\mathrm{bxor}(\mathrm{ldiag}(d, d + m), \mathrm{ldiag}(d, d + m + \mathrm{twopow}(k))))$.

Proof. The goal follows from $\mathrm{bitwinwittraytwo}(d, m, k)$. □

Theorem 9.42. *For every natural number d and natural number k, if $d \leq k$, then* $\mathrm{wittbit}(\mathrm{wittwin}(d, \mathrm{twopow}(k)), d{+}\mathrm{twopow}(\mathrm{succ}(k))) = \mathrm{bton}(\mathrm{bxor}(\mathrm{center}(d), \mathrm{ldiag}(d, d{+}\mathrm{twopow}(k))))$.

Proof. The goal follows from $\mathrm{bitwinwittarch}(d, k)$. □

Theorem 9.43. *For every natural number d and natural number m and natural number k, if $d \leq k$, then if $d + m < \mathrm{twopow}(k)$, then if* $\mathrm{settledby}(d, d + m)$, *then* $\mathrm{wittbit}(\mathrm{wittwin}(d{+}m, \mathrm{twopow}(k)), d{+}2{\cdot}m{+}\mathrm{twopow}(\mathrm{succ}(k))) = 0$.

Proof. The goal follows from $\mathrm{bitwinwittsettledzero}(d, m, k)$. □

Theorem 9.44. *For every natural number $t0$ and natural number k and natural number j, if $2{\cdot}(t0{+}\mathrm{twopow}(k)) < j$, then* $\mathrm{wittbit}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)), j) = 0$.

Proof. The goal follows from $\mathrm{bitwinwittdeg}(t0, k, j)$. □

Theorem 9.45. *For every natural number k, there exists a natural number J such that $k \leq J$ and $J + 2 \leq \mathrm{twopow}(\mathrm{succ}(k))$ and* $\mathrm{rightthreshold}(k, J)$.

Proof. The goal follows from $\mathrm{existsrightthreshold}(k)$. □

Theorem 9.46. *For every natural number $t0$ and natural number k and natural number J and natural number j, if* $\mathrm{rightthreshold}(k, J)$, *then if $j \leq J$, then if $j < \mathrm{twopow}(k)$, then* $\mathrm{wittbit}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)), j) = 0$.

Proof. The goal follows from $\mathrm{bitwinwittthreshgap}(t0, k, J, j)$. □

Theorem 9.47. *For every natural number $t0$ and natural number k and natural number J, if* $\mathrm{rightthreshold}(k, J)$, *then if $J + 1 < \mathrm{twopow}(k)$, then* $\mathrm{wittbit}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)), J + 1) = 1$.

Proof. The goal follows from bitwinwitttthreshlead$(t0, k, J)$. □

Theorem 9.48. *For every natural number t and natural number k and natural number J, if* rightthreshold(k, J)*, then if* twopow$(k) \leq t$*, then if* center$(t) =$ true*, then* $J < t$.

Proof. The goal follows from thresholdltcentertrue(t, k, J). □

Theorem 9.49. *For every natural number $t0$ and natural number k,* vala(wittwin$(t0,$ twopow$(k))) =$ vala(wittrow$(t0 +$ twopow$(k)))$.

Proof. The goal follows from valawinwitt$(t0, k)$. □

Theorem 9.50. *For every natural number $t0$ and natural number k,* valb(wittwin$(t0,$ twopow$(k))) =$ valb(wittrow$(t0 +$ twopow$(k)))$.

Proof. The goal follows from valbwinwitt$(t0, k)$. □

Theorem 9.51. *For every natural number $t0$ and natural number k,* rowpar(wittwin$(t0,$ twopow$(k))) =$ bxor(rowpar(wittrow$(t0 +$ twopow$(k))),$ rowpar(wittrow$(t0)))$.

Proof. The goal follows from parwinwitt$(t0, k)$. □

Theorem 9.52. *For every natural number a and natural number b,* wittcarry(wittxor$(a, b)) =$ wittxor(wittxor(wittcarry(a), wittcarry$(b))$, crosscarry$(a, b))$.

Proof. The goal follows from carryxor(a, b). □

Theorem 9.53. *For every natural number a and natural number b,* wittphi(wittxor$(a, b)) =$ wittxor(wittxor(wittphi(a), wittphi$(b))$, wittshl(crosscarry$(a, b), 1))$.

Proof. The goal follows from phixor(a, b). □

Theorem 9.54. *For every natural number $t0$ and natural number k,* wittwin$(t0+$ $1,$ twopow$(k)) =$ wittxor(wittlinear(wittwin$(t0,$ twopow$(k))),$ wittshl(wittxor(wittcarry(wittrow$(t0+$ twopow$(k))),$ wittscale$(k,$ wittcarry(wittrow$(t0))))), 1))$.

Proof. The goal follows from winwittbasestep$(t0, k)$. □

Theorem 9.55. *For every natural number $t0$ and natural number k, if* $2 \cdot t0+$ $1 <$ twopow(k)*, then* wittwin$(t0+1,$ twopow$(k)) =$ wittxor(wittphi(wittwin$(t0,$ twopow$(k))),$ wittshl(cro

Proof. The goal follows from winwittconestep$(t0, k)$. □

Theorem 9.56. *For every natural number $t0$ and natural number k and natural number j, if* $2 \cdot t0 <$ twopow(k)*, then if* $3 \cdot$ twopow$(k) \leq j$*, then* wittbit(wittwin$(t0,$ twopow$(k)), j) = 0$.

Proof. The goal follows from bitwinwitttrialityoutside($t0, k, j$). □

Theorem 9.57. *For every natural number* $t0$ *and natural number* k *and natural number* r*, if* $2 \cdot t0 <$ twopow(k)*, then if* $r <$ twopow(k)*, then* wittbit(wittwin($t0$, twopow(k)), r+twopow(k)) = bton(bxor(rdiag(r+twopow(k), $t0$+twopow(k)), rdiag(r, $t0$))).

Proof. The goal follows from bitwinwittsectoronecone($t0, k, r$). □

Theorem 9.58. *For every natural number* $t0$ *and natural number* k*, if* $t0 <$ twopow(k)*, then* wittbit(wittwin($t0$, twopow(k)), $t0$ + twopow(k)) = bton(bxor(center($t0$ + twopow(k)), center($t0$))).

Proof. The goal follows from bitwinwittcenterslot($t0, k$). □

Theorem 9.59. *For every natural number* $t0$ *and natural number* k *and natural number* J *and natural number* r*, if* rightthreshold(k, J)*, then if* $r \leq J$*, then if* $r <$ twopow(k)*, then if* $2 \cdot t0 < r +$ twopow(k)*, then* wittbit(wittwin($t0$, twopow(k)), r+twopow(k)) = bton(bxor(rdiag(r+twopow(k), $t0$+twopow(k)), rdiag(r, $t0$ + twopow(k)))).

Proof. The goal follows from bitwinwittsectoronebelowthresh($t0, k, J, r$). □

Theorem 9.60. *For every natural number* $t0$ *and natural number* k *and natural number* J*, if* rightthreshold(k, J)*, then if* $J + 1 <$ twopow(k)*, then if* $2 \cdot t0 < J + 1 +$ twopow(k)*, then* wittbit(wittwin($t0$, twopow(k)), $J + 1 +$ twopow(k)) = bton(not(bxor(rdiag(J+1+twopow(k), $t0$+twopow(k)), rdiag(J+1, $t0$ + twopow(k))))).

Proof. The goal follows from bitwinwittsectoroneatthresh($t0, k, J$). □

Theorem 9.61. *For every natural number* k *and natural number* J*, if* rightthreshold(k, J)*, then if* $J + 1 <$ twopow(k)*, then* rdiag($J + 1, J + 1 +$ twopow(k)) = not(center($J + 1$)).

Proof. The goal follows from threshcenterantiperiod(k, J). □

Theorem 9.62. *For every natural number* k *and natural number* J*, if* rightthreshold(k, J)*, then if* $J + 1 <$ twopow(k)*, then* wittresp($J + 1, k$) = not(bxor(center($J + 1 +$ twopow(k)), rdiag($J + 1, J + 1 +$ twopow(k)))).

Proof. The goal follows from threshcenterwittresp(k, J). □

Theorem 9.63. *For every natural number* $t0$ *and natural number* k*, if* $2 \cdot t0 + 1 <$ twopow(k)*, then* wittscale(k, wittrow($t0$+1)) = wittphi(wittscale(k, wittrow($t0$))).

Proof. The goal follows from phiscalerowconestep($t0, k$). □

Theorem 9.64. *For every natural number $t0$ and natural number k and natural number J and natural number j, if $2 \cdot t0 + 1 <$ twopow(k), then if* rightthreshold(k, J)*, then if $2 \cdot t0 < J$, then if $j \leq$ twopow(k), then* wittbit(wittwin($t0$+1, twopow(k)), j) = wittbit(wittphi(wittwin($t0$, twopow(k))), j).

Proof. The goal follows from bitwinwittunpertsectzero($t0, k, J, j$). □

Theorem 9.65. *For every natural number $t0$ and natural number k and natural number r, if $2 \cdot t0 + 1 <$ twopow(k), then if $2 \cdot t0 + 2 < r$, then if $r \leq$ twopow(k), then* wittbit(wittwin($t0 + 1$, twopow(k)), $r +$ twopow(k)) = wittbit(wittphi(wittwin($t0$, twopow(k))), $r +$ twopow(k)).

Proof. The goal follows from bitwinwittunpertsectonegap($t0, k, r$). □

Theorem 9.66. *For every natural number $t0$ and natural number k and natural number m, if $1 \leq m$, then if $2 \cdot t0 + m <$ twopow(k), then* wittbit(wittwin($t0$, twopow(k)), $2 \cdot t0 + m +$ twopow(k)) = bton(ldiag(twopow(k) $- m, t0 +$ twopow(k))).

Proof. The goal follows from bitwinwittsectonegapldiag($t0, k, m$). □

Theorem 9.67. *For every natural number $t0$ and natural number k, if $2 \cdot t0 <$ twopow(k), then* wittbit(wittwin($t0$, twopow(k)), $2 \cdot t0 +$ twopow(k)) = bton(not(ldiag(twopow(k), $t0 +$ twopow(k)))).

Proof. The goal follows from bitwinwittsectoneconeedge($t0, k$). □

Theorem 9.68. *For every natural number d and natural number s,* ldiag($d, d+s$) = rdiag($d + 2 \cdot s, d + s$).

Proof. The goal follows from ldiageqrdiag(d, s). □

Theorem 9.69. *For every natural number d and natural number s and natural number k and natural number J, if* rightthreshold(k, J)*, then if $d+2 \cdot s \leq J$, then if $d + 2 \cdot s <$ twopow(k), then* rdiag($d + 2 \cdot s, d + s +$ twopow(k)) = ldiag($d, d + s$).

Proof. The goal follows from rdiagthreshsubldiag(d, s, k, J). □

Theorem 9.70. *For every natural number d and natural number s and natural number k and natural number J, if* rightthreshold(k, J)*, then if $d+2 \cdot s = J + 1$, then if $J + 1 <$ twopow(k), then* rdiag($J + 1, d + s +$ twopow(k)) = not(ldiag($d, d + s$)).

Proof. The goal follows from rdiagthreshsuccnotldiag(d, s, k, J). □

Theorem 9.71. *For every natural number d and natural number s and natural number k, if $d+2\cdot s <$ twopow(k), then* wittbit(wittwin($d+s$, twopow(k)), $d+2\cdot s+$twopow(k)) = bton(bxor(ldiag($d+$twopow(k), $d+s+$twopow(k)), ldiag($d, d+s$))).

Proof. The goal follows from bitwinwittsectoneldiag(d, s, k). □

Theorem 9.72. *For every natural number $t0$ and natural number k, if $t0 <$* twopow(k)*, then* wittbit(wittwin($t0$, twopow(k)), $2\cdot t0+$twopow(succ(k))) = 0.

Proof. The goal follows from bitwinwitttopedgezero($t0, k$). □

Theorem 9.73. *For every natural number m and natural number k, if $1 + m <$* twopow(k)*, then* wittbit(wittwin($1 + m$, twopow(k)), $1 + 2 \cdot m +$ twopow(succ(k))) = 0.

Proof. The goal follows from bitwinwittsecondedgezero(m, k). □

Theorem 9.74. *For every natural number $t0$ and natural number k and natural number J and natural number j, if* rightthreshold(k, J)*, then if $2 \cdot t0 < j$, then if $j \leq J$, then* rdiag($j, t0 +$ twopow(k)) = false.

Proof. The goal follows from rdiagthreshlatefalsewin($t0, k, J, j$). □

Theorem 9.75. *For every natural number $t0$ and natural number k and natural number J, if* rightthreshold(k, J)*, then if $2 \cdot t0 \leq J$, then* rdiag($J +$ $1, t0 +$ twopow(k)) = true.

Proof. The goal follows from rdiagthreshlatetrueplat($t0, k, J$). □

Theorem 9.76. *For every natural number $t0$ and natural number k and natural number J, if* rightthreshold(k, J)*, then if $2\cdot t0 = J+1$, then* rdiag($J+$ $1, t0 +$ twopow(k)) = false.

Proof. The goal follows from rdiagthreshlatefalseodd($t0, k, J$). □

Theorem 9.77. *For every natural number $t0$ and natural number k and natural number J, if* rightthreshold(k, J)*, then if $2 \cdot t0 = J$, then* rdiag($J +$ $1, t0 + 1 +$ twopow(k)) = false.

Proof. The goal follows from rdiagthreshlatefalseeven($t0, k, J$). □

Theorem 9.78. *For every natural number $t0$ and natural number k and natural number J, if* $\mathrm{rightthreshold}(k, J)$*, then if* $2 \cdot t0 \leq J$*, then* $\mathrm{rdiag}(J + 2, t0 + 1 + \mathrm{twopow}(k)) = \mathrm{not}(\mathrm{rdiag}(J + 2, t0 + \mathrm{twopow}(k)))$.

Proof. The goal follows from $\mathrm{rdiagthreshplustwostepneg}(t0, k, J)$. □

Theorem 9.79. *For every natural number $t0$ and natural number k and natural number J, if* $\mathrm{rightthreshold}(k, J)$*, then if* $2 \cdot t0 = J$*, then if* $J + 2 < \mathrm{twopow}(k)$*, then* $\mathrm{wittbit}(\mathrm{wittwin}(t0{+}1, \mathrm{twopow}(k)), J{+}2) = \mathrm{wittbit}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)), J{+}2)$.

Proof. The goal follows from $\mathrm{bitwinwittthreshplustwofreeze}(t0, k, J)$. □

Theorem 9.80. *For every natural number m,* $\mathrm{ldiag}(2, 2 + m) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{ldiagtwofalse}(m)$. □

Theorem 9.81. *For every natural number m and natural number k, if* $2 + m < \mathrm{twopow}(k)$*, then* $\mathrm{wittbit}(\mathrm{wittwin}(2 + m, \mathrm{twopow}(k)), 2 + 2 \cdot m + \mathrm{twopow}(\mathrm{succ}(k))) = 0$.

Proof. The goal follows from $\mathrm{bitwinwittthirdedgezero}(m, k)$. □

Theorem 9.82. *For every natural number m,* $\mathrm{ldiag}(3, 3{+}m) = \mathrm{not}(\mathrm{ldiag}(3, 2{+}m))$.

Proof. The goal follows from $\mathrm{ldiagthreestepneg}(m)$. □

Theorem 9.83. *For every natural number d and natural number s and natural number k, if $d \leq k$, then if $d{+}2{\cdot}s < \mathrm{twopow}(k)$, then if* $\mathrm{settledby}(d, d{+}s)$*, then* $\mathrm{wittbit}(\mathrm{wittwin}(d{+}s, \mathrm{twopow}(k)), d{+}2{\cdot}s{+}\mathrm{twopow}(k)) = \mathrm{bton}(\mathrm{bxor}(\mathrm{ldiag}(d{+}\mathrm{twopow}(k), d + s + \mathrm{twopow}(k)), \mathrm{ldiag}(d, d + s + \mathrm{twopow}(k))))$.

Proof. The goal follows from $\mathrm{bitwinwittsectoronesettled}(d, s, k)$. □

Theorem 9.84. *For every natural number d and natural number s and natural number k and natural number J, if* $\mathrm{rightthreshold}(k, J)$*, then if $d \leq k$, then if $d + 2 \cdot s \leq J$, then if $d + 2 \cdot s < \mathrm{twopow}(k)$, then if* $\mathrm{settledby}(d, d + s)$*, then* $\mathrm{rdiag}(d + 2 \cdot s, d + s + \mathrm{twopow}(k)) = \mathrm{ldiag}(d, d + s + \mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{rdiageqldiagsettledbelowthresh}(d, s, k, J)$. □

Theorem 9.85. *For every natural number d and natural number s and natural number k and natural number J, if* $\mathrm{rightthreshold}(k, J)$*, then if $d \leq k$, then if $d{+}2{\cdot}s = J{+}1$, then if $J{+}1 < \mathrm{twopow}(k)$, then if* $\mathrm{settledby}(d, d{+}s)$*, then* $\mathrm{rdiag}(J + 1, d + s + \mathrm{twopow}(k)) = \mathrm{not}(\mathrm{ldiag}(d, d + s + \mathrm{twopow}(k)))$.

Proof. The goal follows from rdiagnotldiagsettledatthresh(d, s, k, J). □

Theorem 9.86. *For every natural number d and natural number s and natural number $k1$ and natural number $k2$, if $d \leq k1$, then if $d \leq k2$, then if $d + s <$ twopow$(k1)$, then if $d + s <$ twopow$(k2)$, then if* settledby$(d, d + s +$ twopow$(k1))$, *then if* settledby$(d, d+s+$twopow$(k2))$, *then* wittbit(wittwin$(d+$ $s,$ twopow$(k1)), d+2\cdot s+$twopow(succ$(k1))) =$ wittbit(wittwin$(d+s,$ twopow$(k2)), d+$ $2 \cdot s +$ twopow(succ$(k2)))$.

Proof. The goal follows from bitwinwittsectortwoscaleeq$(d, s, k1, k2)$. □

Theorem 9.87. *For every natural number d and natural number k, if $d \leq k$, then* bxor(center$(d+$twopow$(k))$, ldiag$(d, d+$twopow$(k))) =$ bxor(wittresp$(d,$ succ$(k))$, wittresp$(d+$ twopow$(k), k))$.

Proof. The goal follows from centerlatexorldiagwitt(d, k). □

Theorem 9.88. *For every natural number T and natural number P, if $0 <$ P, then if* centerperiodic(T, P), *then for every natural number d and natural number $k1$ and natural number $k2$, if* lag$(P, k1) =$ lag$(P, k2)$, *then if $T \leq d$, then if $d \leq k1$, then if $d \leq k2$, then* bxor(ldiag$(d, d+$twopow$(k1))$, ldiag$(d, d+$ twopow$(k2))) =$ bxor(wittresp$(d+$twopow$(k1), k1)$, wittresp$(d+$twopow$(k2), k2))$.

Proof. Assume hP, h. The goal follows from periodicldiagxorlatewittlag(T, P, hP, h). □

Theorem 9.89. *For every natural number T and natural number P, if $0 <$ P, then if* centerperiodic(T, P), *then for every natural number t and natural number k, if $T \leq t$, then if $t + P <$* twopow(k), *then* response$(t + P, k) =$ response(t, k).

Proof. Assume hP, h. The goal follows from periodicresponseshiftp(T, P, hP, h). □

Theorem 9.90. *For every natural number T and natural number P, if $0 <$ P, then if* centerperiodic(T, P), *then for every natural number t and natural number k, if $T \leq t$, then if $t + P <$* twopow(k), *then* wittresp$(t + P, k) =$ wittresp(t, k).

Proof. Assume hP, h. The goal follows from periodicwittshiftp(T, P, hP, h). □

Theorem 9.91. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d + P <$* twopow(k)*, then* wittbit(wittwin$(d + P,$ twopow$(k)), d + P +$ twopow$(k)) =$ wittbit(wittwin$(d,$ twopow$(k)), d +$ twopow$(k))$.

Proof. Assume hP, h. The goal follows from bitperiodicwinwittsectoneshiftp(T, P, hP, h). □

Theorem 9.92. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d \leq k$, then* response$(d+$twopow$(k), k) =$ bxor(bxor(center$(d+$lag$(P,$ succ$(k))),$ center$(d+$lag$(P, k))),$ ldiag$(d, d+$twopow$(k)))$.

Proof. Assume hP, h. The goal follows from periodiclaterespunsettled(T, P, hP, h). □

Theorem 9.93. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d \leq k$, then if* settledat(d, k)*, then* zcentereq$(d,$ bxor(bxor(center$(d+$lag$(P,$ succ$(k))),$ center$(d+$lag$(P, k))),$ response$(d+$ twopow$(k), k)))$.

Proof. Assume hP, h. The goal follows from periodiczcenterfromlaglateresp(T, P, hP, h). □

Theorem 9.94. *For every natural number T and natural number P, if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d+P \leq k$, then if* settledat(d, k)*, then if* settledat$(d+P, k)$*, then* zcenterxoreq$(d+P, d,$ bxor(response$(d+P+$twopow$(k), k),$ response$(d+$ twopow$(k), k)))$.

Proof. Assume h. The goal follows from periodiczcentershiftxorlateresp(T, P, h). □

Theorem 9.95. *For every natural number T and natural number P, if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d+P \leq k$, then if* settledat(d, k)*, then if* settledat$(d+P, k)$*, then* zcenterxoreq$(d+P, d,$ bxor(wittresp$(d+P+$twopow$(k), k),$ wittresp$(d+$ twopow$(k), k)))$.

Proof. Assume h. The goal follows from periodiczcentershiftxorlatewitt(T, P, h). □

Theorem 9.96. *For every natural number $d1$ and natural number $d2$, there exists a natural number k such that $d1 \leq k$ and $d2 \leq k$ and* settledat$(d1, k)$ *and* settledat$(d2, k)$.

Proof. The goal follows from existssettledscalepair$(d1, d2)$. □

Theorem 9.97. *For every natural number T, not* zcenterperiodic$(T, 1)$.

Proof. The goal follows from zcenternotperiodicone(T). □

Theorem 9.98. *For every natural number T, not* zcenterperiodic$(T, 2)$.

Proof. The goal follows from zcenternotperiodictwo(T). □

Theorem 9.99. *For every natural number T and natural number Q, if* zcenterperiodic(T, Q)*, then for every natural number z and natural number m, if* zerodiagonal(z)*, then if $T \leq z$, then not* zerodiagonal$(z + 2 + m \cdot Q)$.

Proof. Assume h. The goal follows from zcenterperiodicnotzerodiagshift(T, Q, h). □

Theorem 9.100. *For every natural number $t0$ and natural number k,* wittwin$(t0+$ twopow(k), twopow$(k))$ = wittxor(wittwin$(t0,$ twopow(succ$(k))),$ wittscale$(k,$ wittwin$(t0,$ twopow$(k))))$.

Proof. The goal follows from winwittscalecocycle$(t0, k)$. □

Theorem 9.101. *For every natural number d and natural number k, if $d \leq k$, then* bitwinwitteq$(d,$ twopow$(k), d +$ twopow(succ$(k))$, bxor(wittresp$(d +$ twopow$(k), k)$, bxor(wittresp$(d,$ succ$(k))$, wittresp$(d, k))))$.

Proof. The goal follows from wittresplatefromearlytriality(d, k). □

Theorem 9.102. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d \leq k$, then* bitwinwitteq$(d,$ twopow$(k), d +$ twopow(succ$(k))$, bxor(wittresp$(d+$twopow$(k), k)$, bxor(center$(d+$lag$(P,$ succ$(k)))$, center$(d+$ lag$(P, k)))))$.

Proof. Assume hP, h. The goal follows from periodiclatewittfromlagsectortwo(T, P, hP, h). □

Theorem 9.103. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d \leq k$, then* maskparity$(d+$twopow$(k), k) =$ bxor(bxor(leftresponse$(d,$ succ$(k))$, leftresponse$(d, k))$, bxor(center$(d+$lag$(P,$ succ$(k)))$, center$(d+$ lag$(P, k))))$.

Proof. Assume hP, h. The goal follows from periodiclatesmaskparitydecomp(T, P, hP, h). □

Theorem 9.104. *For every natural number t,* leftresponse$(t, 0) =$ false.

Proof. The goal follows from leftresponsezero(t). □

Theorem 9.105. *For every natural number t,* response$(t, 0) =$ maskparity$(t, 0)$.

Proof. The goal follows from responsezeromaskparity(t). □

Theorem 9.106. *For every natural number t and natural number k,* bxor(maskparity($t+1, k$), maskparity(t, k)) = bxor(maskparity(t+twopow$(k), 0)$, maskparity$(t, 0)$).

Proof. The goal follows from maskparitystepdiff(t, k). □

Theorem 9.107. *For every natural number s,* maskparity$(s+1, 0) =$ not(rdiag($s+1, s+1$)) and rdiag$(s, s+1)$.

Proof. The goal follows from maskparitysucczerordiag(s). □

Theorem 9.108. *For every natural number T and natural number P, if* centerperiodic(T, P)*, then if* maskedperiodic(T, P)*, then* maskparityzeroperiodic(T, P).

Proof. Assume h, hm. The goal follows from maskparityzeroperiodicofmaskedper(T, P, h, hm). □

Theorem 9.109. *For every natural number T and natural number P, if* centerperiodic(T, P)*, then if* maskparityzeroperiodic(T, P)*, then* maskedperiodic(T, P).

Proof. Assume h, hmp. The goal follows from maskedperofmaskparityzeroperiodic(T, P, h, hmp). □

Theorem 9.110. *For every natural number T and natural number P, if* centerperiodic(T, P)*, then if* maskedperiodic(T, P)*, then for every natural number t and natural number k, if $T + 1 \leq t$, then* maskparity$(t + P, k) =$ maskparity(t, k).

Proof. Assume h, hm. The goal follows from maskparityperiodicofmaskedper(T, P, h, hm). □

Theorem 9.111. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number t and natural number k, if $T \leq t$, then if $t + P <$* twopow(k)*, then* bxor(maskparity($t + P, k$), maskparity(t, k)) = bxor(leftresponse$(t + P, k)$, leftresponse(t, k)).

Proof. Assume hP, h. The goal follows from periodicmaskparityshiftpcone(T, P, hP, h). □

Theorem 9.112. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number t and natural number k, if $T \leq t$, then if $t+1 <$* twopow(k)*, then* bxor(maskparity($t+$ twopow(k), 0), maskparity(t, 0)) = bxor(bxor(leftresponse($t+1$, k), leftresponse(t, k)), bxor(bxor(center($1 +$ lag(P, k)), center($t +$ lag(P, k))), bxor(center($t + 1$), center(t)))).

Proof. Assume hP, h. The goal follows from periodicmaskparityzerodyadicjumpcone(T, P, hP, h). □

Theorem 9.113. *For every natural number T and natural number P, if $0 <$ P, then if* centerperiodic(T, P)*, then for every natural number t and natural number k, if $T \leq t$, then if $t+P+1<$* twopow(k)*, then* bxor(bxor(maskparity($t+$ $P+$twopow(k), 0), maskparity($t+$twopow(k), 0)), bxor(maskparity($t+P$, 0), maskparity(t, 0))) = bxor(bxor(leftresponse($t+P+1$, k), leftresponse($t+1$, k)), bxor(leftresponse($t+$ P, k), leftresponse(t, k))).

Proof. Assume hP, h. The goal follows from periodicmaskparityzerodefectdyadicjumpcone(T, P, hP, h). □

Theorem 9.114. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d+P \leq k$, then* bxor(maskparity($d+P+$ twopow(k), k), maskparity($d +$ twopow(k), k)) = bxor(bxor(leftresponse($d +$ P, succ(k)), leftresponse($d+P$, k)), bxor(leftresponse(d, succ(k)), leftresponse(d, k))).

Proof. Assume hP, h. The goal follows from periodiclatesmaskparityshiftp(T, P, hP, h). □

Theorem 9.115. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d + 1 \leq k$, then* bxor(maskparity($d +$ twopow(succ(k)), 0), maskparity($d+$twopow(k), 0)) = bxor(bxor(bxor(leftresponse($d+$ 1, succ(k)), leftresponse($d+1$, k)), bxor(leftresponse(d, succ(k)), leftresponse(d, k))), bxor(bxor(center($d+$ $1+$lag(P, succ(k))), center($d+1+$lag(P, k))), bxor(center($d+$lag(P, succ(k))), center($d+$ lag(P, k))))).

Proof. Assume hP, h. The goal follows from periodiclatesmaskparityzerodyadicjump(T, P, hP, h). □

Theorem 9.116. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d+P+1 \leq k$, then* bxor(bxor(maskparity($d+P+$ twopow(succ(k)), 0), maskparity($d+$twopow(succ(k)), 0)), bxor(maskparity($d+$ $P+$twopow(k), 0), maskparity($d+$twopow(k), 0))) = bxor(bxor(bxor(leftresponse($d+$ $P+1$, succ(k)), leftresponse($d+1$, succ(k))), bxor(leftresponse($d+P$, succ(k)), leftresponse(d, succ(k)))), b $P+1, k$), leftresponse($d+1, k$)), bxor(leftresponse($d+P, k$), leftresponse(d, k)))).

Proof. Assume hP, h. The goal follows from periodiclatesmaskparityzerodefectdyadicjump(T, P, hP, h). □

Theorem 9.117. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d \leq k$, then* leftresponse$(d +$ twopow$(k), k) =$ bxor(bxor(center($d+$lag$(P,$ succ(k))), center($d+$lag(P, k))), bxor(ldiag$(d, d+$ twopow(k)), maskparity$(d +$ twopow$(k), k)$)).

Proof. Assume hP, h. The goal follows from periodiclateleftresponseunsettleddecomp(T, P, hP, h). □

Theorem 9.118. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$, then if $d \leq k$, then if* settledat(d, k)*, then* zcentereq$(d,$ bxor(bxor(center($d+$lag$(P,$ succ(k))), center($d+$lag(P, k))), bxor(leftresponse($d+$ twopow$(k), k)$, maskparity$(d +$ twopow$(k), k)$))).

Proof. Assume hP, h. The goal follows from periodiczcenterfromlaglateleftmask(T, P, hP, h). □

Theorem 9.119. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number t, if $T \leq t$, then* bxor(maskparity$(t + P, 0)$, maskparity$(t, 0)$) = bxor(cell$(t +$ $P,$ negpow$(0))$, cell$(t,$ negpow$(0))$).

Proof. Assume hP, h. The goal follows from periodicmaskparityzerodefectcolm1defect(T, P, hP, h). □

Theorem 9.120. *For every natural number r and natural number k,* rdiag$(r, r+$ twopow$(k))$ = bxor(wittresp($r+$twopow(k), k), bxor(leftresponse($r+$twopow(k), k), maskparity($r+$ twopow(k), k))).

Proof. The goal follows from rdiaglateeqwittxorleftmask(r, k). □

Theorem 9.121. *For every natural number r and natural number k,* $\mathrm{bxor}(\mathrm{rdiag}(r, r+\mathrm{twopow}(k)), \mathrm{ldiag}(r, r+\mathrm{twopow}(k))) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{center}(r+\mathrm{twopow}(k)+\mathrm{twopow}(k)), \mathrm{center}(r+\mathrm{twopow}(k))), \mathrm{wittresp}(r+\mathrm{twopow}(k), k))$.

Proof. The goal follows from $\mathrm{twohalveswittscale}(r, k)$. □

Theorem 9.122. *For every natural number r and natural number k, if* $r < \mathrm{twopow}(k)$, *then* $\mathrm{ldiag}(r, r+\mathrm{twopow}(k)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{leftresponse}(r, \mathrm{succ}(k)), \mathrm{leftresponse}(r, k)), \mathrm{leftresp}$ $\mathrm{twopow}(k), k))$.

Proof. Assume hr. The goal follows from $\mathrm{ldiagleftresponsecoboundarycone}(r, k, hr)$. □

Theorem 9.123. *For every natural number r and natural number k, if* $r < \mathrm{twopow}(k)$, *then* $\mathrm{bxor}(\mathrm{rdiag}(r, r+\mathrm{twopow}(k)), \mathrm{ldiag}(r, r+\mathrm{twopow}(k))) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{wittresp}(r, \mathrm{succ}(k)), \mathrm{wittresp}(r, k)), \mathrm{wittresp}(r+\mathrm{twopow}(k), k))$.

Proof. Assume hr. The goal follows from $\mathrm{twohalveswittcoboundarycone}(r, k, hr)$. □

Theorem 9.124. *For every natural number T and natural number P, if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number r and natural number k, if* $T \leq r$, *then if* $r < \mathrm{twopow}(k)$, *then* $\mathrm{maskparity}(r + \mathrm{twopow}(k), k) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{leftresponse}(r, \mathrm{succ}(k)), \mathrm{leftresponse}(r, k)), \mathrm{bxor}(\mathrm{center}(r+\mathrm{lag}(P, \mathrm{succ}(k))), \mathrm{center}(r+\mathrm{lag}(P, k))))$.

Proof. Assume hP, h. The goal follows from $\mathrm{periodicmaskparitylatecone}(T, P, hP, h)$. □

Theorem 9.125. *For every natural number T and natural number P, if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number r and natural number k, if* $T \leq r$, *then if* $r < \mathrm{twopow}(k)$, *then* $\mathrm{bxor}(\mathrm{rdiag}(r, r+\mathrm{twopow}(k)), \mathrm{ldiag}(r, r+\mathrm{twopow}(k))) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{center}(r+\mathrm{lag}(P, \mathrm{succ}(k))), \mathrm{center}(r+\mathrm{lag}(P, k))), \mathrm{wittresp}(r+\mathrm{twopow}(k), k))$.

Proof. Assume hP, h. The goal follows from $\mathrm{periodictwohalvesscaleeq}(T, P, hP, h)$. □

Theorem 9.126. *For every natural number T and natural number P, if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number r and natural number k, if* $T \leq r$, *then if* $r < \mathrm{twopow}(k)$, *then* $\mathrm{bxor}(\mathrm{rdiag}(r, r+\mathrm{twopow}(k)), \mathrm{wittresp}(r+\mathrm{twopow}(k), k)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{bxor}(\mathrm{leftresponse}(r, \mathrm{succ}(k)), \mathrm{leftresponse}(r, k)), \mathrm{le}$ $\mathrm{twopow}(k), k)), \mathrm{bxor}(\mathrm{center}(r+\mathrm{lag}(P, \mathrm{succ}(k))), \mathrm{center}(r+\mathrm{lag}(P, k))))$.

Proof. Assume hP, h. The goal follows from periodicrdiaglatefromleftandcenter(T, P, hP, h). □

Theorem 9.127. *For every natural number r and natural number k and natural number J, if* rightthreshold(k, J)*, then if $r \leq J$, then if $r <$* twopow(k)*, then* rdiag$(r, r +$ twopow$(k)) =$ center(r).

Proof. The goal follows from rdiagbelowthresholdeqcenter(r, k, J). □

Theorem 9.128. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number r and natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq r$, then if $r \leq J$, then if $r <$* twopow(k)*, then* bxor(center(r), ldiag$(r, r +$ twopow$(k))) =$ bxor(bxor(center$(r+$lag$(P,$ succ$(k)))$, center$(r+$lag$(P, k)))$, wittresp$(r+$ twopow$(k), k))$.

Proof. Assume hP, h. The goal follows from periodictwohalvesbelowthreshold(T, P, hP, h). □

Theorem 9.129. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number r and natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq r$, then if $r \leq J$, then if $r <$* twopow(k)*, then* bxor(center(r), wittresp$(r +$ twopow$(k), k)) =$ bxor(bxor(bxor(leftresponse$(r,$ succ$(k))$, leftresponse$(r, k))$, leftresponse$(r+$ twopow$(k), k))$, bxor(center$(r +$ lag$(P,$ succ$(k)))$, center$(r +$ lag$(P, k))))$.

Proof. Assume hP, h. The goal follows from periodiccenterxorwittbelowthreshold(T, P, hP, h). □

Theorem 9.130. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq J + 1$, then if $J + 1 <$* twopow(k)*, then* bxor(not(center$(J+1))$, ldiag$(J+1, J+1+$twopow$(k))) =$ bxor(bxor(center$(J+1+$lag$(P,$ succ$(k)))$, center$(J+1+$lag$(P, k)))$, wittresp$(J+$ $1 +$ twopow$(k), k))$.

Proof. Assume hP, h. The goal follows from periodictwohalvesatthreshold(T, P, hP, h). □

Theorem 9.131. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq J + 1$, then if $J + 1 <$* twopow(k)*, then* bxor(not(center$(J + 1))$, wittresp$(J + 1 +$ twopow$(k), k)) =$

bxor(bxor(bxor(leftresponse(J+1, succ(k)), leftresponse(J+1, k)), leftresponse(J+1+twopow(k), k)), bxor(center(J+1+lag(P, succ(k))), center(J+1+lag(P, k)))).

Proof. Assume hP, h. The goal follows from periodicnotcenterxorwittatthreshold(T, P, hP, h). □

Theorem 9.132. *For every natural number k and natural number J, if* rightthreshold(k, J), *then if* $J+1 <$ twopow(k), *then* bxor(leftresponse($J+1$, k), maskparity($J+1$, k)) = not(bxor(center($J+1+$twopow(k)), rdiag($J+1$, $J+1+$twopow(k)))).

Proof. The goal follows from threshcenterleftmaskeqnotrdiag(k, J). □

Theorem 9.133. *For every natural number d and natural number s and natural number k and natural number J, if* rightthreshold(k, J), *then if* $d \leq k$, *then if* $d+2 \cdot s \leq J$, *then if* $d+2 \cdot s <$ twopow(k), *then if* settledby(d, $d+s$), *then* bxor(rdiag($d+2 \cdot s+$twopow(k), $d+s+$twopow(k)), rdiag($d+2 \cdot s$, $d+s+$twopow(k))) = bxor(ldiag($d+$twopow(k), $d+s+$twopow(k)), ldiag(d, $d+s+$twopow(k))).

Proof. The goal follows from sectoronebelowthresholdeqleftsettled(d, s, k, J). □

Theorem 9.134. *For every natural number d and natural number s and natural number k and natural number J, if* rightthreshold(k, J), *then if* $d \leq k$, *then if* $d+2 \cdot s = J+1$, *then if* $J+1 <$ twopow(k), *then if* settledby(d, $d+s$), *then* not(bxor(rdiag($J+1+$twopow(k), $d+s+$twopow(k)), rdiag($J+1$, $d+s+$twopow(k)))) = bxor(ldiag($d+$twopow(k), $d+s+$twopow(k)), ldiag(d, $d+s+$twopow(k))).

Proof. The goal follows from sectoroneatthresholdeqleftsettled(d, s, k, J). □

Theorem 9.135. *For every natural number t and natural number k and natural number J and natural number j, if* rightthreshold(k, J), *then if* $j \leq J$, *then* bxor(bxor(rdiag($j+2$, $t+1+$twopow(k)), rdiag($j+2$, $t+1$)), bxor(rdiag($j+2$, $t+$twopow(k)), rdiag($j+2$, t))) = bxor(rdiag($j+1$, $t+$twopow(k)), rdiag($j+1$, t)) and not(rdiag(j, t)).

Proof. The goal follows from rdiagdefectstepbelowthreshold(t, k, J, j). □

Theorem 9.136. *For every natural number t and natural number k and natural number J, if* rightthreshold(k, J), *then* bxor(bxor(rdiag($J+2$, $t+1+$twopow(k)), rdiag($J+2$, $t+1$)), bxor(rdiag($J+2$, $t+$twopow(k)), rdiag($J+2$, t))) = not(rdiag(J, t)).

Proof. The goal follows from rdiagthresholdplustwodefectstep(t, k, J). □

Theorem 9.137. *For every natural number d and natural number s and natural number k and natural number J, if* rightthreshold(k, J)*, then if* $d + 2 \cdot s = J$*, then* bxor(bxor(rdiag$(J + 2, d + s + 1 +$ twopow$(k))$, rdiag$(J + 2, d + s + 1))$, bxor(rdiag$(J + 2, d + s +$ twopow$(k))$, rdiag$(J + 2, d + s))) =$ not(ldiag$(d, d + s)$).

Proof. The goal follows from rdiagthresholdplustwodefectstepldiag(d, s, k, J). □

Theorem 9.138. *For every natural number k and natural number J, if* rightthreshold(k, J)*, then* rdiag$(J+2, J+2+$twopow$(k)) =$ bxor(bxor(center$(J+2)$, bxor(rdiag$(J+2, J+$twopow$(k))$, rdiag$(J+2, J))$), bxor(center(J), rdiag$(J, J+1))$).

Proof. The goal follows from rdiagthresholdplustwocenterdecomp(k, J). □

Theorem 9.139. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq J + 2$, then if $J + 2 <$* twopow(k)*, then* bxor(bxor(bxor(center$(J + 2)$, bxor(rdiag$(J + 2, J +$ twopow$(k))$, rdiag$(J+2, J))$)$)$, bxor(center(J), rdiag$(J, J+1))$)$)$, ldiag$(J+2, J+2+$twopow$(k))) =$ bxor(bxor(center$(J + 2 +$ lag$(P,$ succ$(k)))$, center$(J + 2 +$ lag$(P, k)))$, wittresp$(J + 2 +$ twopow$(k), k)$).

Proof. Assume hP, h. The goal follows from periodictwohalvesatthresholdplustwo(T, P, hP, h). □

Theorem 9.140. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq J + 2$, then if $J + 2 <$* twopow(k)*, then* bxor(bxor(bxor(center$(J + 2)$, bxor(rdiag$(J + 2, J +$ twopow$(k))$, rdiag$(J + 2, J))$)$)$, bxor(center(J), rdiag$(J, J + 1))$)$)$, wittresp$(J +$ 2+twopow$(k), k)) =$ bxor(bxor(bxor(leftresponse$(J+2,$ succ$(k))$, leftresponse$(J+2, k))$, leftresponse$(J+2+$twopow$(k), k))$, bxor(center$(J+2+$lag$(P,$ succ$(k)))$, center$(J+2 +$ lag$(P, k))))$.

Proof. Assume hP, h. The goal follows from periodiccenterxorwittatthresholdplustwo(T, P, hP, h). □

Theorem 9.141. *For every natural number d and natural number s and natural number k, if $d \leq k$, then if* settledby(d, s)*, then* bxor(ldiag$(d+2, s+1+$ twopow$(k))$, ldiag$(d+2, s+1)) =$ bxor(ldiag$(d+1, s+$twopow$(k))$ or ldiag$(d+2, s+$ twopow$(k))$, ldiag$(d+1, s)$ or ldiag$(d+2, s))$.

Proof. The goal follows from ldiagdamagestepofsettled(d, s, k). □

Theorem 9.142. *For every natural number d and natural number s and natural number k, if $d+1 \leq k$, then if* settledby(d, s)*, then if* settledby$(d+1, s)$*, then* bxor(ldiag$(d+2, s+1+$twopow$(k))$, ldiag$(d+2, s+1)) =$ bxor(ldiag$(d+2, s+$ twopow$(k))$, ldiag$(d+2, s))$ and not(ldiag$(d+1, s))$.

Proof. The goal follows from ldiagdamagestepofsettledpair(d, s, k). □

Theorem 9.143. *For every natural number d and natural number s and natural number k, if $d + 2 \cdot s <$* twopow(k)*, then* bitwinwitteq$(d + s +$ twopow(k), twopow(k), $d + 2 \cdot s +$ twopow(succ$(k))$, bxor(bxor(rdiag$(d + 2 \cdot s, d+s+$twopow$(k))$, ldiag$(d, d+s+$twopow$(k)))$, bxor(ldiag$(d+$twopow(succ$(k))$, $d+ s +$ twopow(succ$(k)))$, ldiag$(d +$ twopow$(k), d + s +$ twopow$(k)))))$.

Proof. The goal follows from twohalvesoffcenterwittscale(d, s, k). □

Theorem 9.144. *For every natural number d and natural number s and natural number k and natural number J, if* rightthreshold(k, J)*, then if $d \leq k$, then if $d + 2 \cdot s \leq J$, then if $d + 2 \cdot s <$* twopow(k)*, then if* settledby$(d, d+s)$*, then* bitwinwitteq$(d+s+$twopow(k), twopow(k), $d+2 \cdot s+$ twopow(succ$(k))$, bxor(ldiag$(d+$twopow(succ$(k))$, $d+s+$twopow(succ$(k)))$, ldiag$(d+$ twopow$(k), d + s +$ twopow$(k))))$.

Proof. The goal follows from offcenterwittscalesettledbelowthreshold(d, s, k, J). □

Theorem 9.145. *For every natural number d and natural number s and natural number k and natural number J, if* rightthreshold(k, J)*, then if $d \leq k$, then if $d + 2 \cdot s = J + 1$, then if $J + 1 <$* twopow(k)*, then if* settledby$(d, d+s)$*, then* bitwinwitteq$(d+s+$twopow(k), twopow(k), $J+1+$ twopow(succ$(k))$, not(bxor(ldiag$(d+$twopow(succ$(k))$, $d+s+$twopow(succ$(k)))$, ldiag$(d+$ twopow$(k), d + s +$ twopow$(k)))))$.

Proof. The goal follows from offcenterwittscalesettledatthreshold(d, s, k, J). □

Theorem 9.146. *For every natural number d and natural number k,* bxor(ldiag(d+ twopow(succ(k)), d + 1 + twopow(succ(k))), ldiag(d + twopow(k), d + 1 + twopow(k))) = bxor(bxor(maskparity(d+1+twopow(succ(k)), 0), maskparity(d+ 1+twopow(k), 0)), bxor(bxor(center(d+1+twopow(succ(k))+1), center(d+ 1 + twopow(succ(k)))), bxor(center(d + 1 + twopow(k) + 1), center(d + 1 + twopow(k))))).

Proof. The goal follows from colm1dyadicjumpeqmaskparityzerojump(d, k). □

Theorem 9.147. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d + 1$, then if $d + 2 \leq k$, then* bxor(ldiag(d + twopow(succ(k)), d + 1 + twopow(succ(k))), ldiag(d + twopow(k), d + 1 + twopow(k))) = bxor(bxor(leftresponse(d+2, succ(k)), leftresponse(d+2, k)), bxor(leftresponse(d+ 1, succ(k)), leftresponse(d + 1, k))).

Proof. Assume hP, h. The goal follows from periodiccolm1dyadicjumpeqleftresponsediff(T, P, hP, h). □

Theorem 9.148. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq d+1$, then if $d+2 \leq k$, then if $d+2 \leq J$, then if* settledby(d, $d+1$)*, then* bitwinwitteq(d+1+twopow(k), twopow(k), d+2+twopow(succ(k)), bxor(bxor(leftresponse(d+ 2, succ(k)), leftresponse(d+2, k)), bxor(leftresponse(d+1, succ(k)), leftresponse(d+ 1, k)))).

Proof. Assume hP, h. The goal follows from periodicoffcenterwittbelowthresholdeqleftresponse(T, P, hP □

Theorem 9.149. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k and natural number J, if* rightthreshold(k, J)*, then if $T \leq d + 1$, then if $d + 2 \leq k$, then if $d + 2 = J + 1$, then if $J + 1 <$* twopow(k)*, then if* settledby(d, $d+1$)*, then* bitwinwitteq($d+1+$twopow(k), twopow(k), $J+1+$ twopow(succ(k)), not(bxor(bxor(leftresponse(d+2, succ(k)), leftresponse(d+ 2, k)), bxor(leftresponse(d + 1, succ(k)), leftresponse(d + 1, k))))).

Proof. Assume hP, h. The goal follows from periodicoffcenterwittatthresholdeqnotleftresponse(T, P, hP □

Theorem 9.150. *For every natural number $t0$ and natural number k and natural number J, if $2 \cdot t0 \leq J+1$, then* $\mathrm{bxor}(\mathrm{rdiag}(J+2, t0+\mathrm{twopow}(k)), \mathrm{rdiag}(J+2, t0)) = \mathrm{rdiag}(J+2, t0+\mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{rdiagthresholdplustwodefectcone}(t0, k, J)$. □

Theorem 9.151. *For every natural number t and natural number k and natural number J, if* $\mathrm{rightthreshold}(k, J)$, *then* $\mathrm{bxor}(\mathrm{bxor}(\mathrm{rdiag}(J+2, t+2+\mathrm{twopow}(k)), \mathrm{rdiag}(J+2, t+2)), \mathrm{bxor}(\mathrm{rdiag}(J+2, t+\mathrm{twopow}(k)), \mathrm{rdiag}(J+2, t))) = \mathrm{bxor}(\mathrm{rdiag}(J, t), \mathrm{rdiag}(J, t+1))$.

Proof. The goal follows from $\mathrm{rdiagthresholdplustwodefecttwostep}(t, k, J)$. □

Theorem 9.152. *For every natural number $t0$ and natural number k,* $\mathrm{wittcarry}(\mathrm{wittrow}(t0+\mathrm{twopow}(k))) = \mathrm{wittxor}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{wittscale}(k, \mathrm{wittrow}(t0))), \mathrm{wittcarry}(\mathrm{wittwin}(t0, \mathrm{twopow}(k)))$

Proof. The goal follows from $\mathrm{carryrowscaledecomp}(t0, k)$. □

Theorem 9.153. *For every natural number $t0$ and natural number k, if $2 \cdot t0+1 < \mathrm{twopow}(k)$, then* $\mathrm{wittcarry}(\mathrm{wittrow}(t0+\mathrm{twopow}(k))) = \mathrm{wittxor}(\mathrm{wittxor}(\mathrm{wittscale}(k, \mathrm{wittcarry}(v$

Proof. The goal follows from $\mathrm{carryrowscalecone}(t0, k)$. □

Theorem 9.154. *For every natural number $t0$ and natural number k,* $\mathrm{wittcarry}(\mathrm{wittwin}(t0+\mathrm{twopow}(k), \mathrm{twopow}(k))) = \mathrm{wittxor}(\mathrm{wittxor}(\mathrm{wittcarry}(\mathrm{wittwin}(t0, \mathrm{twopow}(\mathrm{succ}(k))))), \mathrm{wittcarry}(\mathrm{wittsca$

Proof. The goal follows from $\mathrm{carrywinwittlatedecomp}(t0, k)$. □

Theorem 9.155. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number $t0$ and natural number k, if $T \leq t0$, then if $t0+1 < \mathrm{twopow}(k)$, then* $\mathrm{bitcrosscarryeq}(\mathrm{wittscale}(k, \mathrm{wittrow}(t0)), \mathrm{wittwin}(t$ $1+\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{rdiag}(t0+1, t0)$ and $\mathrm{bxor}(\mathrm{center}(t0+\mathrm{lag}(P, k)), \mathrm{center}(t0)), \mathrm{bxor}(\mathrm{rdiag}(t0+1+\mathrm{twopow}(k), t0+\mathrm{twopow}(k)), \mathrm{rdiag}(t0+1, t0))$ and $\mathrm{center}(t0)))$.

Proof. Assume hP, h. The goal follows from $\mathrm{periodiccrosscarrycenterslotand}(T, P, hP, h)$. □

Theorem 9.156. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number d and natural number k, if $T \leq d+1$, then if $d+2 < \mathrm{twopow}(k)$, then* $\mathrm{bitcrosscarryeq}(\mathrm{wittscale}(k, \mathrm{wittrow}(d+1)), \mathrm{wittwin}(d+1, \mathrm{twopow}(k)), d+2+\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{ldiag}(d, d+1)$ and $\mathrm{bxor}(\mathrm{center}(d+1+\mathrm{lag}(P, k)), \mathrm{center}(d+1)), \mathrm{bxor}(\mathrm{ldiag}(d+\mathrm{twopow}(k), d+1+\mathrm{twopow}(k)), \mathrm{ldiag}(d, d+1))$ and $\mathrm{center}(d+1)))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1and(T, P, hP, h). □

Theorem 9.157. *For every natural number* $t0$ *and natural number* k *and natural number* J*, if* rightthreshold(k, J)*, then if* $J + 1 <$ twopow(k)*, then* bitcrosscarryeq(wittscale$(k,$ wittrow$(t0))$, wittwin$(t0,$ twopow$(k))$, J+1, rdiag$(J, t0)$).

Proof. The goal follows from crosscarryrowscaleatthresholdeqrdiag$(t0, k, J)$. □

Theorem 9.158. *For every natural number* $t0$ *and natural number* k *and natural number* J*, if* rightthreshold(k, J)*, then if* $J + 1 <$ twopow(k)*, then* bitcarryeq(wittrow($t0$+twopow$(k))$, J+1, bxor(rdiag(J+1, $t0$) and rdiag(J, $t0$), rdiag(J, $t0$))).

Proof. The goal follows from carryrowscaleatthreshold$(t0, k, J)$. □

Theorem 9.159. *For every natural number* T *and natural number* P*, if* $0 <$ P*, then if* centerperiodic(T, P)*, then for every natural number* $t0$ *and natural number* k*, if* $T \leq t0$*, then if* $2 \cdot t0+2 <$ twopow(k)*, then* bitwinwitteq($t0$+1, twopow(k), $t0$+2+twopow(k), bxor(bxor(bxor(rdiag($t0$+2+twopow(k), $t0$+twopow$(k))$, rdiag($t0$+2, $t0$)), bxor(bxor(rdiag($t0$+1+twopow(k), $t0$+twopow$(k))$, rdiag($t0$+1, $t0$)), bxor(center($t0$+lag(P, k)), center($t0$)))), bxor(bxor(rdiag($t0$+1+twopow(k), $t0$+twopow$(k))$, rdiag($t0$+1, $t0$)) and center($t0$+lag(P, k)), rdiag($t0$+1, $t0$) and bxor(center($t0$+lag(P, k)), center($t0$))))).

Proof. Assume hP, h. The goal follows from periodicwinwittstepcenterslotexpanded(T, P, hP, h). □

Theorem 9.160. *For every natural number* T *and natural number* P*, if* $0 <$ P*, then if* centerperiodic(T, P)*, then for every natural number* s *and natural number* k*, if* $T \leq s$+1*, then if* s+2 $<$ twopow(k)*, then* bitcrosscarryeq(wittscale$(k,$ wittrow(s+1)), wittwin(s+1, twopow(k)), s+1+twopow(k), bxor(center(s+1) and bxor(rdiag(s+twopow(k), s+1+twopow$(k))$, rdiag(s, s+1)), bxor(center(s+1+lag(P, k)), center(s+1)) and rdiag$(s, s + 1)$)).

Proof. Assume hP, h. The goal follows from periodiccrosscarryatcenterbitand(T, P, hP, h). □

Theorem 9.161. *For every natural number* T *and natural number* P*, if* $0 < P$*, then if* centerperiodic(T, P)*, then for every natural number* s *and natural number* k*, if* $T \leq s+1$*, then if* $s+2 <$ twopow(k)*, then if* center(s+1) = false*, then* bitcrosscarryeq(wittscale$(k,$ wittrow$(s + 1))$, wittwin(s + 1, twopow(k)), s+1+twopow(k), center(s+1+lag(P, k)) and maskparity(s+1, 0)).

Proof. Assume hP, h. The goal follows from periodiccrosscarryatwhitecentereqmaskparity(T, P, hP, h). □

Theorem 9.162. *For every natural number* s, bitcarryeq(wittrow$(s+1), s+1,$ bxor(rdiag$(s, s+1),$ maskparity$(s+1, 0))$).

Proof. The goal follows from carryrowxormaskparityzeroeqcol1(s). □

Theorem 9.163. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number* d *and natural number* k, *if* $T \leq d+1$, *then if* $d+2 <$ twopow(k), *then if* center$(d+1) =$ false, *then* bitcrosscarryeq(wittscale$(k,$ wittrow$(d+1)),$ wittwin$(d+1,$ twopow$(k)), d+2+$twopow$(k),$ ldiag$(d, d+1)$ and center$(d+1+$lag$(P, k)))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1atwhitecenter(T, P, hP, h). □

Theorem 9.164. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number* d *and natural number* $k1$ *and natural number* $k2$, *if* lag$(P, k1) =$ lag$(P, k2)$, *then if* $T \leq d+1$, *then if* $d+2 <$ twopow$(k1)$, *then if* $d+2 <$ twopow$(k2)$, *then if* center$(d+1) =$ false, *then* wittbit(crosscarry(wittscale$(k1,$ wittrow$(d+1)),$ wittwin$(d+1,$ twopow$(k1))), d+2+$twopow$(k1)) =$ wittbit(crosscarry(wittscale$(k2,$ wittrow$(d+1)),$ wittwin$(d+1,$ twopow$(k2))), d+2+$twopow$(k2))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1whitecenterlageq(T, P, hP, h). □

Theorem 9.165. *For every natural number* d *and natural number* k, *if* $d+1 \leq k$, *then* bitcrosscarryeq(wittscale$(k,$ wittrow$(d+1)),$ wittwin$(d+1,$ twopow$(k)), d+2+$twopow(succ$(k)),$ bxor(ldiag$(d, d+1)$ and bxor(center$(d+1),$ ldiag$(d+1, d+1+$ twopow$(k))),$ bxor(ldiag$(d, d+1),$ ldiag$(d, d+1+$ twopow$(k)))$ and center$(d+1)))$.

Proof. The goal follows from crosscarryrowscalesectortwocolm1and(d, k). □

Theorem 9.166. *For every natural number* d *and natural number* k, *if* $d+1 \leq k$, *then if* center$(d+1) =$ false, *then* bitcrosscarryeq(wittscale$(k,$ wittrow$(d+1)),$ wittwin$(d+1,$ twopow$(k)), d+2+$twopow(succ$(k)),$ ldiag$(d, d+1)$ and ldiag$(d+1, d+1+$ twopow$(k)))$.

Proof. The goal follows from crosscarryrowscalesectortwocolm1atwhitecenter(d, k). □

Theorem 9.167. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d + 1$, then if $d + 1 \leq k$, then if $d + 2 <$* twopow(k)*, then* bitcrosscarryxoreq(wittscale$(k,$ wittrow$(d{+}1))$, wittwin$(d{+}1,$ twopow$(k))$, $d + 2 +$ twopow(k), $d + 2 +$ twopow(succ$(k))$, bxor(ldiag$(d, d{+}1)$ and bxor(center$(d{+}1{+}$lag$(P, k))$, ldiag$(d{+}1, d{+}1{+}$twopow$(k)))$, bxor(ldiag$(d{+}$twopow$(k), d + 1 +$ twopow$(k))$, ldiag$(d, d + 1 +$ twopow$(k)))$ and center$(d + 1)))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrysectoronexortwocolm1and(T, P, hP, h). □

Theorem 9.168. *For every natural number d and natural number s and natural number k, if $d + 2 \cdot s \leq k$, then* wittbit(wittscale$(k,$ wittwin$(d + s,$ twopow$(k)))$, $d{+}2{\cdot}s{+}$twopow(succ$(k))) =$ bton(bxor(ldiag$(d{+}$twopow$(k), d{+}s +$ twopow$(k))$, ldiag$(d, d + s +$ twopow$(k))))$.

Proof. The goal follows from phiscalewinwitttrialitylow(d, s, k). □

Theorem 9.169. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d{+}1$, then if $d{+}2 \leq k$, then* bitcrosscarryeq(wittwin$(d{+}1,$ twopow(succ$(k)))$, wittscale$(k,$ wittwin$(d{+}1,$ twopow$(k)))$, $d{+}2{+}$twopow(succ$(k))$, bxor(bxor(ldiag$(d{+}$twopow(succ$(k))$, $d{+}1{+}$twopow(succ$(k)))$, ldiag$(d, d{+}1))$ and bxor(center$(d{+}1{+}$lag$(P, k))$, ldiag$(d{+}1, d{+}1{+}$twopow$(k)))$, bxor(ldiag$(d{+}$twopow$(k), d{+}1{+}$twopow$(k))$, ldiag$(d, d{+}1{+}$twopow$(k)))$ and bxor(center$(d{+}1{+}$lag$(P,$ succ$(k)))$, center$(d{+}1))))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrywinwittcocyclecolm1and(T, P, hP, h). □

Theorem 9.170. *For every natural number k and natural number J, if* rightthreshold(k, J)*, then if $J{+}1 <$* twopow(k)*, then* bitcarryeq(wittrow$(J{+}1 +$ twopow$(k)), J + 1,$ maskparity$(J + 1, 0))$.

Proof. The goal follows from carryrowlateatthresholdcentereqmaskparity(k, J). □

Theorem 9.171. *For every natural number s and natural number k, if $s + 2 <$* twopow(k)*, then* bitwinwitteq$(s{+}1,$ twopow$(k), s{+}$twopow$(k),$ bxor(bxor(maskparity$(s{+}1 +$ twopow$(k), 0),$ maskparity$(s + 1, 0))$, bxor(bxor(rdiag$(s +$ twopow$(k), s{+}1{+}$twopow$(k))$, rdiag$(s, s{+}1))$ and wittresp$(s{+}1, k)$, bxor(center$(s{+}1)$ and bxor(rdiag$(s{+}$twopow$(k), s{+}1{+}$twopow$(k))$, rdiag$(s, s{+}1))$, wittresp$(s{+}1, k)$ and rdiag$(s, s{+}1)))))$.

Proof. The goal follows from maskparityzerodyadicjumpeqwittcarrycrosscarry(s, k).
□

Theorem 9.172. *For every natural number s and natural number k, if* $s + 2 <$ twopow(k), *then* bitcrosscarryeq(wittscale$(k,$ wittrow$(s{+}1))$, wittwin$(s{+}1,$ twopow$(k))$, $s{+}1{+}$twopow(k), bxor(bxor(maskparity$(s{+}1{+}$twopow$(k), 0)$, maskparity$(s{+}1, 0))$, not(wittresp$(s{+}1, k)$) and bxor(rdiag$(s{+}$twopow$(k), s{+}1{+}$twopow$(k))$, rdiag$(s, s{+}1))))$.

Proof. The goal follows from maskparityzerodyadicjumpeqnotwittandxorcrosscarry(s, k).
□

Theorem 9.173. *For every natural number s and natural number k, if* $s + 2 <$ twopow(k), *then if* wittresp$(s{+}1, k) =$ true, *then* bitcrosscarryeq(wittscale$(k,$ wittrow$(s{+}1))$, wittwin$(s + 1,$ twopow$(k))$, $s + 1 +$ twopow(k), bxor(maskparity$(s + 1 +$ twopow$(k), 0)$, maskparity$(s + 1, 0)))$.

Proof. The goal follows from maskparitydyadicjumpeqcrosscarryofwittresptrue(s, k).
□

Theorem 9.174. *For every natural number s and natural number k, if* $s + 2 <$ twopow(k), *then if* wittresp$(s + 1, k) =$ false, *then* bxor(maskparity$(s{+}1{+}$twopow$(k), 0)$, maskparity$(s{+}1, 0)) =$ not(center$(s{+}1))$ and bxor(rdiag$(s{+}$twopow$(k), s + 1 +$ twopow$(k))$, rdiag$(s, s + 1))$.

Proof. The goal follows from maskparitydyadicjumpeqnotcenterandwittofwittrespfalse(s, k).
□

Theorem 9.175. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number s and natural number k, if* $T \leq s{+}1$, *then if* $s{+}2 <$ twopow(k), *then if* wittresp$(s{+}1, k) =$ true, *then* bitcrosscarryeq(wittscale$(k,$ wittrow$(s + 1))$, wittwin$(s + 1,$ twopow$(k))$, $s{+}1{+}$twopow(k), bxor(bxor(leftresponse$(s{+}2, k)$, leftresponse$(s{+}1, k))$, bxor(bxor(center$(s{+}2{+}$lag$(P, k))$, center$(s{+}1{+}$lag$(P, k)))$, bxor(center$(s{+}2)$, center$(s + 1)))))$.

Proof. Assume hP, h. The goal follows from periodicleftresponsestepdiffeqcrosscarryofoutofphase$(T, P,$
□

Theorem 9.176. *For every natural number $t0$ and natural number k and natural number J, if* rightthreshold(k, J), *then if* $J + 2 <$ twopow(k), *then* bitcrosscarryeq(wittwin$(t0,$ twopow(succ$(k)))$, wittscale$(k,$ wittwin$(t0,$ twopow$(k)))$, $J{+}2$, bxor(rdiag$(J + 2, t0 +$ twopow(succ$(k)))$, rdiag$(J + 2, t0)))$.

Proof. The goal follows from crosscarrywinwittcocycleatthresholdsucc($t0, k, J$). □

Theorem 9.177. *For every natural number* $t0$ *and natural number* k, crosscarry(wittscale(k, wittrow($t0$ crosscarry(wittscale(k, wittrow($t0$)), wittrow($t0$ + twopow(k))).

Proof. The goal follows from crosscarryrowscaleeqcrosscarrylaterow($t0, k$). □

Theorem 9.178. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number* d *and natural number* k, *if* $T \leq d{+}1$, *then if* $d{+}2 <$ twopow(k), *then* bitcrosscarryeq(wittscale(k, wittrow($d{+}1$)), wittwin($d{+}1$, twopow(k)), $d{+}2{+}$twopow(k), bxor(ldiag($d, d{+}1$) and center($d{+}1 +$ lag(P, k)), ldiag($d +$ twopow(k), $d + 1 +$ twopow(k)) and center($d + 1$))).

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1wronskian(T, P, hP, h). □

Theorem 9.179. *For every natural number* d *and natural number* k, *if* $d + 1 \leq k$, *then* bitcrosscarryeq(wittscale(k, wittrow($d + 1$)), wittwin($d + 1$, twopow(k)), $d{+}2{+}$twopow(succ(k)), bxor(ldiag($d, d{+}1$) and ldiag($d{+}1, d{+}1 +$ twopow(k)), ldiag($d, d + 1 +$ twopow(k)) and center($d + 1$))).

Proof. The goal follows from crosscarryrowscalesectortwocolm1wronskian(d, k). □

Theorem 9.180. *For every natural number* t, maskparity($t, 0$) and center(t) = false.

Proof. The goal follows from maskparityzeroandcentereqfalse(t). □

Theorem 9.181. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number* d, *if* $T \leq d{+}1$, *then* bxor(ldiag($d + P, d + P + 1$), ldiag($d, d + 1$)) and center($d + 1$) = false.

Proof. Assume hP, h. The goal follows from periodiccolm1defectandcentereqfalse(T, P, hP, h). □

Theorem 9.182. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number* d *and natural number* k, *if* $T \leq d + 1$, *then if* $d + P + 2 <$ twopow(k), *then* bitcrosscarryeq(wittscale(k, wittrow($d{+}P{+}1$)), wittwin($d{+}P{+}1$, twopow(k)), $d{+}P{+}2{+}$twopow(k), bxor(bxor(ldiag($d, d{+}1$) and center($d{+}1{+}$lag(P, k)), ldiag($d{+}$twopow(k), $d{+}1{+}$twopow(k)) and center($d{+}1$)), bxor(bxor(leftresponse($d{+}$

P+2, k), leftresponse(d+2, k)), bxor(leftresponse(d+P+1, k), leftresponse(d+1, k))) and wittresp($d+1, k$))).

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1shiftpeqleftresponseandwitt(T, P, h □

Theorem 9.183. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d + 1$, then if $d + P + 2 <$* twopow(k)*, then if* wittresp($d + 1, k$) = false*, then* wittbit(crosscarry(wittscale(k, wittrow(d + P+1)), wittwin(d+P+1, twopow(k))), d+P+2+twopow(k)) = wittbit(crosscarry(wittscale(k, wittrow(d 1)), wittwin($d + 1$, twopow(k))), $d + 2 +$ twopow(k)).

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1shiftpeqofwittrespfalse(T, P, hP, h) □

Theorem 9.184. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number s and natural number k, if $T \leq s+1$, then if $s+2 <$* twopow(k)*, then if* center(s+ 1) = true*, then if* wittresp($s + 1, k$) = false*, then* bxor(leftresponse(s + 2, k), leftresponse($s + 1, k$)) = wittresp($s + 2, k$).

Proof. Assume hP, h. The goal follows from periodicleftresponsestepdiffeqwittofinphaseblack($T, P, hP,$ □

Theorem 9.185. *For every natural number $t0$ and natural number k and natural number r, if* twopow(k) $\leq r$*, then* wittbit(wittscale(k, wittrow($t0$)), r+ twopow(k)) = bton(bxor(bxor(rdiag(r+twopow(k), $t0$), rdiag($r, t0$)), rdiag(r− twopow(k), $t0$))).

Proof. The goal follows from phiscalerowtestbitoutsidecone($t0, k, r$). □

Theorem 9.186. *For every natural number $t0$ and natural number k and natural number r, if* twopow(k) $\leq r$*, then* bitwinwitteq($t0$, twopow(k), r + twopow(k), bxor(rdiag(r + twopow(k), $t0$ + twopow(k)), bxor(bxor(rdiag(r + twopow(k), $t0$), rdiag($r, t0$)), rdiag(r − twopow(k), $t0$)))).

Proof. The goal follows from winwitttestbitoutsidecone($t0, k, r$). □

Theorem 9.187. *For every natural number T and natural number P, if $0 <$ P, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d$+1, then if* twopow(k) $\leq d$*, then* bitcrosscarryeq(wittscale(k, wittrow(d+ 1)), wittwin($d + 1$, twopow(k)), $d + 2 +$ twopow(k), bxor(bxor(bxor(ldiag(d −

twopow$(k), d{+}1), \mathrm{ldiag}(d, d{+}1)), \mathrm{rdiag}(d{+}2{-}\mathrm{twopow}(k), d{+}1))$ and bxor(center($d{+}1{+}\mathrm{lag}(P, k)), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(d{+}1{-}\mathrm{twopow}(k), d{+}1), \mathrm{center}(d{+}1)), \mathrm{rdiag}(d{+}1{-}\mathrm{twopow}(k), d{+}1))), \mathrm{bxor}(\mathrm{ldiag}(d{+}\mathrm{twopow}(k), d{+}1{+}\mathrm{twopow}(k)), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(d{-}\mathrm{twopow}(k), d{+}1), \mathrm{ldiag}(d, d{+}1)), \mathrm{rdiag}(d{+}2{-}\mathrm{twopow}(k), d{+}1)))$ and bxor(bxor(ldiag($d{+}1 - \mathrm{twopow}(k), d + 1), \mathrm{center}(d + 1)), \mathrm{rdiag}(d + 1 - \mathrm{twopow}(k), d + 1))))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1outsideconeand(T, P, hP, h). □

Theorem 9.188. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number d and natural number k, if* $T \leq d{+}1$, *then if* twopow$(k) \leq d$, *then* bitcrosscarryeq(wittscale$(k, \mathrm{wittrow}(d{+}1)), \mathrm{wittwin}(d + 1, \mathrm{twopow}(k)), d + 2 + \mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(d - \mathrm{twopow}(k), d{+}1), \mathrm{ldiag}(d, d{+}1)), \mathrm{rdiag}(d{+}2{-}\mathrm{twopow}(k), d{+}1))$ and center$(d{+}1{+}\mathrm{lag}(P, k)), \mathrm{ldiag}(d{+}\mathrm{twopow}(k), d{+}1{+}\mathrm{twopow}(k))$ and bxor(bxor(ldiag($d{+}1 - \mathrm{twopow}(k), d + 1), \mathrm{center}(d + 1)), \mathrm{rdiag}(d + 1 - \mathrm{twopow}(k), d + 1))))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1outsideconewronskian(T, P, hP, h). □

Theorem 9.189. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number d and natural number k, if* $T \leq d + 1$, *then if* twopow$(k) \leq d$, *then if* center$(d + 1) =$ false, *then* bitcrosscarryeq(wittscale$(k, \mathrm{wittrow}(d + 1)), \mathrm{wittwin}(d + 1, \mathrm{twopow}(k)), d + 2 + \mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(d - \mathrm{twopow}(k), d + 1), \mathrm{ldiag}(d, d{+}1)), \mathrm{rdiag}(d{+}2{-}\mathrm{twopow}(k), d{+}1))$ and center$(d{+}1{+}\mathrm{lag}(P, k)), \mathrm{ldiag}(d{+}\mathrm{twopow}(k), d{+}1{+}\mathrm{twopow}(k))$ and bxor(ldiag($d{+}1{-}\mathrm{twopow}(k), d{+}1), \mathrm{rdiag}(d + 1 - \mathrm{twopow}(k), d + 1))))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1outsideconeatwhitecenter$(T, P, hP,$ □

Theorem 9.190. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number d and natural number k, if* $T \leq d + 1$, *then if* twopow$(k) \leq d$, *then if* center$(d + 1) =$ false, *then if* center$(d{+}1{+}\mathrm{lag}(P, k)) =$ false, *then* bitcrosscarryeq(wittscale$(k, \mathrm{wittrow}(d{+}1)), \mathrm{wittwin}(d{+}1, \mathrm{twopow}(k)), d{+}2{+}\mathrm{twopow}(k), \mathrm{ldiag}(d{+}\mathrm{twopow}(k), d{+}1{+}\mathrm{twopow}(k))$ and bxor(ldiag($d{+}1{-}\mathrm{twopow}(k), d{+}1), \mathrm{rdiag}(d{+}1{-}\mathrm{twopow}(k), d{+}1)))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1outsideconeatbothwhitecenters$(T,$ □

Theorem 9.191. *For every natural number d and natural number k, if* $\mathrm{center}(d{+}1{+}\mathrm{twopow}(k)) = \mathrm{false}$*, then* $\mathrm{bitcrosscarryeq}(\mathrm{wittscale}(k, \mathrm{wittrow}(d{+}1)), \mathrm{wittwin}(d+1, \mathrm{twopow}(k)), d+2+\mathrm{twopow}(k), \mathrm{ldiag}(d+\mathrm{twopow}(k), d+1+\mathrm{twopow}(k))$ *and* $\mathrm{wittresp}(d+1, k))$.

Proof. The goal follows from $\mathrm{crosscarrycolm1atwhitelatecentereqandwitt}(d, k)$. □

Theorem 9.192. *For every natural number d and natural number k, if* $\mathrm{twopow}(k) \leq d{+}1$*, then if* $\mathrm{center}(d{+}1{+}\mathrm{twopow}(k)) = \mathrm{false}$*, then* $\mathrm{bitcrosscarryeq}(\mathrm{wittscale}(k, \mathrm{wittrow}(d{+}1)), \mathrm{wittwin}(d+1, \mathrm{twopow}(k)), d+2+\mathrm{twopow}(k), \mathrm{ldiag}(d+\mathrm{twopow}(k), d+1{+}\mathrm{twopow}(k))$ *and* $\mathrm{bxor}(\mathrm{response}(d{+}1, k), \mathrm{rdiag}(d{+}1{-}\mathrm{twopow}(k), d{+}1)))$.

Proof. The goal follows from $\mathrm{crosscarrycolm1outsideoneatwhitelatecenterviaresponse}(d, k)$. □

Theorem 9.193. *For every natural number s and natural number k, if* $\mathrm{twopow}(k) \leq s$*, then* $\mathrm{bitcrosscarryeq}(\mathrm{wittscale}(k, \mathrm{wittrow}(s{+}1)), \mathrm{wittwin}(s{+}1, \mathrm{twopow}(k)), s{+}1{+}\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1), \mathrm{center}(s{+}1)), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1))$ *and* $\mathrm{bxor}(\mathrm{rdiag}(s{+}\mathrm{twopow}(k), s{+}1{+}\mathrm{twopow}(k)), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s, s{+}1)), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}1))), \mathrm{wittresp}(s{+}1, k)$ *and* $\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s, s+1)), \mathrm{rdiag}(s-\mathrm{twopow}(k), s+1))))$.

Proof. The goal follows from $\mathrm{crosscarryrowscaleatcenterbitoutsideoneand}(s, k)$. □

Theorem 9.194. *For every natural number s and natural number k, if* $\mathrm{twopow}(k) \leq s$*, then if* $\mathrm{center}(s{+}1) = \mathrm{false}$*, then* $\mathrm{bitcrosscarryeq}(\mathrm{wittscale}(k, \mathrm{wittrow}(s{+}1)), \mathrm{wittwin}(s+1, \mathrm{twopow}(k)), s+1+\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s+1-\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1))$ *and* $\mathrm{bxor}(\mathrm{rdiag}(s{+}\mathrm{twopow}(k), s{+}1{+}\mathrm{twopow}(k)), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s, s{+}1)), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}1))), \mathrm{wittresp}(s{+}1, k)$ *and* $\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s, s+1)), \mathrm{rdiag}(s-\mathrm{twopow}(k), s+1))))$.

Proof. The goal follows from $\mathrm{crosscarryatwhitecenteroutsideconenoncenterand}(s, k)$. □

Theorem 9.195. *For every natural number s and natural number k, if* $\mathrm{twopow}(k) \leq s$*, then* $\mathrm{bitcrosscarryeq}(\mathrm{wittscale}(k, \mathrm{wittrow}(s{+}1)), \mathrm{wittwin}(s{+}1, \mathrm{twopow}(k)), s{+}1{+}\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1), \mathrm{center}(s{+}1)), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1))$ *and* $\mathrm{rdiag}(s{+}\mathrm{twopow}(k), s{+}1{+}\mathrm{twopow}(k)), \mathrm{center}(s{+}1{+}\mathrm{twopow}(k))$ *and* $\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s, s+1)), \mathrm{rdiag}(s-\mathrm{twopow}(k), s+1))))$.

Proof. The goal follows from crosscarryatcenterbitoutsideconewronskian(s, k).
□

Theorem 9.196. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number s and natural number k, if $T \leq s + 1$, then if* twopow$(k) \leq s$*, then if* center$(s + 1) =$ false*, then if* center$(s{+}1{+}\mathrm{lag}(P, k)) =$ false*, then* bitcrosscarryeq(wittscale$(k,$ wittrow$(s{+}1)),$ wittwin$(s{+}1,$ twopow$(k)),$ $s{+}1{+}$twopow$(k),$ bxor(ldiag$(s{+}1{-}$twopow$(k), s{+}1),$ rdiag$(s + 1 -$ twopow$(k), s + 1))$ and maskparity$(s + 1 +$ twopow$(k), 0))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarryatbothwhitecentersoutsideconeeqmaskpa
□

Theorem 9.197. *For every natural number s and natural number k, if* center$(s{+}1{+}$twopow$(k)) =$ false*, then* bitcrosscarryeq(wittscale$(k,$ wittrow$(s{+}1)),$ wittwin$(s{+}1,$ twopow$(k)),$ $s{+}1{+}$twopow$(k),$ wittresp$(s{+}1, k)$ and maskparity$(s{+}1 +$ twopow$(k), 0))$.

Proof. The goal follows from crosscarryatwhitelatecentereqwittandmaskparity(s, k).
□

Theorem 9.198. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number d, if $T \leq d{+}1$, then* bxor(maskparity$(d{+}m{\cdot}P{+}1, 0),$ maskparity$(d{+}1, 0)) =$ bxor(ldiag$(d + m \cdot P, d + m \cdot P + 1),$ ldiag$(d, d + 1))$.

Proof. Assume hP, h. The goal follows from periodicmaskparityzerodefectmuleqcolm1defectmul$(T, P, h$
□

Theorem 9.199. *For every natural number t and natural number k and natural number m and natural number P, if* twopow$(k) \leq t$*, then* bxor(bxor(wittresp$(t{+}m{\cdot}P, k),$ wittresp$(t, k)),$ bxor(response$(t{+}m{\cdot}P, k),$ response$(t, k))) =$ bxor(rdiag$(t{+}m \cdot P -$ twopow$(k), t + m \cdot P),$ rdiag$(t -$ twopow$(k), t))$.

Proof. The goal follows from wittrespxorresponseshiftmuloutsidecone(t, k, m, P).
□

Theorem 9.200. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number t and natural number k and natural number m, if $T \leq t$, then if* twopow$(k) \leq t$*, then* bxor(wittresp$(t + m \cdot P, k),$ wittresp$(t, k)) =$ bxor(bxor(ldiag$(t + m \cdot P -$ twopow$(k), t + m \cdot P),$ ldiag$(t -$ twopow$(k), t)),$ bxor(rdiag$(t + m \cdot P -$ twopow$(k), t + m \cdot P),$ rdiag$(t -$ twopow$(k), t)))$.

Proof. Assume hP, h. The goal follows from periodicwittrespshiftmuloutsidecone(T, P, hP, h). □

Theorem 9.201. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number d and natural number k, if $T \leq d + 1$, then if* center($d + 1 + \mathrm{lag}(P, k)) = \mathrm{false}$*, then* bitcrosscarryeq(wittscale(k, wittrow($d + m \cdot P + 1$)), wittwin(d+m·P+1, twopow(k)), d+m·P+2+twopow(k), bxor(ldiag(d+twopow(k), d+1+twopow(k)) and wittresp(d+1, k), bxor(bxor(maskparity(d+m·P+1+twopow(k), 0), maskparity(d+1+twopow(k), 0)) and wittresp(d+$m \cdot P + 1, k$), ldiag(d + twopow(k), $d + 1$ + twopow(k)) and bxor(wittresp(d+$m \cdot P + 1, k$), wittresp($d + 1, k$))))).

Proof. Assume hP, h. The goal follows from periodiccrosscarrycolm1shiftmulatwhitelatecenter(T, P, hP □

Theorem 9.202. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number s and natural number k, if $T \leq s$+1, then if* center(s+1+lag(P, k)) = false*, then* bitcrosscarryeq(wittscale(k, wittrow(s+m·P+1)), wittwin(s+m·P+1, twopow(k)), s+m·P+1+twopow(k), bxor(wittresp(s+1, k) and maskparity(s+1+twopow(k), 0), bxor(bxor(wittresp(s+m·P+1, k), wittresp(s+1, k)) and maskparity(s+$m \cdot P + 1$ + twopow(k), 0), wittresp($s + 1, k$) and bxor(maskparity($s + m \cdot P +$ 1 + twopow(k), 0), maskparity($s + 1$ + twopow(k), 0))))).

Proof. Assume hP, h. The goal follows from periodiccrosscarrycenterbitshiftmulatwhitelatecenter(T, P, □

Theorem 9.203. *For every natural number $t0$ and natural number k, if $2 \cdot t0 + 1 <$ twopow(k), then* frobdefect(k, wittrow($t0$)) = 0.

Proof. The goal follows from frobdefectrowconeeqzero($t0$, k). □

Theorem 9.204. *For every natural number $t0$ and natural number k,* wittwin($t0$+1, twopow(k)) = wittxor(wittxor(wittphi(wittwin($t0$, twopow(k))), wittshl(crosscarry(wittscale(k, wittr

Proof. The goal follows from winwittstepfrobdefect($t0$, k). □

Theorem 9.205. *For every natural number $t0$ and natural number k and natural number j,* bitwinwitteq($t0$+1, twopow(k), j+1, bxor(bxor(wittbitbool(wittphi(wittwin($t0$, twopo 1), wittbitbool(crosscarry(wittscale(k, wittrow($t0$)), wittwin($t0$, twopow(k))), j)), wittbitbool(frobdefect

Proof. The goal follows from winwittstepfrobdefecttestbitsucc($t0$, k, j). □

Theorem 9.206. *For every natural number* $t0$ *and natural number* k *and natural number* r*, if* $\mathrm{twopow}(k) \leq r$*, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(t0), r + 1 + \mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{rdiag}(r+1+\mathrm{twopow}(k), t0) \text{ and } \mathrm{bxor}(\mathrm{rdiag}(r, t0), \mathrm{rdiag}(r - \mathrm{twopow}(k), t0)), \mathrm{rdiag}(r+1, t0) \text{ and } \mathrm{bxor}(\mathrm{rdiag}(r + \mathrm{twopow}(k), t0), \mathrm{rdiag}(r - \mathrm{twopow}(k), t0))), \mathrm{rdiag}(r+1-\mathrm{twopow}(k), t0) \text{ and } \mathrm{bxor}(\mathrm{rdiag}(r+\mathrm{twopow}(k), t0), \mathrm{rdiag}(r, t0))))$.

Proof. The goal follows from $\mathrm{frobdefectrowtestbitoutsidecone}(t0, k, r)$. □

Theorem 9.207. *For every natural number* s *and natural number* k*, if* $\mathrm{twopow}(k) \leq s$*, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(s+1), s+1+\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s+1-\mathrm{twopow}(k), s+1) \text{ and } \mathrm{bxor}(\mathrm{rdiag}(s, s+1), \mathrm{rdiag}(s-\mathrm{twopow}(k), s+1)), \mathrm{center}(s+1) \text{ and } \mathrm{bxor}(\mathrm{ldiag}(s+2-\mathrm{twopow}(k), s+1), \mathrm{rdiag}(s-\mathrm{twopow}(k), s+1))), \mathrm{rdiag}(s+1-\mathrm{twopow}(k), s+1) \text{ and } \mathrm{bxor}(\mathrm{ldiag}(s+2-\mathrm{twopow}(k), s+1), \mathrm{rdiag}(s, s+1))))$.

Proof. The goal follows from $\mathrm{frobdefectrowcenterslotoutsidecone}(s, k)$. □

Theorem 9.208. *For every natural number* s *and natural number* k*, if* $\mathrm{twopow}(k) \leq s$*, then if* $\mathrm{center}(s+1) = \mathrm{false}$*, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(s+1), s+1+\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{ldiag}(s+1-\mathrm{twopow}(k), s+1) \text{ and } \mathrm{bxor}(\mathrm{maskparity}(s+1, 0), \mathrm{rdiag}(s-\mathrm{twopow}(k), s+1)), \mathrm{rdiag}(s+1-\mathrm{twopow}(k), s+1) \text{ and } \mathrm{bxor}(\mathrm{ldiag}(s+2-\mathrm{twopow}(k), s+1), \mathrm{maskparity}(s+1, 0))))$.

Proof. The goal follows from $\mathrm{frobdefectrowcenterslotatwhitecenteroutsidecone}(s, k)$. □

Theorem 9.209. *For every natural number* d *and natural number* k*, if* $\mathrm{twopow}(k) \leq d$*, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(d+1), d+2+\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(d-\mathrm{twopow}(k), d+1) \text{ and } \mathrm{bxor}(\mathrm{center}(d+1), \mathrm{rdiag}(d+1-\mathrm{twopow}(k), d+1)), \mathrm{ldiag}(d, d+1) \text{ and } \mathrm{bxor}(\mathrm{ldiag}(d+1-\mathrm{twopow}(k), d+1), \mathrm{rdiag}(d+1-\mathrm{twopow}(k), d+1))), \mathrm{rdiag}(d+2-\mathrm{twopow}(k), d+1) \text{ and } \mathrm{bxor}(\mathrm{ldiag}(d+1-\mathrm{twopow}(k), d+1), \mathrm{center}(d+1))))$.

Proof. The goal follows from $\mathrm{frobdefectrowcolm1slotoutsidecone}(d, k)$. □

Theorem 9.210. *For every natural number* d *and natural number* k*, if* $\mathrm{twopow}(k) \leq d$*, then if* $\mathrm{center}(d+1) = \mathrm{false}$*, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(d+1), d+2+\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(d-\mathrm{twopow}(k), d+1) \text{ and } \mathrm{rdiag}(d+1-\mathrm{twopow}(k), d+1), \mathrm{ldiag}(d, d+1) \text{ and } \mathrm{bxor}(\mathrm{ldiag}(d+1-\mathrm{twopow}(k), d+1), \mathrm{rdiag}(d+1-\mathrm{twopow}(k), d+1))), \mathrm{rdiag}(d+2-\mathrm{twopow}(k), d+1) \text{ and } \mathrm{ldiag}(d+1-\mathrm{twopow}(k), d+1)))$.

Proof. The goal follows from $\mathrm{frobdefectrowcolm1slotatwhitecenteroutsidecone}(d, k)$. □

Theorem 9.211. *For every natural number s and natural number k, if* $\mathrm{twopow}(k) \leq s$, *then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(s{+}2), s{+}1{+}\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}3{-}\mathrm{twopow}(k), s{+}2)$ and $\mathrm{bxor}(\mathrm{rdiag}(s, s{+}2), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}2)), \mathrm{rdiag}(s{+}1, s{+}2)$ and $\mathrm{bxor}(\mathrm{rdiag}(s{+}\mathrm{twopow}(k), s{+}2), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}2))), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s+2)$ and $\mathrm{bxor}(\mathrm{rdiag}(s+\mathrm{twopow}(k), s+2), \mathrm{rdiag}(s, s+2))))$.

Proof. The goal follows from $\mathrm{frobdefectrowcol1slotoutsidecone}(s, k)$. □

Theorem 9.212. *For every natural number s and natural number k, if* $\mathrm{twopow}(k) \leq s$, *then if* $\mathrm{center}(s{+}2) = \mathrm{false}$, *then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(s{+}2), s{+}1{+}\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}3{-}\mathrm{twopow}(k), s{+}2)$ and $\mathrm{bxor}(\mathrm{rdiag}(s, s{+}2), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}2)), \mathrm{maskparity}(s{+}2, 0)$ and $\mathrm{bxor}(\mathrm{rdiag}(s{+}\mathrm{twopow}(k), s{+}2), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}2))), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s{+}2)$ and $\mathrm{bxor}(\mathrm{rdiag}(s{+}\mathrm{twopow}(k), s+2), \mathrm{rdiag}(s, s+2))))$.

Proof. The goal follows from $\mathrm{frobdefectrowcol1slotatwhitecenteroutsidecone}(s, k)$. □

Theorem 9.213. *For every natural number T and natural number P, if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number s and natural number k, if* $T \leq s+1$, *then if* $\mathrm{twopow}(k) \leq s$, *then* $\mathrm{wittresp}(s+2, k) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittphi}(\mathrm{wittwin}(s+1, \mathrm{twopow}(k))), s+2+\mathrm{twopow}(k)), \mathrm{bxor}(\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1), \mathrm{center}(s{+}1)), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1))$ and $\mathrm{rdiag}(s{+}\mathrm{twopow}(k), s{+}1{+}\mathrm{twopow}(k)), \mathrm{center}(s{+}1{+}\mathrm{lag}(P, k))$ and $\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s, s{+}1)), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}1))))$, $\mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1)$ and $\mathrm{bxor}(\mathrm{rdiag}(s, s{+}1), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}1)), \mathrm{center}(s{+}1)$ and $\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}1))), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1)$ and $\mathrm{bxor}(\mathrm{ldiag}(s{+}2-\mathrm{twopow}(k), s+1), \mathrm{rdiag}(s, s+1))))$.

Proof. Assume hP, h. The goal follows from $\mathrm{periodicwinwittstepcenterslotoutsidecone}(T, P, hP, h)$. □

Theorem 9.214. *For every natural number T and natural number P, if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number s and natural number k, if* $T \leq s+1$, *then if* $\mathrm{twopow}(k) \leq s$, *then if* $\mathrm{center}(s+1+\mathrm{lag}(P, k)) = \mathrm{false}$, *then* $\mathrm{wittresp}(s{+}2, k) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittphi}(\mathrm{wittwin}(s{+}1, \mathrm{twopow}(k))), s+2+\mathrm{twopow}(k)), \mathrm{wittresp}(s+1, k)$ and $\mathrm{maskparity}(s+1+\mathrm{twopow}(k), 0)), \mathrm{bxor}(\mathrm{bxor}(\mathrm{ldiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1)$ and $\mathrm{bxor}(\mathrm{rdiag}(s, s{+}1), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}1)), \mathrm{center}(s{+}1)$ and $\mathrm{bxor}(\mathrm{ldiag}(s{+}2{-}\mathrm{twopow}(k), s{+}1), \mathrm{rdiag}(s{-}\mathrm{twopow}(k), s{+}1))), \mathrm{rdiag}(s{+}1{-}\mathrm{twopow}(k), s{+}1)$ and $\mathrm{bxor}(\mathrm{ldiag}(s{+}2-\mathrm{twopow}(k), s+1), \mathrm{rdiag}(s, s+1))))$.

Proof. Assume hP, h. The goal follows from periodicwinwittstepcenterslotatwhitelatecenteroutsidecone

□

Theorem 9.215. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number s and natural number k, if $T \leq s + 1$, then if* twopow$(k) \leq s$*, then if* center$(s + 1) =$ false*, then if* center$(s + 1 +$ lag$(P, k)) =$ false*, then* wittresp$(s + 2, k) =$ bxor(bxor(wittbitbool(wittphi(wittwin$(s + 1,$ twopow$(k))), s + 2 +$ twopow$(k))$, bxor(ldiag$(s+1-$twopow$(k), s+1)$, rdiag$(s+1-$twopow$(k), s+1))$ and maskparity$(s+1+$twopow$(k), 0))$, bxor(ldiag$(s+1-$twopow$(k), s+1)$ and bxor(maskparity$(s + 1, 0)$, rdiag$(s -$ twopow$(k), s + 1))$, rdiag$(s + 1 -$ twopow$(k), s + 1)$ and bxor(ldiag$(s + 2 -$ twopow$(k), s + 1)$, maskparity$(s + 1, 0))))$.

Proof. Assume hP, h. The goal follows from periodicwinwittstepcenterslotatbothwhitecentersoutsidecor

□

Theorem 9.216. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d+1$, then if* twopow$(k) \leq d$*, then* bitwinwitteq$(d+2,$ twopow$(k), d+3+$ twopow$(k),$ bxor(bxor(wittbitbool(wittphi(wittwin$(d+1,$ twopow$(k))), d+3+$ twopow$(k))$, bxor(bxor(bxor(ldiag$(d-$twopow$(k), d+1)$, ldiag$(d, d+1))$, rdiag$(d+2-$twopow$(k), d+1))$ and center$(d+1+$lag$(P, k))$, ldiag$(d+$ twopow$(k), d + 1 +$ twopow$(k))$ and bxor(bxor(ldiag$(d + 1 -$ twopow$(k), d + 1)$, center$(d + 1))$, rdiag$(d + 1 -$ twopow$(k), d + 1))))$, bxor(bxor(ldiag$(d -$ twopow$(k), d+1)$ and bxor(center$(d+1)$, rdiag$(d+1-$twopow$(k), d+1))$, ldiag$(d, d+1)$ and bxor(ldiag$(d + 1 -$ twopow$(k), d + 1)$, rdiag$(d + 1 -$ twopow$(k), d + 1)))$, rdiag$(d + 2 -$ twopow$(k), d + 1)$ and bxor(ldiag$(d + 1 -$ twopow$(k), d + 1)$, center$(d + 1)))))$.

Proof. Assume hP, h. The goal follows from periodicwinwittstepcolm1slotoutsidecone(T, P, hP, h).

□

Theorem 9.217. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d + 1$, then if* twopow$(k) \leq d$*, then if* center$(d + 1) =$ false*, then if* center$(d + 1 +$ lag$(P, k)) =$ false*, then* bitwinwitteq$(d + 2,$ twopow$(k), d+3+$ twopow$(k),$ bxor(bxor(wittbitbool(wittphi(wittwin$(d+1,$ twopow$(k))), d+3+$twopow$(k))$, ldiag$(d+$twopow$(k), d+1+$twopow$(k))$ and bxor(ldiag$(d+1-$twopow$(k), d+1)$, rdiag$(d+1-$twopow$(k), d+1)))$, bxor(bxor(ldiag$(d-$twopow$(k), d+1)$ and rdiag$(d+1-$twopow$(k), d+1)$, ldiag$(d, d+1)$ and bxor(ldiag$(d+$

1−twopow(k), d+1), rdiag(d+1−twopow(k), d+1))), rdiag(d+2−twopow(k), d+1) and ldiag(d + 1 − twopow(k), d + 1)))).

Proof. Assume hP, h. The goal follows from periodicwinwittstepcolm1slotatbothwhitecentersoutsidecon □

Theorem 9.218. *For every natural number s and natural number k, if* twopow(k) ≤ s, *then* bitwinwitteq(s+3, twopow(k), s+2+twopow(k), bxor(bxor(wittbitbool(wittphi(wi 2, twopow(k))), s+2+twopow(k)), wittbitbool(crosscarry(wittscale(k, wittrow(s+ 2)), wittwin(s + 2, twopow(k))), s + 1 + twopow(k))), bxor(bxor(ldiag(s + 3 − twopow(k), s+2) and bxor(rdiag(s, s+2), rdiag(s−twopow(k), s+2)), rdiag(s+ 1, s+2) and bxor(rdiag(s+twopow(k), s+2), rdiag(s−twopow(k), s+2))), rdiag(s+ 1 − twopow(k), s + 2) and bxor(rdiag(s + twopow(k), s + 2), rdiag(s, s + 2))))).

Proof. The goal follows from periodicwinwittstepcol1slotoutsidecone(s, k). □

Theorem 9.219. *For every natural number T and natural number P, if* 0 < P, *then if* centerperiodic(T, P), *then for every natural number m and natural number s and natural number k, if* T ≤ s+1, *then if* center(s+1+lag(P, k)) = false, *then* bxor(bxor(bxor(wittresp(s+m·P+2, k), wittbitbool(wittphi(wittwin(s+ m·P+1, twopow(k))), s+m·P+2+twopow(k))), wittbitbool(frobdefect(k, wittrow(s+ m·P+1)), s+m·P+1+twopow(k))), bxor(bxor(wittresp(s+2, k), wittbitbool(wittphi(wittwin(s+ 1, twopow(k))), s + 2 + twopow(k))), wittbitbool(frobdefect(k, wittrow(s + 1)), s+1+twopow(k)))) = bxor(bxor(wittresp(s + m · P + 1, k), wittresp(s+ 1, k)) and maskparity(s+m·P+1+twopow(k), 0), wittresp(s+1, k) and bxor(maskparity(s+ m · P + 1 + twopow(k), 0), maskparity(s + 1 + twopow(k), 0))).

Proof. Assume hP, h. The goal follows from periodicwinwittstepcenterslotshiftmulatwhitelatecenter(T, □

Theorem 9.220. *For every natural number k and natural number a and natural number b,* frobdefect(k, wittxor(a, b)) = wittxor(wittxor(frobdefect(k, a), frobdefect(k, b)), wittxor(

Proof. The goal follows from frobdefectxor(k, a, b). □

Theorem 9.221. *For every natural number $t0$ and natural number k, if* 2 · $t0$ + 1 < twopow(k), *then* frobdefect(k, wittscale(k, wittrow($t0$))) = 0.

Proof. The goal follows from frobdefectphiscalerowconeeqzero($t0$, k). □

Theorem 9.222. *For every natural number $t0$ and natural number k, if* 2 · $t0$ + 1 < twopow(k), *then* frobdefect(k, wittrow($t0$ + twopow(k))) = wittxor(frobdefect(k, wittwin($t0$, twopow(k))), wittxor(crosscarry(wittscale(succ(k), wittrow($t0$)), witts

Proof. The goal follows from frobdefectrowlatefromcone($t0, k$). □

Theorem 9.223. *For every natural number d and natural number k, if $d + 1 \leq k$, then* bitfrobdefecteq(k, wittrow($d + 1 +$ twopow(k)), $d + 1 +$ twopow(k)+twopow(k), bxor(bxor(ldiag(d+1, d+1+twopow(k)) and bxor(rdiag(d+ twopow(k), d+1+twopow(k)), rdiag(d, d+1)), center(d+1+twopow(k)) and bxor(ldiag(d+ 2, d+1+twopow(k)), rdiag(d, d+1))), center(d+1) and bxor(ldiag(d+2, d+ 1 + twopow(k)), rdiag(d + twopow(k), $d + 1 +$ twopow(k))))).

Proof. The goal follows from frobdefectrowlatecenterslotrperiodic(d, k). □

Theorem 9.224. *For every natural number d and natural number k, if $d + 1 \leq k$, then if* settledat($d + 1, k$)*, then if* center($d + 1$) = false*, then if* center(d+1+twopow(k)) = false*, then* zcenterandeq(d+1, bxor(maskparity(d+ 1+twopow(k), 0), maskparity(d+1, 0)), wittbitbool(frobdefect(k, wittrow(d+ $1 +$ twopow(k))), $d + 1 +$ twopow(k) + twopow(k))).

Proof. The goal follows from frobdefectrowlatecenterslotatbothwhitecenterseqzcenterandmaskparityjum □

Theorem 9.225. *For every natural number d and natural number k, if $d + 2 \leq k$, then* bitfrobdefecteq(k, wittrow($d + 1 +$ twopow(k)), $d + 2 +$ twopow(k)+twopow(k), bxor(bxor(ldiag(d, d+1+twopow(k)) and bxor(center(d+ 1+twopow(k)), center(d+1)), ldiag(d+twopow(k), d+1+twopow(k)) and bxor(ldiag(d+ 1, d+1+twopow(k)), center(d+1))), ldiag(d, d+1) and bxor(ldiag(d+1, d+ 1 + twopow(k)), center($d + 1 +$ twopow(k))))).

Proof. The goal follows from frobdefectrowlatecolm1slotrperiodic(d, k). □

Theorem 9.226. *For every natural number d and natural number k, if $d + 2 \leq k$, then if* settledat($d + 1, k$)*, then if* center($d + 1 +$ twopow(k)) = center($d + 1$)*, then* zcenterandeq($d + 1$, bxor(ldiag($d +$ twopow(k), $d + 1 +$ twopow(k)), ldiag($d, d + 1$)), bxor(wittbitbool(frobdefect(k, wittrow($d + 1 +$ twopow(k))), d+2+twopow(k)+twopow(k)), center(d+1) and bxor(ldiag(d+ twopow(k), $d + 1 +$ twopow(k)), ldiag($d, d + 1$)))).

Proof. The goal follows from frobdefectrowlatecolm1slotofinphaseeqcolm1jumpanddamage(d, k). □

Theorem 9.227. *For every natural number d and natural number k, if $d + 2 \leq k$, then if* settledat($d + 1, k$)*, then if* center($d + 1$) = false*, then if* center($d + 1 +$ twopow(k)) = false*, then* zcenterandeq($d + 1$, bxor(center(d+ 2+twopow(k)), center(d+2)), bxor(wittbitbool(frobdefect(k, wittrow(d+1+

twopow(k))), d+2+twopow(k)+twopow(k)), wittbitbool(frobdefect(k, wittrow(d+ 1 + twopow(k))), d + 1 + twopow(k) + twopow(k)))).

Proof. The goal follows from frobdefectrowlatecolm1xorcenterslotatbothwhitecenters(d, k). □

Theorem 9.228. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d + 1$, then if $d + 1 \leq k$, then if* settledat(d + 1, k)*, then if* center(d + 1) = false*, then if* center(d + 1 + lag(P, k)) = false*, then if* center(d + 1 + lag(P, succ(k))) = false*, then* zcenterandeq(d + 1, bxor(maskparity(d + 1 + twopow(succ(k)), 0), bxor(maskparity(d + 1 + twopow(k), 0), maskparity(d+1, 0))), bxor(wittresp(d+2+twopow(k), k), wittbitbool(wittphi(wittwin(d 1 + twopow(k), twopow(k))), d + 2 + twopow(k) + twopow(k)))).

Proof. Assume hP, h. The goal follows from periodicwinwittsteplatecenterslotatwhitecenterseqzcenterar □

Theorem 9.229. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d + 1$, then if $d + 2 \leq k$, then if* settledat(d + 1, k)*, then if* center(d + 1) = false*, then if* center(d + 1 + lag(P, k)) = false*, then if* center(d + 1 + lag(P, succ(k))) = false*, then* zcenterandeq(d + 1, bxor(ldiag(d + twopow(succ(k)), d + 1 + twopow(succ(k))), bxor(ldiag(d + twopow(k), d+1+twopow(k)), ldiag(d, d+1))), bxor(wittbitbool(wittwin(d+ 2+twopow(k), twopow(k)), d+3+twopow(k)+twopow(k)), wittbitbool(wittphi(wittwin(d+ 1 + twopow(k), twopow(k))), d + 3 + twopow(k) + twopow(k)))).

Proof. Assume hP, h. The goal follows from periodicwinwittsteplatecolm1slotatwhitecenterseqzcenterar □

Theorem 9.230. *For every natural number d and natural number k, if $d+1 \leq k$, then if* settledat(d+1, k)*, then* zcentereq(d+1, bxor(wittresp(d+ 1+twopow(k), k), bxor(center(d+1+twopow(succ(k))), bxor(center(d+1+ twopow(k)), center(d + 1))))).

Proof. The goal follows from wittresplateeqzcenterxorcentertriality(d, k). □

Theorem 9.231. *For every natural number d and natural number k, if $d + 1 \leq k$, then if* settledat(d + 1, k)*, then if* center(d + 1) = false*, then if* center(d+1+twopow(k)) = false*, then if* center(d+1+twopow(succ(k))) = false*, then* zcentereq(d + 1, wittresp(d + 1 + twopow(k), k)).

Proof. The goal follows from wittresplateatwhitecenterseqzcenter(d, k). □

Theorem 9.232. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number d and natural number k, if $T \leq d + 1$, then if $d + 1 \leq k$, then if* settledat$(d + 1, k)$*, then* zcentereq$(d + 1,$ bxor(bxor(wittresp$(d + m \cdot P + 1 +$ twopow$(k), k)$, wittresp$(d + 1 +$ twopow$(k), k))$, bxor(ldiag$(d + m \cdot P + 1, d + m \cdot P + 1 +$ twopow$(k))$, bxor(rdiag$(d + m \cdot P + 1, d + m \cdot P + 1 +$ twopow$(k))$, center$(d + 1)))))$.

Proof. Assume hP, h. The goal follows from periodicwittresplateshiftmuleqzcenterxor(T, P, hP, h). □

Theorem 9.233. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number d and natural number k, if $T \leq d + 1$, then if $d + 1 \leq k$, then if* settledat$(d + 1, k)$*, then if* center$(d + 1) =$ false*, then if* center$(d +$ $1 +$ lag$(P, k)) =$ false*, then if* center$(d + 1 +$ lag$(P,$ succ$(k))) =$ false*, then* zcenterandeq$(d$+1, maskparity$(d$+1+twopow(succ$(k))$, 0), bxor(bxor(wittbitbool(crosscarry(wittscale(k $m \cdot P$+1+twopow$(k)))$, wittwin$(d$+$m \cdot P$+1+twopow(k), twopow$(k)))$, d+$m \cdot$ P+1+twopow(k)+twopow$(k))$, wittbitbool(crosscarry(wittscale$(k,$ wittrow$(d$+ 1+twopow$(k)))$, wittwin$(d$+1+twopow(k), twopow$(k)))$, d+1+twopow(k)+ twopow$(k)))$, wittresp$(d + m \cdot P + 1 +$ twopow$(k), k)$ and maskparity$(d + m \cdot$ $P + 1 +$ twopow(succ$(k)), 0)))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrylatecenterbitshiftmulatwhitecenters$(T, P$ □

Theorem 9.234. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number d and natural number k, if $T \leq d + 1$, then if $d + 1 \leq k$, then if* settledat$(d + 1, k)$*, then if* center$(d + 1) =$ false*, then if* center$(d +$ $1 +$ lag$(P, k)) =$ false*, then if* center$(d + 1 +$ lag$(P,$ succ$(k))) =$ false*, then* zcenterandeq$(d$+1, ldiag$(d$+twopow(succ$(k))$, d+1+twopow(succ$(k)))$, bxor(bxor(wittbitbool(crosscarry $m \cdot P$+1+twopow$(k)))$, wittwin$(d$+$m \cdot P$+1+twopow(k), twopow$(k)))$, d+$m \cdot$ P+2+twopow(k)+twopow$(k))$, wittbitbool(crosscarry(wittscale$(k,$ wittrow$(d$+ 1+twopow$(k)))$, wittwin$(d$+1+twopow(k), twopow$(k)))$, d+2+twopow(k)+ twopow$(k)))$, bxor(bxor(maskparity$(d$+$m \cdot P$+1+twopow(succ$(k))$, 0), maskparity$(d$+ 1+twopow(succ$(k))$, 0)) and wittresp$(d$+$m \cdot P$+1+twopow$(k), k)$, ldiag$(d$+ twopow(succ$(k))$, d + 1 + twopow(succ$(k))$) and wittresp$(d + m \cdot P + 1 +$ twopow$(k), k))))$.

Proof. Assume hP, h. The goal follows from periodiccrosscarrylatecolm1shiftmulatwhitecenters(T, P, hI □

Theorem 9.235. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number d and natural number k, if $T \leq d + 1$, then if $d + 1 \leq k$, then if* settledat($d + 1, k$)*, then if* center($d + 1$) = false*, then if* center($d + 1 + \mathrm{lag}(P, k)$) = false*, then if* center($d + 1 + \mathrm{lag}(P, \mathrm{succ}(k))$) = false*, then* zcenterandeq(d+1, bxor(maskparity(d+1+twopow(succ(k)), 0), bxor(maskparity(d+ 1 + twopow(k), 0), maskparity($d + 1$, 0))), bxor(bxor(wittresp($d + m \cdot P + 2 +$ twopow(k), k), wittresp(d+2+twopow(k), k)), bxor(bxor(wittbitbool(wittphi(wittwin(d+ m·P+1+twopow(k), twopow(k))), d+m·P+2+twopow(k)+twopow(k)), wittbitbool(wittphi(wittwin(d- 1+twopow(k), twopow(k))), d+2+twopow(k)+twopow(k))), bxor(wittresp(d+ m·P+1+twopow(k), k) and maskparity(d+m·P+1+twopow(succ(k)), 0), wittbitbool(frobdefect(k, witt $m \cdot P + 1 +$ twopow(k))), $d + m \cdot P + 1 +$ twopow(k) + twopow(k)))))).

Proof. Assume hP, h. The goal follows from periodicwinwittsteplatecenterslotshiftmulatwhitecenters(T □

Theorem 9.236. *For every natural number k and natural number j and natural number u and natural number s, if $j \leq k$, then* rdiag($j, s + u \cdot$ twopow(k)) = rdiag(j, s).

Proof. The goal follows from rdiagrestart(k, j, u, s). □

Theorem 9.237. *For every natural number k and natural number J and natural number j and natural number u and natural number s, if* rightthreshold(k, J)*, then if $j \leq J$, then* rdiag($j, s + u \cdot$ twopow(k)) = rdiag(j, s).

Proof. The goal follows from rdiagrestartofthreshold(k, J, j, u, s). □

Theorem 9.238. *For every natural number k and natural number s and natural number u, if $s \leq k$, then* rdiag($s, s + u \cdot$ twopow(k)) = center(s).

Proof. The goal follows from centerrestartrdiag(k, s, u). □

Theorem 9.239. *For every natural number $t0$ and natural number k and natural number j, if $j <$* twopow(k)*, then* bitfrobdefecteq(k, wittrow($t0$), j, false).

Proof. The goal follows from frobdefectrowlttwopow($t0, k, j$). □

Theorem 9.240. *For every natural number $t0$ and natural number k and natural number r, if r+1 <* twopow(k)*, then* bitfrobdefecteq(k, wittrow($t0$), r+ 1 + twopow(k), bxor(rdiag($r + 1 +$ twopow(k), $t0$) and rdiag($r, t0$), rdiag($r +$ 1, $t0$) and rdiag($r +$ twopow(k), $t0$))).

Proof. The goal follows from frobdefectrowsectorone$(t0, k, r)$. □

Theorem 9.241. *For every natural number d and natural number k and natural number u, if $d + 1 \leq k$, then* bitfrobdefecteq$(k,$ wittrow$(d + 1 + u \cdot$ twopow$(k)),$ $d + 1 +$ twopow$(k),$ bxor(rdiag$(d + 1 +$ twopow$(k), d + 1 + u \cdot$ twopow$(k))$ and rdiag$(d, d+1),$ center$(d+1)$ and rdiag$(d+$twopow$(k), d+1+$ $u \cdot$ twopow$(k))))$.

Proof. The goal follows from frobdefectrowsectorerestart(d, k, u). □

Theorem 9.242. *For every natural number d and natural number k and natural number u, if $d + 1 \leq k$, then* bitfrobdefecteq$(k,$ wittrow$(d + 1 +$ $u \cdot$ twopow$(k)), d + 1 +$ twopow$(k) +$ twopow$(k),$ bxor(bxor(rdiag$(d + 1 +$ twopow$(\text{succ}(k)), d + 1 + u \cdot$ twopow$(k))$ and bxor(rdiag$(d +$ twopow$(k), d +$ $1 + u \cdot$ twopow$(k)),$ rdiag$(d, d + 1)),$ rdiag$(d + 1 +$ twopow$(k), d + 1 + u \cdot$ twopow$(k))$ and bxor(rdiag$(d+$twopow$(\text{succ}(k)), d+1+u\cdot$twopow$(k)),$ rdiag$(d, d+$ $1))),$ center$(d+1)$ and bxor(rdiag$(d+$twopow$(\text{succ}(k)), d+1+u\cdot$twopow$(k)),$ rdiag$(d+$ twopow$(k), d + 1 + u \cdot$ twopow$(k)))))$.

Proof. The goal follows from frobdefectrowsectortworestart(d, k, u). □

Theorem 9.243. *For every natural number $t0$ and natural number k and natural number r, if $r+1 \leq k$, then* bitfrobdefecteq$(k,$ wittwin$(t0,$ twopow$(k)),$ $r+$ $1+$twopow$(\text{succ}(k)),$ bxor(wittbitbool(wittwin$(t0,$ twopow$(k)),$ $r+1+$twopow$(\text{succ}(k)))$ and wittbitbool(twopow$(k)),$ wittbitbool(wittwin$(t0,$ twopow$(k)),$ $r+1+$twopow$(k))$ and wittbitbool(wittwin$(t0,$ twopow twopow$(\text{succ}(k)))))$.

Proof. The goal follows from frobdefectwinwittsectortwowronskian$(t0, k, r)$. □

Theorem 9.244. *For every natural number $t0$ and natural number k and natural number r, if $2 \cdot t0 + 1 <$* twopow(k)*, then if $r + 1 \leq k$, then* wittbitbool(crosscarry(wittscale(succ$(k),$ wittrow$(t0)),$ wittscale$(k,$ wittwin$(t0,$ twopow$(k)))),$ $r+$ $1+$twopow$(\text{succ}(k))) =$ wittbitbool(wittscale$(k,$ crosscarry(wittscale$(k,$ wittrow$(t0)),$ wittwin$(t0,$ twopow $1 +$ twopow$(\text{succ}(k)))$.

Proof. The goal follows from crosscarryscalerowwinwittsectortwoconeeq$(t0, k, r)$. □

Theorem 9.245. *For every natural number $t0$ and natural number k and natural number r, if $2 \cdot t0 + 1 <$* twopow(k)*, then if $r + 1 \leq k$, then* bitfrobdefecteq$(k,$ wittrow$(t0+$twopow$(k)), r+1+$twopow$(\text{succ}(k)),$ wittbitbool(frobdefect$(k,$ wittwin$(t0$ $1 +$ twopow$(\text{succ}(k))))$.

Proof. The goal follows from frobdefectrowlateeqfrobdefectwinwittsectortwo($t0, k, r$). □

Theorem 9.246. *For every natural number $t0$ and natural number k and natural number r, if $2 \cdot t0 + 1 <$ twopow(k), then if $r + 1 \leq k$, then* bitfrobdefecteq(k, wittrow($t0$+twopow(k)), r+1+twopow(succ(k)), bxor(wittbitbool(wittwin($t0$, twopow
1+twopow(succ(k))) and wittbitbool(wittwin($t0$, twopow(k)), r+twopow(k)), wittbitbool(wittwin($t0$, tw
1+twopow(k)) and wittbitbool(wittwin($t0$, twopow(k)), r+twopow(succ(k))))).

Proof. The goal follows from frobdefectrowlateeqwinwittsectortwowronskian($t0, k, r$). □

Theorem 9.247. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* twopow(k) $\leq t$*, then* bxor(ldiag($t + m \cdot P -$ twopow(k), $t + m \cdot P$), ldiag($t -$ twopow(k), t)) = bxor(response($t + m \cdot P, k$), response(t, k)).

Proof. Assume hP, h. The goal follows from periodicldiagshiftmuleqresponseshiftmul(T, P, hP, h). □

Theorem 9.248. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number $T1$ and natural number m, if $0 < m$, then not* maskparityzeroperiodic($T1, m \cdot P$).

Proof. Assume hP, h, $T1$, m, hm. The goal follows from notperiodicmaskparityzeromul($T, P, hP, h, T1,$ □

Theorem 9.249. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number $T1$ and natural number m, if $0 < m$, then not* colnegoneperiodic($T1, m \cdot P$).

Proof. Assume hP, h, $T1$, m, hm. The goal follows from notperiodiccolm1mul($T, P, hP, h, T1, m, hm$). □

Theorem 9.250. *For every natural number $t0$ and natural number k and natural number j, if $2 \cdot t0 + 1 <$ twopow(k), then if $j \leq k$, then* wittbitbool(wittscale(k, wittrow($t0$+ twopow(k))), j+twopow(k)) = wittbitbool(wittwin($t0$, twopow(k)), j+twopow(k)).

Proof. The goal follows from phiscalerowlatesectoroneeqwinwittcone($t0, k, j$). □

Theorem 9.251. *For every natural number $t0$ and natural number k and natural number r, if $2 \cdot t0 + 1 < \mathrm{twopow}(k)$, then if $r + 1 \leq k$, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(t0{+}\mathrm{twopow}(k)), r{+}1{+}\mathrm{twopow}(k), \mathrm{wittbitbool}(\mathrm{crosscarry}(\mathrm{wittscale}(k, \mathrm{wittrow}$ $1 + \mathrm{twopow}(k)))$.

Proof. The goal follows from $\mathrm{frobdefectrowlatesectoroneeqcrosscarrycone}(t0, k, r)$. □

Theorem 9.252. *For every natural number d and natural number k, if $d + 1 \leq k$, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(d + 1 + \mathrm{twopow}(k)), d + 1 + \mathrm{twopow}(k), \mathrm{bxor}(\mathrm{center}(d + 1 + \mathrm{twopow}(k)) \text{ and } \mathrm{rdiag}(d, d + 1), \mathrm{center}(d + 1) \text{ and } \mathrm{rdiag}(d + \mathrm{twopow}(k), d + 1 + \mathrm{twopow}(k))))$.

Proof. The goal follows from $\mathrm{frobdefectrowlatesectoronecenterslotwronskian}(d, k)$. □

Theorem 9.253. *For every natural number d and natural number k, if $d + 1 \leq k$, then if* $\mathrm{settledat}(d + 1, k)$*, then if* $\mathrm{center}(d + 1) = \mathrm{false}$*, then* $\mathrm{zcenterandeq}(d{+}1, \mathrm{maskparity}(d{+}1, 0), \mathrm{wittbitbool}(\mathrm{crosscarry}(\mathrm{wittscale}(k, \mathrm{wittrow}(d{+}1)), \mathrm{wittwin}(d + 1, \mathrm{twopow}(k))), d + 1 + \mathrm{twopow}(\mathrm{succ}(k))))$.

Proof. The goal follows from $\mathrm{crosscarryrowwinwittsectortwocenterslot}(d, k)$. □

Theorem 9.254. *For every natural number k and natural number j and natural number v and natural number t, if $j \leq k$, then if* $\mathrm{settledby}(j, t)$*, then* $\mathrm{ldiag}(j, t + v \cdot \mathrm{twopow}(k)) = \mathrm{ldiag}(j, t)$.

Proof. The goal follows from $\mathrm{ldiagsettledrestart}(k, j, v, t)$. □

Theorem 9.255. *For every natural number k and natural number d and natural number u, if $d \leq k$, then if $1 \leq u$, then if* $\mathrm{settledat}(d, k)$*, then* $\mathrm{zcentereq}(d, \mathrm{ldiag}(d, d + u \cdot \mathrm{twopow}(k)))$.

Proof. The goal follows from $\mathrm{zcenterrestartldiag}(k, d, u)$. □

Theorem 9.256. *For every natural number d and natural number k, if $d + 1 \leq k$, then if* $\mathrm{settledat}(d + 1, k)$*, then if* $\mathrm{zcentereq}(d + 1, \mathrm{true})$*, then if* $\mathrm{center}(d + 1) = \mathrm{false}$*, then if* $\mathrm{center}(d + 1 + \mathrm{twopow}(k)) = \mathrm{false}$*, then* $\mathrm{bitfrobdefecteq}(k, \mathrm{wittrow}(d{+}1{+}\mathrm{twopow}(k)), d{+}1{+}\mathrm{twopow}(k){+}\mathrm{twopow}(k), \mathrm{bxor}(\mathrm{maskparity}(d{+}1 + \mathrm{twopow}(k), 0), \mathrm{maskparity}(d + 1, 0)))$.

Proof. The goal follows from $\mathrm{frobdefectrowlatecenterslotofzcentertrue}(d, k)$. □

Theorem 9.257. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d + 1$, then if $d + 1 \leq k$, then if* settledat$(d + 1, k)$*, then if* zcentereq$(d + 1, \text{true})$*, then if* center$(d + 1) =$ false*, then if* center$(d+1+\text{lag}(P, k)) =$ false*, then if* center$(d+1+\text{lag}(P, \text{succ}(k))) =$ false*, then* bxor(maskparity$(d + 1 + \text{twopow}(\text{succ}(k)), 0)$, bxor(maskparity$(d + 1 +$ twopow$(k), 0)$, maskparity$(d+1, 0))) =$ bxor(wittresp$(d+2+\text{twopow}(k), k)$, wittbitbool(wittphi(wittwin$1 + \text{twopow}(k), \text{twopow}(k))), d + 2 + \text{twopow}(k) + \text{twopow}(k)))$.

Proof. Assume hP, h. The goal follows from periodicwinwittsteplatecenterslotofzcentertrue(T, P, hP, h). □

Theorem 9.258. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number d and natural number k, if $T \leq d+1$, then if $d+2 \leq k$, then if* settledat$(d+1, k)$*, then if* zcentereq$(d+1, \text{true})$*, then if* center$(d+1) =$ false*, then if* center$(d+1 + \text{lag}(P, k)) =$ false*, then if* center$(d + 1 + \text{lag}(P, \text{succ}(k))) =$ false*, then* bxor(ldiag$(d + \text{twopow}(\text{succ}(k)), d + 1 + \text{twopow}(\text{succ}(k)))$, bxor(ldiag$(d +$ twopow$(k), d+1+\text{twopow}(k))$, ldiag$(d, d+1))) =$ bxor(wittbitbool(wittwin$(d+$ $2+\text{twopow}(k), \text{twopow}(k)), d+3+\text{twopow}(k)+\text{twopow}(k))$, wittbitbool(wittphi(wittwin$(d+$ $1 + \text{twopow}(k), \text{twopow}(k))), d + 3 + \text{twopow}(k) + \text{twopow}(k)))$.

Proof. Assume hP, h. The goal follows from periodicwinwittsteplatecolm1slotofzcentertrue(T, P, hP, h). □

Theorem 9.259. *For every natural number t and natural number Q and natural number a and natural number L and natural number $a1$ and natural number $L1$, if $a \leq a1$, then if $a1 + L1 \leq a + L$, then if* rowagree(t, Q, a, L)*, then* rowagree$(t, Q, a1, L1)$.

Proof. The goal follows from rowagreemono$(t, Q, a, L, a1, L1)$. □

Theorem 9.260. *For every natural number t and natural number k and natural number Q and natural number a and natural number L and natural number $a1$ and natural number $L1$, if $a \leq a1$, then if $a1 + L1 \leq a + L$, then if* winwittagree(t, k, Q, a, L)*, then* winwittagree$(t, k, Q, a1, L1)$.

Proof. The goal follows from winwittagreemono$(t, k, Q, a, L, a1, L1)$. □

Theorem 9.261. *For every natural number t and natural number Q and natural number a and natural number L, if* rowagree$(t, Q, a, L + 2)$*, then* rowagree$(t + 1, Q, a + 2, L)$.

Proof. The goal follows from rowagreestep(t, Q, a, L). □

Theorem 9.262. *For every natural number t and natural number Q and natural number a and natural number L and natural number s, if* rowagree$(t, Q, a, L+2 \cdot s)$*, then* rowagree$(t + s, Q, a + 2 \cdot s, L)$.

Proof. The goal follows from rowagreesteps(t, Q, a, L, s). □

Theorem 9.263. *For every natural number t and natural number k and natural number Q and natural number a and natural number L and natural number i, if* twopow(succ$(k)) \leq a$*, then if $i \leq L$, then if* rowagree$(t, Q, a -$ twopow(succ$(k)), L)$*, then if* rowagree$(t, Q, a-$twopow$(k), L)$*, then if* rowagree(t, Q, a, L)*, then* wittbitbool(wittscale$(k,$ wittrow$(t+Q)), a+i+Q) =$ wittbitbool(wittscale$(k,$ wittrow$(t)), a+i)$.

Proof. The goal follows from phiscalerowtestbiteqofrowagree(t, k, Q, a, L, i). □

Theorem 9.264. *For every natural number t and natural number k and natural number Q and natural number a and natural number L and natural number i, if* twopow(succ$(k)) \leq a$*, then if $i \leq L$, then if* rowagree$(t, Q, a -$ twopow(succ$(k)), L + 1)$*, then if* rowagree$(t, Q, a -$ twopow$(k), L + 1)$*, then if* rowagree$(t, Q, a, L + 1)$*, then* bitfrobdefecteq$(k,$ wittrow$(t + Q), a + i + 1 + Q,$ wittbitbool(frobdefect$(k,$ wittrow$(t)), a + i + 1))$.

Proof. The goal follows from frobdefectrowtestbiteqofrowagree(t, k, Q, a, L, i). □

Theorem 9.265. *For every natural number t and natural number k and natural number Q and natural number a and natural number L and natural number i, if* twopow(succ$(k)) \leq a$*, then if $i \leq L$, then if* winwittagree$(t, k, Q, a, L+1)$*, then if* rowagree$(t, Q, a-$twopow(succ$(k)), L+1)$*, then if* rowagree$(t, Q, a-$twopow$(k), L+1)$*, then if* rowagree$(t, Q, a, L+1)$*, then* bitcrosscarryeq(wittscale$(k,$ wittrow$(t+Q))$, wittwin$(t+Q,$ twopow$(k)), a+i+1+Q,$ wittbitbool(crosscarry(wittscale$(k,$ wittrow$(t))$, wittwin$(t,$ tw$i + 1))$.

Proof. The goal follows from crosscarrytestbiteqofrowagreewinwittagree(t, k, Q, a, L, i). □

Theorem 9.266. *For every natural number t and natural number k and natural number Q and natural number a and natural number L, if* twopow(succ$(k)) \leq a$*, then if* winwittagree$(t, k, Q, a, L+2)$*, then if* rowagree$(t, Q, a-$twopow(succ$(k)), L+1)$*, then if* rowagree$(t, Q, a -$ twopow$(k), L + 1)$*, then if* rowagree$(t, Q, a, L + 1)$*, then* winwittagree$(t + 1, k, Q, a + 2, L)$.

Proof. The goal follows from winwittagreestep(t, k, Q, a, L). □

Theorem 9.267. *For every natural number t and natural number k and natural number Q and natural number a and natural number L and natural number s, if* twopow(succ$(k)) \leq a$, *then if* winwittagree$(t, k, Q, a, L + 2 \cdot s)$, *then if* rowagree$(t, Q, a-$twopow(succ$(k)), L+2 \cdot s)$, *then if* rowagree$(t, Q, a-$twopow$(k), L+2 \cdot s)$, *then if* rowagree$(t, Q, a, L+2 \cdot s)$, *then* winwittagree$(t+s, k, Q, a + 2 \cdot s, L)$.

Proof. The goal follows from winwittagreesteps(t, k, Q, a, L, s). □

Theorem 9.268. *For every natural number $t0$ and natural number k and natural number Q and natural number s, if* twopow$(k) + s \leq t0$, *then if* winwittagree$(t0, k, Q, t0 +$ twopow$(k) - s, 2 \cdot s)$, *then if* rowagree$(t0, Q, t0 +$ twopow$(k)-s-$twopow(succ$(k)), 2 \cdot s)$, *then if* rowagree$(t0, Q, t0+$twopow$(k)-s-$ twopow$(k), 2 \cdot s)$, *then if* rowagree$(t0, Q, t0 +$ twopow$(k) - s, 2 \cdot s)$, *then* wittresp$(t0 + s + Q, k) =$ wittresp$(t0 + s, k)$.

Proof. The goal follows from wittrespeqofwinwittagreesteps$(t0, k, Q, s)$. □

Theorem 9.269. *For every natural number $t0$ and natural number Q and natural number s, if $s \leq t0 + 2$, then if* rowagree$(t0 + 1, Q, t0 + 2 - s, 2 \cdot s)$, *then* ldiag$(t0 + s + Q, t0 + s + 1 + Q) =$ ldiag$(t0 + s, t0 + s + 1)$.

Proof. The goal follows from colm1eqofrowagreesteps$(t0, Q, s)$. □

Theorem 9.270. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number $t0$ and natural number s, if $T \leq t0+s+1$, then if $s \leq t0+2$, then if* rowagree$(t0 + 1, m \cdot P, t0 + 2 - s, 2 \cdot s)$, *then* maskparity$(t0 + s + 1 + m \cdot P, 0) =$ maskparity$(t0 + s + 1, 0)$.

Proof. Assume hP, $hper$. The goal follows from maskparityzeroeqofrowagreesteps$(T, P, hP, hper)$. □

Theorem 9.271. *For every natural number k and natural number d and natural number U and natural number Q, if $1 \leq U$, then if* sectorrowagree(k, d, U, Q), *then* colnegoneperiodic$(d + U \cdot$ twopow$(k), Q)$.

Proof. The goal follows from colnegoneperiodicofsectorrowagree(k, d, U, Q). □

Theorem 9.272. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U and natural number m, if $1 \leq U$, then if $0 < m$, then not* sectorrowagree$(k, d, U, m \cdot P)$.

Proof. Assume hP, h, k, d, U, m, hU, hm. The goal follows from notsectorrowagreemul$(T, P, hP, h, k, d,$ □

Theorem 9.273. *For every natural number P and natural number k and natural number j and natural number t, if $j \leq k$, then* rdiag$(j, t+$lcmperiod$(P, k)) =$ rdiag(j, t).

Proof. The goal follows from rdiaglcmperiodeq(P, k, j, t). □

Theorem 9.274. *For every natural number P and natural number k and natural number j and natural number t, if $j \leq k$, then if* settledby(j, t)*, then* ldiag$(j, t + \text{lcmperiod}(P, k)) = \text{ldiag}(j, t)$.

Proof. The goal follows from ldiaglcmperiodeq(P, k, j, t). □

Theorem 9.275. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number t, if $T \leq t$, then* center$(t + \text{lcmperiod}(P, k)) = \text{center}(t)$.

Proof. Assume hP, h. The goal follows from centerlcmperiodeq(T, P, hP, h). □

Theorem 9.276. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U, if $1 \leq U$, then not* sectorrowagree$(k, d, U, \text{lcmperiod}(P, k))$.

Proof. Assume hP, h, k, d, U, hU. The goal follows from notsectorrowagreelcmperiod$(T, P, hP, h, k, d, U$ □

Theorem 9.277. *For every natural number t and natural number Q and natural number a and natural number L, if* rowagree$(t, Q, a, L + 1)$*, then if* wittbitbool(wittrow$(t + 1 + Q), a + L + 2 + Q) =$ wittbitbool(wittrow$(t + 1), a + L + 2)$*, then* rowagree$(t, Q, a, L + 2)$.

Proof. The goal follows from rowagreeextendright(t, Q, a, L). □

Theorem 9.278. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then* rowagree$(t, m \cdot P, t, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicrowagreecenter$(T, P, hP, hper)$. □

Theorem 9.279. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t+1$, then if* center$(t+1) =$ true*, then* rowagree$(t+1, m \cdot P, t+1, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodicrowagreeblackcenter$(T, P, hP, hper)$. □

Theorem 9.280. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number R and natural number t, if $T \leq t$, then if $R \leq t$, then if* rowagree$(t, m \cdot P, t - R, R)$*, then* rowagree$(t, m \cdot P, t - R, 2 \cdot R)$.

Proof. Assume hP, $hper$. The goal follows from periodicrowagreesymmofright$(T, P, hP, hper)$. □

Theorem 9.281. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number R and natural number t, if $T \leq t$, then if $R \leq t$, then if* rowagree$(t, \text{lcmperiod}(P, k), t - R, R)$*, then* rowagree$(t, \text{lcmperiod}(P, k), t - R, 2 \cdot R)$.

Proof. Assume hP, $hper$. The goal follows from periodicrowagreelcmperiodsymmofright$(T, P, hP, hper)$ □

Theorem 9.282. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U and natural number m, if $1 \leq U$, then if $1 \leq d$, then if $T \leq d + U \cdot$* twopow(k)*, then if* sectorrightrowagree$(k, d, U, m \cdot P)$*, then* colnegoneperiodic$(d + U \cdot \text{twopow}(k), m \cdot P)$.

Proof. Assume hP, $hper$. The goal follows from colnegoneperiodicofsectorrightrowagree$(T, P, hP, hper)$. □

Theorem 9.283. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U, if $1 \leq U$, then if $1 \leq d$, then if $T \leq d + U \cdot$* twopow(k)*, then not* sectorrightrowagree$(k, d, U, \text{lcmperiod}(P, k))$.

Proof. Assume hP, $hper$, k, d, U, hU, hd, ht. The goal follows from notsectorrightrowagreelcmperiod$(T, P, hP, hper, k, d, U, hU, hd, ht)$. □

Theorem 9.284. *For every natural number t and natural number k and natural number Q and natural number a and natural number L, if* $\mathrm{twopow}(\mathrm{succ}(k)) \leq a$, *then if* $\mathrm{winwittagree}(t, k, Q, a, L+1)$, *then if* $\mathrm{rowagree}(t, Q, a - \mathrm{twopow}(\mathrm{succ}(k)), L+1)$, *then if* $\mathrm{rowagree}(t, Q, a - \mathrm{twopow}(k), L+1)$, *then if* $\mathrm{rowagree}(t, Q, a, L+1)$, *then if* $\mathrm{wittbitbool}(\mathrm{wittwin}(t+1+Q, \mathrm{twopow}(k)), a+L+2+Q) = \mathrm{wittbitbool}(\mathrm{wittwin}(t+1, \mathrm{twopow}(k)), a+L+2)$, *then* $\mathrm{winwittagree}(t, k, Q, a, L+2)$.

Proof. The goal follows from $\mathrm{winwittagreeextendright}(t, k, Q, a, L)$. □

Theorem 9.285. *For every natural number t and natural number k and natural number Q and natural number a and natural number L, if* $\mathrm{twopow}(\mathrm{succ}(k)) \leq a$, *then if* $\mathrm{rowagree}(t, Q, a - \mathrm{twopow}(\mathrm{succ}(k)), L)$, *then if* $\mathrm{rowagree}(t, Q, a - \mathrm{twopow}(k), L)$, *then if* $\mathrm{rowagree}(t, Q, a, L)$, *then if* $\mathrm{rowagree}(t+\mathrm{twopow}(k), Q, a, L)$, *then* $\mathrm{winwittagree}(t, k, Q, a, L)$.

Proof. The goal follows from $\mathrm{winwittagreeofrowagreethreeandlate}(t, k, Q, a, L)$. □

Theorem 9.286. *For every natural number t and natural number k and natural number Q and natural number a and natural number L, if* $\mathrm{twopow}(\mathrm{succ}(k)) \leq a$, *then if* $\mathrm{winwittagree}(t, k, Q, a, L)$, *then if* $\mathrm{rowagree}(t, Q, a - \mathrm{twopow}(\mathrm{succ}(k)), L)$, *then if* $\mathrm{rowagree}(t, Q, a - \mathrm{twopow}(k), L)$, *then if* $\mathrm{rowagree}(t, Q, a, L)$, *then* $\mathrm{rowagree}(t + \mathrm{twopow}(k), Q, a, L)$.

Proof. The goal follows from $\mathrm{rowagreelateofwinwittagreeandthree}(t, k, Q, a, L)$. □

Theorem 9.287. *For every natural number k and natural number d and natural number U and natural number Q, if* $2 \leq U$, *then if* $\mathrm{sectorwinwittagree}(k, d, U, Q)$, *then for every natural number t, if* $d + U \cdot \mathrm{twopow}(k) \leq t$, *then* $\mathrm{wittresp}(t + Q, k) = \mathrm{wittresp}(t, k)$.

Proof. The goal follows from $\mathrm{wittrespboolperiodicofsectorwinwittagree}(k, d, U, Q)$. □

Theorem 9.288. *For every natural number P and natural number k and natural number j and natural number t, if* $j \leq k$, *then* $\mathrm{wittbitbool}(\mathrm{wittrow}(t+\mathrm{lcmperiod}(P, k)), j) = \mathrm{wittbitbool}(\mathrm{wittrow}(t), j)$.

Proof. The goal follows from $\mathrm{rightboundaryrowlcmperiodeq}(P, k, j, t)$. □

Theorem 9.289. *For every natural number P and natural number k and natural number j and natural number t, if $j \leq k$, then if $j \leq t$, then if* settledby(j, t), *then* wittbitbool(wittrow(t+lcmperiod(P, k)), 2·(t+lcmperiod(P, k))−j) = wittbitbool(wittrow(t), $2 \cdot t - j$).

Proof. The goal follows from leftboundaryrowlcmperiodeq(P, k, j, t). □

Theorem 9.290. *For every natural number P and natural number d and natural number k and natural number u, if d+1 $\leq k$, then* bxor(wittbitbool(frobdefect(k, wittrow(d+1+u·twopow(k)+lcmperiod(P, k))), d+1+twopow(k)), wittbitbool(frobdefect(k, wittrow(d+1+u·twopow(k))), d+1+twopow(k))) = bxor(bxor(rdiag(d+1+twopow(k), d+1 + $u \cdot$ twopow(k) + lcmperiod(P, k)), rdiag($d + 1 +$ twopow(k), $d + 1 + u \cdot$ twopow(k))) and rdiag(d, d+1), center(d+1) and bxor(rdiag(d+twopow(k), d+1+u·twopow(k)+lcmperiod(P, k)), rdiag(d+twopow(k), d+1+u·twopow(k)))).

Proof. The goal follows from frobdefectrowsectoronelcmperiodxor(P, d, k, u). □

Theorem 9.291. *For every natural number t and natural number p, if $p \leq t$, then* rhcenter(t, p) = rdiag$(t - p, t)$.

Proof. The goal follows from rhcentereqrdiag(t, p). □

Theorem 9.292. *For every natural number t and natural number p, if $p \leq t$, then* rhcenter(t, p) = wittbitbool(wittrow(t), $t - p$).

Proof. The goal follows from rhcentereqrowtestbit(t, p). □

Theorem 9.293. *For every natural number t and natural number p, if $p \leq t$, then* lhcenter(t, p) = ldiag$(t - p, t)$.

Proof. The goal follows from lhcentereqldiag(t, p). □

Theorem 9.294. *For every natural number t and natural number p, if $p \leq t$, then* lhcenter(t, p) = wittbitbool(wittrow(t), $t + p$).

Proof. The goal follows from lhcentereqrowtestbit(t, p). □

Theorem 9.295. *For every natural number t,* rhcenter$(t, 0)$ = center(t).

Proof. The goal follows from rhcenterzeroeqcenter(t). □

Theorem 9.296. *For every natural number t,* rhcenter(t, t) = true.

Proof. The goal follows from rhcenterselfeqtrue(t). □

Theorem 9.297. *For every natural number t and natural number p, if $t < p$, then* rhcenter$(t, p) =$ false.

Proof. The goal follows from rhcentergteqfalse(t, p). □

Theorem 9.298. *For every natural number t,* phicenter$(t) =$ bxor(lhcenter$(t, 1)$, center(t) or rhcenter$(t,$

Proof. The goal follows from phicentereqlhcenterrhcenter(t). □

Theorem 9.299. *For every natural number t,* lhcenter$(t, 1) =$ bxor(center$(t+$ $1)$, center(t) or rhcenter$(t, 1)$).

Proof. The goal follows from lhcenteroneeqofcenterfixed(t). □

Theorem 9.300. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then* bxor(lhcenter$(t + m \cdot P, 1)$, lhcenter$(t, 1)) =$ not(center(t)) and bxor(rhcenter$(t + m \cdot P, 1)$, rhcenter$(t, 1)$).

Proof. Assume hP, $hper$. The goal follows from periodiclhcenteronedefecteqnotcenterandrhcenteronedef

□

Theorem 9.301. *For every natural number t and natural number p,* rhcenter$(t+$ $1, p + 1) =$ bxor(rhcenter(t, p), rhcenter$(t, p + 1)$ or rhcenter$(t, p + 2)$).

Proof. The goal follows from rhcenterstepsucc(t, p). □

Theorem 9.302. *For every natural number t and natural number p,* rhcenter$(t, p) =$ bxor(rhcenter$(t + 1, p + 1)$, rhcenter$(t, p + 1)$ or rhcenter$(t, p + 2)$).

Proof. The goal follows from rhcenterinwardstep(t, p). □

Theorem 9.303. *For every natural number t and natural number Q and natural number $p0$ and natural number L, if* rhbandagree$(t, Q, p0{+}1, L{+}1)$*, then if* rhcenter$(t{+}1{+}Q, p0{+}1) =$ rhcenter$(t{+}1, p0{+}1)$*, then* rhbandagree$(t, Q, p0, L+$ $2)$.

Proof. The goal follows from rhbandagreeextendinward$(t, Q, p0, L)$. □

Theorem 9.304. *For every natural number t and natural number Q and natural number $p0$ and natural number L, if* rhbandagree$(t, Q, p0, L + 2)$*, then* rhbandagree$(t + 1, Q, p0 + 1, L)$.

Proof. The goal follows from rhbandagreestep$(t, Q, p0, L)$. □

Theorem 9.305. *For every natural number t and natural number Q and natural number $p0$ and natural number L, if $p0+L \leq t$, then if* rhbandagree$(t, Q, p0, L)$, *then* rowagree$(t, Q, t-(p0+L), L)$.

Proof. The goal follows from rowagreerightofrhbandagree$(t, Q, p0, L)$. □

Theorem 9.306. *For every natural number t and natural number Q and natural number $p0$ and natural number L, if $p0+L \leq t$, then if* rowagree$(t, Q, t-(p0+L), L)$, *then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreeofrowagreeright$(t, Q, p0, L)$. □

Theorem 9.307. *For every natural number k and natural number d and natural number U and natural number Q, if $1 \leq U$, then if $1 \leq d$, then if* sectorrhbandagree(k, d, U, Q), *then* sectorrightrowagree(k, d, U, Q).

Proof. The goal follows from sectorrightrowagreeofrhbandagree(k, d, U, Q). □

Theorem 9.308. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number k and natural number d and natural number U, if $1 \leq U$, then if $1 \leq d$, then if $T \leq d+U \cdot$ twopow(k), then not* sectorrhbandagree$(k, d, U,$ lcmperiod$(P, k))$.

Proof. Assume hP, $hper$, k, d, U, hU, hd, ht. The goal follows from notsectorrhbandagreelcmperiod$(T, P, hP, hper, k, d, U, hU, hd, ht)$. □

Theorem 9.309. *For every natural number P and natural number k and natural number j and natural number t, if $j \leq k$, then if $j \leq t$, then* rhcenter$(t +$ lcmperiod$(P, k), t +$ lcmperiod$(P, k) - j) =$ rhcenter$(t, t - j)$.

Proof. The goal follows from rhcenterrightedgelcmperiodeq(P, k, j, t). □

Theorem 9.310. *For every natural number k and natural number d and natural number U and natural number Q, if $2 \leq U$, then if* sectorwinwittagree(k, d, U, Q), *then* colnegoneperiodic$(d + (U+1) \cdot$ twopow$(k), Q)$.

Proof. The goal follows from colnegoneperiodicofsectorwinwittagree(k, d, U, Q). □

Theorem 9.311. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number k and natural number d and natural number U, if $2 \leq U$, then not* sectorwinwittagree$(k, d, U,$ lcmperiod$(P, k))$.

Proof. Assume hP, $hper$, k, d, U, hU. The goal follows from notsectorwinwittagreelcmperiod(T, P, hP, h □

Theorem 9.312. *For every natural number k and natural number d and natural number U and natural number Q, if $2 \leq U$, then if* sectorwiderowagree(k, d, U, Q), *then* sectorwinwittagree(k, d, U, Q).

Proof. The goal follows from sectorwinwittagreeofsectorwiderowagree(k, d, U, Q). □

Theorem 9.313. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number k and natural number d and natural number U and natural number m, if $2 \leq U$, then if $T \leq d + U \cdot$ twopow(k), then if* sectorwiderightrowagree$(k, d, U, m \cdot P)$, *then* sectorwinwittagree$(k, d, U, m \cdot P)$.

Proof. Assume hP, $hper$. The goal follows from sectorwinwittagreeofsectorwiderightrowagree(T, P, hP, h □

Theorem 9.314. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number k and natural number d and natural number U, if $2 \leq U$, then if $T \leq d + U \cdot$ twopow(k), then not* sectorwiderightrowagree$(k, d, U,$ lcmperiod$(P, k))$.

Proof. Assume hP, $hper$, k, d, U, hU, ht. The goal follows from notsectorwiderightrowagreelcmperiod(T □

Theorem 9.315. *For every natural number t,* lhcenter$(t, 0) =$ center(t).

Proof. The goal follows from lhcenterzeroeqcenter(t). □

Theorem 9.316. *For every natural number t,* lhcenter$(t, t) =$ true.

Proof. The goal follows from lhcenterselfeqtrue(t). □

Theorem 9.317. *For every natural number t and natural number p, if $t < p$, then* lhcenter$(t, p) =$ false.

Proof. The goal follows from lhcentergteqfalse(t, p). □

Theorem 9.318. *For every natural number t and natural number p,* lhcenter$(t+1, p+1) =$ bxor(lhcenter$(t, p+2)$, lhcenter$(t, p+1)$ or lhcenter(t, p)).

Proof. The goal follows from lhcenterstepsucc(t, p). □

Theorem 9.319. *For every natural number t and natural number p,* $\mathrm{lhcenter}(t, p+2) = \mathrm{bxor}(\mathrm{lhcenter}(t+1, p+1), \mathrm{lhcenter}(t, p+1)$ or $\mathrm{lhcenter}(t, p))$.

Proof. The goal follows from $\mathrm{lhcenteroutwardstep}(t, p)$. □

Theorem 9.320. *For every natural number t,* $\mathrm{center}(t) = \mathrm{bxor}(\mathrm{rhcenter}(t+1, 1), \mathrm{rhcenter}(t, 1)$ or $\mathrm{rhcenter}(t, 2))$.

Proof. The goal follows from $\mathrm{centereqrhcenterinwardzero}(t)$. □

Theorem 9.321. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t, if $T \leq t$, then if* $(\mathrm{rhcenter}(t + m \cdot P, 1)$ or $\mathrm{rhcenter}(t + m \cdot P, 2)) = (\mathrm{rhcenter}(t, 1)$ or $\mathrm{rhcenter}(t, 2))$*, then* $\mathrm{rhcenter}(t + 1 + m \cdot P, 1) = \mathrm{rhcenter}(t + 1, 1)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicrhcentersteponeeqoforeq}(T, P, hP, hper)$. □

Theorem 9.322. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t, if $T \leq t$, then if* $\mathrm{rhcenter}(t+1+m \cdot P, 1) = \mathrm{rhcenter}(t+1, 1)$*, then* $(\mathrm{rhcenter}(t+m \cdot P, 1)$ or $\mathrm{rhcenter}(t+m \cdot P, 2)) = (\mathrm{rhcenter}(t, 1)$ or $\mathrm{rhcenter}(t, 2))$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicrhcenteroreqofsteponeeq}(T, P, hP, hper)$. □

Theorem 9.323. *For every natural number t and natural number Q and natural number $p0$ and natural number L and natural number s, if* $\mathrm{rhbandagree}(t, Q, p0, L+2 \cdot s)$*, then* $\mathrm{rhbandagree}(t + s, Q, p0 + s, L)$.

Proof. The goal follows from $\mathrm{rhbandagreesteps}(t, Q, p0, L, s)$. □

Theorem 9.324. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number L, if $T \leq t$, then if* $\mathrm{rhbandagree}(t, m \cdot P, 1, L)$*, then* $\mathrm{rhbandagree}(t, m \cdot P, 0, L + 1)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicrhbandagreezeroofone}(T, P, hP, hper)$. □

Theorem 9.325. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number L, if $T \leq t$, then if* $\mathrm{rhbandagree}(t, m \cdot P, 0, L + 1)$*, then* $\mathrm{rhbandagree}(t + 1, m \cdot P, 0, L)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreezerostep$(T, P, hP, hper)$. □

Theorem 9.326. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number L and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 0, L + s)$*, then* rhbandagree$(t + s, m \cdot P, 0, L)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreezerosteps$(T, P, hP, hper)$. □

Theorem 9.327. *For every natural number t and natural number Q and natural number $p0$ and natural number S and natural number L, if* rhbandagree$(t, Q, p0+S, L+1)$*, then if* rhbandagree$(t+1, Q, p0+1, S)$*, then* rhbandagree$(t, Q, p0, L+1+S)$.

Proof. The goal follows from rhbandagreeextendinwardsteps$(t, Q, p0, S, L)$. □

Theorem 9.328. *For every natural number t and natural number Q and natural number S and natural number L, if* rhbandagree$(t, Q, S, L+1)$*, then if* rhbandagree$(t+1, Q, 1, S)$*, then* rhbandagree$(t, Q, 0, L+1+S)$.

Proof. The goal follows from rhbandagreezeroofedgeandsucc(t, Q, S, L). □

Theorem 9.329. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S and natural number L, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, S+1, L+1)$*, then if* rhbandagree$(t+1, m \cdot P, 2, S)$*, then* rhbandagree$(t, m \cdot P, 0, L+2+S)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreebouncezero$(T, P, hP, hper)$. □

Theorem 9.330. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S and natural number L, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, S+1, L+1)$*, then if* rhbandagree$(t+1, m \cdot P, 2, S)$*, then* rhbandagree$(t+1, m \cdot P, 0, L+1+S)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreebouncesucc$(T, P, hP, hper)$. □

Theorem 9.331. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U and natural number m, if $T \leq d + U \cdot$ twopow(k), then if* sectorrhbandagree$(k, d, U, m \cdot P)$*, then for every natural number t, if $d + U \cdot$ twopow$(k) + 1 \leq t$, then* rhcenter$(t + m \cdot P, 1) =$ rhcenter$(t, 1)$.

Proof. Assume hP, $hper$. The goal follows from rhcenteroneperiodicofsectorrhbandagree$(T, P, hP, hper)$. □

Theorem 9.332. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U and natural number m, if $T \leq d + U \cdot$ twopow(k), then if* sectorrhbandagree$(k, d, U, m \cdot P)$*, then for every natural number t, if $d + U \cdot$ twopow$(k) + 1 \leq t$, then* lhcenter$(t + m \cdot P, 1) =$ lhcenter$(t, 1)$.

Proof. Assume hP, $hper$. The goal follows from lhcenteroneperiodicofsectorrhbandagree$(T, P, hP, hper)$. □

Theorem 9.333. *For every natural number k and natural number t and natural number p and natural number v, if $p \leq t$, then if $t - p \leq k$, then* rhcenter$(t + v \cdot$ twopow$(k), p + v \cdot$ twopow$(k)) =$ rhcenter(t, p).

Proof. The goal follows from rhcenterdiagshifttwopoweq(k, t, p, v). □

Theorem 9.334. *For every natural number j and natural number k and natural number u, if $j \leq k$, then* rhcenter$(j + u \cdot$ twopow$(k), u \cdot$ twopow$(k)) =$ center(j).

Proof. The goal follows from rhcentersectordiageqcenter(j, k, u). □

Theorem 9.335. *For every natural number t and natural number k and natural number p, if $p +$* twopow(succ$(k)) \leq t$*, then* rhcenter$(t +$ twopow$(k), p +$ twopow$(k)) =$ bxor(bxor(bxor(rhcenter(t, p), rhcenter$(t, p+$twopow$(k)))$, rhcenter$(t, p+$ twopow(succ$(k))))$, wittbitbool(wittwin$(t,$ twopow$(k)), t - p)$.

Proof. The goal follows from rhcenterscalestep(t, k, p). □

Theorem 9.336. *For every natural number t and natural number k and natural number p, if $p+$*twopow(succ$(k)) \leq t$*, then* rhcenter$(t, p) =$ bxor(bxor(bxor(rhcenter$(t+$ twopow$(k), p+$twopow$(k))$, rhcenter$(t, p+$twopow$(k)))$, rhcenter$(t, p+$twopow(succ$(k))))$, wittbitbool(wi $p))$.

Proof. The goal follows from rhcenterscaleinwardstep(t, k, p). □

Theorem 9.337. *For every natural number t and natural number Q and natural number $p0$ and natural number L and natural number $p0p$ and natural number Lp, if $p0 \leq p0p$, then if $p0p+Lp \leq p0+L$, then if* rhbandagree$(t, Q, p0, L)$, *then* rhbandagree$(t, Q, p0p, Lp)$.

Proof. The goal follows from rhbandagreemono$(t, Q, p0, L, p0p, Lp)$. □

Theorem 9.338. *For every natural number t and natural number Q and natural number $p0$ and natural number S and natural number $s0$ and natural number Sp, if $s0+Sp \leq S$, then if* rhcolagree$(t, Q, p0, S)$, *then* rhcolagree$(t+s0, Q, p0, Sp)$.

Proof. The goal follows from rhcolagreemono$(t, Q, p0, S, s0, Sp)$. □

Theorem 9.339. *For every natural number t and natural number Q and natural number $p0$ and natural number S and natural number $s0$ and natural number Sp, if $s0 + Sp \leq S$, then if* rhoragree$(t, Q, p0, S)$, *then* rhoragree$(t + s0, Q, p0, Sp)$.

Proof. The goal follows from rhoragreemono$(t, Q, p0, S, s0, Sp)$. □

Theorem 9.340. *For every natural number t and natural number Q and natural number $p0$ and natural number S, if* rhcolagree$(t, Q, p0, S)$, *then if for every natural number s, if $s < S$, then* rhcenter$(t + (s + 1) + Q, p0 + 1) =$ rhcenter$(t + (s + 1), p0 + 1)$, *then* rhoragree$(t, Q, p0 + 1, S)$.

Proof. The goal follows from rhoragreeofcolagree$(t, Q, p0, S)$. □

Theorem 9.341. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number S, if $T \leq t$, then if for every natural number s, if $s < S$, then* rhcenter$(t+(s+1)+m \cdot P, 1) =$ rhcenter$(t+(s+1), 1)$, *then* rhoragree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhoragreeoneofcolone$(T, P, hP, hper)$. □

Theorem 9.342. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number S, if $T \leq t$, then if* rhoragree$(t, m \cdot P, 1, S)$, *then for every natural number s, if $s < S$, then* rhcenter$(t+(s+1)+m \cdot P, 1) =$ rhcenter$(t + (s + 1), 1)$.

Proof. Assume hP, *hper*. The goal follows from periodiccoloneofrhoragreeone$(T, P, hP, hper)$. □

Theorem 9.343. *For every natural number* t *and natural number* Q *and natural number* $p0$ *and natural number* S*, if* rhcolagree$(t, Q, p0 + 1, S + 1)$*, then if* rhoragree$(t, Q, p0 + 1, S + 1)$*, then* rhcolagree$(t, Q, p0, S)$ *and* rhoragree$(t, Q, p0, S)$.

Proof. The goal follows from rhcenterinwarddeterminestep$(t, Q, p0, S)$. □

Theorem 9.344. *For every natural number* t *and natural number* Q *and natural number* $p0$ *and natural number* S *and natural number* K*, if* rhcolagree$(t, Q, p0 + K, S + K)$*, then if* rhoragree$(t, Q, p0 + K, S + K)$*, then* rhcolagree$(t, Q, p0, S)$ *and* rhoragree$(t, Q, p0, S)$.

Proof. The goal follows from rhcenterinwarddetermine$(t, Q, p0, S, K)$. □

Theorem 9.345. *For every natural number* t *and natural number* Q *and natural number* $p0$ *and natural number* L*, if* rhcolagree$(t, Q, p0 + L, L)$*, then if* rhoragree$(t, Q, p0 + L, L)$*, then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreeofinwarddetermine$(t, Q, p0, L)$. □

Theorem 9.346. *For every natural number* t *and natural number* Q *and natural number* $p0$ *and natural number* L*, if* $t + Q < p0$*, then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreeoutsidecone$(t, Q, p0, L)$. □

Theorem 9.347. *For every natural number* t *and natural number* k *and natural number* Q *and natural number* $p0$ *and natural number* L*, if* $p0 + L +$ twopow(succ$(k)) \leq t$*, then if* rhbandagree$(t, Q, p0, L)$*, then if* rhbandagree$(t, Q, p0 +$ twopow$(k), L)$*, then if* rhbandagree$(t, Q, p0 +$ twopow(succ$(k)), L)$*, then if* winwittagree$(t, k, Q, t-(p0+L), L)$*, then* rhbandagree$(t+$twopow$(k), Q, p0+$ twopow$(k), L)$.

Proof. The goal follows from rhbandagreescalestep$(t, k, Q, p0, L)$. □

Theorem 9.348. *For every natural number* t *and natural number* k *and natural number* Q *and natural number* $p0$ *and natural number* L*, if* $p0 + L +$ twopow(succ$(k)) \leq t$*, then if* rhbandagree$(t+$twopow$(k), Q, p0+$twopow$(k), L)$*, then if* rhbandagree$(t, Q, p0 +$ twopow$(k), L)$*, then if* rhbandagree$(t, Q, p0 +$ twopow(succ$(k)), L)$*, then if* winwittagree$(t, k, Q, t-(p0+L), L)$*, then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreescaleinwardstep$(t, k, Q, p0, L)$. □

Theorem 9.349. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $p0 + L +$ twopow(succ(k)) $\leq t$, then if* rhbandagree($t, Q, p0, L$)*, then if* rhbandagree($t, Q, p0+$ twopow(k), L)*, then if* rhbandagree($t, Q, p0 +$ twopow(succ(k)), L)*, then if* rhbandagree($t+$twopow(k), $Q, p0+$twopow(k), L)*, then* winwittagree($t, k, Q, t-(p0 + L), L$).

Proof. The goal follows from winwittagreeofrhbandagreescale($t, k, Q, p0, L$). □

Theorem 9.350. *For every natural number k and natural number d and natural number U and natural number Q, if for every natural number u, if $U \leq u$, then* rhcolagree($d+u \cdot$twopow(k)$+1, Q,$ twopow(k)$+1,$ twopow(k)$+1$)*, then if for every natural number u, if $U \leq u$, then* rhoragree($d + u \cdot$ twopow(k)$+1, Q,$ twopow(k)$+1,$ twopow(k)$+1$)*, then* sectorrhbandagree(k, d, U, Q).

Proof. The goal follows from sectorrhbandagreeofinwarddetermine(k, d, U, Q). □

Theorem 9.351. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U, if $1 \leq U$, then if $1 \leq d$, then if $T \leq d + U \cdot$ twopow(k)*, then if for every natural number u, if $U \leq u$, then* rhcolagree($d+u \cdot$twopow(k)$+1,$ lcmperiod(P, k)$+1,$ twopow(k)$+1,$ twopow(k)$+1$)*, then not for every natural number u, if $U \leq u$, then* rhoragree($d + u \cdot$ twopow(k) $+ 1,$ lcmperiod(P, k)$,$ twopow(k) $+ 1,$ twopow(k) $+ 1$).

Proof. Assume hP, $hper$, k, d, U, hU, hd, ht, $hcol$. The goal follows from notsectorinwarddeterminelcmperiod($T, P, hP, hper, k, d, U, hU, hd, ht, hcol$). □

Theorem 9.352. *For every natural number t and natural number k and natural number p, if $p \leq t$, then* rhcenter($t +$ twopow(k)$, p +$ twopow(k)$) =$ bxor(bxor(bxor(rhcenter(t, p), rhcenter($t, p+$twopow(k))), rhcenter($t, p+$twopow(succ(k)))), wittbitbool p)).

Proof. The goal follows from rhcenterscalesteple(t, k, p). □

Theorem 9.353. *For every natural number t and natural number k and natural number p, if $p \leq t$, then* rhcenter(t, p) = bxor(bxor(bxor(rhcenter($t+$ twopow(k)$, p+$twopow(k)), rhcenter($t, p+$twopow(k))), rhcenter($t, p+$twopow(succ(k)))), wittbitbool(wi p)).

Proof. The goal follows from rhcenterscaleinwardsteple(t, k, p). □

Theorem 9.354. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $p0 + L \leq t$, then if* rhbandagree$(t, Q, p0, L)$*, then if* rhbandagree$(t, Q, p0+\text{twopow}(k), L)$*, then if* rhbandagree$(t, Q, p0+\text{twopow}(\text{succ}(k)), L)$*, then if* winwittagree$(t, k, Q, t-(p0 + L), L)$*, then* rhbandagree$(t + \text{twopow}(k), Q, p0 + \text{twopow}(k), L)$.

Proof. The goal follows from rhbandagreescalesteple$(t, k, Q, p0, L)$. □

Theorem 9.355. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $p0 + L \leq t$, then if* rhbandagree$(t + \text{twopow}(k), Q, p0 + \text{twopow}(k), L)$*, then if* rhbandagree$(t, Q, p0+\text{twopow}(k), L)$*, then if* rhbandagree$(t, Q, p0+\text{twopow}(\text{succ}(k)), L)$*, then if* winwittagree$(t, k, Q, t - (p0 + L), L)$*, then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreescaleinwardsteple$(t, k, Q, p0, L)$. □

Theorem 9.356. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $p0 + L \leq t$, then if* rhbandagree$(t, Q, p0, L)$*, then if* rhbandagree$(t, Q, p0+\text{twopow}(k), L)$*, then if* rhbandagree$(t, Q, p0 + \text{twopow}(\text{succ}(k)), L)$*, then if* rhbandagree$(t + \text{twopow}(k), Q, p0+\text{twopow}(k), L)$*, then* winwittagree$(t, k, Q, t-(p0+L), L)$.

Proof. The goal follows from winwittagreeofrhbandagreescalele$(t, k, Q, p0, L)$. □

Theorem 9.357. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number M, if $p0 + M \cdot \text{twopow}(k) + L \leq t$, then if* rhbandagree$(t, Q, p0 + M \cdot \text{twopow}(k), L)$*, then if* rhbandagree$(t, Q, p0 + (M + 1) \cdot \text{twopow}(k), L)$*, then if for every natural number m, if $m < M$, then* rhbandagree$(t + \text{twopow}(k), Q, p0 + (m+1) \cdot \text{twopow}(k), L)$*, then if for every natural number m, if $m < M$, then* winwittagree$(t, k, Q, t - (p0 + m \cdot \text{twopow}(k) + L), L)$*, then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreescaleinwardsteps$(t, k, Q, p0, L, M)$. □

Theorem 9.358. *For every natural number t and natural number p, if* rhcenter$(t, p + 1) = \text{true}$*, then* rhcenter$(t, p) = \text{not}(\text{rhcenter}(t + 1, p + 1))$.

Proof. The goal follows from rhcenterinwardofblackleft(t, p). □

Theorem 9.359. *For every natural number t and natural number p, if* rhcenter$(t, p + 2) = \text{true}$*, then* rhcenter$(t, p) = \text{not}(\text{rhcenter}(t + 1, p + 1))$.

Proof. The goal follows from rhcenterinwardofblackright(t, p). □

Theorem 9.360. *For every natural number t and natural number Q and natural number p, if* rhcenter$(t + Q, p + 1) =$ true*, then if* rhcenter$(t, p + 1) =$ true*, then if* rhcenter$(t + 1 + Q, p + 1) =$ rhcenter$(t + 1, p + 1)$*, then* rhcenter$(t + Q, p) =$ rhcenter(t, p).

Proof. The goal follows from rhcenteragreeofblackleft(t, Q, p). □

Theorem 9.361. *For every natural number t and natural number p, if* rhcenter$(t + 1, p + 1) =$ rhcenter(t, p)*, then* rhcenter$(t, p + 1) =$ false *and* rhcenter$(t, p + 2) =$ false.

Proof. The goal follows from rhcenterwhitepairofdiageq(t, p). □

Theorem 9.362. *For every natural number t and natural number p, if* rhcenter$(t, p+1) =$ false*, then if* rhcenter$(t, p+2) =$ false*, then* rhcenter$(t + 1, p + 1) =$ rhcenter(t, p).

Proof. The goal follows from rhcenterdiageqofwhitepair(t, p). □

Theorem 9.363. *For every natural number k and natural number J and natural number p and natural number s and natural number u, if* rightthreshold(k, J)*, then if $s \leq J$, then* rhcenter$(p + s + u \cdot$ twopow$(k), p + u \cdot$ twopow$(k)) =$ rhcenter$(p + s, p)$.

Proof. The goal follows from rhcenterdiagshifttwopoweqofthreshold(k, J, p, s, u). □

Theorem 9.364. *For every natural number k and natural number J and natural number s and natural number u, if* rightthreshold(k, J)*, then if $s \leq J$, then* rhcenter$(s + u \cdot$ twopow$(k), u \cdot$ twopow$(k)) =$ center(s).

Proof. The goal follows from rhcentersectordiageqcenterofthreshold(k, J, s, u). □

Theorem 9.365. *For every natural number k and natural number J and natural number u and natural number j, if* rightthreshold(k, J)*, then if $1 \leq j$, then if $j \leq J$, then if $j \leq u \cdot$ twopow(k), then* rhcenter$(u \cdot$ twopow$(k), u \cdot$ twopow$(k) - j) =$ false.

Proof. The goal follows from rhcenterzerogapofthreshold(k, J, u, j). □

Theorem 9.366. *For every natural number k and natural number J and natural number u and natural number j, if* rightthreshold(k, J)*, then if* $j \leq J$*, then if* $j \leq (u+1) \cdot \text{twopow}(k) - 1$*, then* rhcenter$((u+1) \cdot \text{twopow}(k) - 1, (u+1) \cdot \text{twopow}(k) - 1 - j) = \text{true}$.

Proof. The goal follows from rhcenterlastoneofthreshold(k, J, u, j). □

Theorem 9.367. *For every natural number t and natural number k and natural number p,* rhcenter$(t+\text{twopow}(k), p+\text{twopow}(k)) = \text{bxor}(\text{bxor}(\text{bxor}(\text{rhcenter}(t, p), \text{rhcenter}(t, p+\text{twopow}(k))), \text{rhcenter}(t, p+\text{twopow}(\text{succ}(k)))), \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t-p))$.

Proof. The goal follows from rhcenterscalestepuncond(t, k, p). □

Theorem 9.368. *For every natural number t and natural number k and natural number p,* rhcenter$(t, p) = \text{bxor}(\text{bxor}(\text{bxor}(\text{rhcenter}(t + \text{twopow}(k), p + \text{twopow}(k)), \text{rhcenter}(t, p+\text{twopow}(k))), \text{rhcenter}(t, p+\text{twopow}(\text{succ}(k)))), \text{wittbitbool}(\text{wittwin}(t, \text{twopo}$ $p))$.

Proof. The goal follows from rhcenterscaleinwardstepuncond(t, k, p). □

Theorem 9.369. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $p0 + L \leq t$, then if* winwittagree$(t, k, Q, t-(p0+L), L)$*, then* rhwinwittagree$(t, k, Q, p0, L)$.

Proof. The goal follows from rhwinwittagreeofwinwittagree$(t, k, Q, p0, L)$. □

Theorem 9.370. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q \leq p0$, then* rhwinwittagree$(t, k, Q, p0, L)$.

Proof. The goal follows from rhwinwittagreeoutsidecone$(t, k, Q, p0, L)$. □

Theorem 9.371. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* rhbandagree$(t+\text{twopow}(k), Q, p0+\text{twopow}(k), L)$*, then if* rhbandagree$(t, Q, p0+\text{twopow}(k), L)$*, then if* rhbandagree$(t, Q, p0+\text{twopow}(\text{succ}(k)), L)$*, then if* rhwinwittagree$(t, k, Q, p0, L)$*, then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreescaleinwardstepuncond$(t, k, Q, p0, L)$. □

Theorem 9.372. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number M, if* $\mathrm{rhbandagree}(t, Q, p0+M\cdot\mathrm{twopow}(k), L)$*, then if* $\mathrm{rhbandagree}(t, Q, p0+(M+1)\cdot\mathrm{twopow}(k), L)$*, then if for every natural number m, if $m < M$, then* $\mathrm{rhbandagree}(t+\mathrm{twopow}(k), Q, p0+(m+1)\cdot\mathrm{twopow}(k), L)$*, then if for every natural number m, if $m < M$, then* $\mathrm{rhwinwittagree}(t, k, Q, p0+m\cdot\mathrm{twopow}(k), L)$*, then* $\mathrm{rhbandagree}(t, Q, p0, L)$.

Proof. The goal follows from $\mathrm{rhbandagreescaleinwardstepsuncond}(t, k, Q, p0, L, M)$. □

Theorem 9.373. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number M, if $t+Q < p0+M\cdot\mathrm{twopow}(k)$, then if for every natural number m, if $m < M$, then* $\mathrm{rhbandagree}(t+\mathrm{twopow}(k), Q, p0+(m+1)\cdot\mathrm{twopow}(k), L)$*, then if for every natural number m, if $m < M$, then* $\mathrm{rhwinwittagree}(t, k, Q, p0+m\cdot\mathrm{twopow}(k), L)$*, then* $\mathrm{rhbandagree}(t, Q, p0, L)$.

Proof. The goal follows from $\mathrm{rhbandagreeofscaleinwardcone}(t, k, Q, p0, L, M)$. □

Theorem 9.374. *For every natural number t and natural number p,* $\mathrm{bxor}(\mathrm{rhcenter}(t+1, p+1), \mathrm{rhcenter}(t, p)) = (\mathrm{rhcenter}(t, p+1)\ \mathrm{or}\ \mathrm{rhcenter}(t, p+2))$.

Proof. The goal follows from $\mathrm{rhcenterdiagdiffeqor}(t, p)$. □

Theorem 9.375. *For every natural number t and natural number Q and natural number p,* $\mathrm{bxor}(\mathrm{bxor}(\mathrm{rhcenter}(t+1+Q, p+1), \mathrm{rhcenter}(t+1, p+1)), \mathrm{bxor}(\mathrm{rhcenter}(t+Q, p), \mathrm{rhcenter}(t, p))) = \mathrm{bxor}(\mathrm{rhcenter}(t+Q, p+1)\ \mathrm{or}\ \mathrm{rhcenter}(t+Q, p+2), \mathrm{rhcenter}(t, p+1)\ \mathrm{or}\ \mathrm{rhcenter}(t, p+2))$.

Proof. The goal follows from $\mathrm{rhcenterdefectdiagstep}(t, Q, p)$. □

Theorem 9.376. *For every natural number t and natural number Q and natural number p, if* $\mathrm{rhcenter}(t+Q, p+1) = \mathrm{rhcenter}(t, p+1)$*, then* $\mathrm{bxor}(\mathrm{bxor}(\mathrm{rhcenter}(t+1+Q, p+1), \mathrm{rhcenter}(t+1, p+1)), \mathrm{bxor}(\mathrm{rhcenter}(t+Q, p), \mathrm{rhcenter}(t, p))) = \mathrm{not}(\mathrm{rhcenter}(t, p+1))\ \mathrm{and}\ \mathrm{bxor}(\mathrm{rhcenter}(t+Q, p+2), \mathrm{rhcenter}(t, p+2))$.

Proof. The goal follows from $\mathrm{rhcenterdefectdiagstepofcoleq}(t, Q, p)$. □

Theorem 9.377. *For every natural number t and natural number Q and natural number p, if* $\mathrm{rhcenter}(t+Q, p+1) = \mathrm{rhcenter}(t, p+1)$*, then if* $\mathrm{rhcenter}(t+1+Q, p+1) = \mathrm{rhcenter}(t+1, p+1)$*, then* $\mathrm{bxor}(\mathrm{rhcenter}(t+Q, p), \mathrm{rhcenter}(t, p)) = \mathrm{not}(\mathrm{rhcenter}(t, p+1))\ \mathrm{and}\ \mathrm{bxor}(\mathrm{rhcenter}(t+Q, p+2), \mathrm{rhcenter}(t, p+2))$.

Proof. The goal follows from rhcenterdefectinwardofcoleqtwo(t, Q, p). □

Theorem 9.378. *For every natural number t and natural number p, if* rhcenter$(t, p+1) =$ false*, then* rhcenter$(t, p+2) =$ bxor(rhcenter$(t+1, p+1)$, rhcenter(t, p))*.*

Proof. The goal follows from rhcenteroutwardofwhitemid(t, p). □

Theorem 9.379. *For every natural number t and natural number Q and natural number p, if* rhcenter$(t, p+1) =$ false*, then if* rhcenter$(t+Q, p) =$ rhcenter(t, p)*, then if* rhcenter$(t+Q, p+1) =$ rhcenter$(t, p+1)$*, then if* rhcenter$(t+1+Q, p+1) =$ rhcenter$(t+1, p+1)$*, then* rhcenter$(t+Q, p+2) =$ rhcenter$(t, p+2)$*.*

Proof. The goal follows from rhcenteragreeoutwardofwhitemid(t, Q, p). □

Theorem 9.380. *For every natural number t and natural number Q and natural number p, if* rhcenter$(t, p+1) =$ false*, then if* rhcenter$(t+Q, p+1) =$ rhcenter$(t, p+1)$*, then if* rhcenter$(t+1+Q, p+1) =$ rhcenter$(t+1, p+1)$*, then* bxor(rhcenter$(t+Q, p)$, rhcenter(t, p)) = bxor(rhcenter$(t+Q, p+2)$, rhcenter$(t, p+2)$)*.*

Proof. The goal follows from rhcenterdefectinwardofwhitemid(t, Q, p). □

Theorem 9.381. *For every natural number t and natural number Q and natural number p, if* rhcenter$(t+Q, p+1) =$ rhcenter$(t, p+1)$*, then if* rhcenter$(t+1+Q, p+1) =$ rhcenter$(t+1, p+1)$*, then if* not(rhcenter$(t, p+1)$) and bxor(rhcenter$(t+Q, p+2)$, rhcenter$(t, p+2)$) = false*, then* rhcenter$(t+Q, p) =$ rhcenter(t, p)*.*

Proof. The goal follows from rhcenteragreeinwardofcoleqtwo(t, Q, p). □

Theorem 9.382. *For every natural number t and natural number Q and natural number L, if $0 < Q$, then if $t + Q \leq L$, then not* lhbandagree$(t, Q, 0, L)$*.*

Proof. The goal follows from notlhbandagreeofltadd(t, Q, L). □

Theorem 9.383. *For every natural number t and natural number Q and natural number $p0$ and natural number S, if* lhcolagree$(t, Q, p0, S+1)$*, then if* lhcolagree$(t, Q, p0+1, S+1)$*, then* lhcolagree$(t, Q, p0+2, S)$*.*

Proof. The goal follows from lhcenteroutwarddeterminestep$(t, Q, p0, S)$. □

Theorem 9.384. *For every natural number t and natural number Q and natural number $p0$ and natural number S and natural number K, if* lhcolagree$(t, Q, p0, S+K)$*, then if* lhcolagree$(t, Q, p0+1, S+K)$*, then* lhcolagree$(t, Q, p0+K, S)$ *and* lhcolagree$(t, Q, p0+K+1, S)$.

Proof. The goal follows from lhcenteroutwarddetermine$(t, Q, p0, S, K)$. □

Theorem 9.385. *For every natural number t and natural number Q and natural number $p0$ and natural number L, if* lhcolagree$(t, Q, p0, L)$*, then if* lhcolagree$(t, Q, p0+1, L)$*, then* lhbandagree$(t, Q, p0, L+1)$.

Proof. The goal follows from lhbandagreeofoutwarddetermine$(t, Q, p0, L)$. □

Theorem 9.386. *For every natural number t and natural number Q and natural number S, if $0 < Q$, then if $t+Q \leq S+1$, then if* lhcolagree$(t, Q, 0, S)$*, then not* lhcolagree$(t, Q, 1, S)$.

Proof. The goal follows from notlhcolagreezerooneofpos(t, Q, S). □

Theorem 9.387. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then* lhcolagree$(t, m \cdot P, 0, S)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhcolagreezero$(T, P, hP, hper)$. □

Theorem 9.388. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $0 < m$, then if $T \leq t$, then if $t + m \cdot P \leq S + 1$, then not* lhcolagree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, $hper$. The goal follows from notperiodiclhcolagreeonehorizon$(T, P, hP, hper)$. □

Theorem 9.389. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if* rhcolagree$(t, m \cdot P, 1, S)$*, then* lhcolagree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhcolagreeoneofrhcolagreeone$(T, P, hP, hper)$. □

Theorem 9.390. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $0 < m$, then if $T \leq t$, then if $t + m \cdot P \leq S + 1$, then not* rhcolagree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, $hper$. The goal follows from notperiodicrhcolagreeonehorizon$(T, P, hP, hper)$. □

Theorem 9.391. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 0, S + 1)$*, then* rhcolagree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhcolagreeoneofrhbandagreezero$(T, P, hP, hper)$ □

Theorem 9.392. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U and natural number M and natural number m, if $1 \leq M$, then if $T \leq d + U \cdot$ twopow(k), then if for every natural number j, if $j < M$, then* rhbandagree$(d + (U + j) \cdot \text{twopow}(k) + 1, m \cdot P, 0, \text{twopow}(k) + 1)$*, then* rhcolagree$(d + U \cdot \text{twopow}(k) + 1, m \cdot P, 1, M \cdot \text{twopow}(k))$.

Proof. Assume hP, $hper$. The goal follows from periodicrhcolagreeoneoffinitesectors$(T, P, hP, hper)$. □

Theorem 9.393. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U and natural number M and natural number m, if $1 \leq M$, then if $0 < m$, then if $T \leq d + U \cdot \text{twopow}(k)$, then if $d + U \cdot \text{twopow}(k) + m \cdot P \leq M \cdot \text{twopow}(k)$, then not for every natural number j, if $j < M$, then* rhbandagree$(d + (U + j) \cdot \text{twopow}(k) + 1, m \cdot P, 0, \text{twopow}(k) + 1)$.

Proof. Assume hP, $hper$. The goal follows from notperiodicfinitesectorrhbandagreemul$(T, P, hP, hper)$. □

Theorem 9.394. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k and natural number d and natural number U and natural number M, if $1 \leq M$, then if $T \leq d + U \cdot \text{twopow}(k)$, then if $d + U \cdot \text{twopow}(k) + \text{lcmperiod}(P, k) \leq M \cdot \text{twopow}(k)$, then not for every natural number j, if $j < M$, then* rhbandagree$(d + (U + j) \cdot \text{twopow}(k) + 1, \text{lcmperiod}(P, k), 0, \text{twopow}(k) + 1)$.

Proof. Assume hP, *hper*. The goal follows from notperiodicfinitesectorrhbandagreelcmperiod(T, P, hP, h □

Theorem 9.395. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then* bxor(lhcenter$(t + m \cdot P, 1)$ or lhcenter$(t + m \cdot P, 0)$, lhcenter$(t, 1)$ or lhcenter$(t, 0)$) = bxor(lhcenter$(t+m \cdot P, 1)$, lhcenter$(t, 1)$).

Proof. Assume hP, *hper*. The goal follows from periodiclhcenterorzerodefecteq($T, P, hP, hper$). □

Theorem 9.396. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then* bxor(lhcenter$(t + m \cdot P, 2)$, lhcenter$(t, 2)$) = bxor(bxor(lhcenter$(t + 1 + m \cdot P, 1)$, lhcenter$(t + 1, 1)$), bxor(lhcenter$(t + m \cdot P, 1)$, lhcenter$(t, 1)$)).

Proof. Assume hP, *hper*. The goal follows from periodiclhcentertwodefecteq($T, P, hP, hper$). □

Theorem 9.397. *For every natural number t and natural number p,* lhcenter(t, p) = wittbitbool(wittrow$(t), t + p$).

Proof. The goal follows from lhcentereqrowtestbituncond(t, p). □

Theorem 9.398. *For every natural number t and natural number k and natural number p,* lhcenter$(t+$twopow$(k), p+$twopow$(k))$ = bxor(bxor(bxor(lhcenter(t, p), lhcenter$(t, p+$ twopow$(k)))$, lhcenter$(t, p+$twopow$(k+1)))$, wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $p +$ twopow$(k + 1)))$.

Proof. The goal follows from lhcenterscalestepuncond(t, k, p). □

Theorem 9.399. *For every natural number t and natural number k and natural number p,* lhcenter$(t, p+$twopow$(k+1))$ = bxor(bxor(bxor(lhcenter$(t+$ twopow$(k), p+$twopow$(k))$, lhcenter$(t, p))$, lhcenter$(t, p+$twopow$(k)))$, wittbitbool(wittwin$(t,$ twopow(k) $p +$ twopow$(k + 1)))$.

Proof. The goal follows from lhcenterscaleoutwardstepuncond(t, k, p). □

Theorem 9.400. *For every natural number t and natural number k and natural number r, if $r \leq$* twopow(k)*, then* lhcenter$(t +$ twopow$(k),$ twopow$(k) -$ $r)$ = bxor(bxor(bxor(rhcenter(t, r), lhcenter$(t,$ twopow$(k)-r))$, lhcenter$(t,$ twopow$(k+$ $1) - r))$, wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k + 1) - r))$.

Proof. The goal follows from lhcenterscalecrossstep(t, k, r). □

Theorem 9.401. *For every natural number t and natural number k and natural number r, if $r \leq$ twopow(k), then* rhcenter$(t, r) =$ bxor(bxor(bxor(lhcenter$(t+$ twopow(k), twopow$(k)-r)$, lhcenter$(t,$ twopow$(k)-r))$, lhcenter$(t,$ twopow$(k+$ $1)-r))$, wittbitbool(wittwin$(t,$ twopow$(k))$, $t +$ twopow$(k+1)-r))$.

Proof. The goal follows from rhcentereqlhcenterscalecross(t, k, r). □

Theorem 9.402. *For every natural number t and natural number k and natural number Q and natural number L, if $L \leq$ twopow(k), then if* lhbandagree$(t+$ twopow$(k), Q,$ twopow$(k)-L, L)$, *then if* lhbandagree$(t, Q,$ twopow$(k)-L, L)$, *then if* lhbandagree$(t, Q,$ twopow$(k+1)-L, L)$, *then if* winwittagree$(t, k, Q, t+$ twopow$(k+1)-L, L)$, *then* rhbandagree$(t, Q, 0, L)$.

Proof. The goal follows from rhbandagreeoflhbandagreescalecross(t, k, Q, L). □

Theorem 9.403. *For every natural number t and natural number k and natural number Q and natural number L, if $L \leq$ twopow(k), then if* rhbandagree$(t, Q, 0, L)$, *then if* lhbandagree$(t+$twopow$(k), Q,$ twopow$(k)-L, L)$, *then if* lhbandagree$(t, Q,$ twopow$(k)-$ $L, L)$, *then if* lhbandagree$(t, Q,$ twopow$(k+1)-L, L)$, *then* winwittagree$(t, k, Q, t+$ twopow$(k+1)-L, L)$.

Proof. The goal follows from winwittagreeofrhbandagreelhbanscalecross(t, k, Q, L). □

Theorem 9.404. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* lhcolagree$(t, m \cdot P, 1,$ twopow$(k+1))$, *then if* winwittagree$(t, k, m \cdot P, t+$twopow$(k),$ twopow$(k))$, *then* rhbandagree$(t, m \cdot P, 0,$ twopow$(k))$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreeoflhcolagreeoneandwinwittagree(T, □

Theorem 9.405. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* lhcolagree$(t, m \cdot P, 1,$ twopow$(k+1))$, *then if* rhbandagree$(t, m \cdot P, 0,$ twopow$(k))$, *then* winwittagree$(t, k, m \cdot P, t +$ twopow$(k),$ twopow$(k))$.

Proof. Assume hP, $hper$. The goal follows from periodicwinwittagreeoflhcolagreeoneandrhbandagree(T, □

Theorem 9.406. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* $\mathrm{lhbandagree}(t, Q, p0, L)$, *then if* $\mathrm{lhbandagree}(t, Q, p0 + \mathrm{twopow}(k), L)$, *then if* $\mathrm{lhbandagree}(t, Q, p0 + \mathrm{twopow}(k+1), L)$, *then if* $\mathrm{winwittagree}(t, k, Q, t + p0 + \mathrm{twopow}(k+1), L)$, *then* $\mathrm{lhbandagree}(t + \mathrm{twopow}(k), Q, p0 + \mathrm{twopow}(k), L)$.

Proof. The goal follows from $\mathrm{lhbandagreescalestepuncond}(t, k, Q, p0, L)$. □

Theorem 9.407. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* $\mathrm{lhbandagree}(t + \mathrm{twopow}(k), Q, p0 + \mathrm{twopow}(k), L)$, *then if* $\mathrm{lhbandagree}(t, Q, p0, L)$, *then if* $\mathrm{lhbandagree}(t, Q, p0 + \mathrm{twopow}(k), L)$, *then if* $\mathrm{winwittagree}(t, k, Q, t + p0 + \mathrm{twopow}(k+1), L)$, *then* $\mathrm{lhbandagree}(t, Q, p0 + \mathrm{twopow}(k+1), L)$.

Proof. The goal follows from $\mathrm{lhbandagreescaleoutwardstepuncond}(t, k, Q, p0, L)$. □

Theorem 9.408. *For every natural number t and natural number p,* $\mathrm{bxor}(\mathrm{lhcenter}(t+1, p+1), \mathrm{lhcenter}(t, p+2)) = (\mathrm{lhcenter}(t, p+1) \operatorname{or} \mathrm{lhcenter}(t, p))$.

Proof. The goal follows from $\mathrm{lhcenterdiagdiffeqor}(t, p)$. □

Theorem 9.409. *For every natural number t and natural number Q and natural number p,* $\mathrm{bxor}(\mathrm{lhcenter}(t+Q, p+2), \mathrm{lhcenter}(t, p+2)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{lhcenter}(t+1+Q, p+1), \mathrm{lhcenter}(t+1, p+1)), \mathrm{bxor}(\mathrm{lhcenter}(t+Q, p+1) \operatorname{or} \mathrm{lhcenter}(t+Q, p), \mathrm{lhcenter}(t, p+1) \operatorname{or} \mathrm{lhcenter}(t, p)))$.

Proof. The goal follows from $\mathrm{lhcenterdefectoutwardstep}(t, Q, p)$. □

Theorem 9.410. *For every natural number t and natural number Q and natural number p, if* $\mathrm{lhcenter}(t+Q, p) = \mathrm{lhcenter}(t, p)$, *then* $\mathrm{bxor}(\mathrm{lhcenter}(t+Q, p+2), \mathrm{lhcenter}(t, p+2)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{lhcenter}(t+1+Q, p+1), \mathrm{lhcenter}(t+1, p+1)), \mathrm{not}(\mathrm{lhcenter}(t, p)) \operatorname{and} \mathrm{bxor}(\mathrm{lhcenter}(t+Q, p+1), \mathrm{lhcenter}(t, p+1)))$.

Proof. The goal follows from $\mathrm{lhcenterdefectoutwardofcoleq}(t, Q, p)$. □

Theorem 9.411. *For every natural number t and natural number Q and natural number p, if* $\mathrm{lhcenter}(t + Q, p) = \mathrm{true}$, *then if* $\mathrm{lhcenter}(t, p) = \mathrm{true}$, *then* $\mathrm{bxor}(\mathrm{lhcenter}(t + Q, p + 2), \mathrm{lhcenter}(t, p + 2)) = \mathrm{bxor}(\mathrm{lhcenter}(t + 1 + Q, p + 1), \mathrm{lhcenter}(t + 1, p + 1))$.

Proof. The goal follows from $\mathrm{lhcenterdefectoutwardofblack}(t, Q, p)$. □

Theorem 9.412. *For every natural number t and natural number k,* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k+1)) = \mathrm{false}$.

Proof. The goal follows from winwitttestbitleftedge(t, k). □

Theorem 9.413. *For every natural number t and natural number k and natural number p, if $t < p$, then* wittbitbool(wittwin$(t,$ twopow$(k)), t + p +$ twopow$(k + 1)) =$ false.

Proof. The goal follows from winwitttestbitgtleftedge(t, k, p). □

Theorem 9.414. *For every natural number t and natural number k and natural number p, if $t \leq p$, then* wittbitbool(wittwin$(t,$ twopow$(k)), t + p +$ twopow$(k + 1)) =$ false.

Proof. The goal follows from winwitttestbitgeleftedge(t, k, p). □

Theorem 9.415. *For every natural number t and natural number k and natural number j, if $j \leq k$, then if $j \leq t$, then if* settledby(j, t), *then* wittbitbool(wittwin$(t,$ twopow$(k)), t + (t - j) +$ twopow$(k + 1)) =$ false.

Proof. The goal follows from winwitttestbitsetledleft(t, k, j). □

Theorem 9.416. *For every natural number t and natural number Q and natural number $p0$ and natural number L, if $t+Q < p0$, then* lhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from lhbandagreeoutsidecone$(t, Q, p0, L)$. □

Theorem 9.417. *For every natural number t and natural number Q and natural number $p0$ and natural number L and natural number $p0p$ and natural number Lp, if $p0 \leq p0p$, then if $p0p+Lp \leq p0+L$, then if* lhbandagree$(t, Q, p0, L)$, *then* lhbandagree$(t, Q, p0p, Lp)$.

Proof. The goal follows from lhbandagreemono$(t, Q, p0, L, p0p, Lp)$. □

Theorem 9.418. *For every natural number t and natural number Q and natural number $p0$ and natural number S and natural number $s0$ and natural number Sp, if $s0+Sp \leq S$, then if* lhcolagree$(t, Q, p0, S)$, *then* lhcolagree$(t+s0, Q, p0, Sp)$.

Proof. The goal follows from lhcolagreemono$(t, Q, p0, S, s0, Sp)$. □

Theorem 9.419. *For every natural number t and natural number Q and natural number $p0$ and natural number L, if* lhbandagree$(t, Q, p0, L + 2)$, *then* lhbandagree$(t + 1, Q, p0 + 1, L)$.

Proof. The goal follows from lhbandagreestep$(t, Q, p0, L)$. □

Theorem 9.420. *For every natural number t and natural number Q and natural number $p0$ and natural number L and natural number s, if* lhbandagree$(t, Q, p0, L+2 \cdot s)$*, then* lhbandagree$(t+s, Q, p0+s, L)$.

Proof. The goal follows from lhbandagreesteps$(t, Q, p0, L, s)$. □

Theorem 9.421. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number L, if $T \leq t$, then if* lhbandagree$(t, m \cdot P, 1, L)$*, then* lhbandagree$(t, m \cdot P, 0, L+1)$.

Proof. Assume hP, *hper*. The goal follows from periodiclhbandagreezeroofone$(T, P, hP, hper)$. □

Theorem 9.422. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* rhcolagree$(t, m \cdot P, 1, \text{twopow}(k+1))$*, then if* winwittagree$(t, k, m \cdot P, t+\text{twopow}(k), \text{twopow}(k))$*, then* rhbandagree$(t, m \cdot P, 0, \text{twopow}(k))$.

Proof. Assume hP, *hper*. The goal follows from periodicrhbandagreeofrhcolagreeoneandwinwittagree(T □

Theorem 9.423. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* rhcolagree$(t, m \cdot P, 1, \text{twopow}(k+1))$*, then if* rhbandagree$(t, m \cdot P, 0, \text{twopow}(k))$*, then* winwittagree$(t, k, m \cdot P, t + \text{twopow}(k), \text{twopow}(k))$.

Proof. Assume hP, *hper*. The goal follows from periodicwinwittagreeofrhcolagreeoneandrhbandagree(T □

Theorem 9.424. *For every natural number t and natural number k and natural number p,* lhcenter$(t, p) =$ bxor(bxor(bxor(lhcenter$(t + \text{twopow}(k), p + \text{twopow}(k))$, lhcenter$(t, p+\text{twopow}(k)))$, lhcenter$(t, p+\text{twopow}(k+1)))$, wittbitbool(wittwin$(t, \text{twopow}(k$ $p + \text{twopow}(k+1)))$.

Proof. The goal follows from lhcenterscaleinwardstepuncond(t, k, p). □

Theorem 9.425. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* lhbandagree$(t+\text{twopow}(k), Q, p0+\text{twopow}(k), L)$*, then if* lhbandagree$(t, Q, p0+\text{twopow}(k), L)$*, then if* lhbandagree$(t, Q, p0+\text{twopow}(k+1), L)$*, then if* winwittagree$(t, k, Q, t+p0 + \text{twopow}(k+1), L)$*, then* lhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from lhbandagreescaleinwardstepuncond$(t, k, Q, p0, L)$. □

Theorem 9.426. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q \leq p0$, then* winwittagree$(t, k, Q, t + p0 + \text{twopow}(k + 1), L)$.

Proof. The goal follows from winwittagreeoutsideleftcone$(t, k, Q, p0, L)$. □

Theorem 9.427. *For every natural number t and natural number p,* rdiag$(t + p, t) = \text{lhcenter}(t, p)$.

Proof. The goal follows from rdiagaddeqlhcenteruncond(t, p). □

Theorem 9.428. *For every natural number t and natural number k and natural number p,* wittbitbool(wittwin$(t, \text{twopow}(k)), t + p + \text{twopow}(k + 1)) =$ bxor(bxor(bxor(lhcenter$(t+\text{twopow}(k), p+\text{twopow}(k))$, lhcenter$(t, p))$, lhcenter$(t, p+$ twopow$(k)))$, lhcenter$(t, p + \text{twopow}(k + 1)))$.

Proof. The goal follows from winwitttestbiteqlhcenterscaleuncond(t, k, p). □

Theorem 9.429. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* lhbandagree$(t+$ twopow$(k), Q, p0 + \text{twopow}(k), L)$, *then if* lhbandagree$(t, Q, p0, L)$, *then if* lhbandagree$(t, Q, p0+\text{twopow}(k), L)$, *then if* lhbandagree$(t, Q, p0+\text{twopow}(k+$ $1), L)$, *then* winwittagree$(t, k, Q, t + p0 + \text{twopow}(k + 1), L)$.

Proof. The goal follows from winwittagreeoflhbandagreescaleuncond$(t, k, Q, p0, L)$. □

Theorem 9.430. *For every natural number t and natural number k and natural number p,* lhcenter$(t, p + \text{twopow}(k)) =$ bxor(bxor(bxor(lhcenter$(t +$ twopow$(k), p+\text{twopow}(k))$, lhcenter$(t, p))$, lhcenter$(t, p+\text{twopow}(k+1)))$, wittbitbool(wittwin$(t,$ twopow $p + \text{twopow}(k + 1)))$.

Proof. The goal follows from lhcenterscalemidstepuncond(t, k, p). □

Theorem 9.431. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* lhbandagree$(t+$ twopow$(k), Q, p0 + \text{twopow}(k), L)$, *then if* lhbandagree$(t, Q, p0, L)$, *then if* lhbandagree$(t, Q, p0 + \text{twopow}(k + 1), L)$, *then if* winwittagree$(t, k, Q, t +$ $p0 + \text{twopow}(k + 1), L)$, *then* lhbandagree$(t, Q, p0 + \text{twopow}(k), L)$.

Proof. The goal follows from lhbandagreescalemidstepuncond$(t, k, Q, p0, L)$. □

Theorem 9.432. *For every natural number t and natural number k and natural number p, if $t < p + \text{twopow}(k)$, then* lhcenter$(t + \text{twopow}(k), p + \text{twopow}(k)) = \text{bxor}(\text{lhcenter}(t, p), \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t + p + \text{twopow}(k + 1)))$.

Proof. The goal follows from lhcenterscalestepouterstrip(t, k, p). □

Theorem 9.433. *For every natural number t and natural number k and natural number p, if $t < p + \text{twopow}(k)$, then* lhcenter$(t, p) = \text{bxor}(\text{lhcenter}(t + \text{twopow}(k), p + \text{twopow}(k)), \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t + p + \text{twopow}(k + 1)))$.

Proof. The goal follows from lhcenterscaleinwardouterstrip(t, k, p). □

Theorem 9.434. *For every natural number t and natural number k and natural number p, if $t < p + \text{twopow}(k)$, then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t + p + \text{twopow}(k+1)) = \text{bxor}(\text{lhcenter}(t + \text{twopow}(k), p + \text{twopow}(k)), \text{lhcenter}(t, p))$.

Proof. The goal follows from winwitttestbitouterleftstrip(t, k, p). □

Theorem 9.435. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t < p + \text{twopow}(k)$, then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t + p + \text{twopow}(k + 1)) = \text{bxor}(\text{ldiag}(t - p, t + \text{twopow}(k)), \text{ldiag}(t - p, t))$.

Proof. The goal follows from winwitttestbitouterleftstripeqldiagxor(t, k, p). □

Theorem 9.436. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q < p0 + \text{twopow}(k)$, then if* lhbandagree$(t + \text{twopow}(k), Q, p0 + \text{twopow}(k), L)$*, then if* winwittagree$(t, k, Q, t + p0 + \text{twopow}(k+1), L)$*, then* lhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from lhbandagreescaleinwardouterstrip$(t, k, Q, p0, L)$. □

Theorem 9.437. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q < p0 + \text{twopow}(k)$, then if* lhbandagree$(t, Q, p0, L)$*, then if* winwittagree$(t, k, Q, t + p0 + \text{twopow}(k+1), L)$*, then* lhbandagree$(t + \text{twopow}(k), Q, p0 + \text{twopow}(k), L)$.

Proof. The goal follows from lhbandagreescalestepouterstrip$(t, k, Q, p0, L)$. □

Theorem 9.438. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q < p0 + \mathrm{twopow}(k)$, then if* $\mathrm{lhbandagree}(t + \mathrm{twopow}(k), Q, p0 + \mathrm{twopow}(k), L)$, *then if* $\mathrm{lhbandagree}(t, Q, p0, L)$, *then* $\mathrm{winwittagree}(t, k, Q, t+p0+\mathrm{twopow}(k+1), L)$.

Proof. The goal follows from $\mathrm{winwittagreeouterleftstrip}(t, k, Q, p0, L)$. □

Theorem 9.439. *For every natural number t and natural number k and natural number Q and natural number L, if $0 < Q$, then if $t + Q \leq L$, then if $t + Q < \mathrm{twopow}(k)$, then if* $\mathrm{lhbandagree}(t + \mathrm{twopow}(k), Q, \mathrm{twopow}(k), L)$, *then not* $\mathrm{winwittagree}(t, k, Q, t + \mathrm{twopow}(k+1), L)$.

Proof. The goal follows from $\mathrm{notlhbandagreeandwinwittagreeoflttwopow}(t, k, Q, L)$. □

Theorem 9.440. *For every natural number t and natural number k and natural number p,* $\mathrm{wittbitbool}(\mathrm{frobdefect}(k, \mathrm{wittrow}(t)), t + p + 1 + \mathrm{twopow}(k+1)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{lhcenter}(t, p + 1 + \mathrm{twopow}(k+1)) \text{ and } \mathrm{bxor}(\mathrm{lhcenter}(t, p + \mathrm{twopow}(k)), \mathrm{lhcenter}(t, p)), \mathrm{lhcenter}(t, p+1+\mathrm{twopow}(k)) \text{ and } \mathrm{bxor}(\mathrm{lhcenter}(t, p+\mathrm{twopow}(k+1)), \mathrm{lhcenter}(t, p))), \mathrm{lhcenter}(t, p+1) \text{ and } \mathrm{bxor}(\mathrm{lhcenter}(t, p+\mathrm{twopow}(k+1)), \mathrm{lhcenter}(t, p + \mathrm{twopow}(k))))$.

Proof. The goal follows from $\mathrm{frobdefectrowtestbitlhcenteruncond}(t, k, p)$. □

Theorem 9.441. *For every natural number t and natural number k and natural number p, if $t < p+\mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{frobdefect}(k, \mathrm{wittrow}(t)), t + p + 1 + \mathrm{twopow}(k+1)) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{frobdefectrowtestbitlhcenterouterstripeqfalse}(t, k, p)$. □

Theorem 9.442. *For every natural number t and natural number k and natural number p,* $\mathrm{wittbitbool}(\mathrm{wittscale}(k, \mathrm{wittrow}(t)), t+p+\mathrm{twopow}(k+1)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{lhcenter}(t, p+\mathrm{twopow}(k+1)), \mathrm{lhcenter}(t, p+\mathrm{twopow}(k))), \mathrm{lhcenter}(t, p))$.

Proof. The goal follows from $\mathrm{phiscalerowtestbitlhcenteruncond}(t, k, p)$. □

Theorem 9.443. *For every natural number t and natural number k and natural number p, if $t < p+\mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{wittscale}(k, \mathrm{wittrow}(t)), t + p + \mathrm{twopow}(k+1)) = \mathrm{lhcenter}(t, p)$.

Proof. The goal follows from $\mathrm{phiscalerowtestbitlhcenterouterstrip}(t, k, p)$. □

Theorem 9.444. *For every natural number t and natural number k and natural number p, if $t < p+\mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{crosscarry}(\mathrm{wittscale}(k, \mathrm{wittrow}(t)), \mathrm{wittwin}(t, \mathrm{twop}$ $p+1+\mathrm{twopow}(k+1)) = \mathrm{bxor}(\mathrm{lhcenter}(t, p+1)$ and $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+$ $p+\mathrm{twopow}(k+1)), \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+(p+1)+\mathrm{twopow}(k+$ $1))$ and $\mathrm{lhcenter}(t, p))$.

Proof. The goal follows from $\mathrm{crosscarryrowscalelhcenterouterstrip}(t, k, p)$. □

Theorem 9.445. *For every natural number t and natural number k and natural number p, if $t < p+\mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t+1, \mathrm{twopow}(k)), t+$ $1+(p+1)+\mathrm{twopow}(k+1)) = \mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittphi}(\mathrm{wittwin}(t, \mathrm{twopow}(k))), t+$ $1+(p+1)+\mathrm{twopow}(k+1)), \mathrm{bxor}(\mathrm{lhcenter}(t, p+1)$ and $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+$ $p+\mathrm{twopow}(k+1)), \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+(p+1)+\mathrm{twopow}(k+$ $1))$ and $\mathrm{lhcenter}(t, p)))$.

Proof. The goal follows from $\mathrm{winwittstepouterleftstrip}(t, k, p)$. □

Theorem 9.446. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t+Q < p0+$ $\mathrm{twopow}(k)$, then if* $\mathrm{lhbandagree}(t, Q, p0, L+1)$, *then if* $\mathrm{winwittagree}(t, k, Q, t+$ $p0 + \mathrm{twopow}(k+1), L+2)$, *then* $\mathrm{winwittagree}(t+1, k, Q, t+1+(p0+1)+$ $\mathrm{twopow}(k+1), L)$.

Proof. The goal follows from $\mathrm{winwittagreestepouterleftstrip}(t, k, Q, p0, L)$. □

Theorem 9.447. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number s, if $t + Q < p0 + \mathrm{twopow}(k)$, then if* $\mathrm{lhbandagree}(t, Q, p0, L+$ $2 \cdot s)$, *then if* $\mathrm{winwittagree}(t, k, Q, t + p0 + \mathrm{twopow}(k+1), L + 2 \cdot s)$, *then* $\mathrm{winwittagree}(t + s, k, Q, t + s + (p0 + s) + \mathrm{twopow}(k+1), L)$.

Proof. The goal follows from $\mathrm{winwittagreestepsouterleftstrip}(t, k, Q, p0, L, s)$. □

Theorem 9.448. *For every natural number t and natural number k and natural number p, if $t < p+\mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+$ $(p+2)+\mathrm{twopow}(k+1)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittwin}(t+1, \mathrm{twopow}(k)), t+$ $1 + (p+1) + \mathrm{twopow}(k+1)), \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + (p +$ $1)+\mathrm{twopow}(k+1))$ or $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+p+\mathrm{twopow}(k+$ $1))), \mathrm{bxor}(\mathrm{lhcenter}(t, p+1)$ and $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+p+\mathrm{twopow}(k+$ $1)), \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+(p+1)+\mathrm{twopow}(k+1))$ and $\mathrm{lhcenter}(t, p)))$.

Proof. The goal follows from winwittoutwardstepouterleftstrip(t, k, p). □

Theorem 9.449. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t+Q < p0+$ twopow(k), then if* lhbandagree$(t, Q, p0, L+1)$*, then if* winwittagree$(t, k, Q, t+p0+$twopow$(k+1), L+1)$*, then if* wittbitbool(wittwin$(t+1+Q,$ twopow$(k)), t+1+(p0+L+1)+$twopow$(k+1)+Q) =$ wittbitbool(wittwin$(t+1,$ twopow$(k)), t+1+(p0+L+1)+$twopow$(k+1))$*, then* winwittagree$(t, k, Q, t+p0+$twopow$(k+1), L+2)$.

Proof. The goal follows from winwittagreeextendouterleftstrip$(t, k, Q, p0, L)$. □

Theorem 9.450. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t+Q < p0+$ twopow(k), then if* lhbandagree$(t, Q, p0, L+1)$*, then if* winwittagree$(t, k, Q, t+p0+$twopow$(k+1), L+1)$*, then if* winwittagree$(t+1, k, Q, t+1+(p0+1)+$twopow$(k+1), L)$*, then* winwittagree$(t, k, Q, t+(p0+2)+$twopow$(k+1), L)$.

Proof. The goal follows from winwittagreeoutwardstepouterleftstrip$(t, k, Q, p0, L)$. □

Theorem 9.451. *For every natural number t and natural number k and natural number r, if $r \leq$ twopow(k), then if $t + r <$ twopow$(k+1)$, then* rhcenter$(t, r) =$ bxor(bxor(lhcenter$(t+$twopow$(k),$ twopow$(k)-r),$ lhcenter$(t,$ twopow$(k)-r)),$ wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k+1) - r))$.

Proof. The goal follows from rhcentereqlhcenterscalecrossouter(t, k, r). □

Theorem 9.452. *For every natural number t and natural number k and natural number Q and natural number L, if $L \leq$ twopow(k), then if $t+Q+L <$ twopow$(k+1)$, then if* lhbandagree$(t+$twopow$(k), Q,$ twopow$(k)-L, L)$*, then if* lhbandagree$(t, Q,$ twopow$(k) - L, L)$*, then if* winwittagree$(t, k, Q, t+$ twopow$(k+1) - L, L)$*, then* rhbandagree$(t, Q, 0, L)$.

Proof. The goal follows from rhbandagreeoflhbandagreescalecrossouter(t, k, Q, L). □

Theorem 9.453. *For every natural number t and natural number k and natural number Q and natural number L, if $L \leq$ twopow(k), then if $t+Q+L <$ twopow$(k+1)$, then if* rhbandagree$(t, Q, 0, L)$*, then if* lhbandagree$(t+$ twopow$(k), Q,$ twopow$(k)-L, L)$*, then if* lhbandagree$(t, Q,$ twopow$(k)-L, L)$*, then* winwittagree$(t, k, Q, t+$ twopow$(k+1) - L, L)$.

Proof. The goal follows from winwittagreeofrhbandagreelhbaragreescalecrossouter(t, k, Q, L). □

Theorem 9.454. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P <$* twopow(k)*, then if* lhcolagree$(t, m \cdot P, 1,$ twopow$(k))$*, then if* lhcolagree$(t+$twopow$(k), m \cdot P, 1,$ twopow$(k))$*, then if* winwittagree$(t, k, m \cdot P, t+$twopow$(k),$ twopow$(k))$*, then* rhbandagree$(t, m \cdot P, 0,$ twopow$(k))$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreeoflhcolagreeoneandwinwittagreeout □

Theorem 9.455. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P <$* twopow(k)*, then if* lhcolagree$(t, m \cdot P, 1,$ twopow$(k))$*, then if* lhcolagree$(t +$ twopow$(k), m \cdot P, 1,$ twopow$(k))$*, then if* rhbandagree$(t, m \cdot P, 0,$ twopow$(k))$*, then* winwittagree$(t, k, m \cdot P, t +$ twopow$(k),$ twopow$(k))$.

Proof. Assume hP, $hper$. The goal follows from periodicwinwittagreeoflhcolagreeoneandrhbandagreeout □

Theorem 9.456. *For every natural number t and natural number k and natural number r, if $t + r <$* twopow(k)*, then* rhcenter$(t, r) =$ bxor(lhcenter$(t +$ twopow$(k),$ twopow$(k)-r),$ wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+ 1) - r))$.

Proof. The goal follows from rhcentereqlhcenterscalecrosscone(t, k, r). □

Theorem 9.457. *For every natural number t and natural number k and natural number r, if $t+r <$* twopow(k)*, then* lhcenter$(t+$twopow$(k),$ twopow$(k)- r) =$ bxor(rhcenter$(t, r),$ wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k + 1) - r))$.

Proof. The goal follows from lhcenterscalecrosscone(t, k, r). □

Theorem 9.458. *For every natural number t and natural number k and natural number r, if $t+r <$* twopow(k)*, then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1)-r) =$ bxor(lhcenter$(t+$twopow$(k),$ twopow$(k)-r),$ rhcenter$(t, r))$.

Proof. The goal follows from winwitttestbitscalecrosscone(t, k, r). □

Theorem 9.459. *For every natural number t and natural number k and natural number r, if $t < r$, then if $t+r <$ twopow(k), then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1) - r) =$ lhcenter$(t +$ twopow$(k),$ twopow$(k) - r)$.

Proof. The goal follows from winwitttestbitscalecrosszerogap(t, k, r). □

Theorem 9.460. *For every natural number t and natural number k and natural number p, if $t < p$, then if $t+p <$ twopow(k), then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $p +$ twopow$(k)) =$ lhcenter$(t +$ twopow$(k), p)$.

Proof. The goal follows from winwitttestbitzerogapeqlhcenterlate(t, k, p). □

Theorem 9.461. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q < p0$, then if $t + Q + p0 + L <$ twopow(k), then if* winwittagree$(t, k, Q, t + p0 +$ twopow$(k), L)$, *then* lhbandagree$(t +$ twopow$(k), Q, p0, L)$.

Proof. The goal follows from lhbandagreeofwinwittagreezerogap$(t, k, Q, p0, L)$. □

Theorem 9.462. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q < p0$, then if $t + Q + p0 + L <$* twopow(k), *then if* lhbandagree$(t +$ twopow$(k), Q, p0, L)$, *then* winwittagree$(t, k, Q, t + p0 +$ twopow$(k), L)$.

Proof. The goal follows from winwittagreeoflhbandagreezerogap$(t, k, Q, p0, L)$. □

Theorem 9.463. *For every natural number t and natural number k and natural number Q and natural number L, if $t + Q + L <$* twopow(k), *then if* lhbandagree$(t+$twopow$(k), Q,$ twopow$(k)-L, L)$, *then if* winwittagree$(t, k, Q, t+$ twopow$(k+1) - L, L)$, *then* rhbandagree$(t, Q, 0, L)$.

Proof. The goal follows from rhbandagreeoflhbandagreeandwinwittagreecrosscone(t, k, Q, L). □

Theorem 9.464. *For every natural number t and natural number k and natural number Q and natural number L, if $t + Q + L <$* twopow(k), *then if* rhbandagree$(t, Q, 0, L)$, *then if* lhbandagree$(t +$ twopow$(k), Q,$ twopow$(k) - L, L)$, *then* winwittagree$(t, k, Q, t +$ twopow$(k+1) - L, L)$.

Proof. The goal follows from winwittagreeofrhbandagreeandlhbandagreecrosscone(t, k, Q, L). □

Theorem 9.465. *For every natural number t and natural number k and natural number Q and natural number L, if $t + Q + L <$ twopow(k), then if* rhbandagree$(t, Q, 0, L)$*, then if* winwittagree$(t, k, Q, t+$twopow$(k+1)-L, L)$*, then* lhbandagree$(t +$ twopow$(k), Q,$ twopow$(k) - L, L)$.

Proof. The goal follows from lhbandagreeofrhbandagreeandwinwittagreecrosscone(t, k, Q, L). □

Theorem 9.466. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number L, if $T \leq t$, then if $t + m \cdot P + L <$ twopow(k), then if* lhcolagree$(t +$ twopow$(k), m \cdot P, 1,$ twopow$(k))$*, then if* winwittagree$(t, k, m \cdot P, t +$ twopow$(k + 1) - L, L)$*, then* rhbandagree$(t, m \cdot P, 0, L)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreeoflhcolagreelateandwinwittagreecor □

Theorem 9.467. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number L, if $T \leq t$, then if $t+m \cdot P + L <$ twopow(k), then if* lhcolagree$(t +$ twopow$(k), m \cdot P, 1,$ twopow$(k))$*, then if* rhbandagree$(t, m \cdot P, 0, L)$*, then* winwittagree$(t, k, m \cdot P, t+$twopow$(k+1) - L, L)$.

Proof. Assume hP, $hper$. The goal follows from periodicwinwittagreeoflhcolagreelateandrhbandagreecor □

Theorem 9.468. *For every natural number t and natural number k and natural number p, if $t < p+$*twopow(k)*, then* lhcenter$(t+1+$twopow$(k), p+1+$ twopow$(k)) =$ bxor(bxor(lhcenter$(t, p+2)$, wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $(p+2)+$twopow$(k+1))),$ bxor(lhcenter$(t, p+1)$, wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $(p+1)+$twopow$(k+1)))$ or bxor(lhcenter(t, p), wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $p +$ twopow$(k + 1))))$.

Proof. The goal follows from lhcenterstepouterstripfromwinwitt(t, k, p). □

Theorem 9.469. *For every natural number t and natural number k and natural number p,* wittbitbool(wittwin$(t + 1,$ twopow$(k)), t + 1 + (p + 1) +$ twopow$(k + 1)) =$ bxor(wittbitbool(wittwin$(t,$ twopow$(k)), t + (p + 2) +$ twopow$(k+1)),$ bxor(bxor(bxor(lhcenter$(t+$twopow$(k), p+1+$twopow$(k))$ or lhcenter$(t+$ twopow$(k), p+$twopow$(k))$, lhcenter$(t, p+1)$ or lhcenter$(t, p))$, lhcenter$(t, p+$ $1+$twopow$(k))$ or lhcenter$(t, p+$twopow$(k)))$, lhcenter$(t, p+1+$twopow$(k+$ $1))$ or lhcenter$(t, p +$ twopow$(k + 1))))$.

Proof. The goal follows from winwittsteplhcenterdefectuncond(t, k, p). □

Theorem 9.470. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* lhbandagree$(t+$ twopow$(k), Q, p0 +$ twopow$(k), L + 1)$*, then if* lhbandagree$(t, Q, p0, L + 1)$*, then if* lhbandagree$(t, Q, p0+$twopow$(k), L+1)$*, then if* lhbandagree$(t, Q, p0+$ twopow$(k+1), L+1)$*, then if* winwittagree$(t, k, Q, t+(p0+2)+$twopow$(k+$ $1), L)$*, then* winwittagree$(t + 1, k, Q, t + 1 + (p0 + 1) +$ twopow$(k + 1), L)$.

Proof. The goal follows from winwittagreestepoflhbandagreescaleuncond$(t, k, Q, p0, L)$. □

Theorem 9.471. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if* lhbandagree$(t+$ twopow$(k), Q, p0 +$ twopow$(k), L + 1)$*, then if* lhbandagree$(t, Q, p0, L + 1)$*, then if* lhbandagree$(t, Q, p0+$twopow$(k), L+1)$*, then if* lhbandagree$(t, Q, p0+$ twopow$(k + 1), L + 1)$*, then if* winwittagree$(t + 1, k, Q, t + 1 + (p0 + 1) +$ twopow$(k+1), L)$*, then* winwittagree$(t, k, Q, t+(p0+2)+$twopow$(k+1), L)$.

Proof. The goal follows from winwittagreeoutwardoflhbandagreescaleuncond$(t, k, Q, p0, L)$. □

Theorem 9.472. *For every natural number t and natural number k and natural number p, if $t+p <$* twopow(k)*, then* wittbitbool(wittscale$(k,$ wittrow$(t)), t+$ $p +$ twopow$(k)) =$ lhcenter(t, p).

Proof. The goal follows from phiscalerowtestbitlhcentersectorone(t, k, p). □

Theorem 9.473. *For every natural number t and natural number k and natural number p, if $t + p <$* twopow(k)*, then* lhcenter$(t +$ twopow$(k), p) =$ bxor(lhcenter(t, p), wittbitbool(wittwin$(t,$ twopow$(k)), t + p +$ twopow$(k)))$.

Proof. The goal follows from lhcenterscalesamecolcone(t, k, p). □

Theorem 9.474. *For every natural number t and natural number k and natural number p, if $t + p <$* twopow(k)*, then* lhcenter$(t, p) =$ bxor(lhcenter$(t +$ twopow$(k), p)$, wittbitbool(wittwin$(t,$ twopow$(k)), t + p +$ twopow$(k)))$.

Proof. The goal follows from lhcenterscaleinwardsamecolcone(t, k, p). □

Theorem 9.475. *For every natural number t and natural number k and natural number p, if $t+p <$* twopow(k)*, then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $p +$ twopow$(k)) =$ bxor(lhcenter$(t +$ twopow$(k), p)$, lhcenter$(t, p))$.

Proof. The goal follows from winwitttestbitsectoronelhcenter(t, k, p). □

Theorem 9.476. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q + p0 + L <$ twopow(k), then if* lhbandagree$(t, Q, p0, L)$*, then if* winwittagree$(t, k, Q, t + p0 + $twopow$(k), L)$*, then* lhbandagree$(t + $twopow$(k), Q, p0, L)$.

Proof. The goal follows from lhbandagreescalesamecolcone$(t, k, Q, p0, L)$. □

Theorem 9.477. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q + p0 + L <$ twopow(k), then if* lhbandagree$(t + $twopow$(k), Q, p0, L)$*, then if* winwittagree$(t, k, Q, t + p0 + $twopow$(k), L)$*, then* lhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from lhbandagreescaleinwardsamecolcone$(t, k, Q, p0, L)$. □

Theorem 9.478. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q + p0 + L <$ twopow(k), then if* lhbandagree$(t + $twopow$(k), Q, p0, L)$*, then if* lhbandagree$(t, Q, p0, L)$*, then* winwittagree$(t, k, Q, t + p0 + $twopow$(k), L)$.

Proof. The goal follows from winwittagreesectoronelhcenter$(t, k, Q, p0, L)$. □

Theorem 9.479. *For every natural number t and natural number k and natural number Q and natural number L, if $0 < Q$, then if $t + Q \leq L$, then if $t + Q + L <$ twopow(k), then if* lhbandagree$(t + $twopow$(k), Q, 0, L)$*, then not* winwittagree$(t, k, Q, t + $twopow$(k), L)$.

Proof. The goal follows from notlhbandagreelateandwinwittagreesectorone(t, k, Q, L). □

Theorem 9.480. *For every natural number t and natural number k and natural number p, if $t+p+1 <$ twopow(k), then* wittbitbool(frobdefect$(k,$ wittrow$(t)), t + p + 1 + $twopow$(k)) =$ false.

Proof. The goal follows from frobdefectrowtestbitlhcentersectoroneeqfalse(t, k, p). □

Theorem 9.481. *For every natural number t and natural number k and natural number p, if $t+p+1 <$ twopow(k), then* wittbitbool(crosscarry(wittscale$(k,$ wittrow$(t))$, wittwin$(t, $tw $p+1+$twopow$(k)) =$ bxor(lhcenter$(t, p+1)$ and wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $p+$twopow$(k))$, wittbitbool(wittwin$(t,$ twopow$(k)), t+(p+1)+$twopow$(k))$ and lhcenter$(t, p))$.

Proof. The goal follows from crosscarryrowscalelhcentersectorone(t, k, p). □

Theorem 9.482. *For every natural number t and natural number k and natural number p, if $t + p + 1 <$ twopow(k), then* wittbitbool(wittwin$(t +$ $1,$ twopow$(k)), t{+}1{+}(p{+}1){+}$twopow$(k))$ = bxor(wittbitbool(wittphi(wittwin$(t,$ twopow$(k))), t+$ $1{+}(p{+}1){+}$twopow$(k))$, bxor(lhcenter$(t, p{+}1)$ and wittbitbool(wittwin$(t,$ twopow$(k)), t+$ $p{+}$twopow$(k))$, wittbitbool(wittwin$(t,$ twopow$(k)), t{+}(p{+}1){+}$twopow$(k))$ and lhcenter$(t, p)))$.

Proof. The goal follows from winwittstepsectoronelhcenter(t, k, p). □

Theorem 9.483. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t +$ $Q + p0 + L + 1 <$ twopow(k), then if* lhbandagree$(t, Q, p0, L + 1)$*, then if* winwittagree$(t, k, Q, t + p0 +$ twopow$(k), L + 2)$*, then* winwittagree$(t +$ $1, k, Q, t + 1 + (p0 + 1) +$ twopow$(k), L)$.

Proof. The goal follows from winwittagreestepsectoronelhcenter$(t, k, Q, p0, L)$. □

Theorem 9.484. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t +$ $Q + p0 + L + 1 <$ twopow(k), then if* lhbandagree$(t, Q, p0, L + 1)$*, then if* winwittagree$(t, k, Q, t + p0 +$ twopow$(k), L + 1)$*, then if* winwittagree$(t +$ $1, k, Q, t + 1 + (p0 + 1) +$ twopow$(k), L)$*, then* winwittagree$(t, k, Q, t + p0 +$ twopow$(k), L + 2)$.

Proof. The goal follows from winwittagreeoutwardstepsectoronelhcenter$(t, k, Q, p0, L)$. □

Theorem 9.485. *For every natural number T and natural number P, if $0 <$ P, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P <$* twopow(k)*, then* wittbitbool(wittwin$(t + m \cdot P,$ twopow$(k)), t + m \cdot P +$ twopow$(k)) =$ wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k))$.

Proof. Assume hP, $hper$. The goal follows from periodicwinwittsectoronezeroeq$(T, P, hP, hper)$. □

Theorem 9.486. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P + 2 <$* twopow(k)*, then* bxor(wittbitbool(wittwin$(t + m \cdot P,$ twopow$(k)), t + m \cdot P +$ $2 +$ twopow$(k))$, wittbitbool(wittwin$(t,$ twopow$(k)), t + 2 +$ twopow$(k))) =$

bxor(bxor(wittbitbool(wittwin($t+1+m\cdot P$, twopow(k)), $t+1+m\cdot P+1+$ twopow(k)), wittbitbool(wittwin(t+1, twopow(k)), t+1+1+twopow(k))), bxor(wittbitbool(wittwin(t+ $m\cdot P$, twopow(k)), t+$m\cdot P$+1+twopow(k)), wittbitbool(wittwin(t, twopow(k)), t+ $1+$ twopow(k)))).

Proof. Assume hP, *hper*. The goal follows from periodicwinwittsectoronetwodefecteq($T, P, hP, hper$). □

Theorem 9.487. *For every natural number t and natural number Q and natural number L, if $0 < Q$, then if $t+Q \leq L$, then not* rhbandagree($t, Q, 0, L$).

Proof. The goal follows from notrhbandagreeofltadd(t, Q, L). □

Theorem 9.488. *For every natural number t and natural number k and natural number p, if t+$p <$* twopow(k), *then* wittbitbool(wittscale(k, wittrow(t)), t+ twopow(k) $- p$) = rhcenter(t, p).

Proof. The goal follows from phiscalerowtestbitrhcentercone(t, k, p). □

Theorem 9.489. *For every natural number t and natural number k and natural number p, if $t+p <$* twopow(k), *then* rhcenter($t+$ twopow(k), p) = bxor(rhcenter(t, p), wittbitbool(wittwin(t, twopow(k)), $t+$ twopow(k) $- p$)).

Proof. The goal follows from rhcenterscalesamecolcone(t, k, p). □

Theorem 9.490. *For every natural number t and natural number k and natural number p, if $t+p <$* twopow(k), *then* rhcenter(t, p) = bxor(rhcenter($t+$ twopow(k), p), wittbitbool(wittwin(t, twopow(k)), $t+$ twopow(k) $- p$)).

Proof. The goal follows from rhcenterscaleinwardsamecolcone(t, k, p). □

Theorem 9.491. *For every natural number t and natural number k and natural number p, if t+$p <$* twopow(k), *then* wittbitbool(wittwin(t, twopow(k)), t+ twopow(k) $- p$) = bxor(rhcenter($t+$ twopow(k), p), rhcenter(t, p)).

Proof. The goal follows from winwitttestbitrhcentersamecolcone(t, k, p). □

Theorem 9.492. *For every natural number t and natural number k and natural number p, if $t < p$, then if t+$p <$* twopow(k), *then* wittbitbool(wittwin(t, twopow(k)), t+ twopow(k) $- p$) = rhcenter($t+$ twopow(k), p).

Proof. The goal follows from winwitttestbitzerogapeqrhcenterlate(t, k, p). □

Theorem 9.493. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t+Q+p0+L <$ twopow(k), then if* rhbandagree$(t, Q, p0, L)$*, then if* winwittagree$(t, k, Q, t+$ twopow$(k) - (p0 + L), L)$*, then* rhbandagree$(t +$ twopow$(k), Q, p0, L)$.

Proof. The goal follows from rhbandagreescalesamecolcone$(t, k, Q, p0, L)$. □

Theorem 9.494. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q + p0 + L <$ twopow(k), then if* rhbandagree$(t +$ twopow$(k), Q, p0, L)$*, then if* winwittagree$(t, k, Q, t+$twopow$(k)-(p0+L), L)$*, then* rhbandagree$(t, Q, p0, L)$.

Proof. The goal follows from rhbandagreescaleinwardsamecolcone$(t, k, Q, p0, L)$. □

Theorem 9.495. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q + p0 + L <$ twopow(k), then if* rhbandagree$(t +$ twopow$(k), Q, p0, L)$*, then if* rhbandagree$(t, Q, p0, L)$*, then* winwittagree$(t, k, Q, t +$ twopow$(k) - (p0 + L), L)$.

Proof. The goal follows from winwittagreerhcentersamecolcone$(t, k, Q, p0, L)$. □

Theorem 9.496. *For every natural number t and natural number k and natural number Q and natural number L, if $0 < Q$, then if $t + Q \leq L$, then if $t + Q + L <$ twopow(k), then if* rhbandagree$(t +$ twopow$(k), Q, 0, L)$*, then not* winwittagree$(t, k, Q, t +$ twopow$(k) - L, L)$.

Proof. The goal follows from notrhbandagreelateandwinwittagreecone(t, k, Q, L). □

Theorem 9.497. *For every natural number t and natural number Q and natural number R, if $R \leq t$, then if* rowagree$(t, Q, t-R, 2 \cdot R)$*, then* lhbandagree$(t, Q, 0, R)$.

Proof. The goal follows from lhbandagreeofrowagreesymm(t, Q, R). □

Theorem 9.498. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number R, if $T \leq t$, then if $R \leq t$, then if* rhbandagree$(t, m \cdot P, 0, R)$*, then* lhbandagree$(t, m \cdot P, 0, R)$.

Proof. Assume hP, *hper*. The goal follows from periodiclhbandagreezeroofrhbandagreezero$(T, P, hP, hp$ □

Theorem 9.499. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number R, if $T \leq t$, then if $R \leq$* twopow(k)*, then if* rhbandagree$(t+$twopow$(k), m \cdot P, 0, R)$*, then* lhbandagree$(t +$ twopow$(k), m \cdot P, 0, R)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhbandagreelateofrhbandagreelate$(T, P, hP, hpe$
□

Theorem 9.500. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number L, if $0 < m$, then if $T \leq t$, then if $t + m \cdot P \leq L$, then if $t + m \cdot P + L <$* twopow(k)*, then if* rhbandagree$(t+$twopow$(k), m \cdot P, 0, L)$*, then not* winwittagree$(t, k, m \cdot P, t +$ twopow$(k), L)$.

Proof. Assume hP, $hper$. The goal follows from notperiodicrhbandagreelateandwinwittagreesectorone$(T$
□

Theorem 9.501. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number R, if $T \leq t$, then if $R \leq t$, then if $t+m \cdot P+R <$* twopow(k)*, then if* rhbandagree$(t, m \cdot P, 0, R)$*, then if* winwittagree$(t, k, m \cdot P, t +$ twopow$(k) - R, R)$*, then* winwittagree$(t, k, m \cdot P, t +$ twopow$(k), R)$.

Proof. Assume hP, $hper$. The goal follows from periodicwinwittagreesectoroneleftofright$(T, P, hP, hper)$
□

Theorem 9.502. *For every natural number t and natural number k and natural number b, if $2 \cdot t + 1 < b$, then if $b <$* twopow$(k) +$ twopow(k)*, then* wittbitbool(frobdefect$(k,$ wittrow$(t)), b) =$ false.

Proof. The goal follows from frobdefectrowtestbitgttwomuleqfalse(t, k, b). □

Theorem 9.503. *For every natural number t and natural number k and natural number p, if $t+p+1 <$* twopow(k)*, then* wittbitbool(frobdefect$(k,$ wittrow$(t)), t+$ twopow$(k) - p) =$ false.

Proof. The goal follows from frobdefectrowtestbitrhcenterconeeqfalse(t, k, p).
□

Theorem 9.504. *For every natural number t and natural number k and natural number p, if $t+p+1 < \mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{crosscarry}(\mathrm{wittscale}(k, \mathrm{wittrow}(t)), \mathrm{wittwin}(t, \mathrm{twopow}(k)-p) = \mathrm{bxor}(\mathrm{rhcenter}(t,p)$ and $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k) - (p+1)), \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k) - p)$ and $\mathrm{rhcenter}(t, p+1))$.

Proof. The goal follows from $\mathrm{crosscarryrowscalerhcentercone}(t, k, p)$. □

Theorem 9.505. *For every natural number t and natural number k and natural number p, if $t + p + 1 < \mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t + 1, \mathrm{twopow}(k)), t+1+\mathrm{twopow}(k)-p) = \mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittphi}(\mathrm{wittwin}(t, \mathrm{twopow}(k))), t+1+\mathrm{twopow}(k)-p), \mathrm{bxor}(\mathrm{rhcenter}(t,p)$ and $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k) - (p+1)), \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k) - p)$ and $\mathrm{rhcenter}(t, p+1)))$.

Proof. The goal follows from $\mathrm{winwittsteprhcentercone}(t, k, p)$. □

Theorem 9.506. *For every natural number t and natural number k and natural number p, if $t < p$, then if $t+p+1 < \mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t+1, \mathrm{twopow}(k)), t+1+\mathrm{twopow}(k)-p) = \mathrm{wittbitbool}(\mathrm{wittphi}(\mathrm{wittwin}(t, \mathrm{twopow}(k))), t+1 + \mathrm{twopow}(k) - p)$.

Proof. The goal follows from $\mathrm{winwittstepzerogapeqphi}(t, k, p)$. □

Theorem 9.507. *For every natural number t and natural number k and natural number p, if $t < p$, then if $t+p+1 < \mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t+1, \mathrm{twopow}(k)), t+1+(p+1)+\mathrm{twopow}(k)) = \mathrm{wittbitbool}(\mathrm{wittphi}(\mathrm{wittwin}(t, \mathrm{twopow}(k))), t+1 + (p+1) + \mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{winwittsteplhcenterzerogapeqphi}(t, k, p)$. □

Theorem 9.508. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q + p0 + L + 1 < \mathrm{twopow}(k)$, then if* $\mathrm{rhbandagree}(t, Q, p0, L+1)$, *then if* $\mathrm{winwittagree}(t, k, Q, t+\mathrm{twopow}(k)-(p0+L+1), L+2)$, *then* $\mathrm{winwittagree}(t+1, k, Q, t+1+\mathrm{twopow}(k)-(p0+L), L)$.

Proof. The goal follows from $\mathrm{winwittagreesteprhcentercone}(t, k, Q, p0, L)$. □

Theorem 9.509. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q + p0 + L + 1 < \mathrm{twopow}(k)$, then if* $\mathrm{rhbandagree}(t, Q, p0, L+1)$, *then if* $\mathrm{winwittagree}(t, k, Q, t+\mathrm{twopow}(k)-(p0+L+1), L+1)$, *then if* $\mathrm{winwittagree}(t+1, k, Q, t+1+\mathrm{twopow}(k)-(p0+L), L)$, *then* $\mathrm{winwittagree}(t, k, Q, t+\mathrm{twopow}(k)-(p0+L+1), L+2)$.

Proof. The goal follows from winwittagreeinwardsteprhcentercone$(t, k, Q, p0, L)$. □

Theorem 9.510. *For every natural number t and natural number k, if* $t + 1 <$ twopow(k), *then* wittbitbool(wittwin$(t + 1,$twopow$(k)), t + 1 +$ twopow$(k))$ = bxor(wittbitbool(wittwin$(t,$twopow$(k)), t$+1+twopow$(k))$, bxor(wittbitbool(wittwin$(t,$ tw twopow$(k))$ or wittbitbool(wittwin$(t,$twopow$(k)), t$+twopow$(k)-1)$, bxor(center(t) and wittbitbool(witt twopow$(k)-1)$, wittbitbool(wittwin$(t,$twopow$(k)), t$+twopow$(k))$ and rhcenter$(t, 1))))$.

Proof. The goal follows from winwittstepsectoronecenterslot(t, k). □

Theorem 9.511. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* $t + m \cdot P + 1 <$ twopow(k), *then* bxor(wittbitbool(wittwin$(t + m \cdot P,$twopow$(k)), t + m \cdot P +$ $1 +$ twopow$(k))$, wittbitbool(wittwin$(t,$twopow$(k)), t + 1 +$ twopow$(k))) =$ bxor(wittbitbool(wittwin$(t,$twopow$(k)), t$+twopow$(k))$ and bxor(rhcenter$(t+$ $m \cdot P, 1)$, rhcenter$(t, 1))$, not(center$(t$+twopow$(k)))$ and bxor(wittbitbool(wittwin$(t+$ $m \cdot P,$twopow$(k)), t$+$m \cdot P$+twopow$(k)-1)$, wittbitbool(wittwin$(t,$twopow$(k)), t+$ twopow$(k) - 1)))$.

Proof. Assume hP, $hper$. The goal follows from periodicwinwittsectoroneonedefecteqlate$(T, P, hP, hper$ □

Theorem 9.512. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* $t + m \cdot P + 1 <$ twopow(k), *then* bxor(wittbitbool(wittwin$(t + m \cdot P,$twopow$(k)), t + m \cdot P +$ $1 +$ twopow$(k))$, wittbitbool(wittwin$(t,$twopow$(k)), t + 1 +$ twopow$(k))) =$ bxor(wittbitbool(wittwin$(t,$twopow$(k)), t$+twopow$(k))$ and bxor(rhcenter$(t+$ twopow$(k)+m \cdot P, 1)$, rhcenter$(t+$twopow$(k), 1))$, not(center$(t))$ and bxor(wittbitbool(wittwin$(t+$ $m \cdot P,$twopow$(k)), t$+$m \cdot P$+twopow$(k)-1)$, wittbitbool(wittwin$(t,$twopow$(k)), t+$ twopow$(k) - 1)))$.

Proof. Assume hP, $hper$. The goal follows from periodicwinwittsectoroneonedefecteqearly$(T, P, hP,$ hpe □

Theorem 9.513. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number k, if $T \leq t$, then if* $t + m \cdot P +$ $1 <$ twopow(k), *then if* wittbitbool(wittwin$(t,$twopow$(k)), t+$twopow$(k)) =$ false, *then* bxor(wittbitbool(wittwin$(t + m \cdot P,$twopow$(k)), t + m \cdot P + 1 +$

twopow(k)), wittbitbool(wittwin(t, twopow(k)), t+1+twopow(k))) = not(center(t)) and bxor(wittbitboo
m·P, twopow(k)), t+m·P+twopow(k)−1), wittbitbool(wittwin(t, twopow(k)), t+
twopow(k) − 1)).

Proof. Assume hP, *hper*. The goal follows from periodicwinwittsectoroneonedefectofwittzerofalse(T, P, h
□

Theorem 9.514. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P + 1 <$* twopow(k)*, then if* wittbitbool(wittwin($t + m \cdot P$, twopow(k)), $t + m \cdot P$ + twopow(k) − 1) = wittbitbool(wittwin(t, twopow(k)), t + twopow(k) − 1)*, then* bxor(wittbitbool(wittwin($t + m \cdot P$, twopow(k)), $t + m \cdot P + 1$ + twopow(k)), wittbitbool(wittwin(t, twopow(k)), t+1+twopow(k))) = wittbitbool(wittwin(t, twopow(k)) twopow(k)) and bxor(rhcenter($t + m \cdot P$, 1), rhcenter(t, 1)).

Proof. Assume hP, *hper*. The goal follows from periodicwinwittsectoroneonedefectofwittrighteq(T, P, h
□

Theorem 9.515. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P + 1 <$* twopow(k)*, then if* rhcenter($t + m \cdot P$, 1) = rhcenter(t, 1)*, then if* wittbitbool(wittwin($t + m \cdot P$, twopow(k)), $t + m \cdot P$ + twopow(k) − 1) = wittbitbool(wittwin(t, twopow(k)), t+twopow(k)−1)*, then* winwittagree(t, k, m· P, t + twopow(k), 1).

Proof. Assume hP, *hper*. The goal follows from periodicwinwittagreesectoronetwoofrightone(T, P, hP, h
□

Theorem 9.516. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P + 2 <$* twopow(k)*, then if* rhcenter(t+m·P, 1) = rhcenter(t, 1)*, then if* rhcenter(t+1+m·P, 1) = rhcenter($t + 1$, 1)*, then if* wittbitbool(wittwin($t + m \cdot P$, twopow(k)), $t + m \cdot P$+twopow(k)−1) = wittbitbool(wittwin(t, twopow(k)), t+twopow(k)−1)*, then if* wittbitbool(wittwin(t+1+m·P, twopow(k)), t+1+m·P+twopow(k)− 1) = wittbitbool(wittwin($t + 1$, twopow(k)), $t + 1$ + twopow(k) − 1)*, then* winwittagree(t, k, $m \cdot P$, t + twopow(k), 2).

Proof. Assume hP, *hper*. The goal follows from periodicwinwittagreesectoronethreeofrightonestep(T, P,
□

Theorem 9.517. *For every natural number t and natural number k and natural number b, if $1 \leq b$, then* wittbitbool(wittwin(t, twopow(k)), $b+1$) = bxor(wittbitbool(wittwin(t+1, twopow(k)), b+1), bxor(wittbitbool(wittwin(t, twopow(k)), b) or wittbitb 1), bxor(wittbitbool(crosscarry(wittscale(k, wittrow(t)), wittwin(t, twopow(k))), b), wittbitbool(frobdefe

Proof. The goal follows from winwittinwardstepfrobdefecttestbitsucc(t, k, b). □

Theorem 9.518. *For every natural number t and natural number k and natural number b, if $2 \cdot t + 1 < b$, then if $b <$ twopow(k) + twopow(k), then* wittbitbool(wittwin(t, twopow(k)), $b + 1$) = bxor(wittbitbool(wittwin(t + 1, twopow(k)), b+1), bxor(wittbitbool(wittwin(t, twopow(k)), b) or wittbitbool(wittwin(t, twopow(k)), b- 1), wittbitbool(crosscarry(wittscale(k, wittrow(t)), wittwin(t, twopow(k))), b))).

Proof. The goal follows from winwittinwardstepgttwomul(t, k, b). □

Theorem 9.519. *For every natural number t and natural number k, if $t +$ 1 < twopow(k), then* wittbitbool(wittwin(t, twopow(k)), t+1+twopow(k)) = bxor(wittbitbool(wittwin(t+1, twopow(k)), t+1+twopow(k)), bxor(wittbitbool(wittwin(t, twopow(k)), twopow(k)) or wittbitbool(wittwin(t, twopow(k)), t+twopow(k)−1), bxor(center(t) and wittbitbool(witt twopow(k)−1), wittbitbool(wittwin(t, twopow(k)), t+twopow(k)) and rhcenter(t, 1)))).

Proof. The goal follows from winwittsectoroneonefromcenterandright(t, k). □

Theorem 9.520. *For every natural number t and natural number k and natural number p, if t+p+1 < twopow(k), then* wittbitbool(wittwin(t, twopow(k)), t+ ($p + 2$) + twopow(k)) = bxor(wittbitbool(wittwin($t + 1$, twopow(k)), $t + 1 +$ ($p + 1$) + twopow(k)), bxor(wittbitbool(wittwin(t, twopow(k)), $t + (p + 1) +$ twopow(k)) or wittbitbool(wittwin(t, twopow(k)), t+p+twopow(k)), bxor(lhcenter(t, p+ 1) and wittbitbool(wittwin(t, twopow(k)), t+p+twopow(k)), wittbitbool(wittwin(t, twopow(k)), t+ ($p + 1$) + twopow(k)) and lhcenter(t, p)))).

Proof. The goal follows from winwittinwardstepsectoronelhcenter(t, k, p). □

Theorem 9.521. *For every natural number t and natural number k and natural number p, if t+p+2 < twopow(k), then* wittbitbool(wittwin(t, twopow(k)), t+ twopow(k)−p) = bxor(wittbitbool(wittwin(t+1, twopow(k)), t+1+twopow(k)− (p+1)), bxor(wittbitbool(wittwin(t, twopow(k)), t+twopow(k)−(p+1)) or wittbitbool(wittwin(t, twopow twopow(k)−(p+2)), bxor(rhcenter(t, p+1) and wittbitbool(wittwin(t, twopow(k)), t+ twopow(k)−(p+2)), wittbitbool(wittwin(t, twopow(k)), t+twopow(k)−(p+ 1)) and rhcenter($t, p + 2$)))).

Proof. The goal follows from winwittinwardsteprhcentercone(t, k, p). □

Theorem 9.522. *For every natural number t and natural number k and natural number p, if $t \leq p$, then if $t+p+2 <$ twopow(k), then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k)-p) =$ bxor(wittbitbool(wittwin$(t+1,$ twopow$(k)), t+1+$twopow$(k)-(p+1)),$ wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k)-(p+1))$ or wittbitbool(wittwin$(t,$ twopow$(k)),$ twopow$(k) - (p + 2))$).

Proof. The goal follows from winwittinwardstepzerogap(t, k, p). □

Theorem 9.523. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q <$ $p0$, then if $t + Q + p0 + L + 1 <$* twopow(k)*, then if* winwittagree$(t, k, Q, t +$ twopow$(k) - (p0 + L + 1), L + 2)$*, then* winwittagree$(t + 1, k, Q, t + 1 +$ twopow$(k) - (p0 + L), L)$.

Proof. The goal follows from winwittagreestepzerogap$(t, k, Q, p0, L)$. □

Theorem 9.524. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q <$ $p0$, then if $t + Q + p0 + L + 1 <$* twopow(k)*, then if* winwittagree$(t, k, Q, t +$ twopow$(k) - (p0 + L + 1), L + 1)$*, then if* winwittagree$(t + 1, k, Q, t + 1 +$ twopow$(k) - (p0 + L), L)$*, then* winwittagree$(t, k, Q, t +$ twopow$(k) - (p0 +$ $L + 1), L + 2)$.

Proof. The goal follows from winwittagreeinwardstepzerogap$(t, k, Q, p0, L)$. □

Theorem 9.525. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q <$ $p0$, then if $t + Q + p0 + L + 1 <$* twopow(k)*, then if* winwittagree$(t, k, Q, t +$ $p0 +$ twopow$(k), L + 2)$*, then* winwittagree$(t + 1, k, Q, t + 1 + (p0 + 1) +$ twopow$(k), L)$.

Proof. The goal follows from winwittagreesteplhcenterzerogap$(t, k, Q, p0, L)$. □

Theorem 9.526. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q <$ $p0$, then if $t + Q + p0 + L + 1 <$* twopow(k)*, then if* winwittagree$(t, k, Q, t +$ $p0 +$ twopow$(k), L + 1)$*, then if* winwittagree$(t + 1, k, Q, t + 1 + (p0 + 1) +$ twopow$(k), L)$*, then* winwittagree$(t, k, Q, t + p0 +$ twopow$(k), L + 2)$.

Proof. The goal follows from winwittagreeoutwardsteplhcenterzerogap$(t, k, Q, p0, L)$. □

Theorem 9.527. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number s, if $t+Q+p0+L+2\cdot s <$ twopow(k), then if* rhbandagree$(t, Q, p0, L+2\cdot s)$*, then if* winwittagree$(t, k, Q, t+$twopow$(k)-(p0+L+2\cdot s), L+2\cdot s)$*, then* winwittagree$(t+s, k, Q, t+s+$twopow$(k)-(p0+s+L), L)$.

Proof. The goal follows from winwittagreestepsrhcentercone$(t, k, Q, p0, L, s)$. □

Theorem 9.528. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number s, if $t+Q+p0+L+2\cdot s <$ twopow(k), then if* rhbandagree$(t, Q, p0, L+2\cdot s)$*, then if* winwittagree$(t, k, Q, t+$twopow$(k)-(p0+L+2\cdot s), L+2\cdot s)$*, then* rhbandagree$(t+s+$twopow$(k), Q, p0+s, L)$.

Proof. The goal follows from rhbandagreelatestepscone$(t, k, Q, p0, L, s)$. □

Theorem 9.529. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number s, if $t+Q < p0$, then if $t+Q+p0+L+2\cdot s <$ twopow(k), then if* winwittagree$(t, k, Q, t+$twopow$(k)-(p0+L+2\cdot s), L+2\cdot s)$*, then* winwittagree$(t+s, k, Q, t+s+$twopow$(k)-(p0+s+L), L)$.

Proof. The goal follows from winwittagreestepszerogap$(t, k, Q, p0, L, s)$. □

Theorem 9.530. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number s, if $t+Q+p0+L+2\cdot s <$ twopow(k), then if* lhbandagree$(t, Q, p0, L+2\cdot s)$*, then if* winwittagree$(t, k, Q, t+p0+$twopow$(k), L+2\cdot s)$*, then* winwittagree$(t+s, k, Q, t+s+(p0+s)+$twopow$(k), L)$.

Proof. The goal follows from winwittagreestepssectoronelhcenter$(t, k, Q, p0, L, s)$. □

Theorem 9.531. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L and natural number s, if $t+Q+p0+L+2\cdot s <$ twopow(k), then if* lhbandagree$(t, Q, p0, L+2\cdot s)$*, then if* winwittagree$(t, k, Q, t+p0+$twopow$(k), L+2\cdot s)$*, then* lhbandagree$(t+s+$twopow$(k), Q, p0+s, L)$.

Proof. The goal follows from lhbandagreelatestepssectorone$(t, k, Q, p0, L, s)$. □

Theorem 9.532. *For every natural number t and natural number k and natural number b, if $2{\cdot}t < b$, then if $b <$* twopow(k)*, then* wittbitbool(wittscale$(k,$ wittrow$(t)), b) =$ false.

Proof. The goal follows from phiscalerowtestbitinterconegapeqfalse(t, k, b). □

Theorem 9.533. *For every natural number t and natural number k and natural number b, if $2{\cdot}t < b$, then if $b <$* twopow(k)*, then* wittbitbool(wittwin$(t,$ twopow$(k)), b) =$ wittbitbool(wittrow$(t +$ twopow$(k)), b)$.

Proof. The goal follows from winwitttestbitinterconegapeqrowlate(t, k, b). □

Theorem 9.534. *For every natural number t and natural number k and natural number b, if $2{\cdot}t{+}1 < b$, then if $b <$* twopow(k)*, then* wittbitbool(crosscarry(wittscale$(k,$ wittrow$(t))$, false.

Proof. The goal follows from crosscarryrowscaleinterconegapeqfalse(t, k, b). □

Theorem 9.535. *For every natural number t and natural number k and natural number b, if $2{\cdot}t{+}1 < b$, then if $b <$* twopow(k)*, then* wittbitbool(wittwin$(t+$ $1,$ twopow$(k)), b + 1) =$ wittbitbool(wittphi(wittwin$(t,$ twopow$(k))), b + 1)$.

Proof. The goal follows from winwittstepinterconegapeqphi(t, k, b). □

Theorem 9.536. *For every natural number t and natural number k and natural number b, if $2{\cdot}t{+}1 < b$, then if $b <$* twopow(k)*, then* wittbitbool(wittwin$(t,$ twopow$(k)), b+$ $1) =$ bxor(wittbitbool(wittwin$(t{+}1,$ twopow$(k)), b{+}1)$, wittbitbool(wittwin$(t,$ twopow$(k)), b)$ or wittbitb $1))$.

Proof. The goal follows from winwittinwardstepinterconegap(t, k, b). □

Theorem 9.537. *For every natural number t and natural number k and natural number d, if $2 \cdot t <$* twopow(k)*, then if $d <$* twopow(k)*, then* wittbitbool(wittscale$(k,$ wittrow$(t)), d+$twopow$(k)) =$ wittbitbool(wittrow$(t), d)$.

Proof. The goal follows from phiscalerowtestbitsectoroneeqrow(t, k, d). □

Theorem 9.538. *For every natural number t and natural number k and natural number d, if $2{\cdot}t <$* twopow(k)*, then* wittbitbool(wittscale$(k,$ wittrow$(t)), d+$ twopow$(k + 1)) =$ wittbitbool(wittrow$(t), d)$.

Proof. The goal follows from phiscalerowtestbitsectortwoeqrow(t, k, d). □

Theorem 9.539. *For every natural number t and natural number k and natural number d, if $2 \cdot t <$ twopow(k), then if $d <$ twopow(k), then* wittbitbool(wittwin$(t,$ twopow$(k)),$ $d+$twopow$(k)) =$ bxor(wittbitbool(wittrow$(t+$ twopow$(k)), d +$ twopow$(k)),$ wittbitbool(wittrow$(t), d))$.

Proof. The goal follows from winwitttestbitsectoroneeqrowxor(t, k, d). □

Theorem 9.540. *For every natural number t and natural number k and natural number d, if $2 \cdot t <$* twopow(k)*, then* wittbitbool(wittwin$(t,$ twopow$(k)),$ $d+$ twopow$(k+1)) =$ bxor(wittbitbool(wittrow$(t+$twopow$(k)), d+$twopow$(k+$ $1)),$ wittbitbool(wittrow$(t), d))$.

Proof. The goal follows from winwitttestbitsectortwoeqrowxor(t, k, d). □

Theorem 9.541. *For every natural number t and natural number k and natural number p, if $t+p <$* twopow(k)*, then* bxor(wittbitbool(wittwin$(t,$ twopow$(k)),$ $t+$ twopow$(k)-p),$ wittbitbool(wittwin$(t,$ twopow$(k)),$ $t+$twopow$(k+1)-p)) =$ bxor(rhcenter$(t +$ twopow$(k), p),$ lhcenter$(t +$ twopow$(k),$ twopow$(k) - p))$.

Proof. The goal follows from winwittsectoronexorsectortworhcentercone(t, k, p). □

Theorem 9.542. *For every natural number t and natural number k and natural number p, if $t+p <$* twopow(k)*, then* bxor(wittbitbool(wittwin$(t,$ twopow$(k)),$ $t+$ $p+$twopow$(k)),$ wittbitbool(wittwin$(t,$ twopow$(k)),$ $t+p+$twopow$(k+1))) =$ bxor(lhcenter$(t +$ twopow$(k), p),$ lhcenter$(t +$ twopow$(k), p +$ twopow$(k)))$.

Proof. The goal follows from winwittsectoronexorsectortwolhcentercone(t, k, p). □

Theorem 9.543. *For every natural number t and natural number k and natural number Q and natural number $p0$ and natural number L, if $t + Q +$ $p0 + L <$* twopow(k)*, then if* lhbandagree$(t +$ twopow$(k), Q, p0, L)$*, then if* winwittagree$(t, k, Q, t + p0 +$ twopow$(k), L)$*, then if* winwittagree$(t, k, Q, t +$ $p0+$twopow$(k+1), L)$*, then* lhbandagree$(t+$twopow$(k), Q, p0+$twopow$(k), L)$.

Proof. The goal follows from lhbandagreelateshiftofwinwittagreesectors$(t, k, Q, p0, L)$. □

Theorem 9.544. *For every natural number t and natural number k and natural number Q and natural number L, if $t + Q + L < \mathrm{twopow}(k)$, then if* $\mathrm{lhbandagree}(t+\mathrm{twopow}(k), Q, \mathrm{twopow}(k)-L, L)$*, then if* $\mathrm{winwittagree}(t, k, Q, t+\mathrm{twopow}(k)-L, L)$*, then if* $\mathrm{winwittagree}(t, k, Q, t+\mathrm{twopow}(k+1)-L, L)$*, then* $\mathrm{rhbandagree}(t+\mathrm{twopow}(k), Q, 0, L)$.

Proof. The goal follows from $\mathrm{rhbandagreelateoflhbandagreelateandwinwittagreesectors}(t, k, Q, L)$. □

Theorem 9.545. *For every natural number t and natural number k, if $2 \cdot t < \mathrm{twopow}(k)$, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), \mathrm{twopow}(k)) = \mathrm{not}(\mathrm{rhcenter}(t+\mathrm{twopow}(k), t))$.

Proof. The goal follows from $\mathrm{winwitttestbittwopoweqnotrhcenterlate}(t, k)$. □

Theorem 9.546. *For every natural number t and natural number k, if $2 \cdot t + 2 < \mathrm{twopow}(k)$, then* $\mathrm{rhcenter}(t+\mathrm{twopow}(k), t) = \mathrm{bxor}(\mathrm{rhcenter}(t+1+\mathrm{twopow}(k), t+1), \mathrm{rhcenter}(t+\mathrm{twopow}(k), t+1)\ \mathrm{or}\ \mathrm{rhcenter}(t+\mathrm{twopow}(k), t+2))$.

Proof. The goal follows from $\mathrm{rhcenterlateselffromzerogapstep}(t, k)$. □

Theorem 9.547. *For every natural number t and natural number k and natural number p, if $p \leq t$, then* $\mathrm{lhcenter}(t+\mathrm{twopow}(k), p+\mathrm{twopow}(k)) = \mathrm{ldiag}(t-p, t+\mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{lhcenterlateshifteqldiag}(t, k, p)$. □

Theorem 9.548. *For every natural number t and natural number k and natural number r, if $r \leq \mathrm{twopow}(k)$, then* $\mathrm{lhcenter}(t+\mathrm{twopow}(k), \mathrm{twopow}(k)-r) = \mathrm{ldiag}(t+r, t+\mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{lhcenterlatecrosseqldiag}(t, k, r)$. □

Theorem 9.549. *For every natural number t and natural number k, if $t \leq k$, then if* $\mathrm{settledat}(t, k)$*, then* $\mathrm{zcentereq}(t, \mathrm{lhcenter}(t+\mathrm{twopow}(k), \mathrm{twopow}(k)))$.

Proof. The goal follows from $\mathrm{lhcenterlatetwopoweqzcenter}(t, k)$. □

Theorem 9.550. *For every natural number t and natural number k, if $t \leq k$, then if* $\mathrm{settledat}(t, k)$*, then* $\mathrm{zcentereq}(t, \mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)), \mathrm{center}(t)))$.

Proof. The goal follows from winwitttestbitsectortwocenterslotzcenter(t, k). □

Theorem 9.551. *For every natural number t and natural number k, if $t \leq k$, then if* settledat(t, k)*, then* zcentereq$(t,$ bxor$($center$(t+$twopow$(k))$, bxor$($wittbitbool$($wittwin$(t,$ twopow$($twopow$(k))$, wittbitbool$($wittwin$(t,$ twopow$(k))), t +$ twopow$(k+1)))))$.

Proof. The goal follows from winwittsectoronexorsectortwocenterslotzcenter(t, k). □

Theorem 9.552. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t-p \leq k$, then if* zerodiagonal$(t-p)$*, then if* settledby$(t-p, t+$twopow$(k))$*, then* lhcenter$(t+$twopow$(k), p+$twopow$(k)) =$ false.

Proof. The goal follows from lhcenterlateshiftofzerodiag(t, k, p). □

Theorem 9.553. *For every natural number t and natural number k and natural number r, if $t+r \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r, t +$ twopow$(k))$*, then* lhcenter$(t +$ twopow$(k),$ twopow$(k) - r) =$ false.

Proof. The goal follows from lhcenterlatecrossofzerodiag(t, k, r). □

Theorem 9.554. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t-p \leq k$, then if* zerodiagonal$(t-p)$*, then if* settledby$(t-p, t+$twopow$(k))$*, then* wittbitbool$($wittwin$(t,$ twopow$(k)), t+ p +$ twopow$(k+1)) =$ lhcenter(t, p).

Proof. The goal follows from winwitttestbitouterleftstripofzerodiag(t, k, p). □

Theorem 9.555. *For every natural number t and natural number k and natural number r, if $t+r \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r, t +$ twopow$(k))$*, then* wittbitbool$($wittwin$(t,$ twopow$(k)), t +$ twopow$(k+1) - r) =$ rhcenter(t, r).

Proof. The goal follows from winwitttestbitscalecrossofzerodiag(t, k, r). □

Theorem 9.556. *For every natural number t and natural number k and natural number r, if $t < r$, then if $t+r \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r, t+$twopow$(k))$*, then* wittbitbool$($wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1) - r) =$ false.

Proof. The goal follows from winwitttestbitscalecrosszerogapofzerodiag(t, k, r). □

Theorem 9.557. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t + p <$ twopow(k), then if $t - p \leq k$, then if* zerodiagonal$(t - p)$, *then if* settledby$(t - p, t +$ twopow$(k))$, *then* bxor(wittbitbool(wittwin$(t,$ twopow$(k)), t+p+$twopow$(k))$, wittbitbool(wittwin$(t,$ twopow$(k)), t+p+$ twopow$(k+1))) =$ lhcenter$(t +$ twopow$(k), p)$.

Proof. The goal follows from winwittsectoronexorsectortwolhcenterofzerodiag(t, k, p). □

Theorem 9.558. *For every natural number t and natural number k and natural number r, if $t + r \leq k$, then if* zerodiagonal$(t + r)$, *then if* settledby$(t +$ $r, t+$twopow$(k))$, *then* bxor(wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k)-$ $r)$, wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+1)-r)) =$ rhcenter$(t+$ twopow$(k), r)$.

Proof. The goal follows from winwittsectoronexorsectortworhcenterofzerodiag(t, k, r). □

Theorem 9.559. *For every natural number z and natural number k, if $z \leq k$, then if* zerodiagonal(z), *then if* settledat(z, k), *then* lhcenter$(z +$ twopow$(k),$ twopow$(k)) =$ false.

Proof. The goal follows from lhcenterlatetwopowofzerodiag(z, k). □

Theorem 9.560. *For every natural number z and natural number k, if $z \leq k$, then if* zerodiagonal(z), *then if* settledat(z, k), *then* wittbitbool(wittwin$(z,$ twopow$(k)), z+$ twopow$(k + 1)) =$ center(z).

Proof. The goal follows from winwitttestbitsectortwocenterslotofzerodiag(z, k). □

Theorem 9.561. *For every natural number z and natural number k, if $z \leq k$, then if* zerodiagonal(z), *then if* settledat(z, k), *then* bxor(wittbitbool(wittwin$(z,$ twopow$(k)), z+$ twopow$(k))$, wittbitbool(wittwin$(z,$ twopow$(k)), z+$twopow$(k+1))) =$ center$(z+$ twopow$(k))$.

Proof. The goal follows from winwittsectoronexorsectortwocenterslotofzerodiag(z, k). □

Theorem 9.562. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t - p + 2 \leq k$, then if* zerodiagonal$(t -$ $p)$, *then if* settledby$(t - p + 2, t + 2 +$ twopow$(k))$, *then* lhcenter$(t + 2 +$ twopow$(k), p +$ twopow$(k)) =$ true.

Proof. The goal follows from lhcenterlateshiftafterzerodiag(t, k, p). □

Theorem 9.563. *For every natural number t and natural number k and natural number r, if $t + r + 2 \leq k$, then if* zerodiagonal$(t + r)$*, then if* settledby$(t+r+2, t+$twopow$(k))$*, then* lhcenter$(t+$twopow$(k),$ twopow$(k)-(r + 2)) =$ true.

Proof. The goal follows from lhcenterlatecrossafterzerodiag(t, k, r). □

Theorem 9.564. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t - p + 2 \leq k$, then if* zerodiagonal$(t - p)$*, then if* settledby$(t - p + 2, t + 2 +$ twopow$(k))$*, then* wittbitbool(wittwin$(t +$ $2,$ twopow$(k)), t + 2 + p +$ twopow$(k + 1)) =$ not(lhcenter$(t + 2, p)$).

Proof. The goal follows from winwitttestbitouterleftstripafterzerodiag(t, k, p). □

Theorem 9.565. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+$ $r+2, t+$twopow$(k))$*, then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+$ $1) - (r + 2)) =$ not(rhcenter$(t, r + 2)$).

Proof. The goal follows from winwitttestbitscalecrossafterzerodiag(t, k, r). □

Theorem 9.566. *For every natural number t and natural number k and natural number r, if $t < r+2$, then if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+$twopow$(k))$*, then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k + 1) - (r + 2)) =$ true.

Proof. The goal follows from winwitttestbitscalecrosszerogapafterzerodiag(t, k, r). □

Theorem 9.567. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t + 2 + p <$* twopow(k)*, then if $t - p + 2 \leq k$, then if* zerodiagonal$(t - p)$*, then if* settledby$(t - p + 2, t + 2 +$ twopow$(k))$*, then* bxor(wittbitbool(wittwin$(t + 2,$ twopow$(k)), t + 2 + p +$ twopow$(k))$, wittbitbool(wittwin$(t + 2,$ twopow$(k)), t + 2 + p +$ twopow$(k +$ $1))) =$ not(lhcenter$(t + 2 +$ twopow$(k), p)$).

Proof. The goal follows from winwittsectoronexorsectortwolhcenterafterzerodiag(t, k, p). □

Theorem 9.568. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+\text{twopow}(k))$*, then* bxor(wittbitbool(wittwin$(t, \text{twopow}(k)), t+\text{twopow}(k)-(r+2)), \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+2))) = \text{not}(\text{rhcenter}(t+\text{twopow}(k), r+2))$.

Proof. The goal follows from winwittsectoronexorsectortworhcenterafterzerodiag(t, k, r). □

Theorem 9.569. *For every natural number z and natural number k, if $z+2 \leq k$, then if* zerodiagonal(z)*, then if* settledat$(z+2, k)$*, then* lhcenter$(z+2+\text{twopow}(k), \text{twopow}(k)) = \text{true}$.

Proof. The goal follows from lhcenterlatetwopowafterzerodiag(z, k). □

Theorem 9.570. *For every natural number z and natural number k, if $z+2 \leq k$, then if* zerodiagonal(z)*, then if* settledat$(z+2, k)$*, then* wittbitbool(wittwin$(z+2, \text{twopow}(k)), z+2+\text{twopow}(k+1)) = \text{not}(\text{center}(z+2))$.

Proof. The goal follows from winwitttestbitsectortwocenterslotafterzerodiag(z, k). □

Theorem 9.571. *For every natural number z and natural number k, if $z+2 \leq k$, then if* zerodiagonal(z)*, then if* settledat$(z+2, k)$*, then* bxor(wittbitbool(wittwin$(z+2, \text{twopow}(k)), z+2+\text{twopow}(k)), \text{wittbitbool}(\text{wittwin}(z+2, \text{twopow}(k)), z+2+\text{twopow}(k+1))) = \text{not}(\text{center}(z+2+\text{twopow}(k)))$.

Proof. The goal follows from winwittsectoronexorsectortwocenterslotafterzerodiag(z, k). □

Theorem 9.572. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+\text{twopow}(k))$*, then* lhcenter$(t+1+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{not}(\text{lhcenter}(t+\text{twopow}(k), \text{twopow}(k)-(r+1)))$.

Proof. The goal follows from lhcenterlatecrossresetafterzerodiag(t, k, r). □

Theorem 9.573. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t-p+2 \leq k$, then if* zerodiagonal$(t-p)$*, then if* settledby$(t-p+2, t+2+\text{twopow}(k))$*, then* lhcenter$(t+3+\text{twopow}(k), p+1+\text{twopow}(k)) = \text{not}(\text{lhcenter}(t+2+\text{twopow}(k), p+2+\text{twopow}(k)))$.

Proof. The goal follows from lhcenterlateshiftresetafterzerodiag(t, k, p). □

Theorem 9.574. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+\text{twopow}(k))$*, then* bxor(wittbitbool(wittwin$(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1))$, wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1))) = \text{not}(\text{bxor}(\text{rhcenter}(t, r+1), \text{rhcenter}(t+1, r+2)))$.

Proof. The goal follows from winwittsectortwostepcomplementafterzerodiag(t, k, r). □

Theorem 9.575. *For every natural number t and natural number k and natural number r, if $t \leq r$, then if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+\text{twopow}(k))$*, then* wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)))$.

Proof. The goal follows from winwittsectortwostepflipzerogapafterzerodiag(t, k, r). □

Theorem 9.576. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+\text{twopow}(k))$*, then* bxor(bxor(wittbitbool(wittwin$(t, \text{twopow}(k)), t+\text{twopow}(k)-(r+1))$, wittbitbool(wittwin$(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1))), \text{bxor}(\text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k)-(r+1))$, wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1))))) = \text{not}(\text{bxor}(\text{rhcenter}(t+\text{twopow}(k), r+1), \text{rhcenter}(t+1+\text{twopow}(k), r+2)))$.

Proof. The goal follows from winwittsectoronexorsectortwostepcomplementafterzerodiag(t, k, r). □

Theorem 9.577. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+1+\text{twopow}(k))$*, then* lhcenter$(t+1+\text{twopow}(k), \text{twopow}(k)-(r+1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossssteponeofzerodiag(t, k, r). □

Theorem 9.578. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+1+\text{twopow}(k))$*, then* wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-r) = \text{not}(\text{rhcenter}(t+1, r+1))$.

Proof. The goal follows from winwitttestbitscalecrossssteponeofzerodiag(t, k, r). □

Theorem 9.579. *For every natural number t and natural number k and natural number r, if $t < r$, then if $t + r + 2 \leq k$, then if* zerodiagonal$(t + r)$*, then if* settledby$(t + r + 2, t + 1 + \text{twopow}(k))$*, then* wittbitbool(wittwin$(t + 1, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) = \text{true}$.

Proof. The goal follows from winwitttestbitscalecrosszerogapsteponeofzerodiag(t, k, r). □

Theorem 9.580. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r, t + \text{twopow}(k))$*, then if* settledby$(t + r + 2, t + 1 + \text{twopow}(k))$*, then* bxor(wittbitbool(wittwin$(t, \text{twopow}(k)), t+\text{twopow}(k+1)-r)$, wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-r)) = \text{not}(\text{bxor}(\text{rhcenter}(t, r), \text{rhcenter}(t+1, r+1)))$.

Proof. The goal follows from winwittsectortwozerotoblackstepofzerodiag(t, k, r). □

Theorem 9.581. *For every natural number t and natural number k and natural number r, if $t + r + 2 \leq k$, then if* zerodiagonal$(t + r)$*, then if* settledby$(t + r + 2, t + 1 + \text{twopow}(k))$*, then* bxor(wittbitbool(wittwin$(t + 1, \text{twopow}(k)), t+\text{twopow}(k)-r)$, wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k + 1) - r)) = \text{not}(\text{rhcenter}(t + 1 + \text{twopow}(k), r + 1))$.

Proof. The goal follows from winwittsectoronexorsectortworhcentersteponeofzerodiag(t, k, r). □

Theorem 9.582. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r, t + \text{twopow}(k))$*, then if* settledby$(t + r + 2, t + 1 + \text{twopow}(k))$*, then* bxor(bxor(wittbitbool(wittwin$(t, \text{twopow}(k)), t+\text{twopow}(k)-r)$, wittbitbool(wittwin$(t, \text{twopow}(k)), t+\text{twopow}(k+1)-r)$), bxor(wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k)-r)$, wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-r))) = \text{not}(\text{bxor}(\text{rhcenter}(t+\text{twopow}(k), r), \text{rhcenter}(t + 1 + \text{twopow}(k), r + 1)))$.

Proof. The goal follows from winwittsectoronexorsectortwozerotoblackstepofzerodiag(t, k, r). □

Theorem 9.583. *For every natural number t and natural number k and natural number r, if $t+r < \text{twopow}(k)$, then* wittbitbool(wittscale$(k, \text{wittrow}(t)), t+\text{twopow}(k + 1) - r) = \text{rhcenter}(t, r)$.

Proof. The goal follows from phiscalerowtestbitscalecrosscone(t, k, r). □

Theorem 9.584. *For every natural number t and natural number k and natural number r, if $t < r$, then if $t+r <$ twopow(k), then* wittbitbool(wittscale$(k,$ wittrow$(t)), t+$ twopow$(k+1)-r) =$ false.

Proof. The goal follows from phiscalerowtestbitscalecrosszerogapeqfalse(t, k, r). □

Theorem 9.585. *For every natural number t and natural number k and natural number r, if $t \leq r$, then if $t+r+2 <$ twopow(k), then* wittbitbool(crosscarry(wittscale$(k,$ wittrow(t) twopow$(k+1)-(r+1)) =$ false.

Proof. The goal follows from crosscarryrowscalecrosszerogapeqfalse(t, k, r). □

Theorem 9.586. *For every natural number t and natural number k and natural number r, if $t \leq r$, then if $t+r+2 <$ twopow(k), then* wittbitbool(wittwin$(t+1,$ twopow$(k)), t+$twopow$(k+1)-r) =$ wittbitbool(wittphi(wittwin$(t,$ twopow$(k))), t+$ twopow$(k+1)-r)$.

Proof. The goal follows from winwittstepscalecrosszerogapeqphi(t, k, r). □

Theorem 9.587. *For every natural number t and natural number k and natural number r, if $t \leq r$, then if $t+r+2 <$ twopow(k), then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1)-r) =$ bxor(wittbitbool(wittwin$(t+1,$ twopow$(k)), t+$twopow$(k+1)-r)$, wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+1)-(r+1))$ or wittbitbool(wittwin$(t,$ twopow(k) twopow$(k+1)-(r+2)))$.

Proof. The goal follows from winwittinwardstepscalecrosszerogap(t, k, r). □

Theorem 9.588. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r+1)$*, then if* settledby$(t+r+1, t+$twopow$(k))$*, then* lhcenter$(t+1+$twopow$(k),$ twopow$(k)-(r+1)) =$ bxor(lhcenter$(t+$twopow$(k),$ twopow$(k)-r),$ lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+2)))$.

Proof. The goal follows from lhcenterlatecrosslinearsteponebeforezerodiag(t, k, r). □

Theorem 9.589. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r+2)$*, then if* settledby$(t+r+2, t+$twopow$(k))$*, then* lhcenter$(t+1+$twopow$(k),$ twopow$(k)-(r+1)) =$ bxor(lhcenter$(t+$twopow$(k),$ twopow$(k)-r),$ lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+1)))$.

Proof. The goal follows from lhcenterlatecrosslinearsteptwobeforezerodiag(t, k, r). □

Theorem 9.590. *For every natural number t and natural number k and natural number r, if $t \leq r$, then if $t+r+2 \leq k$, then if* zerodiagonal$(t+r+1)$, *then if* settledby$(t + r + 1, t + \text{twopow}(k))$, *then* wittbitbool(wittwin$(t + 1, \text{twopow}(k)), t+\text{twopow}(k+1)-r) = \text{bxor}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+ \text{twopow}(k+1) - r), \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+2)))$.

Proof. The goal follows from winwittsectortwolinearsteponebeforezerodiag(t, k, r). □

Theorem 9.591. *For every natural number t and natural number k and natural number r, if $t \leq r$, then if $t+r+2 \leq k$, then if* zerodiagonal$(t+r+2)$, *then if* settledby$(t + r + 2, t + \text{twopow}(k))$, *then* wittbitbool(wittwin$(t + 1, \text{twopow}(k)), t+\text{twopow}(k+1)-r) = \text{bxor}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+ \text{twopow}(k+1) - r), \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+1)))$.

Proof. The goal follows from winwittsectortwolinearsteptwobeforezerodiag(t, k, r). □

Theorem 9.592. *For every natural number t and natural number k and natural number r, if $t + r + 2 \leq k$, then if* zerodiagonal$(t + r + 2)$, *then if* settledby$(t+r+2, t+\text{twopow}(k))$, *then* lhcenter$(t+\text{twopow}(k), \text{twopow}(k)- r) = \text{lhcenter}(t + \text{twopow}(k), \text{twopow}(k) - (r+1))$.

Proof. The goal follows from lhcenterlatecrosstwinbeforezerodiag(t, k, r). □

Theorem 9.593. *For every natural number t and natural number k and natural number p, if $p \leq t$, then if $t - p + 2 \leq k$, then if* zerodiagonal$(t - p + 2)$, *then if* settledby$(t - p + 2, t + 2 + \text{twopow}(k))$, *then* lhcenter$(t + 2 + \text{twopow}(k), p + 2 + \text{twopow}(k)) = \text{lhcenter}(t + 2 + \text{twopow}(k), p + 1 + \text{twopow}(k))$.

Proof. The goal follows from lhcenterlateshifttwinbeforezerodiag(t, k, p). □

Theorem 9.594. *For every natural number t and natural number k and natural number r, if $t + r + 2 \leq k$, then if* zerodiagonal$(t + r + 2)$, *then if* settledby$(t+r+2, t+\text{twopow}(k))$, *then* bxor(wittbitbool(wittwin$(t, \text{twopow}(k)), t+ \text{twopow}(k+1) - r), \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+1))) = \text{bxor}(\text{rhcenter}(t, r), \text{rhcenter}(t, r+1))$.

Proof. The goal follows from winwittsectortwotwinbeforezerodiag(t, k, r). □

Theorem 9.595. *For every natural number t and natural number k and natural number r, if $t < r$, then if $t+r+2 \leq k$, then if* zerodiagonal$(t+r+2)$, *then if* settledby$(t+r+2, t+$twopow$(k))$, *then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1)-r) =$ wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+1)-$ $(r+1))$.

Proof. The goal follows from winwittsectortwotwinzerogapbeforezerodiag(t, k, r). □

Theorem 9.596. *For every natural number t and natural number k and natural number r, if $t + r + 2 \leq k$, then if* zerodiagonal$(t + r + 2)$, *then if* settledby$(t+r+2, t+$twopow$(k))$, *then* bxor(bxor(wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k)-r)$, wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+1)-r))$, bxor(wittbitbool(wittwin$(t,$ tw twopow$(k)-(r+1))$, wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+1)-$ $(r+1))))) =$ bxor(rhcenter$(t+$twopow$(k), r)$, rhcenter$(t+$twopow$(k), r+1))$.

Proof. The goal follows from winwittsectoronexorsectortwotwinbeforezerodiag(t, k, r). □

Theorem 9.597. *For every natural number P and natural number t and natural number k and natural number p, if $p \leq t$, then if $t - p + 1 \leq k$, then if* zerodiagonal$(t-p)$, *then if* doublesat$(P, t-p)$, *then if* settledby$(t-p+1, t+$ $1+$twopow$(k))$, *then* lhcenter$(t+1+P+$twopow$(k), p+P+$twopow$(k)) =$ not(lhcenter$(t + 1 +$ twopow$(k), p +$ twopow$(k)))$.

Proof. The goal follows from lhcenterlateshiftantiperiodicofdoubles(P, t, k, p). □

Theorem 9.598. *For every natural number P and natural number t and natural number k and natural number r, if $t + P + r + 1 \leq k$, then if* zerodiagonal$(t+P+r)$, *then if* doublesat$(P, t+P+r)$, *then if* settledby$(t+P+$ $r+1, t+$twopow$(k))$, *then* lhcenter$(t+P+$twopow$(k),$ twopow$(k)-(r+1)) =$ not(lhcenter$(t +$ twopow$(k),$ twopow$(k) - (r + P + 1)))$.

Proof. The goal follows from lhcenterlatecrossantiperiodicofdoubles(P, t, k, r). □

Theorem 9.599. *For every natural number P and natural number t and natural number k and natural number p, if $p \leq t$, then if $t + 1 < p +$* twopow(k), *then if $t - p + 1 \leq k$, then if* zerodiagonal$(t - p)$, *then if* doublesat$(P, t - p)$, *then if* settledby$(t - p + 1, t + 1 +$ twopow$(k))$, *then*

bxor(wittbitbool(wittwin(t+1, twopow(k)), t+1+p+twopow(k+1)), wittbitbool(wittwin(t+1+P, twopow(k)), t+1+P+(p+P)+twopow(k+1))) = not(bxor(lhcenter(t+1, p), lhcenter(t + 1 + P, p + P))).

Proof. The goal follows from winwittouterleftstripantiperiodicofdoubles(P, t, k, p). □

Theorem 9.600. *For every natural number P and natural number t and natural number k and natural number r, if $t + P + r + 1 \leq k$, then if* zerodiagonal($t+P+r$)*, then if* doublesat($P, t+P+r$)*, then if* settledby($t+P+r+1, t+$twopow(k))*, then* bxor(wittbitbool(wittwin(t, twopow(k)), t + twopow($k+1$) − ($r+P+1$)), wittbitbool(wittwin($t+P$, twopow(k)), $t+P$ + twopow($k+1$) − ($r+1$))) = not(bxor(rhcenter($t, r+P+1$), rhcenter(t + $P, r+1$))).

Proof. The goal follows from winwittscalecrossantiperiodicofdoubles(P, t, k, r). □

Theorem 9.601. *For every natural number P and natural number t and natural number k and natural number r, if $t+P \leq r$, then if $t+P+r+1 \leq k$, then if* zerodiagonal($t+P+r$)*, then if* doublesat($P, t+P+r$)*, then if* settledby($t+P+r+1, t+$twopow(k))*, then* wittbitbool(wittwin($t+P$, twopow(k)), t + P + twopow($k+1$) − ($r+1$)) = not(wittbitbool(wittwin(t, twopow(k)), t + twopow($k+1$) − ($r+P+1$))).

Proof. The goal follows from winwittscalecrosszerogapantiperiodicofdoubles(P, t, k, r). □

Theorem 9.602. *For every natural number P and natural number t and natural number k and natural number r, if $t + P + r + 1 \leq k$, then if* zerodiagonal($t+P+r$)*, then if* doublesat($P, t+P+r$)*, then if* settledby($t+P+r+1, t+$twopow(k))*, then* bxor(bxor(wittbitbool(wittwin(t, twopow(k)), t + twopow(k) − ($r+P+1$)), wittbitbool(wittwin(t, twopow(k)), t + twopow($k+1$) − ($r+P+1$))), bxor(wittbitbool(wittwin($t+P$, twopow(k)), $t+P$+twopow(k) − ($r+1$)), wittbitbool(wittwin($t+P$, twopow(k)), $t+P+$ twopow($k+1$) − ($r+1$)))) = not(bxor(rhcenter(t + twopow(k), $r+P+1$), rhcenter($t+P+$ twopow(k), $r+1$))).

Proof. The goal follows from winwittsectoronexorsectortwoantiperiodicofdoubles(P, t, k, r). □

Theorem 9.603. *For every natural number P and natural number z and natural number d and natural number s and natural number k, if* doublesat(P, z)*,*

then if $d \leq z$, *then if* $d \leq k$, *then if* $\mathrm{settledby}(d, s + \mathrm{twopow}(k))$, *then* $\mathrm{ldiag}(d, s + P + \mathrm{twopow}(k)) = \mathrm{ldiag}(d, s + \mathrm{twopow}(k))$.

Proof. The goal follows from $\mathrm{ldiagperiodicofdoublesle}(P, z, d, s, k)$. □

Theorem 9.604. *For every natural number* P *and natural number* z *and natural number* t *and natural number* k *and natural number* r, *if* $t + P + r \leq z$, *then if* $z \leq k$, *then if* $\mathrm{doublesat}(P, z)$, *then if* $\mathrm{settledby}(t + P + r, t + \mathrm{twopow}(k))$, *then* $\mathrm{lhcenter}(t + P + \mathrm{twopow}(k), \mathrm{twopow}(k) - r) = \mathrm{lhcenter}(t + \mathrm{twopow}(k), \mathrm{twopow}(k) - (r + P))$.

Proof. The goal follows from $\mathrm{lhcenterlatecrossperiodicofdoublesle}(P, z, t, k, r)$. □

Theorem 9.605. *For every natural number* P *and natural number* z *and natural number* t *and natural number* k *and natural number* r, *if* $t + P + r \leq z$, *then if* $z \leq k$, *then if* $\mathrm{doublesat}(P, z)$, *then if* $\mathrm{settledby}(t + P + r, t + \mathrm{twopow}(k))$, *then* $\mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k + 1) - (r + P)), \mathrm{wittbitbool}(\mathrm{wittwin}(t + P, \mathrm{twopow}(k)), t + P + \mathrm{twopow}(k + 1) - r)) = \mathrm{bxor}(\mathrm{rhcenter}(t, r + P), \mathrm{rhcenter}(t + P, r))$.

Proof. The goal follows from $\mathrm{winwittscalecrossperiodicofdoublesle}(P, z, t, k, r)$. □

Theorem 9.606. *For every natural number* P *and natural number* z *and natural number* t *and natural number* k *and natural number* r, *if* $t + P < r$, *then if* $t + P + r \leq z$, *then if* $z \leq k$, *then if* $\mathrm{doublesat}(P, z)$, *then if* $\mathrm{settledby}(t + P + r, t + \mathrm{twopow}(k))$, *then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t + P, \mathrm{twopow}(k)), t + P + \mathrm{twopow}(k + 1) - r) = \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t + \mathrm{twopow}(k + 1) - (r + P))$.

Proof. The goal follows from $\mathrm{winwittscalecrosszerogapperiodicofdoublesle}(P, z, t, k, r)$. □

Theorem 9.607. *For every natural number* P *and natural number* t *and natural number* k *and natural number* r, *if* $t + P + r + 2 \leq k$, *then if* $\mathrm{zerodiagonal}(t + P + r)$, *then if* $\mathrm{doublesat}(P, t + P + r)$, *then if* $\mathrm{settledby}(t + P + r + 1, t + \mathrm{twopow}(k))$, *then if* $\mathrm{settledby}(t + P + r + 2, t + \mathrm{twopow}(k))$, *then* $\mathrm{lhcenter}(t + P + \mathrm{twopow}(k), \mathrm{twopow}(k) - (r + 1)) = \mathrm{lhcenter}(t + 1 + \mathrm{twopow}(k), \mathrm{twopow}(k) - (r + P + 2))$.

Proof. The goal follows from $\mathrm{lhcenterlatecrossdoubleseqresetstep}(P, t, k, r)$. □

Theorem 9.608. *For every natural number P and natural number t and natural number k and natural number r, if $t + P + r + 2 \leq k$, then if* zerodiagonal$(t+P+r)$*, then if* doublesat$(P, t+P+r)$*, then if* settledby$(t+P+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+P+r+2, t+\text{twopow}(k))$*, then* lhcenter$(t+1+P+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{not}(\text{lhcenter}(t+1+\text{twopow}(k), \text{twopow}(k)-(r+P+2)))$.

Proof. The goal follows from lhcenterlatecrossresetantiperiodicofdoubles(P, t, k, r). □

Theorem 9.609. *For every natural number P and natural number t and natural number k and natural number r, if $t + P + r + 2 \leq k$, then if* zerodiagonal$(t+P+r)$*, then if* doublesat$(P, t+P+r)$*, then if* settledby$(t+P+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+P+r+2, t+\text{twopow}(k))$*, then* bxor$(\text{wittbitbool}(\text{wittwin}(t+P, \text{twopow}(k)), t+P+\text{twopow}(k+1)-(r+1)), \text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+P+1))) = \text{bxor}(\text{rhcenter}(t+P, r+1), \text{rhcenter}(t+1, r+P+2))$.

Proof. The goal follows from winwittscalecrossdoubleseqresetstep(P, t, k, r). □

Theorem 9.610. *For every natural number P and natural number t and natural number k and natural number r, if $t+P \leq r$, then if $t+P+r+2 \leq k$, then if* zerodiagonal$(t+P+r)$*, then if* doublesat$(P, t+P+r)$*, then if* settledby$(t+P+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+P+r+2, t+\text{twopow}(k))$*, then* wittbitbool$(\text{wittwin}(t+P, \text{twopow}(k)), t+P+\text{twopow}(k+1)-(r+1)) = \text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+P+1))$.

Proof. The goal follows from winwittscalecrosszerogapdoubleseqresetstep(P, t, k, r). □

Theorem 9.611. *For every natural number P and natural number t and natural number k and natural number r, if $t + P + r + 2 \leq k$, then if* zerodiagonal$(t+P+r)$*, then if* doublesat$(P, t+P+r)$*, then if* settledby$(t+P+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+P+r+2, t+\text{twopow}(k))$*, then* bxor$(\text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+P+1)), \text{wittbitbool}(\text{wittwin}(t+1+P, \text{twopow}(k)), t+P+\text{twopow}(k+1)-(r+1))) = \text{not}(\text{bxor}(\text{rhcenter}(t+1, r+P+2), \text{rhcenter}(t+1+P, r+2)))$.

Proof. The goal follows from winwittscalecrossresetantiperiodicofdoubles(P, t, k, r). □

Theorem 9.612. *For every natural number P and natural number t and natural number k and natural number r, if $t+P \leq r$, then if $t+P+r+2 \leq k$, then if* zerodiagonal$(t+P+r)$*, then if* doublesat$(P, t+P+r)$*, then if* settledby$(t+P+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+P+r+2, t+\text{twopow}(k))$*, then* wittbitbool(wittwin$(t+1+P, \text{twopow}(k)), t+P+\text{twopow}(k+1)-(r+1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+P+1)))$.

Proof. The goal follows from winwittscalecrosszerogapresetantiperiodicofdoubles(P, t, k, r). □

Theorem 9.613. *For every natural number P and natural number t and natural number k and natural number r, if $t+P+r+2 \leq k$, then if* zerodiagonal$(t+P+r)$*, then if* doublesat$(P, t+P+r)$*, then if* settledby$(t+P+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+P+r+2, t+\text{twopow}(k))$*, then* bxor(bxor(wittbitbool(wittwin$(t+P, \text{twopow}(k)), t+P+\text{twopow}(k)-(r+1))$, wittbitbool(wittwin$(t+P, \text{twopow}(k)), t+P+\text{twopow}(k+1)-(r+1))$), bxor(wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k)-(r+P+1))$, wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+P+1))$)) = bxor(rhcenter$(t+P+\text{twopow}(k), r+1)$, rhcenter$(t+1+\text{twopow}(k), r+P+2)$).

Proof. The goal follows from winwittsectoronexorsectortwodoubleseqresetstep(P, t, k, r). □

Theorem 9.614. *For every natural number P and natural number t and natural number k and natural number r, if $t+P+r+2 \leq k$, then if* zerodiagonal$(t+P+r)$*, then if* doublesat$(P, t+P+r)$*, then if* settledby$(t+P+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+P+r+2, t+\text{twopow}(k))$*, then* bxor(bxor(wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k)-(r+P+1))$, wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+P+1))$), bxor(wittbitbool(wittwin$(t+1+P, \text{twopow}(k)), t+P+\text{twopow}(k)-(r+1))$, wittbitbool(wittwin$(t+1+P, \text{twopow}(k)), t+P+\text{twopow}(k+1)-(r+1))$)) = not(bxor(rhcenter$(t+1+\text{twopow}(k), r+P+2)$, rhcenter$(t+1+P+\text{twopow}(k), r+2)$)).

Proof. The goal follows from winwittsectoronexorsectortworesetantiperiodicofdoubles(P, t, k, r). □

Theorem 9.615. *For every natural number t and natural number k, if $t+2 \leq k$, then if* zerodiagonal$(t+2)$*, then if* settledby$(t, t+\text{twopow}(k))$*, then if* settledby$(t+2, t+\text{twopow}(k))$*, then* zcentereq$(t, \text{lhcenter}(t+\text{twopow}(k), \text{twopow}(k)-1))$.

Proof. The goal follows from lhcenterlatetwopowsuboneeqzcenterbeforezerodiag(t, k).
□

Theorem 9.616. *For every natural number t and natural number k, if $t + 2 \leq k$, then if* zerodiagonal$(t+2)$*, then if* settledby$(t, t+$twopow$(k))$*, then if* settledby$(t+2, t+$twopow$(k))$*, then* zcentereq$(t,$ bxor$($wittbitbool$($wittwin$(t,$ twopow$(k)),$ rhcenter$(t, 1)))$.

Proof. The goal follows from winwitttestbitsectortwosubonezcenterbeforezerodiag(t, k).
□

Theorem 9.617. *For every natural number t and natural number k, if $t + 2 \leq k$, then if* zerodiagonal$(t + 2)$*, then if* settledby$(t, t +$ twopow$(k))$*, then if* settledby$(t + 2, t +$ twopow$(k))$*, then* zcentereq$(t,$ bxor$($rhcenter$(t +$ twopow$(k), 1),$ bxor$($wittbitbool$($wittwin$(t,$ twopow$(k)), t+$twopow$(k)-1),$ wittbitbool$($wittwin$(t,$ twopow$(k + 1) - 1))))$.

Proof. The goal follows from winwittsectoronexorsectortwosubonezcenterbeforezerodiag(t, k).
□

Theorem 9.618. *For every natural number P and natural number z and natural number k, if $z+1 \leq k$, then if* zerodiagonal(z)*, then if* doublesat(P, z)*, then if* settledby$(z+1, z+1+$twopow$(k))$*, then* zcentereq$(z+1,$ not$($lhcenter$(z+ 1 + P +$ twopow$(k), P +$ twopow$(k))))$.

Proof. The goal follows from lhcenterlateshifttwopowantiperiodiczcenterofdoubles(P, z, k).
□

Theorem 9.619. *For every natural number P and natural number z and natural number k, if $z+1 \leq k$, then if* zerodiagonal(z)*, then if* doublesat(P, z)*, then if* settledby$(z+1, z+1+$twopow$(k))$*, then* zcentereq$(z+1,$ not$($bxor$($wittbitbool$($wittwin$(z+ 1 + P,$ twopow$(k)), z + 1 + P + P +$ twopow$(k+1)),$ lhcenter$(z + 1 + P, P))))$.

Proof. The goal follows from winwittouterleftstripzcenterantiperiodicofdoubles(P, z, k).
□

Theorem 9.620. *For every natural number P and natural number z and natural number k, if $z + 1 + P + P <$* twopow(k)*, then if $z + 1 \leq k$, then if* zerodiagonal(z)*, then if* doublesat(P, z)*, then if* settledby$(z + 1, z + 1 +$ twopow$(k))$*, then* zcentereq$(z+1,$ not$($bxor$($lhcenter$(z+1+P+$twopow$(k), P),$ bxor$($wittbitbool$($wittwin$(1 + P,$ twopow$(k)), z + 1 + P + P +$ twopow$(k)),$ wittbitbool$($wittwin$(z + 1 + P,$ twopow$(k)), z + 1 + P + P +$ twopow$(k + 1))))))$.

Proof. The goal follows from winwittsectoronexorsectortwozcenterantiperiodicofdoubles(P, z, k). □

Theorem 9.621. *For every natural number P and natural number z and natural number d and natural number k, if* doublesat(P, z)*, then if $d \leq z$, then if $d \leq k$, then if* settledby(d, d+twopow(k))*, then* zcentereq(d, lhcenter(d+P + twopow(k), P + twopow(k))).

Proof. The goal follows from lhcenterlateshifttwopowperiodiczcenterofdoublesle(P, z, d, k). □

Theorem 9.622. *For every natural number z and natural number k, if z+2 $\leq k$, then if* zerodiagonal(z)*, then if* settledby(z+2, z+1+twopow(k))*, then* lhcenter(z + 2 + twopow(k), twopow(k) − 1) = not(lhcenter(z + 1 + twopow(k), twopow(k))).

Proof. The goal follows from lhcenterlatetwopowsuboneresetafterzerodiag(z, k). □

Theorem 9.623. *For every natural number z and natural number k, if z+2 $\leq k$, then if* zerodiagonal(z)*, then if* settledby(z+2, z+1+twopow(k))*, then* bxor(wittbitbool(wittwin(z+1, twopow(k)), z+1+twopow(k+1)), wittbitbool(wittwin(z+2, twopow(k)), z+2+twopow(k+1)−1)) = not(bxor(center(z+1), rhcenter(z+2, 1))).

Proof. The goal follows from winwittsectortwocentertosubonestepcomplementafterzerodiag(z, k). □

Theorem 9.624. *For every natural number z and natural number k, if z+2 $\leq k$, then if* zerodiagonal(z)*, then if* settledby(z+2, z+1+twopow(k))*, then* bxor(bxor(wittbitbool(wittwin(z+1, twopow(k)), z+1+twopow(k)), wittbitbool(wittwin(z+1, twopow(k)), z+1+twopow(k+1))), bxor(wittbitbool(wittwin(z+2, twopow(k)), z+2+twopow(k)−1), wittbitbool(wittwin(z+2, twopow(k)), z+2+twopow(k+1)−1))) = not(bxor(center(z+1+twopow(k)), rhcenter(z+2+twopow(k), 1))).

Proof. The goal follows from winwittsectoronexorsectortwocentertosubonestepcomplementafterzerodiag □

Theorem 9.625. *For every natural number z and natural number k, if z+2 $\leq k$, then if* zerodiagonal(z)*, then if* settledby(z+1, z+1+twopow(k))*, then if* settledby(z+2, z+1+twopow(k))*, then* zcentereq(z+1, not(lhcenter(z+2 + twopow(k), twopow(k) − 1))).

Proof. The goal follows from lhcenterlatetwopowsuboneeqnotzcenterafterzerodiag(z, k). □

Theorem 9.626. *For every natural number z and natural number k, if* $z+2 \leq k$, *then if* zerodiagonal(z), *then if* settledby($z+1, z+1+$twopow(k)), *then if* settledby($z+2, z+1+$twopow(k)), *then* zcentereq($z+1$, not(bxor(wittbitbool(wittwin($z+2$, twopow(k)), $z+2+$twopow($k+1$) $-$ 1), rhcenter($z+2, 1$)))).

Proof. The goal follows from winwitttestbitsectortwosubonenotzcenterafterzerodiag(z, k). □

Theorem 9.627. *For every natural number z and natural number k, if* $z+2 \leq k$, *then if* zerodiagonal(z), *then if* settledby($z+1, z+1+$twopow(k)), *then if* settledby($z+2, z+1+$twopow(k)), *then* zcentereq($z+1$, not(bxor(rhcenter($z+2+$twopow(k), 1), bxor(wittbitbool(wittwin($z+2$, twopow(k)), $z+2+$twopow(k)$-$ 1), wittbitbool(wittwin($z+2$, twopow(k)), $z+2+$twopow($k+1$) $-$ 1))))).

Proof. The goal follows from winwittsectoronexorsectortwosubonenotzcenterafterzerodiag(z, k). □

Theorem 9.628. *For every natural number P and natural number z and natural number k, if* $z+2 \leq k$, *then if* zerodiagonal(z), *then if* doublesat(P, z), *then if* settledby($z+1, z+1+$twopow(k)), *then if* settledby($z+2, z+1+$twopow(k)), *then* lhcenter($z+1+P+$twopow(k), $P+$twopow(k)) = lhcenter($z+2+$twopow(k), twopow(k) $-$ 1).

Proof. The goal follows from lhcenterlateshifttwopowdoubleseqafterzerodiagsubone(P, z, k). □

Theorem 9.629. *For every natural number z and natural number k, if* $z+2 \leq k$, *then if* zerodiagonal(z), *then if* settledby($z+2, z+2+$twopow(k)), *then* lhcenter($z+3+$twopow(k), twopow(k)) = not(lhcenter($z+2+$twopow(k), $1+$ twopow(k))).

Proof. The goal follows from lhcenterlatetwopowresetthreeafterzerodiag(z, k). □

Theorem 9.630. *For every natural number z and natural number k, if* $z+2 \leq k$, *then if* zerodiagonal(z), *then if* settledby($z+2, z+2+$twopow(k)), *then* bxor(wittbitbool(wittwin($z+2$, twopow(k)), $z+2+1+$twopow($k+1$)), wittbitbool(wittwin($z+3$, twopow(k)), $z+3+$twopow($k+1$))) = not(bxor(lhcenter($z+2, 1$), center($z+3$))).

Proof. The goal follows from winwittsectortwoonetocenterstepcomplementafterzerodiag(z, k). □

Theorem 9.631. *For every natural number z and natural number k, if $z+2 \leq k$, then if* zerodiagonal(z)*, then if* settledby$(z+2, z+2+\text{twopow}(k))$*, then* bxor(bxor(wittbitbool(wittwin$(z+2, \text{twopow}(k))$, $z+2+1+\text{twopow}(k))$, wittbitbool(wittwin$(z+2, \text{twopow}(k))$, $z+2+1+\text{twopow}(k+1)))$, bxor(wittbitbool(wittwin$(z+3, \text{twopow}(k))$, $z+3+\text{twopow}(k))$, wittbitbool(wittwin$(z+3, \text{twopow}(k))$, $z+3+\text{twopow}(k+1))))) = \text{not}(\text{bxor}(\text{lhcenter}(z+2+\text{twopow}(k), 1), \text{center}(z+3+\text{twopow}(k))))$.

Proof. The goal follows from winwittsectoronexorsectortwoonetocenterstepcomplementafterzerodiag$(z,$ □

Theorem 9.632. *For every natural number t and natural number k and natural number Q and natural number r and natural number a and natural number L, if $t+Q < r$, then if $t+Q+r < \text{twopow}(k)$, then if $a \leq t+\text{twopow}(k+1)-r$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$*, then* lhcenter$(t+Q+\text{twopow}(k), \text{twopow}(k)-r) = \text{lhcenter}(t+\text{twopow}(k), \text{twopow}(k)-r)$.

Proof. The goal follows from lhcenterlatecrosseqofwinwittagreezerogap(t, k, Q, r, a, L). □

Theorem 9.633. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+2 < r$, then if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r, t+\text{twopow}(k))$*, then if* settledby$(t+r+2, t+2+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-r$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then not* winwittagree$(t, k, 2, a, L)$.

Proof. The goal follows from notwinwittagreetwoscalecrosszerogapofzerodiag(t, k, r, a, L). □

Theorem 9.634. *For every natural number t and natural number k and natural number Q and natural number r and natural number z and natural number a and natural number L, if $t+Q < r$, then if $t+Q+r \leq k$, then if* zerodiagonal$(t+r)$*, then if* zerodiagonal(z)*, then if $t+Q+r = z+2$, then if* settledby$(t+r, t+\text{twopow}(k))$*, then if* settledby$(z+2, t+Q+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-r$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then not* winwittagree$(t, k, Q, a, L)$.

Proof. The goal follows from notwinwittagreescalecrosszerogapofzeroandafterzerodiag(t, k, Q, r, z, a, L).
□

Theorem 9.635. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 1 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if $a \leq t + \text{twopow}(k + 1) - (r + 1)$, then if $t + \text{twopow}(k + 1) - (r + 1) \leq a + L$, then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)),$ $\text{twopow}(k + 1) - (r + Q + 1)))$.

Proof. The goal follows from winwittscalecrosszerogapshiftflipofdoublesandwinwittagree(Q, t, k, r, a, L).
□

Theorem 9.636. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+1 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* doublesat$(Q, t+Q+r)$, *then if* settledby$(t+Q+r+1, t+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+1)$, then if $t+\text{twopow}(k+1)-(r+1) \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t+\text{twopow}(k), \text{twopow}(k) - (r + 1)) = \text{not}(\text{lhcenter}(t + \text{twopow}(k), \text{twopow}(k) - (r + Q + 1)))$.

Proof. The goal follows from lhcenterlatecrossshiftflipofdoublesandwinwittagreezerogap(Q, t, k, r, a, L).
□

Theorem 9.637. *For every natural number t and natural number k and natural number Q and natural number p and natural number a and natural number L, if $t+Q+p < \text{twopow}(k)$, then if $a \leq t+p+\text{twopow}(k+1)$, then if $t+p+\text{twopow}(k+1) \leq a+L$, then if* lhcenter$(t+Q, p) = \text{lhcenter}(t, p)$, *then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t+Q+\text{twopow}(k), p+\text{twopow}(k)) = \text{lhcenter}(t + \text{twopow}(k), p + \text{twopow}(k))$.

Proof. The goal follows from lhcenterlateshifteqofwinwittagreeouterleftstrip(t, k, Q, p, a, L).
□

Theorem 9.638. *For every natural number t and natural number k and natural number p and natural number a and natural number L, if $p \leq t$, then if $t - p + 2 \leq k$, then if* zerodiagonal$(t - p)$, *then if* settledby$(t - p, t + \text{twopow}(k))$, *then if* settledby$(t - p + 2, t + 2 + \text{twopow}(k))$, *then if $a \leq t + p + \text{twopow}(k + 1)$, then if $t + p + \text{twopow}(k + 1) \leq a + L$, then if* lhcenter$(t + 2, p) = \text{lhcenter}(t, p)$, *then not* winwittagree$(t, k, 2, a, L)$.

Proof. The goal follows from notwinwittagreetwoouterleftstripofzerodiag(t, k, p, a, L).
□

Theorem 9.639. *For every natural number t and natural number k and natural number Q and natural number a and natural number L, if $t + Q \leq k$, then if* settledby$(t, t + \text{twopow}(k))$*, then if* settledby$(t + Q, t + Q + \text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)$, then if $t+\text{twopow}(k+1) \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$*, then* zcenterxoreq$(t+Q, t, \text{bxor}(\text{center}(t+Q), \text{center}(t)))$.

Proof. The goal follows from zcentershiftxoreqcenterxorofwinwittagreesectortwocenterslot(t, k, Q, a, L).
□

Theorem 9.640. *For every natural number t and natural number k and natural number Q and natural number a and natural number L, if $t+Q \leq k$, then if* settledby$(t, t+\text{twopow}(k))$*, then if* settledby$(t+Q, t+Q+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)$, then if $t+\text{twopow}(k+1) \leq a+L$, then if* center$(t+Q) = \text{center}(t)$*, then if* winwittagree$(t, k, Q, a, L)$*, then* zcenterxoreq$(t + Q, t, \text{false})$.

Proof. The goal follows from zcentershifteqofwinwittagreesectortwocenterslot(t, k, Q, a, L).
□

Theorem 9.641. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number a and natural number L, if $T \leq t$, then if $t+m \cdot P \leq k$, then if* settledby$(t, t+\text{twopow}(k))$*, then if* settledby$(t+m \cdot P, t+m \cdot P+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)$, then if $t+\text{twopow}(k+1) \leq a+L$, then if* winwittagree$(t, k, m \cdot P, a, L)$*, then* zcenterxoreq$(t + m \cdot P, t, \text{false})$.

Proof. Assume hP, $hper$. The goal follows from periodiczcentershifteqofwinwittagreesectortwocenterslot
□

Theorem 9.642. *For every natural number z and natural number k and natural number a and natural number L, if $z+2 \leq k$, then if* zerodiagonal(z)*, then if* settledby$(z, z+\text{twopow}(k))$*, then if* settledby$(z+2, z+2+\text{twopow}(k))$*, then if $a \leq z + \text{twopow}(k + 1)$, then if $z + \text{twopow}(k + 1) \leq a + L$, then if* center$(z + 2) = \text{center}(z)$*, then not* winwittagree$(z, k, 2, a, L)$.

Proof. The goal follows from notwinwittagreetwosectortwocenterslotofzerodiag(z, k, a, L).
□

Theorem 9.643. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z_1 and natural number z_2 and natural number k and natural number a and natural number L, if $T \leq z_1$, then if $z_1 + m \cdot P \leq k$, then if* zerodiagonal(z_1)*, then if* zerodiagonal(z_2)*, then if $z_1 + m \cdot P = z_2 + 2$, then if* settledby$(z_1, z_1 + \text{twopow}(k))$*, then if* settledby$(z_2 + 2, z_2 + 2 + \text{twopow}(k))$*, then if $a \leq z_1 + \text{twopow}(k+1)$, then if $z_1 + \text{twopow}(k+1) \leq a + L$, then not* winwittagree$(z_1, k, m \cdot P, a, L)$.

Proof. Assume hP, $hper$. The goal follows from notperiodicwinwittagreesectortwocenterslotofzeroandaft

□

Theorem 9.644. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+1 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* doublesat$(Q, t+Q+r)$*, then if* settledby$(t+Q+r+1, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+Q+1)$, then if $t+\text{twopow}(k+1)-(r+Q+1) \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$*, then* wittbitbool(wittwin$(t+Q, \text{twopow}(k)), t+Q+\text{twopow}(k+1)-(r+1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t+Q, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+1)))$.

Proof. The goal follows from winwittscalecrosszerogaplateshiftflipofdoublesandwinwittagree$(Q, t, k, r, a,$

□

Theorem 9.645. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 1 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$*, then if $a \leq t + \text{twopow}(k+1) - (r + Q + 1)$, then if $t + \text{twopow}(k+1) - (r+Q+1) \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$*, then* lhcenter$(t + Q + \text{twopow}(k), \text{twopow}(k) - (r + 1)) = \text{not}(\text{lhcenter}(t + Q + \text{twopow}(k), \text{twopow}(k) - (r + Q + 1)))$.

Proof. The goal follows from lhcenterlatecrosslateshiftflipofdoublesandwinwittagreezerogap$(Q, t, k, r, a,$

□

Theorem 9.646. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 1 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$*, then if $a \leq t + \text{twopow}(k+1) - (r + Q + 1)$, then if $t + \text{twopow}(k+1) - (r+1) \leq a + L$, then if* winwittagree$(t, k, Q, a, L)$*,*

then wittbitbool(wittwin($t + Q$, twopow(k)), t + twopow($k + 1$) − ($r + 1$)) = not(wittbitbool(wittwin(t, twopow(k)), t + twopow($k + 1$) − ($r + 1$))).

Proof. The goal follows from winwittscalecrosszerogapsamebitflipofdoublesandwinwittagree(Q, t, k, r, a,

□

Theorem 9.647. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal($t+Q+r$), *then if* doublesat($Q, t+Q+r$), *then if* settledby($t+Q+r+1, t+$twopow(k)), *then if* settledby($t+Q+r+2, t+$twopow(k)), *then if $a \leq t+$*twopow($k+1$)$-$($r+1$), *then if $t+$*twopow($k+1$)$-$($r+1$) $\leq a+L$, *then if* winwittagree($t+1, k, Q, a, L$), *then* wittbitbool(wittwin($t+1$, twopow(k)), $t+$twopow($k+1$)$-$($r+1$)) = not(wittbitbool(wittwin($t+1$, twopow(k)), t + twopow($k+1$) − ($r+Q+1$))).

Proof. The goal follows from winwittscalecrosszerogapresetshiftflipofdoublesandwinwittagree(Q, t, k, r, a

□

Theorem 9.648. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal($t + Q + r$), *then if* doublesat($Q, t + Q + r$), *then if* settledby($t + Q + r + 1, t +$ twopow(k)), *then if* settledby($t + Q + r + 2, t +$ twopow(k)), *then if $a \leq t+$*twopow($k+1$)$-$($r+1$), *then if $t+$*twopow($k+1$)$-$($r+1$) $\leq a+L$, *then if* winwittagree($t+1, k, Q, a, L$), *then* lhcenter($t+1+$twopow(k), twopow(k)$-$($r+2$)) = not(lhcenter($t + 1 +$ twopow(k), twopow(k) − ($r + Q + 2$))).

Proof. The goal follows from lhcenterlatecrossresetshiftflipofdoublesandwinwittagreezerogap(Q, t, k, r, a

□

Theorem 9.649. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal($t+Q+r$), *then if* doublesat($Q, t+Q+r$), *then if* settledby($t+Q+r+1, t+$twopow(k)), *then if* settledby($t+Q+r+2, t+$twopow(k)), *then if $a \leq t+$*twopow($k+1$)$-$($r+1$), *then if $t+$*twopow($k+1$)$-$($r+1$) $\leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin(t, twopow(k)), $t+$twopow($k+1$)$-$($r+1$)) = wittbitbool(wittwin($t+1$, twopow(k)), t + twopow($k+1$) − ($r+Q+1$)).

Proof. The goal follows from winwittscalecrosszerogapshifteqsteponeofdoublesandwinwittagree(Q, t, k, r

□

Theorem 9.650. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$*, then if $a \leq t +$* twopow$(k + 1) - (r + 1)$*, then if $t +$* twopow$(k + 1) - (r + 1) \leq a + L$*, then if* winwittagree$(t, k, Q, a, L)$*, then if* winwittagree$(t + 1, k, Q, a, L)$*, then* wittbitbool(wittwin$(t + 1,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1)) =$ not(wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1))$).

Proof. The goal follows from winwittscalecrosszerogapstepflipatshiftofdoublesandwinwittagree$(Q, t, k, r,$ □

Theorem 9.651. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$*, then if $a \leq t +$* twopow$(k + 1) - (r + 1)$*, then if $t +$* twopow$(k + 1) - (r + 1) \leq a + L$*, then if* winwittagree$(t, k, Q, a, L)$*, then if* winwittagree$(t + 1, k, Q, a, L)$*, then* lhcenter$(t + 1 +$ twopow$(k),$ twopow$(k) - (r + 2)) =$ not(lhcenter$(t +$ twopow$(k),$ twopow$(k) - (r + 1))$).

Proof. The goal follows from lhcenterlatecrossstepflipatshiftofdoublesandwinwittagreezerogap$(Q, t, k, r,$ □

Theorem 9.652. *For every natural number t and natural number k and natural number r, if $r+3 \leq$* twopow(k)*, then* lhcenter$(t+1+$twopow$(k),$ twopow$(k)-(r+2)) =$ bxor(lhcenter$(t +$ twopow$(k),$ twopow$(k) - (r + 1)),$ lhcenter$(t +$ twopow$(k),$ twopow$(k)-(r+2))$ or lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+3)))$.

Proof. The goal follows from lhcenterlatecrossstep(t, k, r). □

Theorem 9.653. *For every natural number t and natural number k and natural number r, if $r+3 \leq$* twopow(k)*, then if* lhcenter$(t+1+$twopow$(k),$ twopow$(k)-(r+2)) =$ not(lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+1)))$*, then* (lhcenter$(t+$ twopow$(k),$ twopow$(k)-(r+2))$ or lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+3))) =$ true.

Proof. The goal follows from lhcenterlatecrossoreqtrueofstepflip(t, k, r). □

Theorem 9.654. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$*, then if* $a \leq t + \text{twopow}(k+1) - (r+1)$*, then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$*, then if* winwittagree$(t, k, Q, a, L)$*, then if* winwittagree$(t+1, k, Q, a, L)$*, then* (lhcenter$(t+\text{twopow}(k), \text{twopow}(k)-(r+2))$ or lhcenter$(t+\text{twopow}(k), \text{twopow}(k)-(r+3))) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossoreqtrueatshiftofdoublesandwinwittagreezerogap$(Q, t, k, r$ □

Theorem 9.655. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$*, then if* $a \leq t + \text{twopow}(k+1) - (r+1)$*, then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$*, then if* winwittagree$(t, k, Q, a, L)$*, then if* winwittagree$(t+1, k, Q, a, L)$*, then* (wittbitbool(wittwin$(t, \text{twopow}(k))), t + \text{twopow}(k+1) - (r+2))$ or wittbitbool(wittwin$(t, \text{twopow}(k))), t + \text{twopow}(k+1) - (r+3))) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogaporeqtrueatshiftofdoublesandwinwittagree$(Q, t, k, $ □

Theorem 9.656. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* zerodiagonal$(t + r + 2)$*, then if* settledby$(t + r + 2, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$*, then if* $a \leq t + \text{twopow}(k+1) - (r+1)$*, then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$*, then if* winwittagree$(t, k, Q, a, L)$*, then if* winwittagree$(t+1, k, Q, a, L)$*, then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r+3)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinrplusthreeofdoublesandwinwittagreeandzerodiagtwo$(Q$ □

Theorem 9.657. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $1 \leq Q$, then if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$,*

then if zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* zerodiagonal$(t + r + 3)$, *then if* settledby$(t + r + 3, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - (r+1)$, *then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then if* winwittagree$(t + 1, k, Q, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 2)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinrplustwoofdoublesandwinwittagreeandzerodiagthree(Q
□

Theorem 9.658. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $1 \leq Q$, then if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* zerodiagonal$(t + r + 3)$, *then if* settledby$(t + r + 3, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - (r+1)$, *then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then if* winwittagree$(t + 1, k, Q, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinrplusoneofdoublesandwinwittagreeandzerodiagthree(Q
□

Theorem 9.659. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $1 \leq Q$, then if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* zerodiagonal$(t + r + 3)$, *then if* settledby$(t + r + 3, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - (r+1)$, *then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then if* winwittagree$(t + 1, k, Q, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + Q + 1)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspindoublingdiagfalseofdoublesandwinwittagreeandzerodia
□

Theorem 9.660. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 3 \leq r$, then if $t + 3 + r + 2 \leq k$, then if* zerodiagonal$(t + 3 + r)$, *then if* doublesat$(3, t + 3 + r)$, *then if* settledby$(t + 3 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 3 + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + 3 + r + 2, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 1)$, *then if* $t + \text{twopow}(k + 1) - (r + 1) \leq a + L$,

then if winwittagree$(t, k, 3, a, L)$, *then if* winwittagree$(t + 1, k, 3, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinbeforezerodiagofdoublesthreeandwinwittagree$(t, k, r, a,$
□

Theorem 9.661. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if $a \leq t + \text{twopow}(k+1) - (r+1)$, then if $t + \text{twopow}(k+1) - (r+1) \leq a + L$, then if* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 2)) = \text{false}$, *then if* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 3)) = \text{false}$, *then if* winwittagree$(t, k, Q, a, L)$, *then not* winwittagree$(t + 1, k, Q, a, L)$.

Proof. The goal follows from notwinwittagreesteppairatshiftofdoublesandbothfalse(Q, t, k, r, a, L).
□

Theorem 9.662. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if $a \leq t + \text{twopow}(k+1) - (r+Q+1)$, then if $t + \text{twopow}(k+1) - (r+Q+1) \leq a + L$, then if* winwittagree$(t+1, k, Q, a, L)$, *then* wittbitbool(wittwin$(t+1+Q, \text{twopow}(k)), t+Q+\text{twopow}(k+1)-(r+1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t+1+Q, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+1)))$.

Proof. The goal follows from winwittscalecrosszerogapresetlateshiftflipofdoublesandwinwittagree$(Q, t, k$
□

Theorem 9.663. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* doublesat$(Q, t+Q+r)$, *then if* settledby$(t+Q+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+Q+1)$, then if $t+\text{twopow}(k+1)-(r+Q+1) \leq a+L$, then if* winwittagree$(t+1, k, Q, a, L)$, *then* lhcenter$(t + 1 + Q + \text{twopow}(k), \text{twopow}(k) - (r + 2)) = \text{not}(\text{lhcenter}(t + 1 + Q + \text{twopow}(k), \text{twopow}(k) - (r + Q + 2)))$.

Proof. The goal follows from lhcenterlatecrossresetlateshiftflipofdoublesandwinwittagreezerogap$(Q, t, k,$
□

Theorem 9.664. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* doublesat$(Q, t+Q+r)$, *then if* settledby$(t+Q+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+Q+1)$, *then if* $t+\text{twopow}(k+1)-(r+1) \leq a+L$, *then if* winwittagree$(t+1, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t+1+Q, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)))$.

Proof. The goal follows from winwittscalecrosszerogapresetsamebitflipofdoublesandwinwittagree$(Q, t, k,$ □

Theorem 9.665. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* doublesat$(Q, t+Q+r)$, *then if* settledby$(t+Q+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+Q+1)$, *then if* $t+\text{twopow}(k+1)-(r+Q+1) \leq a+L$, *then if* winwittagree$(t, k, Q, a, L)$, *then if* winwittagree$(t+1, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t+1+Q, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t+Q, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)))$.

Proof. The goal follows from winwittscalecrosszerogaplatestepflipatshiftofdoublesandwinwittagree$(Q, t,$ □

Theorem 9.666. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* doublesat$(Q, t+Q+r)$, *then if* settledby$(t+Q+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+Q+1)$, *then if* $t+\text{twopow}(k+1)-(r+Q+1) \leq a+L$, *then if* winwittagree$(t, k, Q, a, L)$, *then if* winwittagree$(t+1, k, Q, a, L)$, *then* lhcenter$(t+1+Q+\text{twopow}(k), \text{twopow}(k)-(r+Q+2)) = \text{not}(\text{lhcenter}(t+Q+\text{twopow}(k), \text{twopow}(k)-(r+Q+1)))$.

Proof. The goal follows from lhcenterlatecrosslatestepflipatshiftofdoublesandwinwittagreezerogap$(Q, t, k$ □

Theorem 9.667. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+$

$Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$, *then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+Q+1)$, *then if* $t+$twopow$(k+1)-(r+Q+1) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then if* winwittagree$(t+1, k, Q, a, L)$, *then* (lhcenter$(t + Q +$ twopow$(k),$ twopow$(k) - (r + Q + 2))$ or lhcenter$(t + Q +$ twopow$(k),$ twopow$(k) - (r + Q + 3))) =$ true.

Proof. The goal follows from lhcenterlatecrosslateoreqtrueatshiftofdoublesandwinwittagreezerogap$(Q, t,$

□

Theorem 9.668. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t +$ $Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* settledby$(t + Q + r +$ $1, t +$ twopow$(k))$, *then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$, *then if* $a \leq t +$ twopow$(k + 1) - (r + Q + 1)$, *then if* $t +$ twopow$(k + 1) - (r +$ $Q + 1) \leq a + L$, *then if* winwittagree(t, k, Q, a, L), *then if* winwittagree$(t +$ $1, k, Q, a, L)$, *then* (wittbitbool(wittwin$(t + Q,$ twopow$(k)), t +$ twopow$(k +$ $1) - (r + 2))$ or wittbitbool(wittwin$(t + Q,$ twopow$(k)), t +$ twopow$(k + 1) -$ $(r + 3))) =$ true.

Proof. The goal follows from winwittscalecrosszerogaplateoreqtrueatshiftofdoublesandwinwittagree$(Q, t$

□

Theorem 9.669. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 2 \leq r$, then if $t+2+r+2 \leq k$, then if* zerodiagonal$(t+2+r)$, *then if* doublesat$(2, t+$ $2 + r)$, *then if* settledby$(t + 2 + r, t +$ twopow$(k))$, *then if* settledby$(t + 2 +$ $r + 1, t +$ twopow$(k))$, *then if* settledby$(t + 2 + r + 2, t +$ twopow$(k))$, *then if* $a \leq t +$ twopow$(k + 1) - (r + 1)$, *then if* $t +$ twopow$(k + 1) - (r + 1) \leq a + L$, *then if* winwittagree$(t, k, 2, a, L)$, *then if* winwittagree$(t + 1, k, 2, a, L)$, *then* lhcenter$(t +$ twopow$(k),$ twopow$(k) - (r + 3)) =$ true.

Proof. The goal follows from lhcenterlatecrosspinrplusthreeofdoublestwoandwinwittagree(t, k, r, a, L).

□

Theorem 9.670. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 2 \leq r$, then if $t+2+r+2 \leq k$, then if* zerodiagonal$(t+2+r)$, *then if* doublesat$(2, t+$ $2 + r)$, *then if* settledby$(t + 2 + r, t +$ twopow$(k))$, *then if* settledby$(t + 2 +$ $r + 1, t +$ twopow$(k))$, *then if* settledby$(t + 2 + r + 2, t +$ twopow$(k))$, *then if* $a \leq t +$ twopow$(k + 1) - (r + 1)$, *then if* $t +$ twopow$(k + 1) - (r + 1) \leq a + L$,

then if winwittagree$(t, k, 2, a, L)$, *then if* winwittagree$(t + 1, k, 2, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 1)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspinrplusonefalseofdoublestwoandwinwittagree(t, k, r, a, L) □

Theorem 9.671. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 2 \leq r$, then if $t+2+r+2 \leq k$, then if* zerodiagonal$(t+2+r)$, *then if* doublesat$(2, t+2+r)$, *then if* settledby$(t + 2 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 2 + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + 2 + r + 2, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 1)$, *then if* $t + \text{twopow}(k + 1) - (r + 1) \leq a + L$, *then if* winwittagree$(t, k, 2, a, L)$, *then if* winwittagree$(t + 1, k, 2, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - r) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspinrfalseofdoublestwoandwinwittagree(t, k, r, a, L). □

Theorem 9.672. *For every natural number Q and natural number z and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r \leq z$, then if $z \leq k$, then if* doublesat(Q, z), *then if* settledby$(t + Q + r, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - r$, *then if* $t + \text{twopow}(k+1) - r \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool(wittwin$(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) = \text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + Q))$.

Proof. The goal follows from winwittscalecrosszerogapshifteqofdoublesleandwinwittagree$(Q, z, t, k, r, a,$ □

Theorem 9.673. *For every natural number Q and natural number z and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r \leq z$, then if $z \leq k$, then if* doublesat(Q, z), *then if* settledby$(t + Q + r, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - r$, *then if* $t + \text{twopow}(k+1) - r \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - r) =$ lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + Q))$.

Proof. The goal follows from lhcenterlatecrossshifteqofdoublesleandwinwittagreezerogap$(Q, z, t, k, r, a, L$ □

Theorem 9.674. *For every natural number Q and natural number z and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r \leq z$, then if*

$z \leq k$, *then if* doublesat(Q, z), *then if* settledby$(t+Q+r, t+$twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+Q)$, *then if* $t+$twopow$(k+1)-r \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin$(t+Q,$ twopow$(k)), t+$ twopow$(k+1)-r) =$ wittbitbool(wittwin$(t,$ twopow$(k)), t+$twopow$(k+1)-r)$.

Proof. The goal follows from winwittscalecrosszerogapsamebiteqofdoublesleandwinwittagree$(Q, z, t, k, r,$
□

Theorem 9.675. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q<r$, then if $t+Q+r \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t+Q+r, t+Q+$twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-r$, *then if* $t+$twopow$(k+1)-r \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* lhcenter$(t+$ twopow$(k),$ twopow$(k)-r) =$ false.

Proof. The goal follows from lhcenterlatecrosseqfalseatshiftofzerodiagandwinwittagreezerogap$(Q, t, k, r,$
□

Theorem 9.676. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r+1$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t+Q+r+2, t+Q+$ twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+2)$, *then if* $t+$twopow$(k+1)-(r+2) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+2)) =$ true.

Proof. The goal follows from lhcenterlatecrosseqtrueatshiftafterzerodiagandwinwittagreezerogap$(Q, t, k,$
□

Theorem 9.677. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r+1$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t+Q+r+2, t+Q+$ twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+2)$, *then if* $t+$twopow$(k+1)-(r+2) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* lhcenter$(t+1+$twopow$(k),$ twopow$(k)-(r+2)) =$ not(lhcenter$(t+$ twopow$(k),$ twopow$(k)-(r+1)))$.

Proof. The goal follows from lhcenterlatecrossstepflipatshiftafterzerodiagofsinglewinwittagreezerogap$(Q$
□

Theorem 9.678. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t \leq r$, then if $t + Q \leq r + 1$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* settledby$(t + Q + r + 2, t + Q +$ twopow$(k))$*, then if $a \leq t +$ twopow$(k + 1) - (r + 2)$, then if $t +$ twopow$(k + 1) - (r + 2) \leq a + L$, then if* winwittagree$(t, k, Q, a, L)$*, then* wittbitbool(wittwin$(t + 1,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1)) =$ not(wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1))$).

Proof. The goal follows from winwittscalecrosszerogapstepflipatshiftafterzerodiagofsinglewinwittagree(Q

□

Theorem 9.679. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q < r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* settledby$(t + Q + r, t + Q +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t + Q +$ twopow$(k))$*, then if $a \leq t +$ twopow$(k + 1) - (r + 2)$, then if $t +$ twopow$(k + 1) - r \leq a + L$, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter$(t + 1 +$ twopow$(k),$ twopow$(k) - (r + 1)) =$ true.

Proof. The goal follows from lhcenterlatecrossteponeeqtrueatshiftofzerodiagandsinglewinwittagreezerog

□

Theorem 9.680. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$*, then if $a \leq t +$ twopow$(k+1) - (r+2)$, then if $t +$ twopow$(k+1) - (r+1) \leq a+L$, then if* winwittagree(t, k, Q, a, L)*, then* wittbitbool(wittwin$(t+1,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1)) =$ not(wittbitbool(wittwin$(t + 1,$ twopow$(k)), t +$ twopow$(k + 1) - (r + Q + 1))$).

Proof. The goal follows from winwittscalecrosszerogapresetshiftflipofdoublesandsinglewinwittagree$(Q, t,$

□

Theorem 9.681. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$*, then if $a \leq t +$ twopow$(k+1) - (r+2)$, then if $t +$ twopow$(k+1) - (r+1) \leq a+L$, then*

if winwittagree(t, k, Q, a, L), *then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + 2)) = \text{not}(\text{lhcenter}(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + Q + 2)))$.

Proof. The goal follows from lhcenterlatecrossresetshiftflipofdoublesandsinglewinwittagreezerogap$(Q, t,$ □

Theorem 9.682. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q < r$, then if $t+Q+r \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t+Q+r, t+Q+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-r$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogapeqfalseatshiftofzerodiagandwinwittagree$(Q, t, k, r$ □

Theorem 9.683. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t \leq r$, then if $t + Q \leq r + 1$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* settledby$(t + Q + r + 2, t + Q + \text{twopow}(k))$, *then if $a \leq t + \text{twopow}(k + 1) - (r + 2)$, then if $t + \text{twopow}(k + 1) - (r + 2) \leq a + L$, then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + 2)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapeqtrueatshiftafterzerodiagandwinwittagree$(Q, t, k$ □

Theorem 9.684. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q < r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t + Q + r, t + Q + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + Q + \text{twopow}(k))$, *then if $a \leq t + \text{twopow}(k + 1) - (r + 2)$, then if $t + \text{twopow}(k + 1) - r \leq a + L$, then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t + 1, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteponeeqtrueatshiftofzerodiagandsinglewinwitta□

Theorem 9.685. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q < r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t+Q+r, t+Q+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+$

$1)-r \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-r) = \text{not}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-r))$.

Proof. The goal follows from winwittscalecrosszerogapstepflipatshiftofzerodiagandsinglewinwittagree(Q
□

Theorem 9.686. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r + 1$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* zerodiagonal$(t+r+2)$, *then if* settledby$(t+r+2, t+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+2) \leq a+L$, then not* winwittagree$(t, k, Q, a, L)$.

Proof. The goal follows from notwinwittagreeatshiftafterzerodiagofzerodiagtwo(Q, t, k, r, a, L).
□

Theorem 9.687. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r + 1$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* zerodiagonal$(t+r+3)$, *then if* settledby$(t+r+3, t+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+2) \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t+\text{twopow}(k), \text{twopow}(k)-(r+1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinrplusoneofsinglewinwittagreeandzerodiagthree($Q, t, k,$
□

Theorem 9.688. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r + 1$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* zerodiagonal$(t+r+3)$, *then if* settledby$(t+r+3, t+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+2) \leq a+L$, then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t+1+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstepponepinrplustwofalseofsinglewinwittagreeandzerodiagt
□

Theorem 9.689. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* zerodiagonal$(t + r + 3)$*, then if* settledby$(t + r + 3, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$*, then if* $a \leq t + \text{twopow}(k+1) - (r+2)$*, then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$*, then if* winwittagree$(t, k, Q, a, L)$*, then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + Q + 1)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspindoublingdiagfalseofdoublesandsinglewinwittagreeandze □

Theorem 9.690. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* zerodiagonal$(t + r + 3)$*, then if* settledby$(t + r + 3, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$*, then if* $a \leq t + \text{twopow}(k+1) - (r+2)$*, then if* $t + \text{twopow}(k+1) - (r+1) \leq a + L$*, then if* winwittagree$(t, k, Q, a, L)$*, then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + Q + 2)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstep onepinresetdiagtrueofdoublesandsinglewinwittagreean □

Theorem 9.691. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 2 \leq r$, then if $t + 3 + r + 2 \leq k$, then if* zerodiagonal$(t + 3 + r)$*, then if* settledby$(t + 3 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 2, t + 3 + \text{twopow}(k))$*, then if* $a \leq t + \text{twopow}(k+1) - (r+2)$*, then if* $t + \text{twopow}(k+1) - (r+2) \leq a + L$*, then if* winwittagree$(t, k, 3, a, L)$*, then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinbeforezerodiagofzerodiagthreeandsinglewinwittagree$(t,$ □

Theorem 9.692. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 2 \leq r$, then if $t + 3 + r + 2 \leq k$, then if* zerodiagonal$(t + 3 + r)$*, then if* settledby$(t + 3 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 2, t + 3 + \text{twopow}(k))$*, then if* $a \leq t + \text{twopow}(k+1) - (r+2)$*, then if* $t + \text{twopow}(k+1) - (r+2) \leq a + L$*, then*

if winwittagree$(t, k, 3, a, L)$, *then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + 2)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossteponepinrplustwofalseofzerodiagthreeandsinglewinwitta
□

Theorem 9.693. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 3 \leq r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$, *then if* doublesat$(3, t+3+r)$, *then if* settledby$(t + 3 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 3 + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + 3 + r + 2, t + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - (r + 1) \leq a + L$, *then if* winwittagree$(t, k, 3, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 4)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspindoublingdiagfalseofdoublesthreeandsinglewinwittagree
□

Theorem 9.694. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 3 \leq r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$, *then if* doublesat$(3, t+3+r)$, *then if* settledby$(t + 3 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 3 + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + 3 + r + 2, t + \text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r+1) \leq a+L$, *then if* winwittagree$(t, k, 3, a, L)$, *then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + 5)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossteponepinresetdiagtrueofdoublesthreeandsinglewinwittag
□

Theorem 9.695. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 1 \leq r$, then if $t+2+r+2 \leq k$, then if* zerodiagonal$(t+2+r)$, *then if* settledby$(t+2+r, t+\text{twopow}(k))$, *then if* settledby$(t+2+r+2, t+2+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r+2) \leq a+L$, *then not* winwittagree$(t, k, 2, a, L)$.

Proof. The goal follows from notwinwittagreetwoatshiftafterzerodiag(t, k, r, a, L).
□

Theorem 9.696. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+1 \leq k$, then if* zerodiagonal$(t+Q+r)$,

then if doublesat$(Q, t+Q+r)$, *then if* settledby$(t+Q+r+1, t+$twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+Q+1)$, *then if* $t+$twopow$(k+1)-(r+1) \leq a+L$, *then if* wittbitbool(wittwin$(t+Q,$ twopow$(k)), t+$ twopow$(k+1)-(r+1)) =$ wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1)-(r+1))$, *then not* winwittagree(t, k, Q, a, L).

Proof. The goal follows from notwinwittagreeofdoublesandsamebiteq(Q, t, k, r, a, L). □

Theorem 9.697. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t \leq r$, then if $t+Q \leq r+1$, then if $t+Q+r+3 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* zerodiagonal$(t+r+3)$, *then if* settledby$(t+r+3, t+$ twopow$(k))$, *then if* settledby$(t+Q+r+2, t+Q+$ twopow$(k))$, *then if* $a \leq t+$ twopow$(k+1)-(r+2)$, *then if* $t+$ twopow$(k+1)-(r+2) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1)-(r+1)) =$ true.

Proof. The goal follows from winwittscalecrosszerogappinrplusoneofsinglewinwittagreeandzerodiagthree □

Theorem 9.698. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t \leq r$, then if $t+Q \leq r+1$, then if $t+Q+r+3 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* zerodiagonal$(t+r+3)$, *then if* settledby$(t+r+3, t+$twopow$(k))$, *then if* settledby$(t+Q+r+2, t+Q+$twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+2)$, *then if* $t+$twopow$(k+1)-(r+2) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin$(t+1,$ twopow$(k)), t+$ twopow$(k+1)-(r+1)) =$ false.

Proof. The goal follows from winwittscalecrosszerogapsteponepinrplusonefalseofsinglewinwittagreeandz □

Theorem 9.699. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+3 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* doublesat$(Q, t+Q+r)$, *then if* zerodiagonal$(t+r+3)$, *then if* settledby$(t+r+3, t+$twopow$(k))$, *then if* settledby$(t+Q+r+1, t+$twopow$(k))$, *then if* settledby$(t+Q+r+2, t+$twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+2)$, *then if* $t+$twopow$(k+1)-(r+1) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k+1)-(r+Q+1)) =$ false.

Proof. The goal follows from winwittscalecrosszerogappindoublingdiagfalseofdoublesandsinglewinwittag
□

Theorem 9.700. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* zerodiagonal$(t + r + 3)$*, then if* settledby$(t + r + 3, t +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t + Q +$ twopow$(k))$*, then if* $a \leq t+$twopow$(k+1)-(r+2)$*, then if* $t+$twopow$(k+1)-(r+1) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter$(t + Q +$ twopow$(k),$ twopow$(k) -$ $(r + 1)) =$ true.

Proof. The goal follows from lhcenterlatecrosslatepindoublingdiagtrueofsinglewinwittagreeandzerodiagt
□

Theorem 9.701. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* zerodiagonal$(t + r + 3)$*, then if* settledby$(t + r + 3, t +$ twopow$(k))$*, then if* settledby$(t + Q + r +$ $1, t +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$*, then if* $a \leq t+$twopow$(k+1)-(r+2)$*, then if* $t+$twopow$(k+1)-(r+1) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* wittbitbool(wittwin$(t+1,$ twopow$(k)), t+$ twopow$(k + 1) - (r + Q + 1)) =$ true.

Proof. The goal follows from winwittscalecrosszerogapsteponepinresetdiagtrueofdoublesandsinglewinwit
□

Theorem 9.702. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 2 \leq r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* settledby$(t+$ $3+r, t+$twopow$(k))$*, then if* settledby$(t+3+r+2, t+3+$twopow$(k))$*, then if* $a \leq t+$twopow$(k+1)-(r+2)$*, then if* $t+$twopow$(k+1)-(r+2) \leq a+L$*, then if* winwittagree$(t, k, 3, a, L)$*, then* wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k + 1) - (r + 1)) =$ true.

Proof. The goal follows from winwittscalecrosszerogappinbeforezerodiagofzerodiagthreeandsinglewinwit
□

Theorem 9.703. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 2 \leq r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* settledby$(t+$

$3+r, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+3+r+2, t+3+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r+2) \leq a+L$, *then if* $\text{winwittagree}(t, k, 3, a, L)$, *then* $\text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+1)) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinbeforezerodiagfalseofzerodiagthreeands

□

Theorem 9.704. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+3 \leq r$, then if $t+3+r+2 \leq k$, then if* $\text{zerodiagonal}(t+3+r)$, *then if* $\text{doublesat}(3, t+3+r)$, *then if* $\text{settledby}(t+3+r, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+3+r+1, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+3+r+2, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r+1) \leq a+L$, *then if* $\text{winwittagree}(t, k, 3, a, L)$, *then* $\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+4)) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogappindoublingdiagfalseofdoublesthreeandsinglewinw

□

Theorem 9.705. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+3 \leq r$, then if $t+3+r+2 \leq k$, then if* $\text{zerodiagonal}(t+3+r)$, *then if* $\text{doublesat}(3, t+3+r)$, *then if* $\text{settledby}(t+3+r, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+3+r+1, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+3+r+2, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r+1) \leq a+L$, *then if* $\text{winwittagree}(t, k, 3, a, L)$, *then* $\text{wittbitbool}(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+4)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinresetdiagtrueofdoublesthreeandsinglew

□

Theorem 9.706. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+1 \leq r$, then if $t+1+r+1 \leq k$, then if* $\text{zerodiagonal}(t+1+r)$, *then if* $\text{doublesat}(1, t+1+r)$, *then if* $\text{settledby}(t+1+r, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+1+r+1, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+1)$, *then if* $t+\text{twopow}(k+1)-(r+1) \leq a+L$, *then if* $\text{winwittagree}(t, k, 1, a, L)$, *then* $\text{lhcenter}(t+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspindoublingdiagtrueofdoublesoneandsinglewinwittagree(t,

□

Theorem 9.707. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+1 \leq r$, then if $t+1+r+1 \leq k$, then if* zerodiagonal$(t+1+r)$*, then if* doublesat$(1, t+1+r)$*, then if* settledby$(t+1+r, t+\text{twopow}(k))$*, then if* settledby$(t+1+r+1, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+1)$, then if $t+\text{twopow}(k+1)-(r+1) \leq a+L$, then if* winwittagree$(t, k, 1, a, L)$*, then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+2)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogappindoublingdiagtrueofdoublesoneandsinglewinwit □

Theorem 9.708. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+1 \leq r$, then if $t+1+r+2 \leq k$, then if* zerodiagonal$(t+1+r)$*, then if* doublesat$(1, t+1+r)$*, then if* settledby$(t+1+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+1+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+1) \leq a+L$, then if* winwittagree$(t, k, 1, a, L)$*, then* lhcenter$(t+1+\text{twopow}(k), \text{twopow}(k)-(r+3)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossteponepinresetdiagfalseofdoublesoneandsinglewinwittagr □

Theorem 9.709. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+1 \leq r$, then if $t+1+r+2 \leq k$, then if* zerodiagonal$(t+1+r)$*, then if* doublesat$(1, t+1+r)$*, then if* settledby$(t+1+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+1+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+1) \leq a+L$, then if* winwittagree$(t, k, 1, a, L)$*, then* wittbitbool$(\text{wittwin}(t+1, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+2)) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinresetdiagfalseofdoublesoneandsinglewi □

Theorem 9.710. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+1 \leq r$, then if $t+1+r+2 \leq k$, then if* zerodiagonal$(t+1+r)$*, then if* doublesat$(1, t+1+r)$*, then if* settledby$(t+1+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+1+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+3)$, then if $t+\text{twopow}(k+1)-(r+1) \leq a+L$, then not* winwittagree$(t, k, 1, a, L)$.

Proof. The goal follows from notwinwittagreeoneofdoubles(t, k, r, a, L). □

Theorem 9.711. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* zerodiagonal$(t + r + 3)$, *then if* zerodiagonal$(t + Q + r + 3)$, *then if* settledby$(t + r + 3, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 3, t + 1 + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - (r + 1) \leq a + L$, *then not* winwittagree(t, k, Q, a, L).

Proof. The goal follows from notwinwittagreeofdoublesandzerodiagthreeandafterthree(Q, t, k, r, a, L). □

Theorem 9.712. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r + 2)$, *then if* settledby$(t + Q + r + 2, t + Q + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 1)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - r) =$ lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 1))$.

Proof. The goal follows from lhcenterlatecrosstwinatshiftofzerodiagandwinwittagreezerogap$(Q, t, k, r, a,$ □

Theorem 9.713. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r + 2 \leq k$, then if* zerodiagonal$(t + Q + r + 2)$, *then if* settledby$(t + Q + r + 2, t + Q + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 1)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool(wittwin$(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) =$ wittbitbool(wittwin$(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + 1))$.

Proof. The goal follows from winwittscalecrosszerogaptwinatshiftofzerodiagandwinwittagree$(Q, t, k, r, a,$ □

Theorem 9.714. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r + 2$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r + 3)$, *then if* settledby$(t + Q + r + 3, t + Q + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 3)$, *then if* $t + \text{twopow}(k + 1) - (r + 3) \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + 2)) =$ bxor(lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 1))$, lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 2)))$.

Proof. The goal follows from lhcenterlatecrosslinearsteponeatshiftofzerodiagandwinwittagreezerogap(Q,
□

Theorem 9.715. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r+1$, then if $t+Q+r+3 \leq k$, then if* zerodiagonal($t+Q+r+2$)*, then if* settledby($t+Q+r+2, t+Q+$ twopow(k))*, then if $a \leq t+$*twopow($k+1$)$-(r+2)$*, then if $t+$*twopow($k+1$)$-(r+2) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter($t+1+$ twopow(k), twopow(k) $-(r+2)$) = bxor(lhcenter($t+$ twopow(k), twopow(k) $-(r+1)$), lhcenter($t+$ twopow(k), twopow(k) $-(r+3)$)).

Proof. The goal follows from lhcenterlatecrosslinearsteptwoatshiftofzerodiagandwinwittagreezerogap(Q,
□

Theorem 9.716. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+3 \leq k$, then if* zerodiagonal($t+Q+r+3$)*, then if* settledby($t+Q+r+3, t+Q+$ twopow(k))*, then if $a \leq t+$*twopow($k+1$)$-(r+3)$*, then if $t+$*twopow($k+1$)$-(r+1) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter($t+1+$ twopow(k), twopow(k) $-(r+2)$) = false.

Proof. The goal follows from lhcenterlatecrossteponeeqfalseatshiftofzerodiagandsinglewinwittagreezero
□

Theorem 9.717. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+3 \leq k$, then if* zerodiagonal($t+Q+r+3$)*, then if* settledby($t+Q+r+3, t+Q+$ twopow(k))*, then if $a \leq t+$*twopow($k+1$)$-(r+3)$*, then if $t+$*twopow($k+1$)$-(r+1) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* wittbitbool(wittwin($t+1$, twopow(k)), $t+$ twopow($k+1$) $-(r+1)$) = false.

Proof. The goal follows from winwittscalecrosszerogapsteponeeqfalseatshiftofzerodiagandsinglewinwitta
□

Theorem 9.718. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r$, then if $t+Q+r+3 <$* twopow(k)*, then if $a \leq t+$*twopow($k+1$)$-(r+3)$*, then if $t+$*twopow($k+1$)$-(r+1) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter($t+1+Q+$ twopow(k), twopow(k) $-(r+2)$) = lhcenter($t+1+$ twopow(k), twopow(k) $-(r+2)$).

Proof. The goal follows from lhcenterlatecrossteponeeqofwinwittagreethreezerogap(Q, t, k, r, a, L). □

Theorem 9.719. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 <$ twopow(k), then if $a \leq t +$ twopow$(k + 1) - (r + 3)$, then if $t +$ twopow$(k + 1) - (r + 1) \leq a + L$, then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool(wittwin$(t + 1 + Q,$ twopow$(k)), t + Q +$ twopow$(k + 1) - (r + 1)) =$ wittbitbool(wittwin$(t + 1,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1))$.

Proof. The goal follows from winwittscalecrosszerogapsteponeeqofwinwittagreethree(Q, t, k, r, a, L). □

Theorem 9.720. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r + 2)$, *then if* doublesat$(Q, t + Q + r + 2)$, *then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$, *then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$, *then if* settledby$(t+Q+r+3, t+$twopow$(k))$, *then if $a \leq t+$*twopow$(k+1)-(r+3)$, *then if $t+$*twopow$(k+1)-(r+1) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* lhcenter$(t+1+$twopow$(k),$ twopow$(k)-(r+Q+2)) =$ not(lhcenter$(t+1+$ twopow$(k),$ twopow$(k) - (r + 2)))$.

Proof. The goal follows from lhcenterlatecrossteponeshiftflipondoublingdiagofdoublesandsinglewinwitta □

Theorem 9.721. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r + 2)$, *then if* doublesat$(Q, t + Q + r + 2)$, *then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$, *then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$, *then if* settledby$(t+Q+r+3, t+$twopow$(k))$, *then if $a \leq t+$*twopow$(k+1)-(r+3)$, *then if $t+$*twopow$(k+1)-(r+1) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin$(t+1,$ twopow$(k)), t+$twopow$(k+1)-(r+Q+1)) =$ not(wittbitbool(wittwin$(t + 1,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1)))$.

Proof. The goal follows from winwittscalecrosszerogapsteponeshiftflipondoublingdiagofdoublesandsingle □

Theorem 9.722. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t < r$, then if $t+Q+r+1 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then*

if settledby$(t+Q+r, t+$twopow$(k))$, *then if* $a \leq t+$twopow$(k+1)-(r+Q)$, *then if* $t+$twopow$(k+1)-(r+Q) \leq a+L$, *then if* winwittagree(t, k, Q, a, L), *then* lhcenter$(t + Q +$ twopow$(k),$ twopow$(k) - (r + Q)) =$ false.

Proof. The goal follows from lhcenterlatecrosslateeqfalseatshiftofzerodiagandwinwittagreezerogap$(Q, t,$
□

Theorem 9.723. *For every natural number* Q *and natural number* t *and natural number* k *and natural number* r *and natural number* a *and natural number* L, *if* $t \leq r+1$, *then if* $t+Q+r+2 \leq k$, *then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$, *then if* $a \leq t +$ twopow$(k + 1) - (r + Q + 2)$, *then if* $t +$ twopow$(k + 1) - (r + Q + 2) \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t + Q +$ twopow$(k),$ twopow$(k) - (r + Q + 2)) =$ true.

Proof. The goal follows from lhcenterlatecrosslateeqtrueatshiftafterzerodiagandwinwittagreezerogap$(Q,$
□

Theorem 9.724. *For every natural number* Q *and natural number* t *and natural number* k *and natural number* r *and natural number* a *and natural number* L, *if* $t \leq r+1$, *then if* $t+Q+r+2 \leq k$, *then if* zerodiagonal$(t+Q+r)$, *then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$, *then if* $a \leq t +$ twopow$(k + 1) - (r + Q + 2)$, *then if* $t +$ twopow$(k + 1) - (r + Q + 2) \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t+1+Q+$twopow$(k),$ twopow$(k)-(r + Q + 2)) =$ not(lhcenter$(t + Q +$ twopow$(k),$ twopow$(k) - (r + Q + 1))).$

Proof. The goal follows from lhcenterlatecrosslatestepflipatshiftafterzerodiagofsinglewinwittagreezeroga
□

Theorem 9.725. *For every natural number* Q *and natural number* t *and natural number* k *and natural number* r *and natural number* a *and natural number* L, *if* $t \leq r$, *then if* $t + Q + r + 2 \leq k$, *then if* zerodiagonal$(t +$ $Q + r)$, *then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$, *then if* $a \leq t +$ twopow$(k + 1) - (r + Q + 2)$, *then if* $t +$ twopow$(k + 1) - (r + Q + 2) \leq$ $a + L$, *then if* winwittagree(t, k, Q, a, L), *then* wittbitbool(wittwin$(t + 1 +$ $Q,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1)) =$ not(wittbitbool(wittwin$(t +$ $Q,$ twopow$(k)), t +$ twopow$(k + 1) - (r + 1))).$

Proof. The goal follows from winwittscalecrosszerogaplatestepflipatshiftafterzerodiagofsinglewinwittagr
□

Theorem 9.726. *For every natural number* Q *and natural number* t *and natural number* k *and natural number* r *and natural number* a *and natural*

number L, *if* $t+Q \leq r$, *then if* $t+Q+r+3 \leq k$, *then if* zerodiagonal$(t+r+1)$, *then if* zerodiagonal$(t+Q+r+3)$, *then if* settledby$(t+r+1+2, t+1+$ twopow$(k))$, *then if* settledby$(t+Q+r+3, t+Q+$ twopow$(k))$, *then if* $a \leq t+$ twopow$(k+1)-(r+3)$, *then if* $t+$ twopow$(k+1)-(r+1) \leq a+L$, *then not* winwittagree(t, k, Q, a, L).

Proof. The goal follows from notwinwittagreeofzerodiagoneandshiftthree(Q, t, k, r, a, L). □

Theorem 9.727. *For every natural number* z, *if* zerodiagonal(z), *then* $2 \leq z$.

Proof. The goal follows from zerodiaggetwo(z). □

Theorem 9.728. *For every natural number* z, *if* zerodiagonal(z), *then not* zerodiagonal$(z+2)$.

Proof. The goal follows from notiszerodiagaddtwo(z). □

Theorem 9.729. *For every natural number* z, *if* zerodiagonal(z), *then not* blackdiagonal$(z+1)$.

Proof. The goal follows from notisblackdiagaddoneofzerodiag(z). □

Theorem 9.730. *For every natural number* z, *if* zerodiagonal(z), *then not* zerodiagonal$(z+3)$.

Proof. The goal follows from notiszerodiagaddthree(z). □

Theorem 9.731. *For every natural number* z, *if* zerodiagonal(z), *then not* blackdiagonal$(z+3)$.

Proof. The goal follows from notisblackdiagaddthreeofzerodiag(z). □

Theorem 9.732. *For every natural number* z, *if* zerodiagonal(z), *then not* zerodiagonal$(z+4)$.

Proof. The goal follows from notiszerodiagaddfour(z). □

Theorem 9.733. *For every natural number* $z1$ *and natural number* $z2$, *if* zerodiagonal$(z1)$, *then if* zerodiagonal$(z2)$, *then if* $z1 < z2$, *then* $z1+5 \leq z2$.

Proof. The goal follows from zerodiaggapgefive$(z1, z2)$. □

Theorem 9.734. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t < r$, then if $t+Q+r+1 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* settledby$(t+Q+r, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+Q)$, then if $t+\text{twopow}(k+1)-(r+Q) \leq a+L$, then if* winwittagree$(t,k,Q,a,L)$*, then* wittbitbool$(\text{wittwin}(t+Q, \text{twopow}(k)), t+\text{twopow}(k+1)-r) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogaplateeqfalseatshiftofzerodiagandwinwittagree$(Q, t,$
□

Theorem 9.735. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t \leq r+1$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* settledby$(t+Q+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+Q+2)$, then if $t+\text{twopow}(k+1)-(r+Q+2) \leq a+L$, then if* winwittagree$(t,k,Q,a,L)$*, then* wittbitbool$(\text{wittwin}(t+Q, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+2)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogaplateeqtrueatshiftafterzerodiagandwinwittagree$(Q$
□

Theorem 9.736. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t < r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* settledby$(t+Q+r, t+\text{twopow}(k))$*, then if* settledby$(t+Q+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+Q+2)$, then if $t+\text{twopow}(k+1)-(r+Q) \leq a+L$, then if* winwittagree$(t,k,Q,a,L)$*, then* lhcenter$(t+1+Q+\text{twopow}(k), \text{twopow}(k)-(r+Q+1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosslatesteponeeqtrueatshiftofzerodiagandsinglewinwittagreez
□

Theorem 9.737. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t < r$, then if $t+Q+r+2 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* settledby$(t+Q+r, t+\text{twopow}(k))$*, then if* settledby$(t+Q+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+Q+2)$, then if $t+\text{twopow}(k+1)-(r+Q) \leq a+L$, then if* winwittagree$(t,k,Q,a,L)$*, then* wittbitbool$(\text{wittwin}(t+1+Q, \text{twopow}(k)), t+\text{twopow}(k+1)-r) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogaplatesteponeeqtrueatshiftofzerodiagandsinglewinwi
□

Theorem 9.738. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+3 \leq r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t+3+r, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+1) \leq a+L$, then if* winwittagree$(t, k, 3, a, L)$*, then* lhcenter$(t+1+\text{twopow}(k), \text{twopow}(k)-(r+3)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosssteponepinrplusthreetrueofdoublesthreeandsinglewinwitta

□

Theorem 9.739. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+3 < r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t+3+r, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 3, a, L)$*, then* lhcenter$(t+2+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstepttwopinrplustwofalseofdoublesthreeandsinglewinwittag

□

Theorem 9.740. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+3 < r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t+3+r, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 3, a, L)$*, then* lhcenter$(t+3+\text{twopow}(k), \text{twopow}(k)-(r+3)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstepthreepinrplusthreetrueofdoublesthreeandsinglewinwit

□

Theorem 9.741. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+3 < r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t+3+r, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+3+r+2, t+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+3)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then not* winwittagree$(t, k, 3, a, L)$.

Proof. The goal follows from notwinwittagreethreeofdoubles(t, k, r, a, L). □

Theorem 9.742. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 3 \leq r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t + 3 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 2, t + \text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+1) \leq a+L$, then if* winwittagree$(t, k, 3, a, L)$*, then* wittbitbool$(\text{wittwin}(t + 1, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + 2)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinrplustwotrueofdoublesthreeandsinglew
□

Theorem 9.743. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 3 < r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t + 3 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 2, t + \text{twopow}(k))$*, then if $a \leq t + \text{twopow}(k + 1) - (r + 2)$, then if $t + \text{twopow}(k + 1) - r \leq a + L$, then if* winwittagree$(t, k, 3, a, L)$*, then* wittbitbool$(\text{wittwin}(t + 2, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogapsteptwopinrfalseofdoublesthreeandsinglewinwitta
□

Theorem 9.744. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 3 < r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t + 3 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t + 3 + r + 2, t + \text{twopow}(k))$*, then if $a \leq t + \text{twopow}(k + 1) - (r + 2)$, then if $t + \text{twopow}(k + 1) - r \leq a + L$, then if* winwittagree$(t, k, 3, a, L)$*, then* wittbitbool$(\text{wittwin}(t + 3, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapstepthreepinrtrueofdoublesthreeandsinglewinwitt
□

Theorem 9.745. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 3 < r$, then if $t+3+r+2 \leq k$, then if* zerodiagonal$(t+3+r)$*, then if* doublesat$(3, t+3+r)$*, then if* settledby$(t + 3 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 3 +$

$r+1, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+3+r+2, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-r \leq a+L$, *then if* $\text{winwittagree}(t,k,3,a,L)$, *then* $\text{wittbitbool}(\text{wittwin}(t+3,\text{twopow}(k)),t+\text{twopow}(k+1)-r) = \text{not}(\text{wittbitbool}(\text{wittwin}(t,\text{twopow}(k)),t+\text{twopow}(k+1)-r))$.

Proof. The goal follows from winwittscalecrosszerogapsamebitrflipthreeofdoublesthreeandsinglewinwitta

□

Theorem 9.746. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q<r$, then if $t+Q+r+2 \leq k$, then if $t+r+4 \leq k$, then if* $\text{zerodiagonal}(t+Q+r)$, *then if* $\text{zerodiagonal}(t+r+4)$, *then if* $\text{settledby}(t+Q+r, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+r+4, t+1+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-r \leq a+L$, *then if* $\text{winwittagree}(t,k,Q,a,L)$, *then* $\text{lhcenter}(t+1+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossteponepinrplustwotrueofsinglewinwittagreeandzerodiagfo

□

Theorem 9.747. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q<r$, then if $t+Q+r+2 \leq k$, then if $t+r+4 \leq k$, then if* $\text{zerodiagonal}(t+Q+r)$, *then if* $\text{zerodiagonal}(t+r+4)$, *then if* $\text{settledby}(t+Q+r, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+r+4, t+1+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-r \leq a+L$, *then if* $\text{winwittagree}(t,k,Q,a,L)$, *then* $\text{lhcenter}(t+\text{twopow}(k), \text{twopow}(k)-(r+1)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspinrplusonefalseofsinglewinwittagreeandzerodiagfour(Q, t

□

Theorem 9.748. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q<r$, then if $t+Q+r+2 \leq k$, then if $t+r+4 \leq k$, then if* $\text{zerodiagonal}(t+Q+r)$, *then if* $\text{doublesat}(Q, t+Q+r)$, *then if* $\text{zerodiagonal}(t+r+4)$, *then if* $\text{settledby}(t+Q+r, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+Q+r+1, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+r+4, t+1+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-r \leq a+L$, *then if* $\text{winwittagree}(t,k,Q,a,L)$, *then* $\text{lhcenter}(t+\text{twopow}(k), \text{twopow}(k)-(r+Q+1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspindoublingdiagtrueofdoublesandsinglewinwittagreeandze
□

Theorem 9.749. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 \leq r+1$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+2) \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* lhcenter$(t+\text{twopow}(k), \text{twopow}(k)-(r+3)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinrplusthreetrueofzerodiagfourandsinglewinwittagree(t, k
□

Theorem 9.750. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* settledby$(t+4+r, t+1+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* lhcenter$(t+1+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstepponepinrplustwotrueofzerodiagfourandsinglewinwittagr
□

Theorem 9.751. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* settledby$(t+4+r, t+1+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* lhcenter$(t+\text{twopow}(k), \text{twopow}(k)-(r+1)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspinrplusonefalseofzerodiagfourandsinglewinwittagree(t, k,
□

Theorem 9.752. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* doublesat$(4, t+4+r)$*, then if* settledby$(t+4+r, t+1+\text{twopow}(k))$*, then if* settledby$(t+4+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$,*

then if winwittagree$(t, k, 4, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 4 + 1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspindoublingdiagtrueofdoublesfourandsinglewinwittagree(t

□

Theorem 9.753. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$, *then if* doublesat$(4, t+4+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - (r+2)$, *then if* $t + \text{twopow}(k+1) - r \leq a + L$, *then if* winwittagree$(t, k, 4, a, L)$, *then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + 4)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstep onepinrplusfourfalseofdoublesfourandsinglewinwittagr

□

Theorem 9.754. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$, *then if* doublesat$(4, t+4+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - (r+2)$, *then if* $t + \text{twopow}(k+1) - r \leq a + L$, *then if* winwittagree$(t, k, 4, a, L)$, *then* lhcenter$(t + 2 + \text{twopow}(k), \text{twopow}(k) - (r + 3)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstep twopinrplusthreetrueofdoublesfourandsinglewinwittag

□

Theorem 9.755. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$, *then if* doublesat$(4, t+4+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+2, t+2+\text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k+1) - (r+2)$, *then if* $t + \text{twopow}(k+1) - r \leq a + L$, *then if* winwittagree$(t, k, 4, a, L)$, *then* lhcenter$(t + 3 + \text{twopow}(k), \text{twopow}(k) - (r + 4)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstep threepinrplusfourfalseofdoublesfourandsinglewinwitta

□

Theorem 9.756. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 \leq r+1$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+3+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+3)$, then if $t+\text{twopow}(k+1)-(r+2) \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* lhcenter$(t+3+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstepthreepinrplustwofalseofzerodiagfourandsinglewinwitta □

Theorem 9.757. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+\text{twopow}(k))$*, then if* lhcenter$(t+\text{twopow}(k), \text{twopow}(k)-(r+3)) = \text{true}$*, then* lhcenter$(t+1+\text{twopow}(k), \text{twopow}(k)-(r+3)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstepabsorbfalseafterzerodiag(t, k, r). □

Theorem 9.758. *For every natural number t and natural number k and natural number r, if $t+r+2 \leq k$, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+\text{twopow}(k))$*, then if* lhcenter$(t+\text{twopow}(k), \text{twopow}(k)-(r+1)) = \text{false}$*, then* lhcenter$(t+2+\text{twopow}(k), \text{twopow}(k)-(r+2)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosstwostepabsorbfalseafterzerodiag(t, k, r). □

Theorem 9.759. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* doublesat$(4, t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+1+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* lhcenter$(t+2+\text{twopow}(k), \text{twopow}(k)-(r+5)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstepttwopinrplusfivetrueofdoublesfourandsinglewinwittagre □

Theorem 9.760. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* doublesat$(4, t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+$

$1, t + \text{twopow}(k))$, *then if* $\text{settledby}(t + 4 + r + 2, t + 1 + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then if* $\text{winwittagree}(t, k, 4, a, L)$, *then* $\text{lhcenter}(t + 3 + \text{twopow}(k), \text{twopow}(k) - (r + 5)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstepthreepinrplusfivefalseofdoublesfourandsinglewinwittag
□

Theorem 9.761. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* $\text{zerodiagonal}(t+4+r)$, *then if* $\text{doublesat}(4, t+4+r)$, *then if* $\text{settledby}(t+4+r, t+\text{twopow}(k))$, *then if* $\text{settledby}(t+4+r+1, t+\text{twopow}(k))$, *then if* $\text{settledby}(t + 4 + r + 2, t + 1 + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then if* $\text{winwittagree}(t, k, 4, a, L)$, *then* $\text{lhcenter}(t + 4 + \text{twopow}(k), \text{twopow}(k) - (r + 4)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstepfourpinrplusfourtrueofdoublesfourandsinglewinwittagr
□

Theorem 9.762. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* $\text{zerodiagonal}(t+4+r)$, *then if* $\text{doublesat}(4, t+4+r)$, *then if* $\text{settledby}(t + 4 + r, t + \text{twopow}(k))$, *then if* $\text{settledby}(t + 4 + r + 1, t + \text{twopow}(k))$, *then if* $\text{settledby}(t+4+r+2, t+1+\text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 4)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then not* $\text{winwittagree}(t, k, 4, a, L)$.

Proof. The goal follows from $\text{notwinwittagreefourofdoubles}(t, k, r, a, L)$. □

Theorem 9.763. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q < r$, then if $t+Q+r+2 \leq k$, then if $t+r+4 \leq k$, then if* $\text{zerodiagonal}(t+Q+r)$, *then if* $\text{zerodiagonal}(t+r+4)$, *then if* $\text{settledby}(t+Q+r, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if* $\text{settledby}(t+r+4, t+1+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-r \leq a+L$, *then if* $\text{winwittagree}(t, k, Q, a, L)$, *then* $\text{wittbitbool}(\text{wittwin}(t + 1, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + 1)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinrplusonetrueofsinglewinwittagreeandze
□

Theorem 9.764. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q<r$, then if $t+Q+r+2\leq k$, then if $t+r+4\leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* zerodiagonal$(t+r+4)$*, then if* settledby$(t+Q+r, t+Q+\text{twopow}(k))$*, then if* settledby$(t+Q+r+2, t+Q+\text{twopow}(k))$*, then if* settledby$(t+r+4, t+1+\text{twopow}(k))$*, then if $a\leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r\leq a+L$, then if* winwittagree$(t,k,Q,a,L)$*, then* wittbitbool$(\text{wittwin}(t,\text{twopow}(k)), t+\text{twopow}(k+1)-(r+1))=\text{false}$.

Proof. The goal follows from winwittscalecrosszerogappinrplusonefalseofsinglewinwittagreeandzerodiagf
□

Theorem 9.765. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q<r$, then if $t+Q+r+2\leq k$, then if $t+r+4\leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* doublesat$(Q, t+Q+r)$*, then if* zerodiagonal$(t+r+4)$*, then if* settledby$(t+Q+r, t+Q+\text{twopow}(k))$*, then if* settledby$(t+Q+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+Q+r+2, t+Q+\text{twopow}(k))$*, then if* settledby$(t+r+4, t+1+\text{twopow}(k))$*, then if $a\leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r\leq a+L$, then if* winwittagree$(t,k,Q,a,L)$*, then* wittbitbool$(\text{wittwin}(t,\text{twopow}(k)), t+\text{twopow}(k+1)-(r+Q+1))=\text{true}$.

Proof. The goal follows from winwittscalecrosszerogappindoublingdiagtrueofdoublesandsinglewinwittag
□

Theorem 9.766. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4\leq r+1$, then if $t+4+r+2\leq k$, then if* zerodiagonal$(t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$*, then if $a\leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r+2)\leq a+L$, then if* winwittagree$(t,k,4,a,L)$*, then* wittbitbool$(\text{wittwin}(t,\text{twopow}(k)), t+\text{twopow}(k+1)-(r+3))=\text{true}$.

Proof. The goal follows from winwittscalecrosszerogappinrplusthreetrueofzerodiagfourandsinglewinwitta
□

Theorem 9.767. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4<r$, then if $t+4+r+2\leq k$, then if* zerodiagonal$(t+4+r)$*, then if* settledby$(t+4+r, t+1+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$*, then if $a\leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r\leq a+L$, then*

if winwittagree$(t, k, 4, a, L)$, *then* wittbitbool(wittwin$(t+1, \text{twopow}(k)), t+$ twopow$(k+1)-(r+1)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinrplusonetrueofzerodiagfourandsinglewi □

Theorem 9.768. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4<r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$, *then if* settledby$(t+4+r, t+1+\text{twopow}(k))$, *then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$, *then* wittbitbool(wittwin$(t, \text{twopow}(k)), t+$ twopow$(k+1)-(r+1)) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogappinrplusonefalseofzerodiagfourandsinglewinwittag □

Theorem 9.769. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4<r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$, *then if* doublesat$(4, t+4+r)$, *then if* settledby$(t+4+r, t+1+\text{twopow}(k))$, *then if* settledby$(t+4+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+2, t+4+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$, *then* wittbitbool(wittwin$(t, \text{twopow}(k)), t+$ twopow$(k+1)-(r+4+1)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogappindoublingdiagtrueofdoublesfourandsinglewinwit □

Theorem 9.770. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4<r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$, *then if* doublesat$(4, t+4+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+2, t+1+\text{twopow}(k))$, *then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$, *then* wittbitbool(wittwin$(t+4, \text{twopow}(k)), t+$ twopow$(k+1)-r) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapstepfourpinrtrueofdoublesfourandsinglewinwittag □

Theorem 9.771. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4<r$,*

then if $t+4+r+2 \leq k$, *then if* zerodiagonal$(t+4+r)$, *then if* doublesat$(4, t+4+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+1, t+\text{twopow}(k))$, *then if* settledby$(t+4+r+2, t+1+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-r \leq a+L$, *then if* winwittagree$(t, k, 4, a, L)$, *then* wittbitbool(wittwin$(t+4, \text{twopow}(k)), t+\text{twopow}(k+1)-r) = \text{not}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-r))$.

Proof. The goal follows from winwittscalecrosszerogapsamebitrflipfourofdoublesfourandsinglewinwittagr □

Theorem 9.772. *For every natural number* z, *if* zerodiagonal(z), *then not* blackdiagonal$(z+4)$.

Proof. The goal follows from notisblackdiagaddfourofzerodiag(z). □

Theorem 9.773. *For every natural number* z, *if* zerodiagonal(z), *then not* blackdiagonal$(z+5)$.

Proof. The goal follows from notisblackdiagaddfiveofzerodiag(z). □

Theorem 9.774. *For every natural number* z, *if* zerodiagonal(z), *then not* blackdiagonal$(z+6)$.

Proof. The goal follows from notisblackdiagaddsixofzerodiag(z). □

Theorem 9.775. *For every natural number* z, *if* zerodiagonal(z), *then if* zerodiagonal$(z+5)$, *then* blackdiagonal$(z-1)$.

Proof. The goal follows from isblackdiagsuboneofzerodiagandaddfive(z). □

Theorem 9.776. *For every natural number* z, *if* zerodiagonal(z), *then if* zerodiagonal$(z+5)$, *then* $z \leq 2$.

Proof. The goal follows from zerodiagletwoofaddfive(z). □

Theorem 9.777. *For every natural number* z, *if* $2 < z$, *then if* zerodiagonal(z), *then not* zerodiagonal$(z+5)$.

Proof. The goal follows from notiszerodiagaddfiveofgttwo(z). □

Theorem 9.778. *For every natural number* $z1$ *and natural number* $z2$, *if* $2 < z1$, *then if* zerodiagonal$(z1)$, *then if* zerodiagonal$(z2)$, *then if* $z1 < z2$, *then* $z1+6 \leq z2$.

Proof. The goal follows from zerodiaggapgesixofgttwo($z1, z2$). □

Theorem 9.779. *For every natural number z, if $2 < z$, then if* zerodiagonal(z), *then not* blackdiagonal($z + 7$).

Proof. The goal follows from notisblackdiagaddsevenofgttwoandzerodiag(z). □

Theorem 9.780. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal($t+4+r$), *then if* doublesat($4, t+4+r$), *then if* settledby($t+4+r, t+$twopow(k)), *then if* settledby($t+4+r+1, t+$twopow(k)), *then if* settledby($t+4+r+2, t+4+$twopow(k)), *then if* $a \leq t+$twopow($k+1$)$-(r+2)$, *then if* $t+$twopow($k+1$)$-r \leq a+L$, *then if* winwittagree($t, k, 4, a, L$), *then* wittbitbool(wittwin($t+1$, twopow(k)), $t+$twopow($k+1$)$-(r+3)$) = false.

Proof. The goal follows from winwittscalecrosszerogapsteponepinrplusthreefalseofdoublesfourandsinglew
□

Theorem 9.781. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal($t+4+r$), *then if* doublesat($4, t+4+r$), *then if* settledby($t+4+r, t+$twopow(k)), *then if* settledby($t+4+r+1, t+$twopow(k)), *then if* settledby($t+4+r+2, t+4+$twopow(k)), *then if* $a \leq t+$twopow($k+1$)$-(r+2)$, *then if* $t+$twopow($k+1$)$-r \leq a+L$, *then if* winwittagree($t, k, 4, a, L$), *then* wittbitbool(wittwin($t+2$, twopow(k)), $t+$twopow($k+1$)$-(r+1)$) = true.

Proof. The goal follows from winwittscalecrosszerogapsteptwopinrplusonetrueofdoublesfourandsinglewi
□

Theorem 9.782. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal($t+4+r$), *then if* doublesat($4, t+4+r$), *then if* settledby($t+4+r, t+$twopow(k)), *then if* settledby($t+4+r+1, t+$twopow(k)), *then if* settledby($t+4+r+2, t+2+$twopow(k)), *then if* $a \leq t+$twopow($k+1$)$-(r+2)$, *then if* $t+$twopow($k+1$)$-r \leq a+L$, *then if* winwittagree($t, k, 4, a, L$), *then* wittbitbool(wittwin($t+3$, twopow(k)), $t+$twopow($k+1$)$-(r+1)$) = false.

Proof. The goal follows from winwittscalecrosszerogapstepthreepinrplusonefalseofdoublesfourandsinglew
□

Theorem 9.783. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* doublesat$(4, t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+1+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* wittbitbool$(\text{wittwin}(t+2, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+3)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteptwopinrplusthreetrueofdoublesfourandsinglew
□

Theorem 9.784. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* doublesat$(4, t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+1+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* wittbitbool$(\text{wittwin}(t+3, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+2)) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogapstepthreepinrplustwofalseofdoublesfourandsinglew
□

Theorem 9.785. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+4 < r$, then if $t+4+r+2 \leq k$, then if* zerodiagonal$(t+4+r)$*, then if* doublesat$(4, t+4+r)$*, then if* settledby$(t+4+r, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+1, t+\text{twopow}(k))$*, then if* settledby$(t+4+r+2, t+1+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-r \leq a+L$, then if* winwittagree$(t, k, 4, a, L)$*, then* wittbitbool$(\text{wittwin}(t+3, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+2)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+2)))$.

Proof. The goal follows from winwittscalecrosszerogapsamebitrplustwoflipthreeofdoublesfourandsinglew
□

Theorem 9.786. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r + 2 \leq k$, then if $t + r + 4 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* doublesat$(Q, t + Q + r)$*, then if* zerodiagonal$(t + r + 4)$*, then if* settledby$(t + Q + r, t + Q + \text{twopow}(k))$*,*

then if settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if* settledby$(t + r + 4, t + 1 + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + Q + 2)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossteponepinresetdiagfalseofdoublesandsinglewinwittagreear
□

Theorem 9.787. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r + 2 \leq k$, then if $t + r + 4 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* zerodiagonal$(t + r + 4)$, *then if* settledby$(t + Q + r, t + Q + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if* settledby$(t + r + 4, t + 1 + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t + 1, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + Q + 1)) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinresetdiagfalseofdoublesandsinglewinwit
□

Theorem 9.788. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r + 2 \leq k$, then if $t + r + 4 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* zerodiagonal$(t + r + 4)$, *then if* settledby$(t + Q + r, t + Q + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t + Q + r + 2, t + \text{twopow}(k))$, *then if* settledby$(t + r + 4, t + 1 + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - r \leq a + L$, *then if* winwittagree$(t, k, Q, a, L)$, *then* wittbitbool$(\text{wittwin}(t + 1, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + Q + 1)) = \text{not}(\text{wittbitbool}(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + Q + 1)))$.

Proof. The goal follows from winwittscalecrosszerogapsteponesamebitdoublingflipofdoublesandsinglewin
□

Theorem 9.789. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q \leq r$, then if $t + Q + r + 3 \leq k$, then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* zerodiagonal$(t + r + 3)$,

then if settledby$(t + r + 3, t +$ twopow$(k))$*, then if* settledby$(t + Q + r +$ $1, t +$ twopow$(k))$*, then if* settledby$(t + Q + r + 2, t +$ twopow$(k))$*, then if* $a \leq t+$twopow$(k+1)-(r+2)$*, then if* $t+$twopow$(k+1)-(r+1) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* wittbitbool(wittwin$(t+1,$ twopow$(k)), t+$ twopow$(k+1) - (r + Q + 1)) =$ not(wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k + 1) - (r + Q + 1)))$.

Proof. The goal follows from winwittscalecrosszerogapsteponesamebitdoublingflipofdoublesandsinglewi

□

Theorem 9.790. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r+1$, then if $t+Q+r+2 \leq k$, then if $t+r+5 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* zerodiagonal$(t+r+5)$*, then if* settledby$(t+$ $Q+r+2, t+Q+$twopow$(k))$*, then if* settledby$(t+r+5, t+$twopow$(k))$*, then if* $a \leq t+$twopow$(k+1)-(r+2)$*, then if* $t+$twopow$(k+1)-(r+2) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter$(t + 1 +$ twopow$(k),$ twopow$(k) -$ $(r + 3)) =$ not(lhcenter$(t +$ twopow$(k),$ twopow$(k) - (r + 3)))$.

Proof. The goal follows from lhcenterlatecrosssteponeatshiftplusthreeeqnotofsinglewinwittagreeandzero

□

Theorem 9.791. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q \leq r+1$, then if $t+Q+r+2 \leq k$, then if $t+r+5 \leq k$, then if* zerodiagonal$(t+Q+r)$*, then if* zerodiagonal$(t+r+5)$*, then if* settledby$(t+$ $Q+r+2, t+Q+$twopow$(k))$*, then if* settledby$(t+r+5, t+$twopow$(k))$*, then if* $a \leq t+$twopow$(k+1)-(r+2)$*, then if* $t+$twopow$(k+1)-(r+2) \leq a+L$*, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter$(t+$twopow$(k),$ twopow$(k) -$ $(r + 1)) =$ lhcenter$(t +$ twopow$(k),$ twopow$(k) - (r + 3))$.

Proof. The goal follows from lhcenterlatecrosseqrplusonerplusthreeofsinglewinwittagreeandzerodiagfive(

□

Theorem 9.792. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t + Q < r$, then if $t + Q + r + 2 \leq k$, then if $t + r + 5 \leq k$, then if* zerodiagonal$(t + Q + r)$*, then if* zerodiagonal$(t + r + 5)$*, then if* settledby$(t + Q + r, t + Q +$ twopow$(k))$*, then if* settledby$(t + Q + r +$ $2, t + Q +$ twopow$(k))$*, then if* settledby$(t + r + 5, t +$ twopow$(k))$*, then if* $a \leq t +$ twopow$(k + 1) - (r + 2)$*, then if* $t +$ twopow$(k + 1) - r \leq a + L$*, then if* winwittagree(t, k, Q, a, L)*, then* lhcenter$(t + 2 +$ twopow$(k),$ twopow$(k) -$ $(r + 1)) =$ lhcenter$(t +$ twopow$(k),$ twopow$(k) - (r + 1))$.

Proof. The goal follows from lhcenterlatecrossteptwoeqrplusoneofsinglewinwittagreeandzerodiagfive(Q,
□

Theorem 9.793. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q<r$, then if $t+Q+r+2\leq k$, then if $t+r+5\leq k$, then if* zerodiagonal($t+Q+r$)*, then if* zerodiagonal($t+r+5$)*, then if* settledby($t+Q+r, t+Q+$twopow(k))*, then if* settledby($t+Q+r+2, t+Q+$twopow(k))*, then if* settledby($t+r+5, t+$twopow(k))*, then if $a\leq t+$*twopow($k+1$)$-(r+2)$*, then if $t+$*twopow($k+1$)$-r\leq a+L$*, then if* winwittagree(t,k,Q,a,L)*, then* lhcenter($t+$twopow(k), twopow(k)$-(r-1)$) = true.

Proof. The goal follows from lhcenterlatecrosspinrsubonetrueofsinglewinwittagreeandzerodiagfive(Q,t,k
□

Theorem 9.794. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+6<r$, then if $t+5+r+2\leq k$, then if* zerodiagonal($t+5+r$)*, then if* doublesat($5, t+5+r$)*, then if* settledby($t+4+r, t+$twopow(k))*, then if* settledby($t+5+r, t+$twopow(k))*, then if* settledby($t+5+r+2, t+5+$twopow(k))*, then if $a\leq t+$*twopow($k+1$)$-(r+2)$*, then if $t+$*twopow($k+1$)$-(r-1)\leq a+L$*, then if* winwittagree($t,k,5,a,L$)*, then* lhcenter($t+$twopow(k), twopow(k)$-(r+4)$) = true.

Proof. The goal follows from lhcenterlatecrosspinrplusfourtrueofzerodiagfiveandsinglewinwittagree($t,k,$
□

Theorem 9.795. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+6<r$, then if $t+5+r+2\leq k$, then if* zerodiagonal($t+5+r$)*, then if* doublesat($5, t+5+r$)*, then if* settledby($t+4+r, t+$twopow(k))*, then if* settledby($t+5+r, t+$twopow(k))*, then if* settledby($t+5+r+2, t+5+$twopow(k))*, then if $a\leq t+$*twopow($k+1$)$-(r+2)$*, then if $t+$*twopow($k+1$)$-(r-1)\leq a+L$*, then if* winwittagree($t,k,5,a,L$)*, then* lhcenter($t+$twopow(k), twopow(k)$-(r+3)$) = true.

Proof. The goal follows from lhcenterlatecrosspinrplusthreetrueofzerodiagfiveandsinglewinwittagree(t,k
□

Theorem 9.796. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+6<r$, then if $t+5+r+2\leq k$, then if* zerodiagonal($t+5+r$)*, then if* doublesat($5, t+$

$5 + r)$, *then if* settledby$(t + 4 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r + 2, t + 5 + \text{twopow}(k))$, *then if* $a \leq t + \text{twopow}(k + 1) - (r + 2)$, *then if* $t + \text{twopow}(k + 1) - (r - 1) \leq a + L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 1)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrosspinrplusonetrueofzerodiagfiveandsinglewinwittagree$(t, k, r$ □

Theorem 9.797. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$, *then if* doublesat$(5, t + 5 + r)$, *then if* settledby$(t + 4 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t + \text{twopow}(k + 1) - (r - 1) \leq a + L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* lhcenter$(t + \text{twopow}(k), \text{twopow}(k) - (r + 6)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrosspindoublingdiagfalseofdoublesfiveandsinglewinwittagree$(t$ □

Theorem 9.798. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$, *then if* doublesat$(5, t + 5 + r)$, *then if* settledby$(t + 4 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t + \text{twopow}(k + 1) - (r - 1) \leq a + L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* lhcenter$(t + 1 + \text{twopow}(k), \text{twopow}(k) - (r + 5)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstep onepinrplusfivetrueofdoublesfiveandsinglewinwittagre □

Theorem 9.799. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$, *then if* doublesat$(5, t + 5 + r)$, *then if* settledby$(t + 4 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r, t + \text{twopow}(k))$, *then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$, *then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t + \text{twopow}(k + 1) - (r - 1) \leq a + L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* lhcenter$(t + 2 + \text{twopow}(k), \text{twopow}(k) - (r + 4)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstepwopinrplusfourtrueofdoublesfiveandsinglewinwittagre
□

Theorem 9.800. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r-1) \leq a+L$, then if* winwittagree$(t, k, 5, a, L)$*, then* lhcenter$(t + 3 + \text{twopow}(k), \text{twopow}(k) - (r + 3)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstepthreepinrplusthreefalseofdoublesfiveandsinglewinwitta
□

Theorem 9.801. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t+5+r+2, t+3+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+4)$, then if $t+\text{twopow}(k+1)-(r-1) \leq a+L$, then not* winwittagree$(t, k, 5, a, L)$.

Proof. The goal follows from notwinwittagreefiveofdoubles(t, k, r, a, L). □

Theorem 9.802. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t+5+r+2, t+3+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r-1) \leq a+L$, then if* winwittagree$(t, k, 5, a, L)$*, then* lhcenter$(t + 4 + \text{twopow}(k), \text{twopow}(k) - (r + 4)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossstepfourpinrplusfourtrueofdoublesfiveandsinglewinwittagr
□

Theorem 9.803. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if*

settledby$(t+5+r+2, t+3+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r-1) \leq a+L$, *then if* winwittagree$(t,k,5,a,L)$, *then* lhcenter$(t+5+\text{twopow}(k), \text{twopow}(k)-(r+4)) = \text{false}$.

Proof. The goal follows from lhcenterlatecrossstepfivepinrplusfourfalseofdoublesfiveandsinglewinwittagr

□

Theorem 9.804. *For every natural number Q and natural number t and natural number k and natural number r and natural number a and natural number L, if $t+Q+1 < r$, then if $t+Q+r+2 \leq k$, then if $t+r+5 \leq k$, then if* zerodiagonal$(t+Q+r)$, *then if* zerodiagonal$(t+r+5)$, *then if* settledby$(t+Q+r, t+Q+\text{twopow}(k))$, *then if* settledby$(t+Q+r+2, t+Q+\text{twopow}(k))$, *then if* settledby$(t+r+5, t+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-r \leq a+L$, *then if* winwittagree$(t,k,Q,a,L)$, *then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r-1)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogappinrsubonetrueofsinglewinwittagreeandzerodiagfi

□

Theorem 9.805. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$, *then if* doublesat$(5, t+5+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r-1) \leq a+L$, *then if* winwittagree$(t,k,5,a,L)$, *then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+4)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogappinrplusfourtrueofdoublesfiveandsinglewinwittagr

□

Theorem 9.806. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t+6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$, *then if* doublesat$(5, t+5+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$, *then if* $a \leq t+\text{twopow}(k+1)-(r+2)$, *then if* $t+\text{twopow}(k+1)-(r-1) \leq a+L$, *then if* winwittagree$(t,k,5,a,L)$, *then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t+\text{twopow}(k+1)-(r+3)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogappinrplusthreetrueofdoublesfiveandsinglewinwittag

□

Theorem 9.807. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 2, t + 5 + \text{twopow}(k))$*, then if $a \leq t + \text{twopow}(k+1) - (r+2)$, then if $t + \text{twopow}(k+1) - (r-1) \leq a + L$, then if* winwittagree$(t, k, 5, a, L)$*, then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+1)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogappinrplusonetrueofdoublesfiveandsinglewinwittagre □

Theorem 9.808. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t + \text{twopow}(k+1) - (r-1) \leq a + L$, then if* winwittagree$(t, k, 5, a, L)$*, then* wittbitbool$(\text{wittwin}(t, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+6)) = \text{false}$.

Proof. The goal follows from winwittscalecrosszerogappindoublingdiagfalseofdoublesfiveandsinglewinwit □

Theorem 9.809. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t+5+r+2, t+5+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t + \text{twopow}(k+1) - (r-1) \leq a + L$, then if* winwittagree$(t, k, 5, a, L)$*, then* wittbitbool$(\text{wittwin}(t + 1, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r + 4)) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteponepinrplusfourtrueofdoublesfiveandsinglewi □

Theorem 9.810. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if*

settledby$(t{+}5{+}r{+}2, t{+}5{+}\text{twopow}(k))$, *then if* $a \leq t{+}\text{twopow}(k{+}1){-}(r{+}2)$, *then if* $t{+}\text{twopow}(k{+}1){-}(r{-}1) \leq a{+}L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* wittbitbool$(\text{wittwin}(t+2, \text{twopow}(k)), t + \text{twopow}(k+1) - (r+2)) =$ true.

Proof. The goal follows from winwittscalecrosszerogapsteptwopinrplustwotrueofdoublesfiveandsinglewin
□

Theorem 9.811. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t{+}5{+}r{+}2 \leq k$, then if* zerodiagonal$(t{+}5{+}r)$, *then if* doublesat$(5, t{+}5+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r+1, t+\text{twopow}(k))$, *then if* settledby$(t{+}5{+}r{+}2, t{+}5{+}\text{twopow}(k))$, *then if* $a \leq t{+}\text{twopow}(k{+}1){-}(r{+}2)$, *then if* $t{+}\text{twopow}(k{+}1){-}(r{-}1) \leq a{+}L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* wittbitbool$(\text{wittwin}(t+3, \text{twopow}(k)), t + \text{twopow}(k+1) - r) =$ false.

Proof. The goal follows from winwittscalecrosszerogapstepthreepinrfalseofdoublesfiveandsinglewinwittag
□

Theorem 9.812. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t{+}5{+}r{+}2 \leq k$, then if* zerodiagonal$(t{+}5{+}r)$, *then if* doublesat$(5, t{+}5+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r+1, t+\text{twopow}(k))$, *then if* settledby$(t{+}5{+}r{+}2, t{+}3{+}\text{twopow}(k))$, *then if* $a \leq t{+}\text{twopow}(k{+}1){-}(r{+}2)$, *then if* $t{+}\text{twopow}(k{+}1){-}(r{-}1) \leq a{+}L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* wittbitbool$(\text{wittwin}(t+4, \text{twopow}(k)), t + \text{twopow}(k+1) - r) =$ true.

Proof. The goal follows from winwittscalecrosszerogapstepfourpinrtrueofdoublesfiveandsinglewinwittagr
□

Theorem 9.813. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t{+}5{+}r{+}2 \leq k$, then if* zerodiagonal$(t{+}5{+}r)$, *then if* doublesat$(5, t{+}5+r)$, *then if* settledby$(t+4+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r, t+\text{twopow}(k))$, *then if* settledby$(t+5+r+1, t+\text{twopow}(k))$, *then if* settledby$(t{+}5{+}r{+}2, t{+}3{+}\text{twopow}(k))$, *then if* $a \leq t{+}\text{twopow}(k{+}1){-}(r{+}2)$, *then if* $t{+}\text{twopow}(k{+}1){-}(r{-}1) \leq a{+}L$, *then if* winwittagree$(t, k, 5, a, L)$, *then* wittbitbool$(\text{wittwin}(t+5, \text{twopow}(k)), t + \text{twopow}(k+1) - (r-1)) =$ false.

Proof. The goal follows from winwittscalecrosszerogapstepfivepinrsubonefalseofdoublesfiveandsinglewinv
□

Theorem 9.814. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 2, t + 5 + \text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r-1) \leq a+L$, then if* winwittagree$(t, k, 5, a, L)$*, then* lhcenter$(t + 2 + \text{twopow}(k), \text{twopow}(k) - (r + 2)) = \text{true}$.

Proof. The goal follows from lhcenterlatecrossteptwopinrplustwotrueofdoublesfiveandsinglewinwittagre
□

Theorem 9.815. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 2, t + 5 + \text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r-1) \leq a+L$, then if* winwittagree$(t, k, 5, a, L)$*, then* wittbitbool(wittwin$(t + 2, \text{twopow}(k)), t + \text{twopow}(k + 1) - r) = \text{true}$.

Proof. The goal follows from winwittscalecrosszerogapsteptwopinrtrueofdoublesfiveandsinglewinwittagr
□

Theorem 9.816. *For every natural number t and natural number k and natural number r and natural number a and natural number L, if $t + 6 < r$, then if $t+5+r+2 \leq k$, then if* zerodiagonal$(t+5+r)$*, then if* doublesat$(5, t+5+r)$*, then if* settledby$(t + 4 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r, t + \text{twopow}(k))$*, then if* settledby$(t + 5 + r + 1, t + \text{twopow}(k))$*, then if* settledby$(t+5+r+2, t+3+\text{twopow}(k))$*, then if $a \leq t+\text{twopow}(k+1)-(r+2)$, then if $t+\text{twopow}(k+1)-(r-1) \leq a+L$, then if* winwittagree$(t, k, 5, a, L)$*, then* wittbitbool(wittwin$(t + 5, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r - 1)) =$ not(wittbitbool(wittwin$(t, \text{twopow}(k)), t + \text{twopow}(k + 1) - (r - 1))$).

Proof. The goal follows from winwittscalecrosszerogapsamebitrsuboneflipfiveofdoublesfiveandsinglewinw
□

Theorem 9.817. *For every natural number T and natural number P and natural number m and natural number t and natural number S, if $0 < P$,*

then if $0 < m$, *then if* $T \leq t$, *then if* $t+m \cdot P \leq S+1$, *then if* rhcolagree$(t, m \cdot P, 1, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhcolagreeonehorizon(T, P, m, t, S). □

Theorem 9.818. *For every natural number* T *and natural number* P *and natural number* m *and natural number* t *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq t$, *then if* $t+m \cdot P \leq S+1$, *then if* lhcolagree$(t, m \cdot P, 1, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhcolagreeonehorizon(T, P, m, t, S). □

Theorem 9.819. *For every natural number* T *and natural number* P *and natural number* m *and natural number* t *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq t$, *then if* $t + m \cdot P \leq S + 1$, *then if* rhbandagree$(t, m \cdot P, 0, S + 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreezerohorizon(T, P, m, t, S). □

Theorem 9.820. *For every natural number* T *and natural number* P *and natural number* m *and natural number* t *and natural number* S *and natural number* K, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq t$, *then if* $t + m \cdot P \leq S + 1$, *then if* rhcolagree$(t, m \cdot P, 1 + K, S + K)$, *then if* rhoragree$(t, m \cdot P, 1 + K, S + K)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhcenterinwarddeterminehorizon(T, P, m, t, S, K). □

Theorem 9.821. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number* m *and natural number* t *and natural number* S, *if* $T \leq t$, *then if* rhcenter$(t + m \cdot P, 1) =$ rhcenter$(t, 1)$, *then if* rhoragree$(t, m \cdot P, 1, S)$, *then* rhcolagree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, *hper*. The goal follows from periodicrhcolagreeoneofrhoragreeone$(T, P, hP, hper)$. □

Theorem 9.822. *For every natural number* T *and natural number* P *and natural number* m *and natural number* t *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq t$, *then if* $t+m \cdot P \leq S+1$, *then if* rhcenter$(t+m \cdot P, 1) =$ rhcenter$(t, 1)$, *then if* rhoragree$(t, m \cdot P, 1, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhoragreeonehorizon(T, P, m, t, S). □

Theorem 9.823. *For every natural number Q and natural number z and natural number d and natural number t, if* doublesat(Q, z)*, then if $d \leq z$, then if* settledby(d, t)*, then* ldiag$(d, t + Q) =$ ldiag(d, t).

Proof. The goal follows from ldiagsettledperiodicofdoublesle(Q, z, d, t). □

Theorem 9.824. *For every natural number Q and natural number z and natural number t, if* zerodiagonal(z)*, then if* doublesat(Q, z)*, then if* settledby$(z+1, t)$*, then* ldiag$(z + 1, t + Q) =$ not(ldiag$(z + 1, t)$).

Proof. The goal follows from ldiagsettledantiperiodicofdoubles(Q, z, t). □

Theorem 9.825. *For every natural number Q and natural number z and natural number t and natural number p, if* doublesat(Q, z)*, then if $t - p \leq z$, then if* settledby$(t - p, t)$*, then* lhcenter$(t + Q, p + Q) =$ lhcenter(t, p).

Proof. The goal follows from lhcenterdiagshiftperiodicofdoublesle(Q, z, t, p). □

Theorem 9.826. *For every natural number Q and natural number t and natural number p, if $p \leq t$, then if* zerodiagonal$(t-p)$*, then if* doublesat$(Q, t-p)$*, then if* settledby$(t - p + 1, t + 1)$*, then* lhcenter$(t + 1 + Q, p + Q) =$ not(lhcenter$(t + 1, p)$).

Proof. The goal follows from lhcenterdiagshiftantiperiodicofdoubles(Q, t, p). □

Theorem 9.827. *For every natural number Q and natural number z and natural number t and natural number p and natural number p_0 and natural number L, if* doublesat(Q, z)*, then if $t-p \leq z$, then if* settledby$(t-p, t)$*, then if $p_0 \leq p + Q$, then if $p + Q \leq p_0 + L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t, p + Q) =$ lhcenter(t, p).

Proof. The goal follows from lhcenterspatialperiodicofdoublesleandlhbandagree(Q, z, t, p, p_0, L). □

Theorem 9.828. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if $p \leq t$, then if* zerodiagonal$(t-p)$*, then if* doublesat$(Q, t-p)$*, then if* settledby$(t-p+1, t+1)$*, then if $p_0 \leq p+Q$, then if $p+Q \leq p_0+L$, then if* lhbandagree$(t+1, Q, p_0, L)$*, then* lhcenter$(t + 1, p + Q) =$ not(lhcenter$(t + 1, p)$).

Proof. The goal follows from lhcenterspatialantiperiodicofdoublesandlhbandagree(Q, t, p, p_0, L). □

Theorem 9.829. *For every natural number t and natural number k and natural number p, if $t-p \leq k$, then if* settledby$(t-p, t)$*, then* wittbitbool(wittwin$(t,$ twopow$(k)), t+p+$ twopow$(k+1)) =$ false.

Proof. The goal follows from winwitttestbitsettledleftge(t, k, p). □

Theorem 9.830. *For every natural number t and natural number k and natural number Q and natural number p_0 and natural number L, if $t + Q - p_0 \leq k$, then if for every natural number d, if $d \leq t + Q - p_0$, then* settledby(d, t)*, then* winwittagree$(t, k, Q, t + p_0 +$ twopow$(k+1), L)$.

Proof. The goal follows from winwittagreesettledleftstrip(t, k, Q, p_0, L). □

Theorem 9.831. *For every natural number t and natural number k and natural number Q and natural number p_0 and natural number L, if $t + Q - p_0 \leq k$, then if for every natural number d, if $d \leq t + Q - p_0$, then* settledby(d, t)*, then if* lhbandagree$(t +$ twopow$(k), Q, p_0 +$ twopow$(k), L)$*, then* lhbandagree(t, Q, p_0, L).

Proof. The goal follows from lhbandagreeinwardofsettledleftstrip(t, k, Q, p_0, L). □

Theorem 9.832. *For every natural number t and natural number k and natural number Q and natural number p_0 and natural number L, if $t + Q - p_0 \leq k$, then if for every natural number d, if $d \leq t + Q - p_0$, then* settledby(d, t)*, then if* lhbandagree(t, Q, p_0, L)*, then* lhbandagree$(t +$ twopow$(k), Q, p_0 +$ twopow$(k), L)$.

Proof. The goal follows from lhbandagreestepofsettledleftstrip(t, k, Q, p_0, L). □

Theorem 9.833. *For every natural number t and natural number k and natural number p, if $t - p \leq k$, then if* settledby$(t - p, t)$*, then* lhcenter$(t +$ twopow$(k), p +$ twopow$(k)) =$ lhcenter(t, p).

Proof. The goal follows from lhcenterscalestepsettledleftge(t, k, p). □

Theorem 9.834. *For every natural number z and natural number t, if* zerodiagonal(z)*, then if* settledby(z, t)*, then* ldiag$(z, t) =$ false.

Proof. The goal follows from ldiagofzerodiag(z, t). □

Theorem 9.835. *For every natural number z and natural number t, if* zerodiagonal(z), *then if* settledby$(z+2,t)$, *then* ldiag$(z+2,t) =$ true.

Proof. The goal follows from ldiagafterzerodiag(z,t). □

Theorem 9.836. *For every natural number t and natural number p, if* zerodiagonal$(t-p)$, *then if* settledby$(t-p,t)$, *then* lhcenter$(t,p) =$ false.

Proof. The goal follows from lhcenterofzerodiag(t,p). □

Theorem 9.837. *For every natural number t and natural number p, if* zerodiagonal$(t-p)$, *then if* settledby$(t-p+2,t+2)$, *then* lhcenter$(t+2,p) =$ true.

Proof. The goal follows from lhcenterafterzerodiag(t,p). □

Theorem 9.838. *For every natural number t and natural number p, if* zerodiagonal$(t-p)$, *then if* settledby$(t-p,t)$, *then* lhcenter$(t,p+2) =$ lhcenter$(t,p+1)$.

Proof. The goal follows from lhcentertwinbeforezerodiag(t,p). □

Theorem 9.839. *For every natural number t and natural number p, if* zerodiagonal$(t-(p+1))$, *then if* settledby$(t-(p+1)+2,t+2)$, *then* lhcenter$(t+3,p+1) =$ not(lhcenter$(t+2,p+2)$).

Proof. The goal follows from lhcenterresetafterzerodiag(t,p). □

Theorem 9.840. *For every natural number t and natural number p, if* zerodiagonal$(t-p)$, *then if* settledby$(t-p,t)$, *then if* settledby$(t-p+2,t+2)$, *then* lhcenter$(t+2,p) =$ not(lhcenter(t,p)).

Proof. The goal follows from lhcentertwostepsamecolflipofzerodiag(t,p). □

Theorem 9.841. *For every natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-p)$, *then if* settledby$(t-p,t)$, *then if* settledby$(t-p+2,t+2)$, *then if* $p_0 \leq p$, *then if* $p \leq p_0 + L$, *then not* lhbandagree$(t,2,p_0,L)$.

Proof. The goal follows from notlhbandagreetwoofzerodiag(t,p,p_0,L). □

Theorem 9.842. *For every natural number Q and natural number t and natural number p and natural number z and natural number p_0 and natural number L, if* zerodiagonal$(t-p)$, *then if* zerodiagonal(z), *then if* $t+Q-p = z+2$, *then if* settledby$(t-p,t)$, *then if* settledby$(z+2,t+Q)$, *then if* $p_0 \leq p$, *then if* $p \leq p_0 + L$, *then not* lhbandagree(t,Q,p_0,L).

Proof. The goal follows from notlhbandagreeofzeroandafterzerodiag(Q, t, p, z, p_0, L).
□

Theorem 9.843. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-p)$*, then if* settledby$(t+Q-p, t+Q)$*, then if $p_0 \leq p$, then if $p \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t, p) =$ true.

Proof. The goal follows from lhcentereqfalseatshiftofzerodiagandlhbandagree(Q, t, p, p_0, L).
□

Theorem 9.844. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-(p+2))$*, then if* settledby$(t+Q-(p+2)+2, t+Q)$*, then if $p_0 \leq p$, then if $p \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t, p) =$ true.

Proof. The goal follows from lhcentereqtrueatshiftafterzerodiagandlhbandagree(Q, t, p, p_0, L).
□

Theorem 9.845. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-(p+3))$*, then if* settledby$(t+Q-(p+3)+2, t+Q)$*, then if $p_0 \leq p+1$, then if $p+1 \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+1, p+1) =$ not(lhcenter$(t, p+2)$).

Proof. The goal follows from lhcenterstepflipatshiftafterzerodiagofsinglelhbandagree(Q, t, p, p_0, L).
□

Theorem 9.846. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-(p+2))$*, then if* settledby$(t+Q-(p+2), t+Q)$*, then if* settledby$(t+Q-(p+2)+2, t+Q)$*, then if $p_0 \leq p$, then if $p+2 \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+1, p+1) =$ true.

Proof. The goal follows from lhcentersteponeeqtrueatshiftofzerodiagandsinglelhbandagree$(Q, t, p, p_0, L$
□

Theorem 9.847. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if* rhcolagree$(t, m \cdot P, 1, S)$*, then* lhbandagree$(t, m \cdot P, 0, S+1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhbandagreezeroofrhcolagreeone$(T, P, hP, hper)$
□

Theorem 9.848. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number L and natural number s, if $T \leq t$, then if* rhcolagree$(t, m \cdot P, 1, L + s)$*, then* lhbandagree$(t + s, m \cdot P, 0, L + 1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhbandagreestepsofrhcolagreeone(T, P, hP, $hper$
□

Theorem 9.849. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-p)$*, then if* settledby$(t+Q-p, t+Q)$*, then if $p_0 \leq p+1$, then if $p+2 \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t, p+2) =$ lhcenter$(t, p+1)$.

Proof. The goal follows from lhcentertwinatshiftofzerodiagandlhbandagree(Q, t, p, p_0, L).
□

Theorem 9.850. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-p)$*, then if* settledby$(t+Q-p, t+Q)$*, then if $p_0 \leq p$, then if $p \leq p_0 + L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+1, p+1) =$ bxor(lhcenter$(t, p+2)$, lhcenter$(t, p+1)$)*.*

Proof. The goal follows from lhcenterlinearsteponeatshiftofzerodiagandlhbandagree(Q, t, p, p_0, L).
□

Theorem 9.851. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-(p+1))$*, then if* settledby$(t+Q-(p+1), t+Q)$*, then if $p_0 \leq p+1$, then if $p+1 \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+1, p+1) =$ bxor(lhcenter$(t, p+2)$, lhcenter(t, p))*.*

Proof. The goal follows from lhcenterlinearsteptwoatshiftofzerodiagandlhbandagree(Q, t, p, p_0, L).
□

Theorem 9.852. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t+Q-p)$*, then if* settledby$(t+Q-p, t+Q)$*, then if $p_0 \leq p$, then if $p+2 \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+1, p+1) =$ false*.*

Proof. The goal follows from lhcentersteponeeqfalseatshiftofzerodiagandsinglelhbandagree$(Q, t, p, p_0, L$
□

Theorem 9.853. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-(p+2))$*, then if* zerodiagonal$(t+Q-p)$*, then if* settledby$(t-(p+2)+2, t+1)$*, then if* settledby$(t+Q-p, t+Q)$*, then if* $p_0 \leq p$*, then if* $p+2 \leq p_0+L$*, then not* lhbandagree(t, Q, p_0, L).

Proof. The goal follows from notlhbandagreeofzerodiagtwoandshiftzerodiag(Q, t, p, p_0, L). □

Theorem 9.854. *For every natural number t and natural number p, if* zerodiagonal$(t-(p+2))$*, then if* settledby$(t-(p+2)+2, t+2)$*, then if* lhcenter$(t+2, p+1) =$ true*, then* lhcenter$(t+3, p+1) =$ false.

Proof. The goal follows from lhcenterstepabsorbfalseafterzerodiag(t, p). □

Theorem 9.855. *For every natural number t and natural number p, if* zerodiagonal$(t-(p+2))$*, then if* settledby$(t-(p+2)+2, t+2)$*, then if* lhcenter$(t+2, p+3) =$ false*, then* lhcenter$(t+4, p+2) =$ false.

Proof. The goal follows from lhcentertwostepabsorbfalseafterzerodiag(t, p). □

Theorem 9.856. *For every natural number t and natural number k and natural number r, if $r \leq$* twopow(k)*, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r, t+$twopow$(k))$*, then* lhcenter$(t+$twopow$(k),$ twopow$(k)-r) =$ false.

Proof. The goal follows from lhcenterlatecrossofzerodiaguncond(t, k, r). □

Theorem 9.857. *For every natural number t and natural number k and natural number r, if $r+2 \leq$* twopow(k)*, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+$twopow$(k))$*, then* lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+2)) =$ true.

Proof. The goal follows from lhcenterlatecrossafterzerodiaguncond(t, k, r). □

Theorem 9.858. *For every natural number t and natural number k and natural number r, if $r+3 \leq$* twopow(k)*, then if* zerodiagonal$(t+r)$*, then if* settledby$(t+r+2, t+$twopow$(k))$*, then* lhcenter$(t+1+$twopow$(k),$ twopow$(k)-(r+2)) =$ not(lhcenter$(t+$twopow$(k),$ twopow$(k)-(r+1)))$.

Proof. The goal follows from lhcenterlatecrossresetafterzerodiaguncond(t, k, r). □

Theorem 9.859. *For every natural number t and natural number k and natural number r, if $r+2 \leq \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+r+2)$*, then if* $\mathrm{settledby}(t+r+2, t+\mathrm{twopow}(k))$*, then* $\mathrm{lhcenter}(t+\mathrm{twopow}(k), \mathrm{twopow}(k)-r) = \mathrm{lhcenter}(t+\mathrm{twopow}(k), \mathrm{twopow}(k)-(r+1))$.

Proof. The goal follows from $\mathrm{lhcenterlatecrosstwinbeforezerodiaguncond}(t, k, r)$. □

Theorem 9.860. *For every natural number Q and natural number z and natural number t and natural number k and natural number r, if $r+Q \leq \mathrm{twopow}(k)$, then if $t+Q+r \leq z$, then if* $\mathrm{doublesat}(Q, z)$*, then if* $\mathrm{settledby}(t+Q+r, t+\mathrm{twopow}(k))$*, then* $\mathrm{lhcenter}(t+Q+\mathrm{twopow}(k), \mathrm{twopow}(k)-r) = \mathrm{lhcenter}(t+\mathrm{twopow}(k), \mathrm{twopow}(k)-(r+Q))$.

Proof. The goal follows from $\mathrm{lhcenterlatecrossperiodicofdoublesleuncond}(Q, z, t, k, r)$. □

Theorem 9.861. *For every natural number Q and natural number t and natural number k and natural number r, if $r+Q+1 \leq \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+Q+r)$*, then if* $\mathrm{doublesat}(Q, t+Q+r)$*, then if* $\mathrm{settledby}(t+Q+r+1, t+\mathrm{twopow}(k))$*, then* $\mathrm{lhcenter}(t+Q+\mathrm{twopow}(k), \mathrm{twopow}(k)-(r+1)) = \mathrm{not}(\mathrm{lhcenter}(t+\mathrm{twopow}(k), \mathrm{twopow}(k)-(r+Q+1)))$.

Proof. The goal follows from $\mathrm{lhcenterlatecrossantiperiodicofdoublesuncond}(Q, t, k, r)$. □

Theorem 9.862. *For every natural number t and natural number k and natural number r, if $t+r < \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+r)$*, then if* $\mathrm{settledby}(t+r, t+\mathrm{twopow}(k))$*, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-r) = \mathrm{rhcenter}(t, r)$.

Proof. The goal follows from $\mathrm{winwitttestbitscalecrossofzerodiaguncond}(t, k, r)$. □

Theorem 9.863. *For every natural number t and natural number k and natural number r, if $t < r$, then if $t+r < \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+r)$*, then if* $\mathrm{settledby}(t+r, t+\mathrm{twopow}(k))$*, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-r) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{winwitttestbitscalecrosszerogapofzerodiaguncond}(t, k, r)$. □

Theorem 9.864. *For every natural number t and natural number k and natural number r, if $t+r+2 < \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+r)$*, then if* $\mathrm{settledby}(t+r+2, t+\mathrm{twopow}(k))$*, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-(r+2)) = \mathrm{not}(\mathrm{rhcenter}(t, r+2))$.

Proof. The goal follows from $\mathrm{winwitttestbitscalecrossafterzerodiaguncond}(t, k, r)$. □

Theorem 9.865. *For every natural number t and natural number k and natural number r, if $t < r+2$, then if $t+r+2 < \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+r)$*, then if* $\mathrm{settledby}(t+r+2, t+\mathrm{twopow}(k))$*, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-(r+2)) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{winwitttestbitscalecrosszerogapafterzerodiaguncond}(t, k, r)$. □

Theorem 9.866. *For every natural number t and natural number k and natural number r, if $t < r$, then if $t+r+2 \leq \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+r+2)$*, then if* $\mathrm{settledby}(t+r+2, t+\mathrm{twopow}(k))$*, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-r) = \mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-(r+1))$.

Proof. The goal follows from $\mathrm{winwitttestbitscalecrosszerogaptwinbeforezerodiaguncond}(t, k, r)$. □

Theorem 9.867. *For every natural number t and natural number k and natural number r, if $t \leq r$, then if $t+r+3 < \mathrm{twopow}(k)$, then if* $\mathrm{zerodiagonal}(t+r)$*, then if* $\mathrm{settledby}(t+r+2, t+\mathrm{twopow}(k))$*, then* $\mathrm{wittbitbool}(\mathrm{wittwin}(t+1, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-(r+1)) = \mathrm{not}(\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-(r+1)))$.

Proof. The goal follows from $\mathrm{winwittsectortwostepflipzerogapafterzerodiaguncond}(t, k, r)$. □

Theorem 9.868. *For every natural number Q and natural number z and natural number t and natural number k and natural number r, if $t+Q+r < \mathrm{twopow}(k)$, then if $t+Q+r \leq z$, then if* $\mathrm{doublesat}(Q, z)$*, then if* $\mathrm{settledby}(t+Q+r, t+\mathrm{twopow}(k))$*, then* $\mathrm{bxor}(\mathrm{wittbitbool}(\mathrm{wittwin}(t, \mathrm{twopow}(k)), t+\mathrm{twopow}(k+1)-(r+Q)), \mathrm{wittbitbool}(\mathrm{wittwin}(t+Q, \mathrm{twopow}(k)), t+Q+\mathrm{twopow}(k+1)-r)) = \mathrm{bxor}(\mathrm{rhcenter}(t, r+Q), \mathrm{rhcenter}(t+Q, r))$.

Proof. The goal follows from $\mathrm{winwittscalecrossperiodicofdoublesleuncond}(Q, z, t, k, r)$. □

Theorem 9.869. *For every natural number Q and natural number z and natural number t and natural number k and natural number r, if $t + Q < r$, then if $t+Q+r <$ twopow(k), then if $t+Q+r \leq z$, then if* doublesat(Q, z), *then if* settledby$(t + Q + r, t +$ twopow$(k))$, *then* wittbitbool(wittwin$(t +$ $Q,$ twopow$(k)), t+Q+$twopow$(k+1)-r) =$ wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k + 1) - (r + Q))$.

Proof. The goal follows from winwittscalecrosszerogapperiodicofdoublesleuncond(Q, z, t, k, r). □

Theorem 9.870. *For every natural number Q and natural number t and natural number k and natural number r, if $t + Q + r + 1 <$* twopow(k), *then if* zerodiagonal$(t+Q+r)$, *then if* doublesat$(Q, t+Q+r)$, *then if* settledby$(t+$ $Q + r + 1, t +$ twopow$(k))$, *then* bxor(wittbitbool(wittwin$(t,$ twopow$(k)), t +$ twopow$(k+1) - (r+Q+1))$, wittbitbool(wittwin$(t+Q,$ twopow$(k)), t+Q+$ twopow$(k + 1) - (r + 1))) =$ not(bxor(rhcenter$(t, r + Q + 1)$, rhcenter$(t +$ $Q, r + 1)))$.

Proof. The goal follows from winwittscalecrossantiperiodicofdoublesuncond(Q, t, k, r). □

Theorem 9.871. *For every natural number Q and natural number t and natural number k and natural number r, if $t + Q \leq r$, then if $t + Q + r + 1 <$* twopow(k), *then if* zerodiagonal$(t + Q + r)$, *then if* doublesat$(Q, t + Q + r)$, *then if* settledby$(t + Q + r + 1, t +$ twopow$(k))$, *then* wittbitbool(wittwin$(t +$ $Q,$ twopow$(k)), t+Q+$twopow$(k+1)-(r+1)) =$ not(wittbitbool(wittwin$(t,$ twopow$(k)), t+$ twopow$(k + 1) - (r + Q + 1)))$.

Proof. The goal follows from winwittscalecrosszerogapantiperiodicofdoublesuncond(Q, t, k, r). □

Theorem 9.872. *For every natural number t and natural number p, if* zerodiagonal$(t-(p+2))$, *then if* settledby$(t-(p+2)+2, t)$, *then* lhcenter$(t, p) =$ true.

Proof. The goal follows from lhcenterattafterzerodiag(t, p). □

Theorem 9.873. *For every natural number t and natural number p, if* zerodiagonal$(t-(p+3))$, *then if* settledby$(t-(p+3)+2, t)$, *then* lhcenter$(t+$ $1, p + 1) =$ not(lhcenter$(t, p + 2)$).

Proof. The goal follows from lhcentersteponeresetafterzerodiag(t, p). □

Theorem 9.874. *For every natural number Q and natural number t and natural number p, if* zerodiagonal$(t - (p + 1))$, *then if* doublesat$(Q, t - (p + 1))$, *then if* settledby$(t - (p + 1) + 1, t)$, *then* lhcenter$(t + Q, p + Q) =$ not(lhcenter(t, p)).

Proof. The goal follows from lhcenterdiagshiftantiperiodicattofdoubles(Q, t, p). □

Theorem 9.875. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(Q, t-(p+1))$, *then if* settledby$(t-(p+1)+1, t)$, *then if $p_0 \leq p + Q$, then if $p + Q \leq p_0 + L$, then if* lhbandagree(t, Q, p_0, L), *then* lhcenter$(t, p + Q) =$ not(lhcenter(t, p)).

Proof. The goal follows from lhcenterspatialantiperiodicattofdoublesandlhbandagree(Q, t, p, p_0, L). □

Theorem 9.876. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-(p+3))$, *then if* doublesat$(Q, t-(p+3))$, *then if* settledby$(t-(p+3)+1, t)$, *then if* settledby$(t-(p+3)+2, t)$, *then if $p_0 \leq p+2+Q$, then if $p+2+Q \leq p_0 + L$, then if* lhbandagree(t, Q, p_0, L), *then* lhcenter$(t, p + 2 + Q) =$ lhcenter$(t + 1, p + 1)$.

Proof. The goal follows from lhcenterspatialdoubleseqsteponeresetoflhbandagree(Q, t, p, p_0, L). □

Theorem 9.877. *For every natural number Q and natural number t and natural number p, if* zerodiagonal$(t-(p+3))$, *then if* doublesat$(Q, t-(p+3))$, *then if* settledby$(t - (p + 3) + 1, t)$, *then if* settledby$(t - (p + 3) + 2, t)$, *then* lhcenter$(t + Q, p + 2 + Q) =$ lhcenter$(t + 1, p + 1)$.

Proof. The goal follows from lhcenterdoubleseqsteponereset(Q, t, p). □

Theorem 9.878. *For every natural number Q and natural number t and natural number p, if* zerodiagonal$(t-(p+3))$, *then if* doublesat$(Q, t-(p+3))$, *then if* settledby$(t - (p + 3) + 1, t)$, *then if* settledby$(t - (p + 3) + 2, t)$, *then* lhcenter$(t + Q + 1, p + Q + 1) =$ not(lhcenter$(t + 1, p + 1)$).

Proof. The goal follows from lhcentersteponeresetantiperiodicofdoubles(Q, t, p). □

Theorem 9.879. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-(p+3))$*, then if* doublesat$(Q, t-(p+3))$*, then if* settledby$(t-(p+3)+1, t)$*, then if* settledby$(t-(p+3)+2, t)$*, then if* $p_0 \leq p+1+Q$*, then if* $p+2+Q \leq p_0+L$*, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+1, p+1+Q) =$ not(lhcenter$(t+1, p+1)$).

Proof. The goal follows from lhcentersteponeresetshiftflipofdoublesandsinglelhbandagree(Q, t, p, p_0, L). □

Theorem 9.880. *For every natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-(p+2))$*, then if* doublesat$(1, t-(p+2))$*, then if* settledby$(t-(p+2), t)$*, then if* settledby$(t-(p+2)+1, t)$*, then if* $p_0 \leq p+2$*, then if* $p+2 \leq p_0+L$*, then if* lhbandagree$(t, 1, p_0, L)$*, then* lhcenter$(t, p+1) =$ true.

Proof. The goal follows from lhcenterpindoublingdiagtrueofdoublesoneandlhbandagree(t, p, p_0, L). □

Theorem 9.881. *For every natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-(p+3))$*, then if* doublesat$(1, t-(p+3))$*, then if* settledby$(t-(p+3), t)$*, then if* settledby$(t-(p+3)+1, t)$*, then if* settledby$(t-(p+3)+2, t)$*, then if* $p_0 \leq p+3$*, then if* $p+3 \leq p_0+L$*, then if* lhbandagree$(t, 1, p_0, L)$*, then* lhcenter$(t+1, p+1) =$ false.

Proof. The goal follows from lhcentersteponepinresetdiagfalseofdoublesoneandlhbandagree(t, p, p_0, L). □

Theorem 9.882. *For every natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal$(t-(p+3))$*, then if* doublesat$(1, t-(p+3))$*, then if* settledby$(t-(p+3), t)$*, then if* settledby$(t-(p+3)+1, t)$*, then if* settledby$(t-(p+3)+2, t)$*, then if* $p_0 \leq p+1$*, then if* $p+3 \leq p_0+L$*, then not* lhbandagree$(t, 1, p_0, L)$.

Proof. The goal follows from notlhbandagreeoneofdoubles(t, p, p_0, L). □

Theorem 9.883. *For every natural number T, if* centerperiodic$(T, 1)$*, then for every natural number t and natural number p and natural number S, if $T \leq t$, then if* zerodiagonal$(t-(p+3))$*, then if* doublesat$(1, t-(p+3))$*, then if* settledby$(t-(p+3), t)$*, then if* settledby$(t-(p+3)+1, t)$*, then if* settledby$(t-(p+3)+2, t)$*, then if* $p+2 \leq S$*, then not* rhcolagree$(t, 1, 1, S)$.

Proof. Assume *hper*. The goal follows from notperiodicrhcolagreeoneofdoublesone(T, *hper*). □

Theorem 9.884. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal($t-p$)*, then if* zerodiagonal($t+Q-(p+3)$)*, then if* settledby($t-p,t$)*, then if* settledby($t+Q-(p+3)+2,t+Q$)*, then if $p_0 \leq p+1$, then if $p+1 \leq p_0+L$, then if* lhbandagree(t,Q,p_0,L)*, then* lhcenter($t,p+2$) = true.

Proof. The goal follows from lhcenterpinplustwotrueoflhbandagreeandzerodiagthree(Q,t,p,p_0,L). □

Theorem 9.885. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal($t-p$)*, then if* zerodiagonal($t+Q-(p+3)$)*, then if* settledby($t-p,t$)*, then if* settledby($t+Q-(p+3)+2,t+Q$)*, then if $p_0 \leq p+1$, then if $p+1 \leq p_0+L$, then if* lhbandagree(t,Q,p_0,L)*, then* lhcenter($t+1,p+1$) = false.

Proof. The goal follows from lhcentersteponepinplusonefalseoflhbandagreeandzerodiagthree(Q,t,p,p_0 □

Theorem 9.886. *For every natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal($t-p$)*, then if* settledby($t-p,t$)*, then if* settledby($t-p+2,t+3$)*, then if $p_0 \leq p+1$, then if $p+1 \leq p_0+L$, then if* lhbandagree($t,3,p_0,L$)*, then* lhcenter($t,p+2$) = true.

Proof. The goal follows from lhcenterpinbeforezerodiagofzerodiagthreeandlhbandagree(t,p,p_0,L). □

Theorem 9.887. *For every natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal($t-(p+1)$)*, then if* doublesat($3,t-(p+1)$)*, then if* settledby($t-(p+1),t$)*, then if* settledby($t-(p+1)+1,t$)*, then if* settledby($t-(p+1)+2,t+3$)*, then if $p_0 \leq p+2$, then if $p+3 \leq p_0+L$, then if* lhbandagree($t,3,p_0,L$)*, then* lhcenter(t,p) = false.

Proof. The goal follows from lhcenterpindoublingdiagfalseofdoublesthreeandlhbandagree(t,p,p_0,L). □

Theorem 9.888. *For every natural number t and natural number p and natural number p_0 and natural number L, if* zerodiagonal($t-(p+1)$)*, then if* doublesat($3,t-(p+1)$)*, then if* settledby($t-(p+1),t$)*, then if*

settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+1)+2,t+3)$, *then if* $p_0 \leq p+2$, *then if* $p+3 \leq p_0+L$, *then if* lhbandagree$(t,3,p_0,L)$, *then* lhcenter$(t+1,p+1) =$ true.

Proof. The goal follows from lhcenterstеponepinplusonetrueofdoublesthreeandlhbandagree(t,p,p_0,L). □

Theorem 9.889. *For every natural number* t *and natural number* p *and natural number* p_0 *and natural number* L, *if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(3,t-(p+1))$, *then if* settledby$(t-(p+1),t)$, *then if* settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+1)+2,t+3)$, *then if* $p_0 \leq p+2$, *then if* $p+4 \leq p_0+L$, *then if* lhbandagree$(t,3,p_0,L)$, *then* lhcenter$(t+2,p+2) =$ false.

Proof. The goal follows from lhcentersteptwopinplustwofalseofdoublesthreeandlhbandagree(t,p,p_0,L) □

Theorem 9.890. *For every natural number* t *and natural number* p *and natural number* p_0 *and natural number* L, *if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(3,t-(p+1))$, *then if* settledby$(t-(p+1),t)$, *then if* settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+1)+2,t+2)$, *then if* $p_0 \leq p+2$, *then if* $p+4 \leq p_0+L$, *then if* lhbandagree$(t,3,p_0,L)$, *then* lhcenter$(t+3,p+1) =$ true.

Proof. The goal follows from lhcenterstepthreepinplusonetrueofdoublesthreeandlhbandagree$(t,p,p_0,L$ □

Theorem 9.891. *For every natural number* t *and natural number* p *and natural number* p_0 *and natural number* L, *if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(3,t-(p+1))$, *then if* settledby$(t-(p+1),t)$, *then if* settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+1)+2,t+2)$, *then if* $p_0 \leq p+1$, *then if* $p+4 \leq p_0+L$, *then not* lhbandagree$(t,3,p_0,L)$.

Proof. The goal follows from notlhbandagreethreeofdoubles(t,p,p_0,L). □

Theorem 9.892. *For every natural number* d *and natural number* t, *if* settledby(d,t), *then* ldiag$(d,t+\text{twopow}(d)) =$ ldiag(d,t).

Proof. The goal follows from ldiagsettledtwopowperiodic(d,t). □

Theorem 9.893. *For every natural number* e *and natural number* d *and natural number* t, *if* $d \leq e$, *then if* settledby(d,t), *then* ldiag$(d,t+\text{twopow}(e)) =$ ldiag(d,t).

Proof. The goal follows from ldiagsettledtwopowperiodicle(e, d, t). □

Theorem 9.894. *For every natural number Q and natural number z and natural number t, if* zerodiagonal(z)*, then if* doublesat(Q, z)*, then if* settledby$(z + 1, t)$*, then* ldiag$(z + 1, t + (Q + Q)) =$ ldiag$(z + 1, t)$.

Proof. The goal follows from ldiagsettledtwomulperiodicofdoubles(Q, z, t). □

Theorem 9.895. *For every natural number Q and natural number z and natural number t, if* zerodiagonal(z)*, then if* doublesat(Q, z)*, then if* settledby$(z + 3, t)$*, then* ldiag$(z + 3, t + Q) =$ not(ldiag$(z + 3, t)$).

Proof. The goal follows from ldiagsettledresetantiperiodicofdoubles(Q, z, t). □

Theorem 9.896. *For every natural number Q and natural number z and natural number d and natural number t, if* zerodiagonal(z)*, then if* doublesat(Q, z)*, then if* $d \leq z + 3$*, then if* settledby(d, t)*, then* ldiag$(d, t + (Q + Q)) =$ ldiag(d, t).

Proof. The goal follows from ldiagsettledtwomulperiodicleaddthreeofdoubles(Q, z, d, t). □

Theorem 9.897. *For every natural number e, there exists a natural number z such that* zerodiagonal(z) *and* doublesat(twopow$(e), z)$.

Proof. The goal follows from existszerodiagdoublesattwopow(e). □

Theorem 9.898. *For every natural number m and natural number P, if $0 < m$, then if $0 < P$, then there exists a natural number z such that* zerodiagonal(z) *and* doublesat$(m \cdot P, z)$.

Proof. The goal follows from existszerodiagdoublesatmul(m, P). □

Theorem 9.899. *For every natural number e and natural number z, if* doublesat(twopow$(e), z)$*, then $e \leq z$.*

Proof. The goal follows from doublesattwopowle(e, z). □

Theorem 9.900. *For every natural number e, there exists a natural number z such that $e \leq z$ and* zerodiagonal$(z) \wedge$ doublesat(twopow$(e), z)$.

Proof. The goal follows from existszerodiagdoublesattwopowge(e). □

Theorem 9.901. *For every natural number Q and natural number t and natural number p, if* zerodiagonal$(t-(p+1))$*, then if* doublesat$(Q, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then* lhcenter$(t+Q+Q, p+Q+Q) =$ lhcenter(t, p).

Proof. The goal follows from lhcentertwoshiftdiagperiodicofdoubles(Q, t, p). □

Theorem 9.902. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if $1 \leq Q$, then if* zerodiagonal$(t-(p+1))$*, then if* doublesat$(Q, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then if* settledby$(t-(p+Q), t)$*, then if $p_0 \leq p+Q$, then if $p+Q \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+Q, p+Q+Q) =$ not(lhcenter(t, p)).

Proof. The goal follows from lhcentershifttwoqflipofdoublesandlhbandagree(Q, t, p, p_0, L). □

Theorem 9.903. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L, if $1 \leq Q$, then if* zerodiagonal$(t-(p+1))$*, then if* doublesat$(Q, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then if* settledby$(t-(p+Q), t)$*, then if $p_0 \leq p+Q$, then if $p+Q \leq p_0+L$, then if* lhbandagree(t, Q, p_0, L)*, then* lhcenter$(t+Q+Q, p+Q+Q) =$ not(lhcenter$(t+Q, p+Q+Q)$).

Proof. The goal follows from lhcentersamecoltwoqflipofdoublesandlhbandagree(Q, t, p, p_0, L). □

Theorem 9.904. *For every natural number Q and natural number t and natural number p and natural number p_0 and natural number L and natural number p_1 and natural number L_1, if $1 \leq Q$, then if* zerodiagonal$(t-(p+1))$*, then if* doublesat$(Q, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then if* settledby$(t-(p+Q), t)$*, then if $p_0 \leq p+Q$, then if $p+Q \leq p_0+L$, then if $p_1 \leq p+Q+Q$, then if $p+Q+Q \leq p_1+L_1$, then if* lhbandagree(t, Q, p_0, L)*, then not* lhbandagree$(t+Q, Q, p_1, L_1)$.

Proof. The goal follows from notlhbandagreestepqofdoubles$(Q, t, p, p_0, L, p_1, L_1)$. □

Theorem 9.905. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number p and natural number S, if $0 < m$, then if $T \leq t$, then if* zerodiagonal$(t-(p+1))$*, then if* doublesat$(m \cdot P, t-(p+1))$*,*

then if settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+m\cdot P),t)$, *then if* $p+3\cdot(m\cdot P)\leq S+1$, *then not* rhcolagree$(t,m\cdot P,1,S)$.

Proof. Assume hP, $hper$. The goal follows from notperiodicrhcolagreeoneofdoubles$(T,P,hP,hper)$. □

Theorem 9.906. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S, if $0<P$, then if $0<m$, then if $T\leq t$, then if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(m\cdot P,t-(p+1))$, *then if* settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+m\cdot P),t)$, *then if* $p+3\cdot(m\cdot P)\leq S+1$, *then if* rhcolagree$(t,m\cdot P,1,S)$, *then not* centerperiodic(T,P).

Proof. The goal follows from notcenterperiodicofrhcolagreeonedoubles(T,P,m,t,p,S). □

Theorem 9.907. *For every natural number T and natural number P, if $0<P$, then if* centerperiodic(T,P), *then for every natural number m and natural number t and natural number p and natural number S, if $0<m$, then if $T\leq t$, then if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(m\cdot P,t-(p+1))$, *then if* settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+m\cdot P),t)$, *then if* $p+3\cdot(m\cdot P)\leq S+1$, *then not* lhcolagree$(t,m\cdot P,1,S)$.

Proof. Assume hP, $hper$. The goal follows from notperiodiclhcolagreeoneofdoubles$(T,P,hP,hper)$. □

Theorem 9.908. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S, if $0<P$, then if $0<m$, then if $T\leq t$, then if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(m\cdot P,t-(p+1))$, *then if* settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+m\cdot P),t)$, *then if* $p+3\cdot(m\cdot P)\leq S+1$, *then if* lhcolagree$(t,m\cdot P,1,S)$, *then not* centerperiodic(T,P).

Proof. The goal follows from notcenterperiodicoflhcolagreeonedoubles(T,P,m,t,p,S). □

Theorem 9.909. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S, if $0<P$, then if $0<m$, then if $T\leq t$, then if* zerodiagonal$(t-(p+1))$, *then if* doublesat$(m\cdot P,t-(p+1))$, *then if* settledby$(t-(p+1)+1,t)$, *then if* settledby$(t-(p+m\cdot P),t)$, *then if* $p+3\cdot(m\cdot P)\leq S+1$, *then if* rhbandagree$(t,m\cdot P,0,S+1)$, *then not* centerperiodic(T,P).

Proof. The goal follows from notcenterperiodicofrhbandagreezerodoubles(T, P, m, t, p, S).
□

Theorem 9.910. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S and natural number K, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if* zerodiagonal$(t - (p + 1))$*, then if* doublesat$(m \cdot P, t - (p + 1))$*, then if* settledby$(t - (p + 1) + 1, t)$*, then if* settledby$(t - (p + m \cdot P), t)$*, then if* $p + 3 \cdot (m \cdot P) \leq S + 1$*, then if* rhcolagree$(t, m \cdot P, 1 + K, S + K)$*, then if* rhoragree$(t, m \cdot P, 1 + K, S + K)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhcenterinwarddeterminedoubles(T, P, m, t, p, S, K).
□

Theorem 9.911. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if* zerodiagonal$(t - (p+1))$*, then if* doublesat$(m \cdot P, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then if* settledby$(t - (p + m \cdot P), t)$*, then if* $p + 3 \cdot (m \cdot P) \leq S + 1$*, then if* rhcenter$(t + m \cdot P, 1) =$ rhcenter$(t, 1)$*, then if* rhoragree$(t, m \cdot P, 1, S)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhoragreeonedoubles(T, P, m, t, p, S).
□

Theorem 9.912. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z and natural number p and natural number S, if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then not* rhcolagree$(z + (p + 1), m \cdot P, 1, S)$.

Proof. Assume hP, $hper$. The goal follows from notperiodicrhcolagreeoneatzerodiagdoubles$(T, P, hP, hp$
□

Theorem 9.913. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m, if $0 < m$, then there exists a natural number z such that there exists a natural number p such that* zerodiagonal(z) *and* doublesat$(m \cdot P, z) \wedge \neg$ rhcolagree$(z + (p + 1), m \cdot P, 1, p + 3 \cdot (m \cdot P))$.

Proof. Assume hP, $hper$. The goal follows from existszerodiagnotperiodicrhcolagreeone$(T, P, hP, hper)$.
□

Theorem 9.914. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m, if $0 < m$, then there exists a natural number z such that there exists a natural number p such that* zerodiagonal(z) *and* doublesat$(m \cdot P, z) \wedge \neg$ lhcolagree$(z + (p + 1), m \cdot P, 1, p + 3 \cdot (m \cdot P))$.

Proof. Assume hP, $hper$. The goal follows from existszerodiagnotperiodiclhcolagreeone$(T, P, hP, hper)$. □

Theorem 9.915. *For every natural number t,* maskparity$(t, 0) =$ not(center(t)) and rhcenter$(t, 1)$.

Proof. The goal follows from maskparityzeroeqnotcenterandrhcenterone(t). □

Theorem 9.916. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* center$(t) =$ true*, then* lhcenter$(t + m \cdot P, 1) =$ lhcenter$(t, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhcenteroneeqofblackcenter$(T, P, hP, hper)$. □

Theorem 9.917. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if if* center$(t) =$ false*, then* rhcenter$(t + m \cdot P, 1) =$ rhcenter$(t, 1)$*, then* lhcenter$(t + m \cdot P, 1) =$ lhcenter$(t, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhcenteroneeqofwhitecenter$(T, P, hP, hper)$. □

Theorem 9.918. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* maskparity$(t+m \cdot P, 0) =$ maskparity$(t, 0)$*, then* lhcenter$(t + m \cdot P, 1) =$ lhcenter$(t, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhcenteroneeqofmaskparityzeroeq$(T, P, hP, hper$ □

Theorem 9.919. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if for every natural number s, if $s \leq S$, then if* center$(t + s) =$ false*, then* rhcenter$(t + s + m \cdot P, 1) =$ rhcenter$(t + s, 1)$*, then* lhbandagree$(t, m \cdot P, 0, S + 1)$.

Proof. Assume hP, *hper*. The goal follows from periodiclhbandagreezeroofwhitecenterrhcenterone($T, P,$
□

Theorem 9.920. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if for every natural number s, if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$*, then* lhbandagree$(t, m \cdot P, 0, S + 1)$.

Proof. Assume hP, *hper*. The goal follows from periodiclhbandagreezeroofmaskparityzeroagree(T, P, hP
□

Theorem 9.921. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if* zerodiagonal$(t - (p+1))$*, then if* doublesat$(m \cdot P, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then if* settledby$(t - (p + m \cdot P), t)$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if for every natural number s, if $s \leq S$, then if* center$(t + s) =$ false*, then* rhcenter$(t + s + m \cdot P, 1) =$ rhcenter$(t + s, 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteronedoubles(T, P, m, t, p, S).
□

Theorem 9.922. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if* zerodiagonal$(t - (p+1))$*, then if* doublesat$(m \cdot P, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then if* settledby$(t - (p + m \cdot P), t)$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if for every natural number s, if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreedoubles(T, P, m, t, p, S).
□

Theorem 9.923. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if $p+3 \cdot (m \cdot P) \leq S+1$, then if for every natural number s, if $s \leq S$, then if* center$(z+(p+1)+s) =$ false*, then* rhcenter$(z + (p + 1) + s + m \cdot P, 1) =$ rhcenter$(z + (p + 1) + s, 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteroneatzerodiagdoubles($T, P, m, z, p,$
□

Theorem 9.924. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat($m \cdot P, z$)*, then if* settledby($z+1, z+(p+1)$)*, then if* settledby($z+1-m \cdot P, z+(p+1)$)*, then if $p+3 \cdot (m \cdot P) \leq S+1$, then if for every natural number s, if $s \leq S$, then* maskparity($z+(p+1)+s+m \cdot P, 0$) = maskparity($z + (p + 1) + s, 0$)*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreeatzerodiagdoubles(T, P, m, z, p, S)
□

Theorem 9.925. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m, if $0 < m$, then there exists a natural number z such that there exists a natural number p such that there exists a natural number s such that* zerodiagonal(z) *and* doublesat($m \cdot P, z$) $\wedge$ $s \leq p + 3 \cdot (m \cdot P)$ $\wedge$ center($z + (p + 1) + s$) = false $\wedge$ rhcenter($z + (p + 1) + s + m \cdot P, 1$) = not(rhcenter($z + (p + 1) + s, 1$)).

Proof. Assume hP, $hper$. The goal follows from existszerodiagwhitecenterrhcenteroneflipofperiodic(T, P
□

Theorem 9.926. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m, if $0 < m$, then there exists a natural number z such that there exists a natural number p such that there exists a natural number s such that* zerodiagonal(z) *and* doublesat($m \cdot P, z$)$\wedge s \leq p+3 \cdot (m \cdot P) \wedge$maskparity($z+(p+1)+s+m \cdot P, 0$) = not(maskparity($z + (p + 1) + s, 0$)).

Proof. Assume hP, $hper$. The goal follows from existszerodiagmaskparityzeroflipofperiodic(T, P, hP, hp
□

Theorem 9.927. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m, if $0 < m$, then there exists a natural number z such that there exists a natural number p such that there exists a natural number s such that* zerodiagonal(z) *and* doublesat($m \cdot P, z$)$\wedge s \leq p+3 \cdot (m \cdot P) \wedge$lhcenter($z+(p+1)+s+m \cdot P, 1$) = not(lhcenter($z + (p + 1) + s, 1$)).

Proof. Assume hP, $hper$. The goal follows from existszerodiaglhcenteroneflipofperiodic($T, P, hP, hper$).
□

Theorem 9.928. *For every natural number t,* rhcenter$(t+1, 1) =$ bxor(center(t), rhcenter$(t, 1)$ or rhcente

Proof. The goal follows from rhcentersteponeeqcenterxoror(t). □

Theorem 9.929. *For every natural number t, if* rhcenter$(t, 1) =$ true*, then* rhcenter$(t + 1, 1) =$ not(center(t)).

Proof. The goal follows from rhcentersteponeeqnotcenterofblackone(t). □

Theorem 9.930. *For every natural number t, if* center$(t) =$ false*, then if* rhcenter$(t, 1) =$ true*, then* rhcenter$(t + 1, 1) =$ true.

Proof. The goal follows from rhcentersteponeeqtrueofwhitecenterandblackone(t). □

Theorem 9.931. *For every natural number t, if* maskparity$(t, 0) =$ true*, then* rhcenter$(t + 1, 1) =$ true.

Proof. The goal follows from rhcentersteponeeqtrueofmaskparityzerotrue(t). □

Theorem 9.932. *For every natural number t, if* maskparity$(t, 0) =$ true*, then* rhcenter$(t + 2, 1) =$ not(center$(t + 1)$).

Proof. The goal follows from rhcentersteptwoeqnotcenterofmaskparityzerotrue(t). □

Theorem 9.933. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* rhcenter$(t + m \cdot P, 1) =$ true*, then if* rhcenter$(t, 1) =$ true*, then* rhcenter$(t + 1 + m \cdot P, 1) =$ rhcenter$(t + 1, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhcentersteponeeqofblackone$(T, P, hP, hper)$. □

Theorem 9.934. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* rhcenter$(t + m \cdot P, 1) =$ true*, then if* rhcenter$(t, 1) =$ true*, then* lhcenter$(t + 1 + m \cdot P, 1) =$ lhcenter$(t + 1, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhcentersteponeeqofblackone$(T, P, hP, hper)$. □

Theorem 9.935. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* maskparity$(t + m \cdot P, 0) =$ true*, then if* maskparity$(t, 0) =$ true*, then* lhcenter$(t + 1 + m \cdot P, 1) =$ lhcenter$(t + 1, 1)$ *and* lhcenter$(t + 2 + m \cdot P, 1) =$ lhcenter$(t + 2, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhcentersteponeandtwoeqofmaskparityzerotrue(□

Theorem 9.936. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if if* center$(t) =$ false*, then* rhcenter$(t + m \cdot P, 1) =$ rhcenter$(t, 1)$*, then if for every natural number s, if $s < S$, then if* center$(t + (s + 1)) =$ false*, then* (rhcenter$(t + s + m \cdot P, 1)$ or rhcenter$(t + s + m \cdot P, 2)) =$ (rhcenter$(t + s, 1)$ or rhcenter$(t + s, 2))$*, then* lhbandagree$(t, m \cdot P, 0, S + 1)$.

Proof. Assume hP, $hper$. The goal follows from periodiclhbandagreezeroofwhitecenterrhoragreeone(T, □

Theorem 9.937. *For every natural number T and natural number P and natural number m and natural number t and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if* zerodiagonal$(t - (p+1))$*, then if* doublesat$(m \cdot P, t-(p+1))$*, then if* settledby$(t-(p+1)+1, t)$*, then if* settledby$(t - (p + m \cdot P), t)$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if if* center$(t) =$ false*, then* rhcenter$(t + m \cdot P, 1) =$ rhcenter$(t, 1)$*, then if for every natural number s, if $s < S$, then if* center$(t + (s + 1)) =$ false*, then* (rhcenter$(t + s + m \cdot P, 1)$ or rhcenter$(t + s + m \cdot P, 2)) =$ (rhcenter$(t + s, 1)$ or rhcenter$(t + s, 2))$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhoragreeonedoubles(T, P, m, t, p, S). □

Theorem 9.938. *For every natural number t, if* rhcenter$(t, 1) =$ false*, then* rhcenter$(t + 1, 1) =$ bxor(center(t), rhcenter$(t, 2))$.

Proof. The goal follows from rhcentersteponeeqcenterxortwoofwhiteone(t). □

Theorem 9.939. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* rhcenter$(t + m \cdot P, 1) =$ false*, then if* rhcenter$(t, 1) =$ false*, then if* rhcenter$(t + m \cdot P, 2) =$ rhcenter$(t, 2)$*, then* rhcenter$(t + 1 + m \cdot P, 1) =$ rhcenter$(t + 1, 1)$.

Proof. Assume hP, *hper*. The goal follows from periodicrhcentersteponeeqofwhiteoneandtwoeq(T, P, hP □

Theorem 9.940. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* rhcenter$(t+m \cdot P, 1) =$ rhcenter$(t, 1)$*, then if if* center$(t+1) =$ false*, then if* rhcenter$(t, 1) =$ false*, then* rhcenter$(t+m \cdot P, 2) =$ rhcenter$(t, 2)$*, then* lhcenter$(t+1+m \cdot P, 1) =$ lhcenter$(t+1, 1)$.

Proof. Assume hP, *hper*. The goal follows from periodiclhcentersteponeeqofwhiteorblackone(T, P, hP, h □

Theorem 9.941. *For every natural number t and natural number k,* cell$(t,$ negpow$(k)) =$ lhcenter$(t,$ twopow$(k))$.

Proof. The goal follows from cellnegpoweqlhcentertwopow(t, k). □

Theorem 9.942. *For every natural number t and natural number k,* center$(t+$ twopow$(k)) =$ bxor(bxor(center(t), lhcenter$(t,$ twopow$(k))),$ response$(t, k))$.

Proof. The goal follows from scalecenterlhcenter(t, k). □

Theorem 9.943. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then* bxor(lhcenter$(t+m \cdot P,$ twopow$(k))$, lhcenter$(t,$ twopow$(k))) =$ bxor(response$(t+m \cdot P, k)$, response$(t, k))$.

Proof. Assume hP, *hper*. The goal follows from periodiclhcentertwopowdefecteqresponsedefect(T, P, hP □

Theorem 9.944. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then* bxor(maskparity$(t+m \cdot P, k)$, maskparity$(t, k)) =$ bxor(bxor(lhcenter$(t+m \cdot P,$ twopow$(k))$, lhcenter$(t,$ twopow$(k))),$ bxor(leftresp $m \cdot P, k)$, leftresponse$(t, k)))$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparitydefecteqlhcentertwopowxorleftrespo □

Theorem 9.945. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P <$* twopow(k)*, then* response$(t + m \cdot P, k) =$ response(t, k).

Proof. Assume hP, $hper$. The goal follows from periodicresponseeqconemul($T, P, hP, hper$). □

Theorem 9.946. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + m \cdot P <$* twopow(k)*, then* bxor(maskparity$(t+m \cdot P, k)$, maskparity(t, k)) = bxor(leftresponse$(t+m \cdot P, k)$, leftresponse(t, k)).

Proof. Assume hP, $hper$. The goal follows from periodicmaskparitydefecteqleftresponsedefectconemul(T □

Theorem 9.947. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$* twopow(k)*, then* response$(t + m \cdot P, k)$ = not(response(t, k)).

Proof. Assume hP, $hper$. The goal follows from periodicresponseflipattwopowmul($T, P, hP, hper$). □

Theorem 9.948. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $0 < m$, then if $T \leq t$, then if $t+m \cdot P =$* twopow(k)*, then* bxor(maskparity$(t+m \cdot P, k)$, maskparity(t, k)) = not(bxor(leftresponse$(t + m \cdot P, k)$, leftresponse(t, k))).

Proof. Assume hP, $hper$. The goal follows from periodicmaskparitydefecteqnotleftresponsedefectattwop □

Theorem 9.949. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then* bxor(lhcenter$(t + m \cdot P, 1)$, lhcenter$(t, 1)$) = bxor(maskparity$(t + m \cdot P, 0)$, maskparity$(t, 0)$).

Proof. Assume hP, $hper$. The goal follows from periodiclhcenteronedefecteqmaskparityzerodefect($T, P,$ □

Theorem 9.950. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* lhcenter$(t+m \cdot P, 1)$ = lhcenter$(t, 1)$*, then* maskparity$(t + m \cdot P, 0)$ = maskparity$(t, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroeqoflhcenteroneeq($T, P, hP,$ *hper* □

Theorem 9.951. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then* bxor(bxor(lhcenter(t+twopow(k)+m·P, 1), lhcenter(t+twopow(k), 1)), bxor(lhcenter(t+m·P, 1), lhcenter(t, 1))) = bxor(bxor(maskparity(t+1+m·P, k), maskparity(t+1, k)), bxor(maskparity(t+$m \cdot P$, k), maskparity(t, k))).

Proof. Assume hP, *hper*. The goal follows from periodiclhcenteronedyadicjumpdefecteqmaskparitystepc □

Theorem 9.952. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then if $t + 1 + m \cdot P <$* twopow(k)*, then* bxor(bxor(lhcenter(t + twopow(k) + $m \cdot P$, 1), lhcenter(t + twopow(k), 1)), bxor(lhcenter(t+m·P, 1), lhcenter(t, 1))) = bxor(bxor(leftresponse(t+1+m·P, k), leftresponse(t+1, k)), bxor(leftresponse(t+m·P, k), leftresponse(t, k))).

Proof. Assume hP, *hper*. The goal follows from periodiclhcenteronedyadicjumpdefecteqleftresponsestep □

Theorem 9.953. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $0 < m$, then if $T \leq t$, then if $t + 1 + m \cdot P =$* twopow(k)*, then* bxor(bxor(lhcenter(t + twopow(k) + $m \cdot P$, 1), lhcenter(t+twopow(k), 1)), bxor(lhcenter(t+m·P, 1), lhcenter(t, 1))) = not(bxor(bxor(leftresponse(t+1+m·P, k), leftresponse(t+1, k)), bxor(leftresponse(t+$m \cdot P$, k), leftresponse(t, k)))).

Proof. Assume hP, *hper*. The goal follows from periodiclhcenteronedyadicjumpdefecteqnotleftresponses □

Theorem 9.954. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k, if $T \leq t$, then* bxor(bxor(lhcenter(t+twopow(k)+m·P, 1), lhcenter(t+twopow(k), 1)), bxor(lhcenter(t+m·P, 1), lhcenter(t, 1))) = bxor(bxor(bxor(lhcenter(t+1+m·P, twopow(k)), lhcenter(t+1, twopow(k))), bxor(lhcenter(t+m·P, twopow(k)), lhcenter(t, twopow(k)))), bxor(bxor(leftresponse(t+1+m·P, k), leftresponse(t+1, k)), bxor(leftresponse(t+m·P, k), leftresponse(t, k)))).

Proof. Assume hP, *hper*. The goal follows from periodiclhcenteronedyadicjumpdefecteqlhcentertwopows □

Theorem 9.955. *For every natural number z and natural number k, if* zerodiagonal(z), *then if* settledby(z, z+twopow(k)), *then* lhcenter(z+twopow(k), twopow(k)) = false.

Proof. The goal follows from lhcentertwopowofzerodiag(z, k). □

Theorem 9.956. *For every natural number z and natural number k, if* zerodiagonal(z), *then if* settledby(z+2, z+2+twopow(k)), *then* lhcenter(z+2 + twopow(k), twopow(k)) = true.

Proof. The goal follows from lhcentertwopowafterzerodiag(z, k). □

Theorem 9.957. *For every natural number z and natural number k, if* zerodiagonal(z), *then if* settledby(z, z+twopow(k)), *then* center(z+twopow(k)+twopow(k)) = bxor(center(z + twopow(k)), response(z + twopow(k), k)).

Proof. The goal follows from scalecenteratzerodiag(z, k). □

Theorem 9.958. *For every natural number z and natural number k, if* zerodiagonal(z), *then if* settledby(z+2, z+2+twopow(k)), *then* center(z+2+twopow(k)+twopow(k)) = not(bxor(center(z+2+twopow(k)), response(z+2 + twopow(k), k))).

Proof. The goal follows from scalecenterafterzerodiag(z, k). □

Theorem 9.959. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number z and natural number k, if* $T \leq z$ + twopow(k), *then if* zerodiagonal(z), *then if* settledby(z, z+twopow(k)), *then* response(z+twopow(k), k) = bxor(center(z+twopow(k) + lag(P, k)), center(z + twopow(k))).

Proof. Assume hP, $hper$. The goal follows from periodicresponseatzerodiag($T, P, hP, hper$). □

Theorem 9.960. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number z and natural number k, if* $T \leq z + 2 +$ twopow(k), *then if* zerodiagonal(z), *then if* settledby($z + 2, z + 2 +$ twopow(k)), *then* response($z + 2 +$ twopow(k), k) = not(bxor(center($z + 2 +$ twopow(k) + lag(P, k)), center($z + 2 +$ twopow(k)))).

Proof. Assume hP, $hper$. The goal follows from periodicresponseafterzerodiag($T, P, hP, hper$). □

Theorem 9.961. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z and natural number k, if $T \leq z +$* twopow(k)*, then if* zerodiagonal(z)*, then if* settledby$(z, z+$twopow$(k))$*, then* bxor(response$(z+$ twopow$(k)+m\cdot P, k)$, response$(z+$twopow$(k), k))$ = lhcenter$(z+$twopow$(k)+$ $m \cdot P,$ twopow$(k))$.

Proof. Assume hP, *hper*. The goal follows from periodicresponsedefectatzerodiageqlhcentertwopow$(T, P$
□

Theorem 9.962. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z and natural number k, if $T \leq z + 2 +$* twopow(k)*, then if* zerodiagonal(z)*, then if* settledby$(z+2, z+2+$twopow$(k))$*, then* bxor(response$(z+$ $2+$twopow$(k)+m\cdot P, k)$, response$(z+2+$twopow$(k), k))$ = not(lhcenter$(z+$ $2 +$ twopow$(k) + m \cdot P,$ twopow$(k)))$.

Proof. Assume hP, *hper*. The goal follows from periodicresponsedefectafterzerodiageqnotlhcentertwopo
□

Theorem 9.963. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number $p0$ and natural number L, if $T \leq t$, then if $p0 \leq$* twopow(k)*, then if* twopow$(k) \leq p0 + L$*, then if* lhbandagree$(t, m\cdot P, p0, L)$*, then* response$(t+m\cdot P, k)$ = response(t, k).

Proof. Assume hP, *hper*. The goal follows from periodicresponseeqoflhbandagreetwopow$(T, P, hP, hper$
□

Theorem 9.964. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number $p0$ and natural number L, if $T \leq t$, then if $p0 \leq$* twopow(k)*, then if* twopow$(k) \leq$ $p0 + L$*, then if* lhbandagree$(t, m \cdot P, p0, L)$*, then* bxor(maskparity$(t + m \cdot$ $P, k)$, maskparity$(t, k))$ = bxor(leftresponse$(t + m \cdot P, k)$, leftresponse$(t, k))$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparitydefecteqleftresponsedefectoflhbanda
□

Theorem 9.965. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if for every natural number s, if $s \leq S$, then if*

center$(t + s) =$ false, *then* rhcenter$(t + s + m \cdot P, 1) =$ rhcenter$(t + s, 1)$, *then* response$(t + m \cdot P, k) =$ response(t, k).

Proof. Assume hP, *hper*. The goal follows from periodicresponseeqofwhitecenterrhcenteronetwopow(T, □

Theorem 9.966. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$, *then if for every natural number s, if $s \leq S$, then if* center$(t + s) =$ false, *then* rhcenter$(t + s + m \cdot P, 1) =$ rhcenter$(t + s, 1)$, *then* bxor(maskparity$(t + m \cdot P, k)$, maskparity$(t, k)) =$ bxor(leftresponse$(t + m \cdot P, k)$, leftresponse(t, k)).

Proof. Assume hP, *hper*. The goal follows from periodicmaskparitydefecteqleftresponsedefectofwhitecer □

Theorem 9.967. *For every natural number t and natural number k,* bxor(bxor(response$(t, k+1)$, response$(t, k))$, response$(t+$twopow$(k), k)) =$ bxor(bxor(lhcenter$(t+$twopow$(k),$ twopow$(k))$, lhcente 1))).

Proof. The goal follows from responsecoboundaryeqlhcentertwopow(t, k). □

Theorem 9.968. *For every natural number t and natural number k,* bxor(bxor(leftresponse$(t, k+1)$, leftresponse$(t, k))$, leftresponse$(t+$twopow$(k), k)) =$ bxor(bxor(lhcenter$(t+$ twopow$(k),$ twopow$(k))$, lhcenter$(t,$ twopow$(k)))$, lhcenter$(t,$ twopow$(k+1)))$.

Proof. The goal follows from leftresponsecoboundaryeqlhcentertwopow(t, k). □

Theorem 9.969. *For every natural number t and natural number k, if $t <$* twopow(k), *then* lhcenter$(t+$twopow$(k),$ twopow$(k)) =$ bxor(bxor(response$(t, k+1)$, response$(t, k))$, response$(t +$ twopow$(k), k))$.

Proof. The goal follows from lhcentertwopoweqresponsecoboundarycone(t, k). □

Theorem 9.970. *For every natural number t and natural number k, if $t <$* twopow(k), *then* lhcenter$(t+$twopow$(k),$ twopow$(k)) =$ bxor(bxor(leftresponse$(t, k+1)$, leftresponse$(t, k))$, leftresponse$(t +$ twopow$(k), k))$.

Proof. The goal follows from lhcentertwopoweqleftresponsecoboundarycone(t, k). □

Theorem 9.971. *For every natural number z and natural number k, if $z <$ twopow(k), then if* zerodiagonal(z)*, then if* settledby$(z, z+\text{twopow}(k))$*, then* bxor(bxor(leftresponse$(z, k+1)$, leftresponse(z, k)), leftresponse$(z+\text{twopow}(k), k)) =$ false.

Proof. The goal follows from leftresponsecoboundaryatzerodiagcone(z, k). □

Theorem 9.972. *For every natural number z and natural number k, if $z + 2 <$ twopow(k), then if* zerodiagonal(z)*, then if* settledby$(z + 2, z + 2 + \text{twopow}(k))$*, then* bxor(bxor(leftresponse$(z + 2, k + 1)$, leftresponse$(z + 2, k)$), leftresponse$(z + 2 + \text{twopow}(k), k)) =$ true.

Proof. The goal follows from leftresponsecoboundaryafterzerodiagcone(z, k). □

Theorem 9.973. *For every natural number z and natural number k, if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if* settledby$(z+1, z+1 + \text{twopow}(k))$*, then* lhcenter$(z + 1 + \text{twopow}(k + 1), \text{twopow}(k + 1)) =$ not(lhcenter$(z + 1 + \text{twopow}(k), \text{twopow}(k))$).

Proof. The goal follows from lhcentertwopowsuccantiperiodicofdoubles(z, k). □

Theorem 9.974. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number z and natural number k, if $T \leq z +$ twopow(k), then if* zerodiagonal(z)*, then if* settledby$(z, z+\text{twopow}(k))$*, then* maskparity$(z+\text{twopow}(k), k) =$ bxor(leftresponse$(z+\text{twopow}(k), k)$, bxor(center$(z+\text{twopow}(k)+\text{lag}(P, k))$, center$(z+\text{twopow}(k))$)).

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityatzerodiag$(T, P, hP, hper)$. □

Theorem 9.975. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number z and natural number k, if $T \leq z + 2 +$ twopow(k), then if* zerodiagonal(z)*, then if* settledby$(z+2, z+2+\text{twopow}(k))$*, then* maskparity$(z+2+\text{twopow}(k), k) =$ not(bxor(leftresponse$(z+2+\text{twopow}(k), k)$, bxor(center$(z+2+\text{twopow}(k)+\text{lag}(P, k))$, center$(z + 2 + \text{twopow}(k))$))).

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityafterzerodiag$(T, P, hP, hper)$. □

Theorem 9.976. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z and natural number k, if $T \leq z + $* twopow$(k)$*, then if* zerodiagonal$(z)$*, then if* settledby$(z, z+$twopow$(k))$*, then* bxor(maskparity$(z+$ twopow$(k) + m \cdot P, k)$, maskparity$(z + $twopow$(k), k)) =$ bxor(lhcenter$(z +$ twopow$(k)+m\cdot P,$ twopow$(k))$, bxor(leftresponse$(z+$twopow$(k)+m\cdot P, k)$, leftresponse$(z+$ twopow$(k), k)))$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparitydefectatzerodiageqlhcentertwopowxo

□

Theorem 9.977. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z and natural number k, if $T \leq z + 2 + $* twopow$(k)$*, then if* zerodiagonal$(z)$*, then if* settledby$(z+2, z+2+$twopow$(k))$*, then* bxor(maskparity$(z+$ $2+$twopow$(k)+m\cdot P, k)$, maskparity$(z+2+$twopow$(k), k)) =$ not(bxor(lhcenter$(z+$ $2 +$ twopow$(k) + m \cdot P,$ twopow$(k))$, bxor(leftresponse$(z + 2 +$ twopow$(k) +$ $m \cdot P, k)$, leftresponse$(z + 2 +$ twopow$(k), k))))$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparitydefectafterzerodiageqnotlhcentertwo

□

Theorem 9.978. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z and natural number k, if $m \cdot P = 2$, then if $T \leq z +$* twopow(k)*, then if* zerodiagonal(z)*, then if* settledby$(z, z+$twopow$(k))$*, then if* settledby$(z+2, z+2+$twopow$(k))$*, then* response$(z+$twopow$(k)+m\cdot P, k) =$ not(response$(z + $twopow$(k), k))$.

Proof. Assume hP, *hper*. The goal follows from periodicresponseflipatzerodiagsteptwo$(T, P, hP,$ *hper*$)$.

□

Theorem 9.979. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number $z1$ and natural number $z2$ and natural number k, if $T \leq$ $z1 +$ twopow(k)*, then if $z1 + m \cdot P = z2 + 2$*, then if* zerodiagonal$(z1)$*, then if* zerodiagonal$(z2)$*, then if* settledby$(z1, z1+$twopow$(k))$*, then if* settledby$(z2+$ $2, z2+2+$twopow$(k))$*, then* response$(z1+$twopow$(k)+m\cdot P, k) =$ not(response$(z1+$ twopow$(k), k))$.

Proof. Assume hP, *hper*. The goal follows from periodicresponseflipoftwozerodiags$(T, P, hP,$ *hper*$)$.

□

Theorem 9.980. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number $z1$ and natural number $z2$ and natural number k, if $T \leq z1 + \text{twopow}(k)$, then if $z1 + m \cdot P = z2$, then if* zerodiagonal$(z1)$*, then if* zerodiagonal$(z2)$*, then if* settledby$(z1, z1{+}\text{twopow}(k))$*, then if* settledby$(z2, z2{+}\text{twopow}(k))$*, then* response$(z1 + \text{twopow}(k) + m \cdot P, k) =$ response$(z1 + \text{twopow}(k), k)$.

Proof. Assume hP, *hper*. The goal follows from periodicresponseeqoftwozerodiags$(T, P, hP, hper)$. □

Theorem 9.981. *For every natural number Q and natural number d and natural number k, if $d{+}Q <$* twopow(k)*, then* bxor(leftresponse$(d{+}\text{twopow}(k){+}Q, k)$, leftresponse$(d{+}\text{twopow}(k), k)) =$ bxor(bxor(lhcenter$(d{+}\text{twopow}(k){+}Q, \text{twopow}(k))$, lhcenter$(d{+}\text{twopow}(k), \text{twopow}(k))$), bxor(bxor(leftresponse$(d{+}Q, k{+}1)$, leftresponse$(d, k{+}1)$), bxor(leftresponse$(d{+}Q, k)$, leftresponse(d, k)))).

Proof. The goal follows from lateleftresponsedefecteqlhcentertwopowdefectcone(Q, d, k). □

Theorem 9.982. *For every natural number t and natural number n,* maskcount$(t, n{+}1) =$ bxor(maskcount(t, n), maskparity$(t + n, 0)$).

Proof. The goal follows from maskcountsucceqxormaskparityzero(t, n). □

Theorem 9.983. *For every natural number t and natural number Q and natural number n, if for every natural number s, if $s < n$, then* maskparity$(t + s{+}Q, 0) =$ maskparity$(t{+}s, 0)$*, then* maskcount$(t{+}Q, n) =$ maskcount(t, n).

Proof. The goal follows from maskcounteqofmaskparityzeroagree(t, Q, n). □

Theorem 9.984. *For every natural number t and natural number Q and natural number k and natural number S, if* twopow$(k) \leq S + 1$*, then if for every natural number s, if $s \leq S$, then* maskparity$(t + s + Q, 0) =$ maskparity$(t + s, 0)$*, then* maskparity$(t + Q, k) =$ maskparity(t, k).

Proof. The goal follows from maskparityeqofmaskparityzeroagree(t, Q, k, S). □

Theorem 9.985. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if for every natural number s, if $s \leq S$, then if* center$(t + s) =$ false*, then* rhcenter$(t + s + m \cdot P, 1) =$ rhcenter$(t + s, 1)$*, then* maskparity$(t + m \cdot P, k) =$ maskparity(t, k).

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityeqofwhitecenterrhcenteronetwopow(□

Theorem 9.986. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if for every natural number s, if $s \leq S$, then if* center$(t + s) =$ false*, then* rhcenter$(t + s + m \cdot P, 1) =$ rhcenter$(t + s, 1)$*, then* leftresponse$(t + m \cdot P, k) =$ leftresponse(t, k)*.*

Proof. Assume hP, *hper*. The goal follows from periodicleftresponseeqofwhitecenterrhcenteronetwopow(□

Theorem 9.987. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if for every natural number s, if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$*, then* leftresponse$(t + m \cdot P, k) =$ leftresponse(t, k)*.*

Proof. Assume hP, *hper*. The goal follows from periodicleftresponseeqofmaskparityzeroagreetwopow(T, □

Theorem 9.988. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if for every natural number s, if $s \leq S$, then* maskparity$(t+s+m\cdot P, 0) =$ maskparity$(t+s, 0)$*, then* response$(t+m\cdot P, k) =$ response(t, k)*.*

Proof. Assume hP, *hper*. The goal follows from periodicresponseeqofmaskparityzeroagreetwopow($T, P,$ □

Theorem 9.989. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if* lhcolagree$(t, m \cdot P, 1, S)$*, then* leftresponse$(t + m \cdot P, k) =$ leftresponse(t, k)*.*

Proof. Assume hP, *hper*. The goal follows from periodicleftresponseeqoflhcolagreeonetwopow($T, P, hP,$ □

Theorem 9.990. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if* lhcolagree$(t, m \cdot P, 1, S)$*, then* maskparity$(t + m \cdot P, k) =$ maskparity(t, k).

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityeqoflhcolagreeonetwopow$(T, P, hP, h$ □

Theorem 9.991. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $T \leq t$, then if* twopow$(k) \leq S + 1$*, then if* lhcolagree$(t, m \cdot P, 1, S)$*, then* response$(t + m \cdot P, k) =$ response(t, k).

Proof. Assume hP, *hper*. The goal follows from periodicresponseeqoflhcolagreeonetwopow$(T, P, hP, hpe$ □

Theorem 9.992. *For every natural number d and natural number k,* lhcenter$(d +$ twopow(k), twopow$(k)) =$ ldiag$(d, d +$ twopow$(k))$.

Proof. The goal follows from lhcentertwopoweqldiag(d, k). □

Theorem 9.993. *For every natural number d and natural number k, if $d \leq k$, then if* settledby$(d, d +$ twopow$(k))$*, then* zcentereq$(d,$ lhcenter$(d +$ twopow$(k),$ twopow$(k)))$.

Proof. The goal follows from lhcentertwopowsettledeqzcenter(d, k). □

Theorem 9.994. *For every natural number d and natural number k, if $d \leq k$, then if* settledby$(d, d +$ twopow$(k))$*, then* lhcenter$(d +$ twopow$(k +$ $1),$ twopow$(k + 1)) =$ lhcenter$(d +$ twopow$(k),$ twopow$(k))$.

Proof. The goal follows from lhcentertwopowsettledscaleinvariant(d, k). □

Theorem 9.995. *For every natural number z and natural number d and natural number k, if* doublesat(twopow$(k), z)$*, then if $d \leq z$, then if* settledby$(d, d +$ twopow$(k))$*, then* lhcenter$(d +$twopow$(k+1),$ twopow$(k+1)) =$ lhcenter$(d +$ twopow$(k),$ twopow$(k))$.

Proof. The goal follows from lhcentertwopowsuccperiodicofdoublesle(z, d, k). □

Theorem 9.996. *For every natural number z and natural number k, if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if* settledby$(z+3, z+3+\text{twopow}(k))$*, then* lhcenter$(z+3+\text{twopow}(k+1), \text{twopow}(k+1)) =$ not(lhcenter$(z+3+\text{twopow}(k), \text{twopow}(k))$).

Proof. The goal follows from lhcentertwopowresetsuccantiperiodicofdoubles(z, k). □

Theorem 9.997. *For every natural number d and natural number k,* lhcenter$(d+\text{twopow}(k), \text{twopow}(k)) =$ bxor(bxor(center$(d+\text{twopow}(k+1))$, center$(d+\text{twopow}(k)))$, response$(d+\text{twopow}(k), k)$).

Proof. The goal follows from lhcentertwopoweqlatescalecenter(d, k). □

Theorem 9.998. *For every natural number z and natural number d and natural number k, if* doublesat(twopow$(k), z)$*, then if* $d \leq z$*, then if* settledby$(d, d+\text{twopow}(k))$*, then* bxor(bxor(center$(d+\text{twopow}(k+2))$, center$(d+\text{twopow}(k+1)))$, response$(d+\text{twopow}(k+1), k+1)) =$ bxor(bxor(center$(d+\text{twopow}(k+1))$, center$(d+\text{twopow}(k)))$, response$(d+\text{twopow}(k), k)$).

Proof. The goal follows from latescalecenterperiodicofdoublesle(z, d, k). □

Theorem 9.999. *For every natural number z and natural number k, if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if* settledby$(z+1, z+1+\text{twopow}(k))$*, then* bxor(bxor(center$(z+1+\text{twopow}(k+2))$, center$(z+1+\text{twopow}(k+1)))$, response$(z+1+\text{twopow}(k+1), k+1)) =$ not(bxor(bxor(center$(z+1+\text{twopow}(k+1))$, center$(z+1+\text{twopow}(k)))$, response$(z+1+\text{twopow}(k), k))$).

Proof. The goal follows from latescalecenterantiperiodicofdoubles(z, k). □

Theorem 9.1000. *For every natural number z and natural number k, if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if* settledby$(z+3, z+3+\text{twopow}(k))$*, then* bxor(bxor(center$(z+3+\text{twopow}(k+2))$, center$(z+3+\text{twopow}(k+1)))$, response$(z+3+\text{twopow}(k+1), k+1)) =$ not(bxor(bxor(center$(z+3+\text{twopow}(k+1))$, center$(z+3+\text{twopow}(k)))$, response$(z+3+\text{twopow}(k), k))$).

Proof. The goal follows from latescalecenterresetantiperiodicofdoubles(z, k). □

Theorem 9.1001. *For every natural number T and natural number P, if* $0 < P$*, then if* centerperiodic(T, P)*, then for every natural number m and natural number d and natural number z and natural number k, if* doublesat(twopow$(k), z)$*, then if* $d + m \cdot P \leq z$*, then if* $T \leq d + \text{twopow}(k)$*, then if* settledby$(d, d+$

twopow$(k))$, *then if* settledby$(d + m \cdot P, d + m \cdot P +$ twopow$(k))$, *then* bxor(response$(d+$twopow$(k+1)+m\cdot P, k+1)$, response$(d+$twopow$(k+1), k+1)) =$ bxor(response$(d +$ twopow$(k) + m \cdot P, k)$, response$(d +$ twopow$(k), k))$.

Proof. Assume hP, *hper*. The goal follows from periodiclateresponsedefecteqofdoublesle$(T, P, hP, hper)$. □

Theorem 9.1002. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number d and natural number z and natural number k, if* zerodiagonal(z), *then if* doublesat(twopow$(k), z)$, *then if $d \leq z$, then if $d + m \cdot P = z + 1$, then if $T \leq d +$* twopow(k), *then if* settledby$(d, d +$ twopow$(k))$, *then if* settledby$(z+1, z+1+$twopow$(k))$, *then* bxor(response$(d+$twopow$(k+1)+ m \cdot P, k + 1)$, response$(d +$ twopow$(k + 1), k + 1)) =$ not(bxor(response$(d +$ twopow$(k) + m \cdot P, k)$, response$(d +$ twopow$(k), k)))$.

Proof. Assume hP, *hper*. The goal follows from periodiclateresponsedefectflipofdoubles$(T, P, hP, hper)$. □

Theorem 9.1003. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number d and natural number z and natural number k, if* zerodiagonal(z), *then if* doublesat(twopow$(k), z)$, *then if $d \leq z$, then if $d + m \cdot P = z + 1$, then if $T \leq d +$* twopow(k), *then if* settledby$(d, d +$ twopow$(k))$, *then if* settledby$(z + 1, z + 1 +$ twopow$(k))$, *then if* response$(d +$ twopow$(k) + m \cdot P, k) =$ response$(d +$ twopow$(k), k)$, *then* response$(d +$ twopow$(k + 1) + m \cdot P, k + 1) =$ not(response$(d +$ twopow$(k + 1), k + 1))$.

Proof. Assume hP, *hper*. The goal follows from periodiclateresponseflipofdoublesandeq$(T, P, hP, hper)$. □

Theorem 9.1004. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$* twopow(k), *then if* twopow$(k) \leq S + 1$, *then if for every natural number s, if $s \leq S$, then if* center$(t + s) =$ false, *then* rhcenter$(t + s + m \cdot P, 1) =$ rhcenter$(t + s, 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteroneattwopow(T, P, m, t, k, S). □

Theorem 9.1005. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural*

number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$ twopow(k), then if twopow$(k) \leq S + 1$, *then if for every natural number s, if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreeattwopow(T, P, m, t, k, S). □

Theorem 9.1006. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if $m \cdot P = 2$, then if $T \leq z +$* twopow(k), *then if* settledby$(z, z +$ twopow$(k))$, *then if* settledby$(z + 2, z + 2 +$ twopow$(k))$, *then if* twopow$(k) \leq S + 1$, *then if for every natural number s, if $s \leq S$, then if* center$(z +$ twopow$(k) + s) =$ false, *then* rhcenter$(z +$ twopow$(k) + s + m \cdot P, 1) =$ rhcenter$(z +$ twopow$(k) + s, 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteroneatzerodiagsteptwo$(T, P, m, z, k,$ □

Theorem 9.1007. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if $m \cdot P = 2$, then if $T \leq z +$* twopow(k), *then if* settledby$(z, z +$ twopow$(k))$, *then if* settledby$(z + 2, z + 2 +$ twopow$(k))$, *then if* twopow$(k) \leq S + 1$, *then if for every natural number s, if $s \leq S$, then* maskparity$(z +$ twopow$(k) + s + m \cdot P, 0) =$ maskparity$(z +$ twopow$(k) + s, 0)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreeatzerodiagsteptwo(T, P, m, z, k, S) □

Theorem 9.1008. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if $2 \leq z + m \cdot P$, then if* zerodiagonal$(z + m \cdot P - 2)$, *then if $T \leq z +$* twopow(k), *then if* settledby$(z, z +$ twopow$(k))$, *then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$, *then if* twopow$(k) \leq S + 1$, *then if for every natural number s, if $s \leq S$, then if* center$(z +$ twopow$(k) + s) =$ false, *then* rhcenter$(z +$ twopow$(k) + s + m \cdot P, 1) =$ rhcenter$(z +$ twopow$(k) + s, 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteroneoftwozerodiags(T, P, m, z, k, S). □

Theorem 9.1009. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z)*, then if $2 \leq z + m \cdot P$, then if* zerodiagonal$(z+m\cdot P-2)$*, then if $T \leq z+$*twopow(k)*, then if* settledby$(z, z+$ twopow$(k))$*, then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$*, then if* twopow$(k) \leq S + 1$*, then if for every natural number s, if $s \leq S$, then* maskparity$(z+$twopow$(k)+s+m\cdot P, 0) =$ maskparity$(z+$twopow$(k)+s, 0)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreeoftwozerodiags(T, P, m, z, k, S). □

Theorem 9.1010. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number $S0$ and natural number $S1$, if $0 < P$, then if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if $d \leq z$, then if $d +$ $m\cdot P = z+1$, then if $T \leq d+$*twopow(k)*, then if* settledby$(d, d+$twopow$(k))$*, then if* settledby$(z+1, z+1+$twopow$(k))$*, then if* twopow$(k) \leq S0+1$*, then if* twopow$(k + 1) \leq S1 + 1$*, then if for every natural number s, if $s \leq S0$, then if* center$(d +$ twopow$(k) + s) =$ false*, then* rhcenter$(d +$ twopow$(k) +$ $s + m \cdot P, 1) =$ rhcenter$(d +$ twopow$(k) + s, 1)$*, then if for every natural number s, if $s \leq S1$, then if* center$(d +$ twopow$(k + 1) + s) =$ false*, then* rhcenter$(d+$twopow$(k+1)+s+m\cdot P, 1) =$ rhcenter$(d+$twopow$(k+1)+s, 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteronescalestepdoubles$(T, P, m, d, z, k$ □

Theorem 9.1011. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number $S0$ and natural number $S1$, if $0 < P$, then if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if $d \leq z$, then if $d +$ $m\cdot P = z+1$, then if $T \leq d+$*twopow(k)*, then if* settledby$(d, d+$twopow$(k))$*, then if* settledby$(z+1, z+1+$twopow$(k))$*, then if* twopow$(k) \leq S0+1$*, then if* twopow$(k+1) \leq S1+1$*, then if for every natural number s, if $s \leq S0$, then* maskparity$(d +$ twopow$(k) + s + m \cdot P, 0) =$ maskparity$(d +$ twopow$(k) +$ $s, 0)$*, then if for every natural number s, if $s \leq S1$, then* maskparity$(d +$ twopow$(k + 1) + s + m \cdot P, 0) =$ maskparity$(d +$ twopow$(k + 1) + s, 0)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreescalestepdoubles$(T, P, m, d, z, k, S$ □

Theorem 9.1012. *For every natural number n, $n <$ twopow(n).*

Proof. The goal follows from lttwopowself(n). □

Theorem 9.1013. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$ twopow(k), then if* twopow(k) $\leq S+1$, *then if* lhcolagree($t, m \cdot P, 1, S$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhcolagreeoneattwopow(T, P, m, t, k, S). □

Theorem 9.1014. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if $m \cdot P = 2$, then if $T \leq z+$*twopow(k), *then if* settledby($z, z+$twopow(k)), *then if* settledby($z+2, z+2+$ twopow(k)), *then if* twopow(k) $\leq S+1$, *then if* lhcolagree($z +$ twopow(k)$, m \cdot P, 1, S$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhcolagreeoneatzerodiagsteptwo(T, P, m, z, k, S). □

Theorem 9.1015. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if $2 \leq z + m \cdot P$, then if* zerodiagonal($z+m \cdot P-2$), *then if $T \leq z+$*twopow(k), *then if* settledby($z, z+$ twopow(k)), *then if* settledby($z + m \cdot P, z + m \cdot P +$ twopow(k)), *then if* twopow(k) $\leq S+1$, *then if* lhcolagree($z +$ twopow(k)$, m \cdot P, 1, S$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhcolagreeoneoftwozerodiags(T, P, m, z, k, S). □

Theorem 9.1016. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number $S0$ and natural number $S1$, if $0 < P$, then if* zerodiagonal(z), *then if* doublesat(twopow(k), z), *then if $d \leq z$, then if $d +$ $m \cdot P = z+1$, then if $T \leq d+$*twopow(k), *then if* settledby($d, d+$twopow(k)), *then if* settledby($z+1, z+1+$twopow(k)), *then if* twopow(k) $\leq S0+1$, *then if* twopow($k+1$) $\leq S1+1$, *then if* lhcolagree($d+$twopow(k)$, m \cdot P, 1, S0$), *then if* lhcolagree($d +$ twopow($k+1$)$, m \cdot P, 1, S1$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhcolagreeonescalestepdoubles$(T, P, m, d, z, k, S0, S1)$. □

Theorem 9.1017. *For every natural number T and natural number P and natural number m and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $T + m \cdot P \leq \text{twopow}(k)$, then if $\text{twopow}(k) \leq S + 1$, then if for every natural number s, if $s \leq S$, then if $\text{center}(\text{twopow}(k) - m \cdot P + s) = \text{false}$, then $\text{rhcenter}(\text{twopow}(k) - m \cdot P + s + m \cdot P, 1) = \text{rhcenter}(\text{twopow}(k) - m \cdot P + s, 1)$, then not $\text{centerperiodic}(T, P)$.*

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteronesubtwopow(T, P, m, k, S). □

Theorem 9.1018. *For every natural number T and natural number P and natural number m and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $T + m \cdot P \leq \text{twopow}(k)$, then if $\text{twopow}(k) \leq S + 1$, then if for every natural number s, if $s \leq S$, then $\text{maskparity}(\text{twopow}(k) - m \cdot P + s + m \cdot P, 0) = \text{maskparity}(\text{twopow}(k) - m \cdot P + s, 0)$, then not $\text{centerperiodic}(T, P)$.*

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreesubtwopow(T, P, m, k, S). □

Theorem 9.1019. *For every natural number T and natural number P, if $0 < P$, then if $\text{centerperiodic}(T, P)$, then for every natural number m and natural number t and natural number k and natural number S, if $0 < m$, then if $T \leq t$, then if $t + m \cdot P = \text{twopow}(k)$, then if $\text{twopow}(k) \leq S + 1$, then there exists a natural number s such that $s \leq S$ and $\text{center}(t + s) = \text{false} \wedge \text{rhcenter}(t + s + m \cdot P, 1) = \text{not}(\text{rhcenter}(t + s, 1))$.*

Proof. Assume hP, $hper$. The goal follows from existswhitecenterrhcenteroneflipattwopowofperiodic$(T,$ □

Theorem 9.1020. *For every natural number T and natural number P, if $0 < P$, then if $\text{centerperiodic}(T, P)$, then for every natural number m and natural number t and natural number k and natural number S, if $0 < m$, then if $T \leq t$, then if $t + m \cdot P = \text{twopow}(k)$, then if $\text{twopow}(k) \leq S + 1$, then there exists a natural number s such that $s \leq S$ and $\text{maskparity}(t + s + m \cdot P, 0) = \text{not}(\text{maskparity}(t + s, 0))$.*

Proof. Assume hP, $hper$. The goal follows from existsmaskparityzeroflipattwopowofperiodic$(T, P, hP, h$ □

Theorem 9.1021. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number S, if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$* twopow(k)*, then if* twopow$(k) \leq S+1$*, then there exists a natural number s such that $s \leq S$ and* lhcenter$(t + s + m \cdot P, 1) =$ not(lhcenter$(t + s, 1)$).

Proof. Assume hP, $hper$. The goal follows from existslhcenteroneflipattwopowofperiodic$(T, P, hP, hper)$ □

Theorem 9.1022. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number k, if $0 < m$, then if $T + m \cdot P \leq$* twopow(k)*, then there exists a natural number s such that $s <$* twopow(k) *and* center(twopow$(k) - m \cdot P +$ $s) =$ false $\wedge$ rhcenter(twopow$(k) - m \cdot P + s + m \cdot P, 1) =$ not(rhcenter(twopow$(k) -$ $m \cdot P + s, 1)$).

Proof. Assume hP, $hper$. The goal follows from existswhitecenterrhcenteroneflipsubtwopowofperiodic$(T$ □

Theorem 9.1023. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number k, if $0 < m$, then if $T + m \cdot P \leq$* twopow(k)*, then there exists a natural number s such that $s <$* twopow(k) *and* maskparity(twopow$(k) -$ $m \cdot P + s + m \cdot P, 0) =$ not(maskparity(twopow$(k) - m \cdot P + s, 0)$).

Proof. Assume hP, $hper$. The goal follows from existsmaskparityzeroflipsubtwopowofperiodic$(T, P, hP,$ □

Theorem 9.1024. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number k, if $0 < m$, then if $T + m \cdot P \leq$* twopow(k)*, then there exists a natural number s such that $s <$* twopow(k) *and* lhcenter(twopow$(k) - m \cdot$ $P + s + m \cdot P, 1) =$ not(lhcenter(twopow$(k) - m \cdot P + s, 1)$).

Proof. Assume hP, $hper$. The goal follows from existslhcenteroneflipsubtwopowofperiodic$(T, P, hP, hper$ □

Theorem 9.1025. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number z and natural number k and natural number S, if* zerodiagonal(z)*, then if $m \cdot P = 2$, then if $T \leq z +$* twopow(k)*, then if* settledby$(z, z +$ twopow$(k))$*, then if* settledby$(z+2, z+2+$twopow$(k))$*, then if* twopow$(k) \leq$

$S + 1$, *then there exists a natural number* s *such that* $s \leq S$ *and* center$(z +$ twopow$(k)+s) =$ false $\wedge$ rhcenter$(z+$twopow$(k)+s+m \cdot P, 1) =$ not(rhcenter$(z+$ twopow$(k) + s, 1))$.

Proof. Assume hP, $hper$. The goal follows from existswhitecenterrhcenteroneflipatzerodiagsteptwoofper □

Theorem 9.1026. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number* m *and natural number* z *and natural number* k *and natural number* S, *if* zerodiagonal(z), *then if* $m \cdot P = 2$, *then if* $T \leq z +$ twopow(k), *then if* settledby$(z, z +$ twopow$(k))$, *then if* settledby$(z+2, z+2+$twopow$(k))$, *then if* twopow$(k) \leq S+1$, *then there exists a natural number* s *such that* $s \leq S$ *and* maskparity$(z+$ twopow$(k) + s + m \cdot P, 0) =$ not(maskparity$(z +$ twopow$(k) + s, 0))$.

Proof. Assume hP, $hper$. The goal follows from existsmaskparityzeroflipatzerodiagsteptwoofperiodic$(T,$ □

Theorem 9.1027. *For every natural number* a *and natural number* b, *if* $a \leq b$, *then* twopow$(a) \leq$ twopow(b).

Proof. The goal follows from twopowmono(a, b). □

Theorem 9.1028. *For every natural number* n *and natural number* k, *if* $n \leq k$, *then* $n <$ twopow(k).

Proof. The goal follows from lttwopowofle(n, k). □

Theorem 9.1029. *For every natural number* T *and natural number* P *and natural number* m *and natural number* k *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T + m \cdot P \leq k$, *then if* twopow$(k) \leq S + 1$, *then if for every natural number* s, *if* $s \leq S$, *then if* center$($twopow$(k)-m \cdot P+s) =$ false, *then* rhcenter$($twopow$(k) - m \cdot P + s + m \cdot P, 1) =$ rhcenter$($twopow$(k) - m \cdot P + s, 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteronesubtwopowge(T, P, m, k, S). □

Theorem 9.1030. *For every natural number* T *and natural number* P *and natural number* m *and natural number* k *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T + m \cdot P \leq k$, *then if* twopow$(k) \leq S + 1$, *then if for every natural number* s, *if* $s \leq S$, *then* maskparity$($twopow$(k)-m \cdot P+s+m \cdot P, 0) =$ maskparity$($twopow$(k)-m \cdot P+s, 0)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreesubtwopowge(T, P, m, k, S). □

Theorem 9.1031. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number k, if $0 < m$, then if $T + m \cdot P \leq k$, then there exists a natural number s such that $s <$* twopow(k) *and* center(twopow(k) $- m \cdot P + s$) = false $\wedge$ rhcenter(twopow(k) $- m \cdot P + s + m \cdot P, 1$) = not(rhcenter(twopow($k$) $- m \cdot P + s, 1$)).

Proof. Assume hP, $hper$. The goal follows from existswhitecenterrhcenteroneflipsubtwopowgeofperiodic □

Theorem 9.1032. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number k, if $0 < m$, then if $T + m \cdot P \leq k$, then there exists a natural number s such that $s <$* twopow(k) *and* maskparity(twopow(k) $- m \cdot P + s + m \cdot P, 0$) = not(maskparity(twopow(k) $- m \cdot P + s, 0$)).

Proof. Assume hP, $hper$. The goal follows from existsmaskparityzeroflipsubtwopowgeofperiodic(T, P, h □

Theorem 9.1033. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number k, if $0 < m$, then if $T + m \cdot P \leq k$, then there exists a natural number s such that $s <$* twopow(k) *and* lhcenter(twopow(k) $- m \cdot P + s + m \cdot P, 1$) = not(lhcenter(twopow($k$) $- m \cdot P + s, 1$)).

Proof. Assume hP, $hper$. The goal follows from existslhcenteroneflipsubtwopowgeofperiodic(T, P, hP, h □

Theorem 9.1034. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k, if $T + P \leq k$, then there exists a natural number s such that $s <$* twopow(k) *and* center(twopow(k) $- P + s$) = false $\wedge$ rhcenter(twopow(k) $- P + s + P, 1$) = not(rhcenter(twopow(k) $- P + s, 1$)).

Proof. Assume hP, $hper$. The goal follows from existswhitecenterrhcenteroneflipsubtwopowpofperiodic(□

Theorem 9.1035. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number k, if $T + P \leq k$, then there exists a natural number s such that $s <$* twopow(k)

and maskparity(twopow(k) $-$ P $+$ s $+$ P, 0) $=$ not(maskparity(twopow(k) $-$ P $+$ s, 0)).

Proof. Assume hP, *hper*. The goal follows from existsmaskparityzeroflipsubtwopowpofperiodic(T, P, hP □

Theorem 9.1036. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number k, if* $T + P \leq k$, *then there exists a natural number s such that* $s <$ twopow(k) *and* lhcenter(twopow(k)$-P+s+P$, 1) $=$ not(lhcenter(twopow(k)$-P+s$, 1)).

Proof. Assume hP, *hper*. The goal follows from existslhcenteroneflipsubtwopowpofperiodic(T, P, hP, hp □

Theorem 9.1037. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number z and natural number k and natural number S, if* zerodiagonal(z), *then if* $m \cdot P = 2$, *then if* $T \leq z +$ twopow(k), *then if* settledby(z, $z +$ twopow(k)), *then if* settledby($z+2$, $z+2+$twopow(k)), *then if* twopow(k) $\leq S+1$, *then there exists a natural number s such that* $s \leq S$ *and* lhcenter($z+$ twopow(k) $+ s + m \cdot P$, 1) $=$ not(lhcenter($z +$ twopow(k) $+ s$, 1)).

Proof. Assume hP, *hper*. The goal follows from existslhcenteroneflipatzerodiagsteptwoofperiodic(T, P, h □

Theorem 9.1038. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number z and natural number k and natural number S, if* zerodiagonal(z), *then if* $2 \leq z + m \cdot P$, *then if* zerodiagonal($z + m \cdot P - 2$), *then if* $T \leq z +$ twopow(k), *then if* settledby(z, $z +$ twopow(k)), *then if* settledby($z + m \cdot P$, $z + m \cdot P +$ twopow(k)), *then if* twopow(k) $\leq S + 1$, *then there exists a natural number s such that* $s \leq S$ *and* center($z +$ twopow(k) $+ s$) $=$ false $\wedge$ rhcenter($z+$twopow(k)$+s+m \cdot P$, 1) $=$ not(rhcenter($z+$twopow(k)$+$ s, 1)).

Proof. Assume hP, *hper*. The goal follows from existswhitecenterrhcenteroneflipoftwozerodiagsandperio □

Theorem 9.1039. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number z and natural number k and natural number S, if* zerodiagonal(z), *then if* $2 \leq z + m \cdot P$, *then if* zerodiagonal($z + m \cdot P - 2$), *then if* $T \leq z +$ twopow(k), *then if* settledby(z, $z +$ twopow(k)), *then if* settledby($z + m \cdot$

$P, z+m\cdot P+\mathrm{twopow}(k))$, *then if* $\mathrm{twopow}(k) \leq S+1$, *then there exists a natural number* s *such that* $s \leq S$ *and* $\mathrm{maskparity}(z+\mathrm{twopow}(k)+s+m\cdot P, 0) = \mathrm{not}(\mathrm{maskparity}(z+\mathrm{twopow}(k)+s, 0))$.

Proof. Assume hP, $hper$. The goal follows from existsmaskparityzeroflipoftwozerodiagsandperiodic(T, P □

Theorem 9.1040. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number* m *and natural number* z *and natural number* k *and natural number* S, *if* $\mathrm{zerodiagonal}(z)$, *then if* $2 \leq z + m \cdot P$, *then if* $\mathrm{zerodiagonal}(z + m \cdot P - 2)$, *then if* $T \leq z+\mathrm{twopow}(k)$, *then if* $\mathrm{settledby}(z, z+\mathrm{twopow}(k))$, *then if* $\mathrm{settledby}(z+m\cdot P, z + m \cdot P + \mathrm{twopow}(k))$, *then if* $\mathrm{twopow}(k) \leq S + 1$, *then there exists a natural number* s *such that* $s \leq S$ *and* $\mathrm{lhcenter}(z+\mathrm{twopow}(k)+s+m\cdot P, 1) = \mathrm{not}(\mathrm{lhcenter}(z + \mathrm{twopow}(k) + s, 1))$.

Proof. Assume hP, $hper$. The goal follows from existslhcenteroneflipoftwozerodiagsandperiodic(T, P, hP □

Theorem 9.1041. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number* m *and natural number* d *and natural number* z *and natural number* k, *if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(\mathrm{twopow}(k), z)$, *then if* $d \leq z$, *then if* $d+m\cdot P = z+1$, *then if* $T \leq d+\mathrm{twopow}(k)$, *then if* $\mathrm{settledby}(d, d+\mathrm{twopow}(k))$, *then if* $\mathrm{settledby}(z+1, z + 1 + \mathrm{twopow}(k))$, *then* $\mathrm{bxor}(\mathrm{bxor}(\mathrm{maskparity}(d + \mathrm{twopow}(k + 1) + m \cdot P, k + 1), \mathrm{maskparity}(d + \mathrm{twopow}(k + 1), k + 1)), \mathrm{bxor}(\mathrm{leftresponse}(d + \mathrm{twopow}(k + 1) + m \cdot P, k + 1), \mathrm{leftresponse}(d + \mathrm{twopow}(k + 1), k + 1))) = \mathrm{not}(\mathrm{bxor}(\mathrm{bxor}(\mathrm{maskparity}(d+\mathrm{twopow}(k)+m\cdot P, k), \mathrm{maskparity}(d+\mathrm{twopow}(k), k)), \mathrm{bxor}(\mathrm{leftresponse}(d+\mathrm{twopow}(k) + m \cdot P, k), \mathrm{leftresponse}(d + \mathrm{twopow}(k), k))))$.

Proof. Assume hP, $hper$. The goal follows from periodiclatemaskparityxorleftresponsedefectflipofdouble □

Theorem 9.1042. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number* m *and natural number* d *and natural number* z *and natural number* k, *if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(\mathrm{twopow}(k), z)$, *then if* $d \leq z$, *then if* $d + m \cdot P = z + 3$, *then if* $T \leq d + \mathrm{twopow}(k)$, *then if* $\mathrm{settledby}(d, d + \mathrm{twopow}(k))$, *then if* $\mathrm{settledby}(z+3, z+3+\mathrm{twopow}(k))$, *then* $\mathrm{bxor}(\mathrm{response}(d+\mathrm{twopow}(k+1)+m \cdot P, k + 1), \mathrm{response}(d + \mathrm{twopow}(k + 1), k + 1)) = \mathrm{not}(\mathrm{bxor}(\mathrm{response}(d + \mathrm{twopow}(k) + m \cdot P, k), \mathrm{response}(d + \mathrm{twopow}(k), k)))$.

Proof. Assume hP, *hper*. The goal follows from periodiclateresponseresetdefectflipofdoubles(T, P, hP, h
□

Theorem 9.1043. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number d and natural number z and natural number k, if* zerodiagonal(z), *then if* doublesat(twopow$(k), z)$, *then if $d \leq z$, then if $d + m \cdot P = z + 3$, then if $T \leq d +$* twopow(k), *then if* settledby$(d, d +$ twopow$(k))$, *then if* settledby$(z + 3, z + 3 +$ twopow$(k))$, *then if* response$(d +$ twopow$(k) + m \cdot P, k) =$ response$(d +$ twopow$(k), k)$, *then* response$(d +$ twopow$(k + 1) + m \cdot P, k + 1) =$ not(response$(d +$ twopow$(k + 1), k + 1))$.

Proof. Assume hP, *hper*. The goal follows from periodiclateresponseresetflipofdoublesandeq(T, P, hP, h
□

Theorem 9.1044. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number d and natural number z and natural number k, if* zerodiagonal(z), *then if* doublesat(twopow$(k), z)$, *then if $d \leq z$, then if $d+m \cdot P = z+3$, then if $T \leq d+$*twopow(k), *then if* settledby$(d, d+$twopow$(k))$, *then if* settledby$(z + 3, z + 3 +$ twopow$(k))$, *then* bxor(bxor(maskparity$(d +$ twopow$(k + 1) + m \cdot P, k + 1)$, maskparity$(d +$ twopow$(k + 1), k + 1))$, bxor(leftresponse$(d +$ twopow$(k + 1) + m \cdot P, k + 1)$, leftresponse$(d +$ twopow$(k + 1), k + 1))) =$ not(bxor(bxor(maskparity$(d+$twopow$(k)+m \cdot P, k)$, maskparity$(d+$twopow$(k), k))$, bxor(leftresponse$(d+$ twopow$(k) + m \cdot P, k)$, leftresponse$(d +$ twopow$(k), k))))$.

Proof. Assume hP, *hper*. The goal follows from periodiclatemaskparityxorleftresponseresetdefectflipofd
□

Theorem 9.1045. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if* doublesat(twopow$(k), z)$, *then if $d \leq z$, then if $d + m \cdot P = z + 1$, then if $T \leq d+$*twopow(k), *then if* settledby$(d, d+$twopow$(k))$, *then if* settledby$(z+1, z+1+$twopow$(k))$, *then if* twopow$(k)+$twopow$(k+1) \leq S+1$, *then if for every natural number s, if $s \leq S$, then if* center$(d +$ twopow$(k) + s) =$ false, *then* rhcenter$(d+$twopow$(k)+s+m \cdot P, 1) =$ rhcenter$(d+$twopow$(k)+s, 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteronethreetwopowdoubles(T, P, m, d,
□

Theorem 9.1046. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if $d \leq z$, then if $d + m \cdot P = z + 1$, then if $T \leq d+$*twopow(k)*, then if* settledby$(d, d+$twopow$(k))$*, then if* settledby$(z+1, z+1+$twopow$(k))$*, then if* twopow$(k) +$ twopow$(k+1) \leq S+1$*, then if for every natural number s, if $s \leq S$, then* maskparity$(d+$twopow$(k)+s+m \cdot P, 0) =$ maskparity$(d+$twopow$(k)+s, 0)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreethreetwopowdoubles$(T, P, m, d, z,$ □

Theorem 9.1047. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if $d \leq z$, then if $d + m \cdot P = z + 1$, then if $T \leq d+$*twopow(k)*, then if* settledby$(d, d+$twopow$(k))$*, then if* settledby$(z+1, z+1+$twopow$(k))$*, then if* twopow$(k) +$ twopow$(k+1) \leq S+1$*, then if* lhcolagree$(d+$twopow$(k), m \cdot P, 1, S)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhcolagreeonethreetwopowdoubles(T, P, m, d, z, k, S). □

Theorem 9.1048. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if $d \leq z$, then if $d + m \cdot P = z + 3$, then if $T \leq d+$*twopow(k)*, then if* settledby$(d, d+$twopow$(k))$*, then if* settledby$(z+3, z+3+$twopow$(k))$*, then if* twopow$(k)+$twopow$(k+1) \leq S+1$*, then if for every natural number s, if $s \leq S$, then if* center$(d+$twopow$(k)+s) =$ false*, then* rhcenter$(d+$twopow$(k)+s+m \cdot P, 1) =$ rhcenter$(d+$twopow$(k)+s, 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofwhitecenterrhcenteronethreetwopowresetdoubles$(T, P,$ □

Theorem 9.1049. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z)*, then if* doublesat(twopow$(k), z)$*, then if $d \leq z$, then if $d + m \cdot P = z + 3$, then if $T \leq d+$*twopow(k)*, then if* settledby$(d, d+$twopow$(k))$*, then if* settledby$(z+3, z+3+$twopow$(k))$*, then if* twopow$(k) +$ twopow$(k+1) \leq S+1$*, then if*

for every natural number s, if $s \leq S$, then $\mathrm{maskparity}(d + \mathrm{twopow}(k) + s + m \cdot P, 0) = \mathrm{maskparity}(d + \mathrm{twopow}(k) + s, 0)$*, then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofmaskparityzeroagreethreetwopowresetdoubles(T, P, m, □

Theorem 9.1050. *For every natural number T and natural number P and natural number m and natural number d and natural number z and natural number k and natural number S, if $0 < P$, then if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(\mathrm{twopow}(k), z)$*, then if $d \leq z$, then if $d + m \cdot P = z + 3$, then if $T \leq d + \mathrm{twopow}(k)$, then if* $\mathrm{settledby}(d, d + \mathrm{twopow}(k))$*, then if* $\mathrm{settledby}(z + 3, z + 3 + \mathrm{twopow}(k))$*, then if* $\mathrm{twopow}(k) + \mathrm{twopow}(k + 1) \leq S + 1$*, then if* $\mathrm{lhcolagree}(d + \mathrm{twopow}(k), m \cdot P, 1, S)$*, then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicoflhcolagreeonethreetwopowresetdoubles(T, P, m, d, z, k, □

Theorem 9.1051. *For every natural number t and natural number k and natural number p,* $\mathrm{rdrv}(t, t, k, p) = \mathrm{rhcenter}(t + k, p)$.

Proof. The goal follows from rdrvdiageqrhcenter(t, k, p). □

Theorem 9.1052. *For every natural number t and natural number p,* $\mathrm{rdrv}(0, 0, t, p) = \mathrm{rhcenter}(t, p)$.

Proof. The goal follows from rdrvzeroeqrhcenter(t, p). □

Theorem 9.1053. *For every natural number td and natural number tx and natural number k and natural number p,* $\mathrm{rdrv}(td, tx, k + 1, p + 1) = \mathrm{bxor}(\mathrm{rdrv}(td, tx, k, p), \mathrm{rdrv}(td, tx, k, p + 1)$ or $\mathrm{rdrv}(td, tx, k, p + 2))$.

Proof. The goal follows from rdrvstepsucc(td, tx, k, p). □

Theorem 9.1054. *For every natural number td and natural number $td2$ and natural number tx and natural number k and natural number p, if $k < p$, then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{rdrv}(td2, tx, k, p)$.

Proof. The goal follows from rdrvdriveindepoflt($td, td2, tx, k, p$). □

Theorem 9.1055. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number td and natural number tx and natural number k and natural number p, if $T \leq td$, then* $\mathrm{rdrv}(td + m \cdot P, tx, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. Assume hP, $hper$. The goal follows from periodicrdrvdriveshifteq$(T, P, hP, hper)$. □

Theorem 9.1056. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number k and natural number p, if $T \leq t$, then* rhcenter$(t + m \cdot P + k, p) =$ rdrv$(t, t + m \cdot P, k, p)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhcentershifteqrdrvsamedrive$(T, P, hP, hper)$. □

Theorem 9.1057. *For every natural number td and natural number tx and natural number k and natural number p, if $k < p$, then* rdrv$(td, tx, k, p) =$ rhcenter$(tx + k, p)$.

Proof. The goal follows from rdrvoutsideconeeqrhcenter(td, tx, k, p). □

Theorem 9.1058. *For every natural number t and natural number a and natural number k and natural number p,* rdrv$(t, t, a + k, p) =$ rdrv$(t + a, t + a, k, p)$.

Proof. The goal follows from rdrvsemigroupadd(t, a, k, p). □

Theorem 9.1059. *For every natural number td and natural number $td2$ and natural number tx and natural number k and natural number p, if for every natural number s, if $s + p \leq k$, then* center$(td + s) =$ center$(td2 + s)$*, then* rdrv$(td, tx, k, p) =$ rdrv$(td2, tx, k, p)$.

Proof. The goal follows from rdrvdrivecausalcone$(td, td2, tx, k, p)$. □

Theorem 9.1060. *For every natural number td and natural number tx and natural number Q and natural number L and natural number k and natural number p, if* rhbandagree$(tx, Q, 1, L)$*, then if $p+k \leq L+1$, then* rdrv$(td, tx+Q, k, p) =$ rdrv(td, tx, k, p).

Proof. The goal follows from rdrvinitcausalcone(td, tx, Q, L, k, p). □

Theorem 9.1061. *For every natural number td and natural number tx and natural number Q and natural number S and natural number s, if* rhbandagree$(tx, Q, 1, S)$*, then if $s \leq S$, then* rdrv$(td, tx+Q, s, 1) =$ rdrv$(td, tx, s, 1)$.

Proof. The goal follows from rdrvcoloneagreeofrhbandagreeone(td, tx, Q, S, s). □

Theorem 9.1062. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, S)$*, then* rhcolagree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, *hper*. The goal follows from periodicrhcolagreeoneofrhbandagreeone$(T, P, hP, hper)$. □

Theorem 9.1063. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, S)$*, then if $s \leq S$, then* maskparity$(t+s+m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofrhbandagreeone$(T, P, hP$ □

Theorem 9.1064. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number s, if $T \leq t$, then if if* center$(t + s) =$ false*, then* rdrv$(t, t + m \cdot P, s, 1) =$ rdrv$(t, t, s, 1)$*, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofrdrvoneagree$(T, P, hP, hp$ □

Theorem 9.1065. *For every natural number td and natural number tx and natural number Q and natural number B and natural number S and natural number s and natural number p, if* rhbandagree$(tx, Q, 1, B)$*, then if for every natural number u, if $u < S$, then* rdrv$(td, tx + Q, u, B + 2) =$ rdrv$(td, tx, u, B + 2)$*, then if $s \leq S$, then if $p \leq B + 1$, then* rdrv$(td, tx + Q, s, p) =$ rdrv(td, tx, s, p).

Proof. The goal follows from rdrvstripagreeofrhbandagreeandcolagree(td, tx, Q, B, S, s, p). □

Theorem 9.1066. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number S and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if* rhcolagree$(t, m \cdot P, B+2, S)$*, then if $s \leq S$, then* maskparity$(t+s+m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofrhbandagreeandrhcolagre
□

Theorem 9.1067. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z+1-m \cdot P, z+(p+1))$, *then if $p+3 \cdot (m \cdot P) \leq S+1$, then if for every natural number s, if $s \leq S$, then if* center$(z+(p+1)+s) =$ false, *then* rdrv$(z+(p+1), z+(p+1)+m \cdot P, s, 1) =$ rdrv$(z+(p+1), z+(p+1), s, 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrdrvoneagreeatzerodiagsteptwo(T, P, m, z, p, S).
□

Theorem 9.1068. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z + 1 - m \cdot P, z + (p + 1))$, *then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if* rhbandagree$(z + (p + 1), m \cdot P, 1, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreeoneatzerodiagsteptwo(T, P, m, z, p, S).
□

Theorem 9.1069. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number B and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z + 1, z + (p + 1))$, *then if* settledby$(z + 1 - m \cdot P, z + (p + 1))$, *then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if* rhbandagree$(z + (p + 1), m \cdot P, 1, B)$, *then if* rhcolagree$(z + (p + 1), m \cdot P, B + 2, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofstripagreeatzerodiagsteptwo(T, P, m, z, p, B, S).
□

Theorem 9.1070. *For every natural number T and natural number P and natural number m and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $m \cdot P \leq$ twopow(k), then if $T + m \cdot P \leq$ twopow(k), then if* twopow$(k) \leq S + 1$, *then if for every natural number s, if $s \leq S$, then if* center$($twopow$(k) - m \cdot P + s) =$ false, *then* rdrv$($twopow$(k) - m \cdot P,$ twopow$(k), s, 1) =$ rdrv$($twopow$(k) - m \cdot P,$ twopow$(k) - m \cdot P, s, 1)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrdrvoneagreesubtwopowge(T, P, m, k, S). □

Theorem 9.1071. *For every natural number T and natural number P and natural number m and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $m \cdot P \leq \mathrm{twopow}(k)$, then if $T + m \cdot P \leq \mathrm{twopow}(k)$, then if $\mathrm{twopow}(k) \leq S+1$, then if* rhbandagree($\mathrm{twopow}(k)-m\cdot P, m\cdot P, 1, S$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreeonesubtwopowge(T, P, m, k, S). □

Theorem 9.1072. *For every natural number td and natural number tx and natural number k and natural number p,* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{bxor}(\mathrm{rdrv}(td, tx, k+1, p+1), \mathrm{rdrv}(td, tx, k, p+1)$ or $\mathrm{rdrv}(td, tx, k, p+2))$.

Proof. The goal follows from rdrvinwardbxor(td, tx, k, p). □

Theorem 9.1073. *For every natural number td and natural number tx and natural number k and natural number p, if* $\mathrm{rdrv}(td, tx, k, p+1) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{not}(\mathrm{rdrv}(td, tx, k+1, p+1))$.

Proof. The goal follows from rdrvinwardofblackleft(td, tx, k, p). □

Theorem 9.1074. *For every natural number td and natural number tx and natural number k and natural number p, if* $\mathrm{rdrv}(td, tx, k, p+2) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{not}(\mathrm{rdrv}(td, tx, k+1, p+1))$.

Proof. The goal follows from rdrvinwardofblackright(td, tx, k, p). □

Theorem 9.1075. *For every natural number td and natural number tx and natural number k,* $\mathrm{rdrv}(td, tx, k+1, 1) = \mathrm{bxor}(\mathrm{center}(td+k), \mathrm{rdrv}(td, tx, k, 1)$ or $\mathrm{rdrv}(td, tx, k, 2))$.

Proof. The goal follows from rdrvsteponeeqcenterxoror(td, tx, k). □

Theorem 9.1076. *For every natural number td and natural number tx and natural number k, if* $\mathrm{rdrv}(td, tx, k, 1) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx, k+1, 1) = \mathrm{not}(\mathrm{center}(td + k))$.

Proof. The goal follows from rdrvsteponeeqnotcenterofblackone(td, tx, k). □

Theorem 9.1077. *For every natural number td and natural number $tx2$ and natural number tx and natural number k,* $\mathrm{bxor}(\mathrm{rdrv}(td, tx2, k+1, 1), \mathrm{rdrv}(td, tx, k+1, 1)) = \mathrm{bxor}(\mathrm{rdrv}(td, tx2, k, 1)$ or $\mathrm{rdrv}(td, tx2, k, 2), \mathrm{rdrv}(td, tx, k, 1)$ or $\mathrm{rdrv}(td, tx, k, 2))$.

Proof. The goal follows from rdrvsteponedefecteqordefect$(td, tx2, tx, k)$. □

Theorem 9.1078. *For every natural number td and natural number $tx2$ and natural number tx and natural number k, if* $\mathrm{rdrv}(td, tx2, k, 1) = \mathrm{true}$*, then if* $\mathrm{rdrv}(td, tx, k, 1) = \mathrm{true}$*, then* $\mathrm{rdrv}(td, tx2, k+1, 1) = \mathrm{rdrv}(td, tx, k+1, 1)$.

Proof. The goal follows from rdrvsteponeeqofblackone$(td, tx2, tx, k)$. □

Theorem 9.1079. *For every natural number td and natural number tx and natural number k and natural number p, if $tx+k < p$, then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{false}$.

Proof. The goal follows from rdrvgteqfalse(td, tx, k, p). □

Theorem 9.1080. *For every natural number td and natural number tx and natural number k, if $0 < tx$, then* $\mathrm{rdrv}(td, tx, k, tx + k) = \mathrm{true}$.

Proof. The goal follows from rdrvselfeqtrue(td, tx, k). □

Theorem 9.1081. *For every natural number td and natural number tx and natural number e and natural number k and natural number p and natural number v, if $k < p$, then if $p \leq tx + k$, then if $tx + k - p \leq e$, then* $\mathrm{rdrv}(td, tx + v \cdot \mathrm{twopow}(e), k, p + v \cdot \mathrm{twopow}(e)) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from rdrvdiagshifttwopoweq(td, tx, e, k, p, v). □

Theorem 9.1082. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if* $\mathrm{rdrv}(td, tx2, k+1, p+1) = \mathrm{rdrv}(td, tx, k+1, p+1)$*, then if* $(\mathrm{rdrv}(td, tx2, k, p+1) \text{ or } \mathrm{rdrv}(td, tx2, k, p+2)) = (\mathrm{rdrv}(td, tx, k, p+1) \text{ or } \mathrm{rdrv}(td, tx, k, p+2))$*, then* $\mathrm{rdrv}(td, tx2, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from rdrvinwardstepagree$(td, tx2, tx, k, p)$. □

Theorem 9.1083. *For every natural number td and natural number $tx2$ and natural number tx and natural number S and natural number p and natural number s, if for every natural number u, if $u \leq S+1$, then* $\mathrm{rdrv}(td, tx2, u, p+1) = \mathrm{rdrv}(td, tx, u, p+1)$*, then if for every natural number u, if $u \leq S$, then* $\mathrm{rdrv}(td, tx2, u, p+2) = \mathrm{rdrv}(td, tx, u, p+2)$*, then if $s \leq S$, then* $\mathrm{rdrv}(td, tx2, s, p) = \mathrm{rdrv}(td, tx, s, p)$.

Proof. The goal follows from rdrvinwardtwocolstepagree$(td, tx2, tx, S, p, s)$. □

Theorem 9.1084. *For every natural number td and natural number $tx2$ and natural number tx and natural number K and natural number $p0$ and natural number S and natural number s, if for every natural number u, if $u \leq S+K$, then* $\mathrm{rdrv}(td, tx2, u, p0+K) = \mathrm{rdrv}(td, tx, u, p0+K)$*, then if for every natural number u, if $u \leq S+K$, then* $\mathrm{rdrv}(td, tx2, u, p0+K+1) = \mathrm{rdrv}(td, tx, u, p0+K+1)$*, then if $s \leq S$, then* $\mathrm{rdrv}(td, tx2, s, p0) = \mathrm{rdrv}(td, tx, s, p0)$.

Proof. The goal follows from $\mathrm{rdrvinwardtwocolpropagate}(td, tx2, tx, K, p0, S, s)$. □

Theorem 9.1085. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number K and natural number S and natural number s, if $T \leq t$, then if for every natural number u, if $u \leq S + K$, then* $\mathrm{rdrv}(t, t + m \cdot P, u, 1 + K) = \mathrm{rdrv}(t, t, u, 1 + K)$*, then if for every natural number u, if $u \leq S+K$, then* $\mathrm{rdrv}(t, t+m \cdot P, u, 1+K+1) = \mathrm{rdrv}(t, t, u, 1+K+1)$*, then if $s \leq S$, then* $\mathrm{maskparity}(t+s+m \cdot P, 0) = \mathrm{maskparity}(t+s, 0)$.

Proof. Assume hP, *hper*. The goal follows from $\mathrm{periodicmaskparityzeroagreeofrdrvtwocolagree}(T, P, hP$ □

Theorem 9.1086. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number K and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(m \cdot P, z)$*, then if* $\mathrm{settledby}(z+1, z+(p+1))$*, then if* $\mathrm{settledby}(z+1-m \cdot P, z+(p+1))$*, then if $p+3 \cdot (m \cdot P) \leq S+1$, then if for every natural number u, if $u \leq S+K$, then* $\mathrm{rdrv}(z+(p+1), z+(p+1)+m \cdot P, u, 1+K) = \mathrm{rdrv}(z+(p+1), z+(p+1), u, 1+K)$*, then if for every natural number u, if $u \leq S+K$, then* $\mathrm{rdrv}(z+(p+1), z+(p+1)+m \cdot P, u, 1+K+1) = \mathrm{rdrv}(z+(p+1), z+(p+1), u, 1+K+1)$*, then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from $\mathrm{notcenterperiodicofrdrvtwocolagreeatzerodiag}(T, P, m, z, p, K, S)$. □

Theorem 9.1087. *For every natural number td and natural number tx and natural number e and natural number J and natural number k and natural number p and natural number v, if* $\mathrm{rightthreshold}(e, J)$*, then if $k < p$, then if $p \leq tx+k$, then if $tx+k-p \leq J$, then* $\mathrm{rdrv}(td, tx+v \cdot \mathrm{twopow}(e), k, p+v \cdot \mathrm{twopow}(e)) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from $\mathrm{rdrvdiagshifttwopoweqofthreshold}(td, tx, e, J, k, p, v)$. □

Theorem 9.1088. *For every natural number td and natural number e and natural number J and natural number u and natural number k and natural number j, if* $\mathrm{rightthreshold}(e, J)$*, then if* $1 \leq j$*, then if* $j \leq J$*, then if* $k+j < u \cdot \mathrm{twopow}(e)$*, then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(e) - k, k, u \cdot \mathrm{twopow}(e) - j) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{rdrvzerogapofthreshold}(td, e, J, u, k, j)$. □

Theorem 9.1089. *For every natural number td and natural number e and natural number J and natural number u and natural number k and natural number j, if* $\mathrm{rightthreshold}(e, J)$*, then if* $j \leq J$*, then if* $k + j < (u + 1) \cdot \mathrm{twopow}(e) - 1$*, then* $\mathrm{rdrv}(td, (u+1) \cdot \mathrm{twopow}(e) - 1 - k, k, (u+1) \cdot \mathrm{twopow}(e) - 1 - j) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{rdrvlastoneofthreshold}(td, e, J, u, k, j)$. □

Theorem 9.1090. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p,* $\mathrm{bxor}(\mathrm{rdrv}(td, tx2, k+1, p+1), \mathrm{rdrv}(td, tx, k+1, p+1)) = \mathrm{bxor}(\mathrm{bxor}(\mathrm{rdrv}(td, tx2, k, p), \mathrm{rdrv}(td, tx, k, p)), \mathrm{bxor}(\mathrm{rdrv}(td, tx2, k, p+1) \text{ or } \mathrm{rdrv}(td, tx2, k, p+2), \mathrm{rdrv}(td, tx, k, p+1) \text{ or } \mathrm{rdrv}(td, tx, k, p+2)))$.

Proof. The goal follows from $\mathrm{rdrvdefectstepsucc}(td, tx2, tx, k, p)$. □

Theorem 9.1091. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p,* $\mathrm{bxor}(\mathrm{rdrv}(td, tx2, k, p), \mathrm{rdrv}(td, tx, k, p))$ $\mathrm{bxor}(\mathrm{bxor}(\mathrm{rdrv}(td, tx2, k+1, p+1), \mathrm{rdrv}(td, tx, k+1, p+1)), \mathrm{bxor}(\mathrm{rdrv}(td, tx2, k, p+1) \text{ or } \mathrm{rdrv}(td, tx2, k, p+2), \mathrm{rdrv}(td, tx, k, p+1) \text{ or } \mathrm{rdrv}(td, tx, k, p+2)))$.

Proof. The goal follows from $\mathrm{rdrvdefectinwardstep}(td, tx2, tx, k, p)$. □

Theorem 9.1092. *For every natural number td and natural number $tx2$ and natural number tx and natural number k, if* $(\mathrm{rdrv}(td, tx2, k, 1) \text{ or } \mathrm{rdrv}(td, tx2, k, 2)) = (\mathrm{rdrv}(td, tx, k, 1) \text{ or } \mathrm{rdrv}(td, tx, k, 2))$*, then* $\mathrm{rdrv}(td, tx2, k+1, 1) = \mathrm{rdrv}(td, tx, k+1, 1)$.

Proof. The goal follows from $\mathrm{rdrvsteponeeqoforeq}(td, tx2, tx, k)$. □

Theorem 9.1093. *For every natural number td and natural number $tx2$ and natural number tx and natural number k, if* $\mathrm{rdrv}(td, tx2, k+1, 1) = \mathrm{rdrv}(td, tx, k+1, 1)$*, then* $(\mathrm{rdrv}(td, tx2, k, 1) \text{ or } \mathrm{rdrv}(td, tx2, k, 2)) = (\mathrm{rdrv}(td, tx, k, 1) \text{ or } \mathrm{rdrv}(td, tx, k, 2))$

Proof. The goal follows from $\mathrm{rdrvoroneeqofsteponeeq}(td, tx2, tx, k)$. □

Theorem 9.1094. *For every natural number td and natural number tx and natural number k and natural number p, if* $\mathrm{rdrv}(td, tx, k, p+1) = \mathrm{false}$*, then if* $\mathrm{rdrv}(td, tx, k, p+2) = \mathrm{false}$*, then* $\mathrm{rdrv}(td, tx, k+1, p+1) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from rdrvdiageqofwhitepair(td, tx, k, p). □

Theorem 9.1095. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if* (rdrv($td, tx2, k, p$+1) or rdrv($td, tx2, k, p$+2)) = (rdrv(td, tx, k, p+1) or rdrv(td, tx, k, p+2)), *then* bxor(rdrv($td, tx2, k$+1, p+1), rdrv(td, tx, k+1, p+1)) = bxor(rdrv($td, tx2, k, p$), rdrv(td, tx, k, p)

Proof. The goal follows from rdrvdefectdiageqoforeq($td, tx2, tx, k, p$). □

Theorem 9.1096. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if* rdrv($td, tx2, k, p + 1$) = true, *then if* rdrv($td, tx, k, p + 1$) = true, *then if* rdrv($td, tx2, k$+1, p+1) = rdrv(td, tx, k+1, p+1), *then* rdrv($td, tx2, k, p$) = rdrv(td, tx, k, p).

Proof. The goal follows from rdrvagreeinwardofblackleft($td, tx2, tx, k, p$). □

Theorem 9.1097. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number S and natural number s, if $T \leq t$, then if if* center(t) = false, *then* rdrv($t, t + m \cdot P, 0, 1$) = rdrv($t, t, 0, 1$), *then if for every natural number u, if $u < S$, then if* center(t+(u+1)) = false, *then* (rdrv(t, t+m·$P, u, 1$) or rdrv(t, t+m·$P, u, 2$)) = (rdrv($t, t, u, 1$) or rdrv($t, t, u, 2$)), *then if $s \leq S$, then* maskparity($t + s + m \cdot P, 0$) = maskparity($t + s, 0$).

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofrdrvoroneagree(T, P, hP, □

Theorem 9.1098. *For every natural number td and natural number $tx2$ and natural number tx and natural number K and natural number $p0$ and natural number S and natural number s, if for every natural number u, if $u \leq S$+K, then* rdrv($td, tx2, u, p0$+K) = rdrv($td, tx, u, p0$+K), *then if for every natural number u, if $u < S + K$, then* (rdrv($td, tx2, u, p0 + K$) or rdrv($td, tx2, u, p0$+K+1)) = (rdrv($td, tx, u, p0$+K) or rdrv($td, tx, u, p0$+K+1)), *then if $s \leq S$, then* rdrv($td, tx2, s, p0$) = rdrv($td, tx, s, p0$).

Proof. The goal follows from rdrvinwardcolorpropagate($td, tx2, tx, K, p0, S, s$). □

Theorem 9.1099. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number K and natural number S and natural number s, if $T \leq t$, then if for every natural number u, if $u \leq S + K$, then*

$\mathrm{rdrv}(t, t + m \cdot P, u, 1 + K) = \mathrm{rdrv}(t, t, u, 1 + K)$, *then if for every natural number* u, *if* $u < S + K$, *then* $(\mathrm{rdrv}(t, t + m \cdot P, u, 1 + K)$ or $\mathrm{rdrv}(t, t + m \cdot P, u, 1 + K + 1)) = (\mathrm{rdrv}(t, t, u, 1 + K)$ or $\mathrm{rdrv}(t, t, u, 1 + K + 1))$, *then if* $s \leq S$, *then* $\mathrm{maskparity}(t + s + m \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofrdrvcoloragree(T, P, hP, h

□

Theorem 9.1100. *For every natural number* T *and natural number* P *and natural number* m *and natural number* z *and natural number* p *and natural number* K *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq z+(p+1)$, *then if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(m \cdot P, z)$, *then if* $\mathrm{settledby}(z+1, z+(p+1))$, *then if* $\mathrm{settledby}(z+1-m \cdot P, z+(p+1))$, *then if* $p+3 \cdot (m \cdot P) \leq S+1$, *then if for every natural number* u, *if* $u \leq S + K$, *then* $\mathrm{rdrv}(z + (p + 1), z + (p+1) + m \cdot P, u, 1+K) = \mathrm{rdrv}(z + (p+1), z + (p+1), u, 1+K)$, *then if for every natural number* u, *if* $u < S + K$, *then* $(\mathrm{rdrv}(z + (p + 1), z + (p + 1) + m \cdot P, u, 1 + K)$ or $\mathrm{rdrv}(z + (p + 1), z + (p + 1) + m \cdot P, u, 1 + K + 1)) = (\mathrm{rdrv}(z+(p+1), z+(p+1), u, 1+K)$ or $\mathrm{rdrv}(z+(p+1), z+(p+1), u, 1+K+1))$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofrdrvcoloragreeatzerodiag(T, P, m, z, p, K, S).

□

Theorem 9.1101. *For every natural number* T *and natural number* P *and natural number* m *and natural number* k *and natural number* K *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $m \cdot P \leq \mathrm{twopow}(k)$, *then if* $T + m \cdot P \leq \mathrm{twopow}(k)$, *then if* $\mathrm{twopow}(k) \leq S+1$, *then if for every natural number* u, *if* $u \leq S+K$, *then* $\mathrm{rdrv}(\mathrm{twopow}(k) - m \cdot P, \mathrm{twopow}(k), u, 1+K) = \mathrm{rdrv}(\mathrm{twopow}(k) - m \cdot P, \mathrm{twopow}(k) - m \cdot P, u, 1+K)$, *then if for every natural number* u, *if* $u < S + K$, *then* $(\mathrm{rdrv}(\mathrm{twopow}(k) - m \cdot P, \mathrm{twopow}(k), u, 1 + K)$ or $\mathrm{rdrv}(\mathrm{twopow}(k) - m \cdot P, \mathrm{twopow}(k), u, 1+K+1)) = (\mathrm{rdrv}(\mathrm{twopow}(k) - m \cdot P, \mathrm{twopow}(k) - m \cdot P, u, 1 + K)$ or $\mathrm{rdrv}(\mathrm{twopow}(k) - m \cdot P, \mathrm{twopow}(k) - m \cdot P, u, 1 + K + 1))$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofrdrvcoloragreesubtwopowge(T, P, m, k, K, S).

□

Theorem 9.1102. *For every natural number* td *and natural number* p, *if* $0 < p$, *then* $\mathrm{rdrv}(td, 0, 0, p) = \mathrm{false}$.

Proof. The goal follows from rdrvwhiteseediniteqfalse(td, p). □

Theorem 9.1103. *For every natural number td and natural number k and natural number p, if $k < p$, then* $\mathrm{rdrv}(td, 0, k, p) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{rdrvwhiteseedgteqfalse}(td, k, p)$. □

Theorem 9.1104. *For every natural number td and natural number k,* $\mathrm{rdrv}(td, 0, k, k) = \mathrm{center}(td)$.

Proof. The goal follows from $\mathrm{rdrvwhiteseeddiageqcenter}(td, k)$. □

Theorem 9.1105. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number td and natural number k and natural number p, if $T \leq td$, then* $\mathrm{rdrv}(td + m \cdot P, 0, k, p) = \mathrm{rdrv}(td, 0, k, p)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicrdrvwhiteseedshifteq}(T, P, hP, hper)$. □

Theorem 9.1106. *For every natural number td and natural number tx and natural number L and natural number k and natural number p, if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(tx, 1 + i) = \mathrm{false}$*, then if $p + k \leq L + 1$, then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{rdrv}(td, 0, k, p)$.

Proof. The goal follows from $\mathrm{rdrveqwhiteseedofinitialwhiteband}(td, tx, L, k, p)$. □

Theorem 9.1107. *For every natural number td and natural number tx and natural number S and natural number s, if for every natural number i, if $i \leq S$, then* $\mathrm{rhcenter}(tx, 1 + i) = \mathrm{false}$*, then if $s \leq S$, then* $\mathrm{rdrv}(td, tx, s, 1) = \mathrm{rdrv}(td, 0, s, 1)$.

Proof. The goal follows from $\mathrm{rdrvcoloneeqwhiteseedofinitialwhiteband}(td, tx, S, s)$. □

Theorem 9.1108. *For every natural number td and natural number tx and natural number k, if* $\mathrm{center}(td + k) = \mathrm{false}$*, then if* $\mathrm{rdrv}(td, tx, k, 1) = \mathrm{true}$*, then* $\mathrm{rdrv}(td, tx, k + 1, 1) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{rdrvsteponeeqtrueofwhitecenterandblackone}(td, tx, k)$. □

Theorem 9.1109. *For every natural number td and natural number tx and natural number k and natural number W and natural number j, if* $\mathrm{rdrv}(td, tx, k, 1) = \mathrm{true}$*, then if for every natural number u, if $u < W$, then* $\mathrm{center}(td + (k + u)) = \mathrm{false}$*, then if $j \leq W$, then* $\mathrm{rdrv}(td, tx, k + j, 1) = \mathrm{true}$.

Proof. The goal follows from rdrvwhitecenterrunblackonepersist(td, tx, k, W, j). □

Theorem 9.1110. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number W and natural number j, if* rdrv$(td, tx2, k, 1) =$ true*, then if* rdrv$(td, tx, k, 1) =$ true*, then if for every natural number u, if $u < W$, then* center$(td + (k + u)) =$ false*, then if $j \leq W + 1$, then* rdrv$(td, tx2, k + j, 1) =$ rdrv$(td, tx, k + j, 1)$.

Proof. The goal follows from rdrvwhitecenterrunblackoneagree$(td, tx2, tx, k, W, j)$. □

Theorem 9.1111. *For every natural number t, if* maskparity$(t, 0) =$ true*, then* rdrv$(t, t, 1, 1) =$ true.

Proof. The goal follows from rdrvsteponeeqtrueofmaskparityzerotrue(t). □

Theorem 9.1112. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t, if $T \leq t$, then if* maskparity$(t + m \cdot P, 0) =$ true*, then* rdrv$(t, t + m \cdot P, 1, 1) =$ true.

Proof. Assume hP, *hper*. The goal follows from periodicrdrvsteponeeqtrueofmaskparityzerotrue$(T, P, h$. □

Theorem 9.1113. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number W and natural number s, if $T \leq t$, then if* maskparity$(t + m \cdot P, 0) =$ true*, then if* maskparity$(t, 0) =$ true*, then if for every natural number u, if $u < W$, then* center$(t + (1 + u)) =$ false*, then if $s \leq W + 2$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeonwhiterunofmaskparityze□

Theorem 9.1114. *For every natural number td and natural number e and natural number J and natural number u and natural number k and natural number j, if* rightthreshold(e, J)*, then if $1 \leq j$, then if $j + 1 \leq J$, then if $k + (j + 1) < u \cdot$ twopow(e), then* rdrv$(td, u \cdot$ twopow$(e) - k, k, u \cdot$ twopow$(e) - (j + 1)) =$ rdrv$(td, 0, k, u \cdot$ twopow$(e) - (j + 1))$ *and* rdrv$(td, u \cdot$ twopow$(e) - k, k, u \cdot$ twopow$(e) - j) =$ rdrv$(td, 0, k, u \cdot$ twopow$(e) - j)$.

Proof. The goal follows from rdrvtwocoleqwhiteseedofzerogap(td, e, J, u, k, j). □

Theorem 9.1115. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number s, if $T \leq t$, then if if* center$(t + s) =$ false*, then* rdrv$(t, t + m \cdot P, s, 1) =$ rdrv$(t, 0, s, 1)$ *and* rdrv$(t, t, s, 1) =$ rdrv$(t, 0, s, 1)$*, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofbotheqwhiteseed$(T, P, h1$
□

Theorem 9.1116. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if $p+3 \cdot (m \cdot P) \leq S+1$, then if for every natural number s, if $s \leq S$, then if* center$(z+(p+1)+s) =$ false*, then* rdrv$(z + (p + 1), z + (p + 1) + m \cdot P, s, 1) =$ rdrv$(z + (p + 1), 0, s, 1)$ *and* rdrv$(z + (p + 1), z + (p + 1), s, 1) =$ rdrv$(z + (p + 1), 0, s, 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofbotheqwhiteseedatzerodiag(T, P, m, z, p, S).
□

Theorem 9.1117. *For every natural number td and natural number k,* rdrv$(td, 0, k + 2, k + 1) =$ bxor(rdrv$(td, 0, k + 1, k)$, center(td)).

Proof. The goal follows from rdrvwhiteseedsubdiagonestep(td, k). □

Theorem 9.1118. *For every natural number td and natural number k,* rdrv$(td, 0, k + 3, k + 2) =$ rdrv$(td, 0, k + 1, k)$.

Proof. The goal follows from rdrvwhiteseedsubdiagoneperiodtwo(td, k). □

Theorem 9.1119. *For every natural number td and natural number u,* rdrv$(td, 0, 2 \cdot u + 1, 2 \cdot u) =$ center$(td + 1)$.

Proof. The goal follows from rdrvwhiteseedsubdiagoneeven(td, u). □

Theorem 9.1120. *For every natural number td and natural number u,* rdrv$(td, 0, 2 \cdot u + 2, 2 \cdot u + 1) =$ bxor(center$(td + 1)$, center(td)).

Proof. The goal follows from rdrvwhiteseedsubdiagoneodd(td, u). □

Theorem 9.1121. *For every natural number td and natural number d and natural number k and natural number u,* rdrv$(td, 0, k+d+u \cdot$ twopow$(d), k+u \cdot$ twopow$(d)) =$ rdrv$(td, 0, k + d, k)$.

Proof. The goal follows from rdrvwhiteseedsubdiagperiodtwopow(td, d, k, u). □

Theorem 9.1122. *For every natural number td and natural number d and natural number k and natural number p and natural number u, if $p \leq k$, then if $k - p \leq d$, then* $\mathrm{rdrv}(td, 0, k + u \cdot \mathrm{twopow}(d), p + u \cdot \mathrm{twopow}(d)) = \mathrm{rdrv}(td, 0, k, p)$.

Proof. The goal follows from rdrvwhiteseeddiagshifttwopoweq(td, d, k, p, u). □

Theorem 9.1123. *For every natural number td and natural number d and natural number u,* $\mathrm{rdrv}(td, 0, d+u\cdot\mathrm{twopow}(d), u\cdot\mathrm{twopow}(d)) = \mathrm{center}(td+d)$.

Proof. The goal follows from rdrvwhiteseedscalemulteqcenter(td, d, u). □

Theorem 9.1124. *For every natural number td and natural number d,* $\mathrm{rdrv}(td, 0, d + \mathrm{twopow}(d), \mathrm{twopow}(d)) = \mathrm{center}(td + d)$.

Proof. The goal follows from rdrvwhiteseedscaleeqcenter(td, d). □

Theorem 9.1125. *For every natural number td and natural number d and natural number k, if $d \leq k$, then* $\mathrm{rdrv}(td, 0, d + \mathrm{twopow}(k), \mathrm{twopow}(k)) = \mathrm{center}(td + d)$.

Proof. The goal follows from rdrvwhiteseedscalegeeqcenter(td, d, k). □

Theorem 9.1126. *For every natural number td and natural number d and natural number k, if $d \leq k$, then* $\mathrm{rdrv}(td, 0, d + \mathrm{twopow}(k + 1), \mathrm{twopow}(k + 1)) = \mathrm{rdrv}(td, 0, d + \mathrm{twopow}(k), \mathrm{twopow}(k))$.

Proof. The goal follows from rdrvwhiteseedscaledoubleeqscale(td, d, k). □

Theorem 9.1127. *For every natural number td and natural number tx and natural number L and natural number d and natural number k, if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(tx, 1+i) = \mathrm{false}$*, then if $d \leq k$, then if $d + 2 \cdot \mathrm{twopow}(k) \leq L + 1$, then* $\mathrm{rdrv}(td, tx, d + \mathrm{twopow}(k), \mathrm{twopow}(k)) = \mathrm{center}(td + d)$.

Proof. The goal follows from rdrvscaleeqcenterofinitialwhiteband(td, tx, L, d, k). □

Theorem 9.1128. *For every natural number t and natural number L and natural number d and natural number k, if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(t, 1+i) = \mathrm{false}$*, then if $d \leq k$, then if $d+2 \cdot \mathrm{twopow}(k) \leq L+1$, then* $\mathrm{rhcenter}(t + (d + \mathrm{twopow}(k)), \mathrm{twopow}(k)) = \mathrm{center}(t+d)$.

Proof. The goal follows from $\mathrm{rhcenterscaleeqcenterofinitialwhiteband}(t, L, d, k)$. □

Theorem 9.1129. *For every natural number td and natural number tx and natural number L and natural number d and natural number k and natural number u, if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(tx, 1+i) = \mathrm{false}$*, then if $k + u \cdot \mathrm{twopow}(d) + (k + d + u \cdot \mathrm{twopow}(d)) \leq L+1$, then* $\mathrm{rdrv}(td, tx, k+d+u \cdot \mathrm{twopow}(d), k+u \cdot \mathrm{twopow}(d)) = \mathrm{rdrv}(td, tx, k+d, k)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagperiodtwopowofinitialwhiteband}(td, tx, L, d, k, u)$. □

Theorem 9.1130. *For every natural number td and natural number $tx2$ and natural number tx and natural number L and natural number k and natural number p, if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(tx2, 1+i) = \mathrm{false}$*, then if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(tx, 1+i) = \mathrm{false}$*, then if $p + k \leq L+1$, then* $\mathrm{rdrv}(td, tx2, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from $\mathrm{rdrveqofbothinitialwhiteband}(td, tx2, tx, L, k, p)$. □

Theorem 9.1131. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number L and natural number k and natural number p, if $T \leq t$, then if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(t + m \cdot P, 1+i) = \mathrm{false}$*, then if for every natural number i, if $i \leq L$, then* $\mathrm{rhcenter}(t, 1+i) = \mathrm{false}$*, then if $p + k \leq L+1$, then* $\mathrm{rhcenter}(t + m \cdot P + k, p) = \mathrm{rhcenter}(t+k, p)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicrhcentershifteqofbothinitialwhiteband}(T, P, hP,$ □

Theorem 9.1132. *For every natural number td and natural number tx and natural number k,* $\mathrm{rdrv}(td, tx, k+1, k+1) = \mathrm{bxor}(\mathrm{rdrv}(td, tx, k, k), \mathrm{rhcenter}(tx + k, k+1) \operatorname{or} \mathrm{rhcenter}(tx+k, k+2))$.

Proof. The goal follows from $\mathrm{rdrvmaindiagsteprhcenter}(td, tx, k)$. □

Theorem 9.1133. *For every natural number $td2$ and natural number td and natural number tx and natural number k,* $\mathrm{bxor}(\mathrm{rdrv}(td2, tx, k, k), \mathrm{rdrv}(td, tx, k, k)) = \mathrm{bxor}(\mathrm{center}(td2), \mathrm{center}(td))$.

Proof. The goal follows from $\mathrm{rdrvmaindiagdrivedefectconst}(td2, td, tx, k)$. □

Theorem 9.1134. *For every natural number td and natural number tx and natural number k,* $\mathrm{rdrv}(td, tx, k, k) = \mathrm{bxor}(\mathrm{rhcenter}(tx+k, k), \mathrm{bxor}(\mathrm{center}(td), \mathrm{center}(tx)))$.

Proof. The goal follows from $\mathrm{rdrvmaindiageqrhcenterxorcenters}(td, tx, k)$. □

Theorem 9.1135. *For every natural number td and natural number tx and natural number k, if* $\mathrm{center}(td) = \mathrm{center}(tx)$, *then* $\mathrm{rdrv}(td, tx, k, k) = \mathrm{rhcenter}(tx + k, k)$.

Proof. The goal follows from $\mathrm{rdrvmaindiageqrhcenterofcentereq}(td, tx, k)$. □

Theorem 9.1136. *For every natural number $td2$ and natural number td and natural number tx and natural number d and natural number p, if for every natural number j, if $j < d$, then* $\mathrm{center}(td2 + j) = \mathrm{center}(td + j)$, *then* $\mathrm{bxor}(\mathrm{rdrv}(td2, tx, p + d, p), \mathrm{rdrv}(td, tx, p + d, p)) = \mathrm{bxor}(\mathrm{center}(td2 + d), \mathrm{center}(td + d))$.

Proof. The goal follows from $\mathrm{rdrvdrivewavefrontdefectconst}(td2, td, tx, d, p)$. □

Theorem 9.1137. *For every natural number td and natural number tx and natural number d and natural number p, if for every natural number j, if $j < d$, then* $\mathrm{center}(td + j) = \mathrm{center}(tx + j)$, *then* $\mathrm{rdrv}(td, tx, p + d, p) = \mathrm{bxor}(\mathrm{rhcenter}(tx + (p + d), p), \mathrm{bxor}(\mathrm{center}(td + d), \mathrm{center}(tx + d)))$.

Proof. The goal follows from $\mathrm{rdrvdrivewavefronteqrhcenterxor}(td, tx, d, p)$. □

Theorem 9.1138. *For every natural number $td2$ and natural number td and natural number tx and natural number d and natural number p, if for every natural number j, if $j < d$, then* $\mathrm{center}(td2 + j) = \mathrm{center}(td + j)$, *then if* $\mathrm{center}(td2 + d) = \mathrm{not}(\mathrm{center}(td + d))$, *then* $\mathrm{rdrv}(td2, tx, p + d, p) = \mathrm{not}(\mathrm{rdrv}(td, tx, p + d, p))$.

Proof. The goal follows from $\mathrm{rdrvdrivewavefrontflipoffirstcenterdiff}(td2, td, tx, d, p)$. □

Theorem 9.1139. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number d, if* $\mathrm{rdrv}(td, tx2, k, d) = \mathrm{rdrv}(td, tx, k, d)$, *then if* $\mathrm{rdrv}(td, tx2, k, d+1) = \mathrm{rdrv}(td, tx, k, d+1)$, *then if* $\mathrm{rdrv}(td, tx, k, d+1) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx2, k+1, d+1) = \mathrm{rdrv}(td, tx, k+1, d+1)$.

Proof. The goal follows from $\mathrm{rdrvinwardagreestepofblackshield}(td, tx2, tx, k, d)$. □

Theorem 9.1140. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number d, if* $\mathrm{rdrv}(td, tx2, k, d) = \mathrm{rdrv}(td, tx, k, d)$, *then if* $\mathrm{rdrv}(td, tx2, k, d+1) = \mathrm{rdrv}(td, tx, k, d+1)$, *then if* $\mathrm{rdrv}(td, tx, k, d+1) = \mathrm{false}$, *then* $\mathrm{bxor}(\mathrm{rdrv}(td, tx2, k+1, d+1), \mathrm{rdrv}(td, tx, k+1, d+1)) = \mathrm{bxor}(\mathrm{rdrv}(td, tx2, k, d+2), \mathrm{rdrv}(td, tx, k, d+2))$.

Proof. The goal follows from $\mathrm{rdrvinwarddefectstepofwhiteshield}(td, tx2, tx, k, d)$. □

Theorem 9.1141. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number d, if* $\mathrm{rdrv}(td, tx2, k, d) = \mathrm{rdrv}(td, tx, k, d)$, *then if* $\mathrm{rdrv}(td, tx2, k, d+1) = \mathrm{rdrv}(td, tx, k, d+1)$, *then if* $\mathrm{rdrv}(td, tx2, k, d+2) = \mathrm{rdrv}(td, tx, k, d+2)$, *then* $\mathrm{rdrv}(td, tx2, k+1, d+1) = \mathrm{rdrv}(td, tx, k+1, d+1)$.

Proof. The goal follows from $\mathrm{rdrvinwardagreestepofthreeagree}(td, tx2, tx, k, d)$. □

Theorem 9.1142. *For every natural number td and natural number tx and natural number Q and natural number L and natural number d and natural number k, if* $\mathrm{rhbandagree}(tx, Q, 1, L)$, *then if* $d + k \leq L$, *then if* $\mathrm{rdrv}(td, tx, k, d+1) = \mathrm{false}$, *then* $\mathrm{bxor}(\mathrm{rdrv}(td, tx+Q, k+1, d+1), \mathrm{rdrv}(td, tx, k+1, d+1)) = \mathrm{bxor}(\mathrm{rdrv}(td, tx+Q, k, d+2), \mathrm{rdrv}(td, tx, k, d+2))$.

Proof. The goal follows from $\mathrm{rdrvinitwavefrontdefectofwhite}(td, tx, Q, L, d, k)$. □

Theorem 9.1143. *For every natural number td and natural number tx and natural number Q and natural number L and natural number d and natural number k, if* $\mathrm{rhbandagree}(tx, Q, 1, L)$, *then if* $d + k \leq L$, *then if* $\mathrm{rdrv}(td, tx, k, d+1) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx+Q, k+1, d+1) = \mathrm{rdrv}(td, tx, k+1, d+1)$.

Proof. The goal follows from $\mathrm{rdrvinitwavefrontabsorbofblack}(td, tx, Q, L, d, k)$. □

Theorem 9.1144. *For every natural number td and natural number tx and natural number Q and natural number L and natural number d and natural number k and natural number j, if* $\mathrm{rhbandagree}(tx, Q, 1, L)$*, then if* $d + (k + j) \leq L + 1$*, then if* $\mathrm{rdrv}(td, tx + Q, k, d + 1 + j) = \mathrm{rdrv}(td, tx, k, d + 1 + j)$*, then* $\mathrm{rdrv}(td, tx + Q, k + j, d + 1) = \mathrm{rdrv}(td, tx, k + j, d + 1)$.

Proof. The goal follows from $\mathrm{rdrvinitwavefrontagreepropagate}(td, tx, Q, L, d, k, j)$. □

Theorem 9.1145. *For every natural number td and natural number tx and natural number Q and natural number L and natural number k, if* $\mathrm{rhbandagree}(tx, Q, 1, L)$*, then if* $k \leq L$*, then if* $\mathrm{rdrv}(td, tx, k, L + 1 - k) = \mathrm{true}$*, then* $\mathrm{rdrv}(td, tx + Q, L + 1, 1) = \mathrm{rdrv}(td, tx, L + 1, 1)$.

Proof. The goal follows from $\mathrm{rdrvinitwavefrontcoloneagreeofblackshield}(td, tx, Q, L, k)$. □

Theorem 9.1146. *For every natural number T and natural number P, if* $0 < P$*, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number L and natural number k and natural number s, if* $T \leq t$*, then if* $\mathrm{rhbandagree}(t, m \cdot P, 1, L)$*, then if* $k \leq L$*, then if* $\mathrm{rdrv}(t, t, k, L + 1 - k) = \mathrm{true}$*, then if* $s \leq L + 1$*, then* $\mathrm{maskparity}(t + s + m \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofrhbandagreeandblackshie □

Theorem 9.1147. *For every natural number td and natural number tx and natural number Q and natural number B and natural number S and natural number s and natural number p, if* $\mathrm{rhbandagree}(tx, Q, 1, B)$*, then if for every natural number u, if $u < S$, then if* $\mathrm{rdrv}(td, tx, u, B + 1) = \mathrm{false}$*, then* $\mathrm{rdrv}(td, tx + Q, u, B + 2) = \mathrm{rdrv}(td, tx, u, B + 2)$*, then if $s \leq S$, then if $p \leq B + 1$, then* $\mathrm{rdrv}(td, tx + Q, s, p) = \mathrm{rdrv}(td, tx, s, p)$.

Proof. The goal follows from $\mathrm{rdrvstripagreeofwhiteboundarycolagree}(td, tx, Q, B, S, s, p)$. □

Theorem 9.1148. *For every natural number td and natural number tx and natural number Q and natural number B and natural number S and natural number s and natural number p, if* $\mathrm{rhbandagree}(tx, Q, 1, B)$*, then if for every natural number u, if $u < S$, then* $\mathrm{rdrv}(td, tx, u, B + 1) = \mathrm{true}$*, then if $s \leq S$, then if $p \leq B + 1$, then* $\mathrm{rdrv}(td, tx + Q, s, p) = \mathrm{rdrv}(td, tx, s, p)$.

Proof. The goal follows from rdrvstripagreeofblackboundarycol(td, tx, Q, B, S, s, p). □

Theorem 9.1149. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number S and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if for every natural number u, if $u < S$, then* rdrv$(t, t, u, B + 1) =$ true*, then if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofblackboundarycol$(T, P, h$ □

Theorem 9.1150. *For every natural number td and natural number tx and natural number k, if* rdrv$(td, tx, k + 1, k + 1) =$ true*, then* rdrv$(td, tx, k + 2, k + 1) =$ not(rdrv$(td, tx, k + 1, k)$).

Proof. The goal follows from rdrvsubdiagonestepofblackmaindiag(td, tx, k). □

Theorem 9.1151. *For every natural number td and natural number tx and natural number k, if* rdrv$(td, tx, k + 1, k + 1) =$ false*, then* rdrv$(td, tx, k + 2, k + 1) =$ bxor(rdrv$(td, tx, k + 1, k)$, rhcenter$(tx + (k + 1), k + 2)$).

Proof. The goal follows from rdrvsubdiagonestepofwhitemaindiag(td, tx, k). □

Theorem 9.1152. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if for every natural number u, if $1 \leq u$, then if $u \leq k$, then* rdrv$(td, tx2, u, u) =$ rdrv(td, tx, u, u)*, then if for every natural number u, if $1 \leq u$, then if $u \leq k$, then if* rdrv$(td, tx, u, u) =$ false*, then* rdrv$(td, tx2, u, u+1) =$ rdrv$(td, tx, u, u+1)$*, then if $p \leq k$, then* rdrv$(td, tx2, k, p) =$ rdrv(td, tx, k, p).

Proof. The goal follows from rdrvdriveconeagreeofmaindiaggate$(td, tx2, tx, k, p)$. □

Theorem 9.1153. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if for every natural number u, if $1 \leq u$, then if $u \leq k$, then* rdrv$(td, tx2, u, u) =$ true*, then if for every natural number u, if $1 \leq u$, then if $u \leq k$, then* rdrv$(td, tx, u, u) =$ true*, then if $p \leq k$, then* rdrv$(td, tx2, k, p) =$ rdrv(td, tx, k, p).

Proof. The goal follows from rdrvdriveconeagreeofblackmaindiag$(td, tx2, tx, k, p)$. □

Theorem 9.1154. *For every natural number td and natural number tx and natural number k and natural number p, if* center$(td) =$ true*, then if for every natural number u, if $1 \leq u$, then if $u \leq k$, then* rdrv$(td, tx, u, u) =$ true*, then if $p \leq k$, then* rdrv$(td, tx, k, p) =$ rdrv$(td, 0, k, p)$.

Proof. The goal follows from rdrvdriveconeeqwhiteseedofblackmaindiag(td, tx, k, p). □

Theorem 9.1155. *For every natural number td and natural number tx and natural number k and natural number p, if* center$(td) =$ true*, then if* center$(tx) =$ true*, then if for every natural number u, if $1 \leq u$, then if $u \leq k$, then* rhcenter$(tx+u, u) =$ true*, then if $p \leq k$, then* rdrv$(td, tx, k, p) =$ rdrv$(td, 0, k, p)$.

Proof. The goal follows from rdrvdriveconeeqwhiteseedofblackcenterandrhcenterdiag(td, tx, k, p). □

Theorem 9.1156. *For every natural number t and natural number k and natural number p, if* center$(t) =$ true*, then if for every natural number u, if $1 \leq u$, then if $u \leq k$, then* rhcenter$(t + u, u) =$ true*, then if $p \leq k$, then* rhcenter$(t + k, p) =$ rdrv$(t, 0, k, p)$.

Proof. The goal follows from rhcenterdriveconeeqwhiteseedofblackdiag(t, k, p). □

Theorem 9.1157. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number s, if $T \leq t$, then if $1 \leq s$, then if for every natural number u, if $1 \leq u$, then if $u \leq s$, then* rdrv$(t, t+m \cdot P, u, u) =$ rdrv(t, t, u, u)*, then if for every natural number u, if $1 \leq u$, then if $u \leq s$, then if* rdrv$(t, t, u, u) =$ false*, then* rdrv$(t, t+m \cdot P, u, u+1) =$ rdrv$(t, t, u, u+1)$*, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofmaindiaggate$(T, P, hP, h$ □

Theorem 9.1158. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number s, if $T \leq t$, then if $1 \leq s$, then if for every natural number u, if $1 \leq u$, then if $u \leq s$, then* rdrv$(t, t+m \cdot P, u, u) =$

true, *then if for every natural number* u, *if* $1 \leq u$, *then if* $u \leq s$, *then* $\mathrm{rdrv}(t, t, u, u) = \mathrm{true}$, *then* $\mathrm{maskparity}(t+s+m \cdot P, 0) = \mathrm{maskparity}(t+s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofblackmaindiag($T, P, hP,$ [illegible]
□

Theorem 9.1159. *For every natural number* T *and natural number* P *and natural number* m *and natural number* z *and natural number* p *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq z+p$, *then if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(m \cdot P, z)$, *then if* $\mathrm{settledby}(z+1, z+(p+1))$, *then if* $\mathrm{settledby}(z+1-m \cdot P, z+(p+1))$, *then if* $p + 3 \cdot (m \cdot P) \leq S + 1$, *then if for every natural number* u, *if* $1 \leq u$, *then if* $u \leq S+1$, *then* $\mathrm{rdrv}(z+p, z+p+m \cdot P, u, u) = \mathrm{rdrv}(z+p, z+p, u, u)$, *then if for every natural number* u, *if* $1 \leq u$, *then if* $u \leq S+1$, *then if* $\mathrm{rdrv}(z+p, z+p, u, u) = \mathrm{false}$, *then* $\mathrm{rdrv}(z+p, z+p+m \cdot P, u, u+1) = \mathrm{rdrv}(z+p, z+p, u, u+1)$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofmaindiaggateatzerodiag(T, P, m, z, p, S).
□

Theorem 9.1160. *For every natural number* T *and natural number* P *and natural number* m *and natural number* z *and natural number* p *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq z+p$, *then if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(m \cdot P, z)$, *then if* $\mathrm{settledby}(z+1, z+(p+1))$, *then if* $\mathrm{settledby}(z+1-m \cdot P, z+(p+1))$, *then if* $p + 3 \cdot (m \cdot P) \leq S + 1$, *then if for every natural number* u, *if* $1 \leq u$, *then if* $u \leq S+1$, *then* $\mathrm{rdrv}(z+p, z+p+m \cdot P, u, u) = \mathrm{true}$, *then if for every natural number* u, *if* $1 \leq u$, *then if* $u \leq S+1$, *then* $\mathrm{rdrv}(z+p, z+p, u, u) = \mathrm{true}$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofblackmaindiagatzerodiag(T, P, m, z, p, S).
□

Theorem 9.1161. *For every natural number* T *and natural number* P *and natural number* m *and natural number* z *and natural number* p *and natural number* B *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq z+(p+1)$, *then if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(m \cdot P, z)$, *then if* $\mathrm{settledby}(z+1, z+(p+1))$, *then if* $\mathrm{settledby}(z+1-m \cdot P, z+(p+1))$, *then if* $p+3 \cdot (m \cdot P) \leq S+1$, *then if* $\mathrm{rhbandagree}(z+(p+1), m \cdot P, 1, B)$, *then if for every natural number* u, *if* $u < S$, *then* $\mathrm{rdrv}(z+(p+1), z+(p+1), u, B+1) = \mathrm{true}$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofblackboundarycolatzerodiag(T, P, m, z, p, B, S). □

Theorem 9.1162. *For every natural number td and natural number $tx2$ and natural number tx and natural number k, if for every natural number u, if $u < k$, then* (rhcenter$(tx2 + u, u + 1)$ or rhcenter$(tx2 + u, u + 2)$) = (rhcenter$(tx + u, u + 1)$ or rhcenter$(tx + u, u + 2)$)*, then* rdrv$(td, tx2, k, k)$ = rdrv(td, tx, k, k).

Proof. The goal follows from rdrvmaindiagagreeofcorridororagree$(td, tx2, tx, k)$. □

Theorem 9.1163. *For every natural number td and natural number tx and natural number k, if for every natural number u, if $u < k$, then* rhcenter$(tx + u, u + 1)$ = false *and* rhcenter$(tx + u, u + 2)$ = false*, then* rdrv(td, tx, k, k) = center(td).

Proof. The goal follows from rdrvmaindiageqcenterofwhitecorridor(td, tx, k). □

Theorem 9.1164. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if for every natural number u, if $u \leq k$, then* rhcenter$(tx2 + u, u + 1)$ = rhcenter$(tx + u, u + 1)$*, then if for every natural number u, if $u \leq k$, then if* rhcenter$(tx + u, u + 1)$ = false*, then* rhcenter$(tx2 + u, u + 2)$ = rhcenter$(tx + u, u + 2)$*, then if $p \leq k$, then* rdrv$(td, tx2, k, p)$ = rdrv(td, tx, k, p).

Proof. The goal follows from rdrvdriveconeagreeofcorridorgate$(td, tx2, tx, k, p)$. □

Theorem 9.1165. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if for every natural number u, if $u \leq k$, then* rhcenter$(tx2 + u, u + 1)$ = true*, then if for every natural number u, if $u \leq k$, then* rhcenter$(tx + u, u + 1)$ = true*, then if $p \leq k$, then* rdrv$(td, tx2, k, p)$ = rdrv(td, tx, k, p).

Proof. The goal follows from rdrvdriveconeagreeofblackfirstsuperdiag$(td, tx2, tx, k, p)$. □

Theorem 9.1166. *For every natural number td and natural number tx and natural number k and natural number p, if for every natural number u, if $u \leq k$, then* rhcenter$(tx + u, u + 1)$ = false *and* rhcenter$(tx + u, u + 2)$ = false*, then if $p \leq k$, then* rdrv(td, tx, k, p) = rdrv$(td, 0, k, p)$.

Proof. The goal follows from rdrvdriveconeeqwhiteseedofwhitecorridor(td, tx, k, p). □

Theorem 9.1167. *For every natural number t and natural number k and natural number p, if for every natural number u, if $u \leq k$, then* rhcenter$(t + u, u + 1) =$ false *and* rhcenter$(t + u, u + 2) =$ false*, then if $p \leq k$, then* rhcenter$(t + k, p) =$ rdrv$(t, 0, k, p)$.

Proof. The goal follows from rhcenterdriveconeeqwhiteseedofwhitecorridor(t, k, p). □

Theorem 9.1168. *For every natural number td and natural number tx and natural number d and natural number k, if* center$(td) =$ true*, then if $d \leq k$, then if for every natural number u, if $1 \leq u$, then if $u \leq d +$* twopow(k)*, then* rdrv$(td, tx, u, u) =$ true*, then* rdrv$(td, tx, d +$ twopow$(k),$ twopow$(k)) =$ center$(td + d)$.

Proof. The goal follows from rdrvscaleeqcenterofblackmaindiag(td, tx, d, k). □

Theorem 9.1169. *For every natural number t and natural number d and natural number k, if* center$(t) =$ true*, then if $d \leq k$, then if for every natural number u, if $1 \leq u$, then if $u \leq d+$*twopow(k)*, then* rhcenter$(t+u, u) =$ true*, then* rhcenter$(t + (d +$ twopow$(k)),$ twopow$(k)) =$ center$(t + d)$.

Proof. The goal follows from rhcenterscaleeqcenterofblackdiag(t, d, k). □

Theorem 9.1170. *For every natural number td and natural number tx and natural number d and natural number k, if $d \leq k$, then if for every natural number u, if $u \leq d +$* twopow(k)*, then* rhcenter$(tx + u, u + 1) =$ false *and* rhcenter$(tx+u, u+2) =$ false*, then* rdrv$(td, tx, d+$twopow$(k),$ twopow$(k)) =$ center$(td + d)$.

Proof. The goal follows from rdrvscaleeqcenterofwhitecorridor(td, tx, d, k). □

Theorem 9.1171. *For every natural number t and natural number d and natural number k, if $d \leq k$, then if for every natural number u, if $u \leq d+$*twopow(k)*, then* rhcenter$(t+u, u+1) =$ false *and* rhcenter$(t+u, u+2) =$ false*, then* rhcenter$(t + (d +$ twopow$(k)),$ twopow$(k)) =$ center$(t + d)$.

Proof. The goal follows from rhcenterscaleeqcenterofwhitecorridor(t, d, k). □

Theorem 9.1172. *For every natural number td and natural number tx and natural number d and natural number k and natural number u, if* $\mathrm{center}(td) = \mathrm{true}$*, then if for every natural number j, if $1 \leq j$, then if $j \leq k + d + u \cdot \mathrm{twopow}(d)$, then* $\mathrm{rdrv}(td, tx, j, j) = \mathrm{true}$*, then* $\mathrm{rdrv}(td, tx, k + d + u \cdot \mathrm{twopow}(d), k + u \cdot \mathrm{twopow}(d)) = \mathrm{rdrv}(td, tx, k + d, k)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagperiodtwopowofblackmaindiag}(td, tx, d, k, u)$.
□

Theorem 9.1173. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number s, if $T \leq t$, then if $1 \leq s$, then if for every natural number u, if $u \leq s$, then* $\mathrm{rhcenter}(t + m \cdot P + u, u + 1) = \mathrm{rhcenter}(t + u, u + 1)$*, then if for every natural number u, if $u \leq s$, then if* $\mathrm{rhcenter}(t + u, u + 1) = \mathrm{false}$*, then* $\mathrm{rhcenter}(t + m \cdot P + u, u + 2) = \mathrm{rhcenter}(t+u, u+2)$*, then* $\mathrm{maskparity}(t+s+m \cdot P, 0) = \mathrm{maskparity}(t+s, 0)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicmaskparityzeroagreeofcorridorgate}(T, P, hP, hp$
□

Theorem 9.1174. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number s, if $T \leq t$, then if $1 \leq s$, then if for every natural number u, if $u \leq s$, then* $\mathrm{rhcenter}(t + m \cdot P + u, u + 1) = \mathrm{true}$*, then if for every natural number u, if $u \leq s$, then* $\mathrm{rhcenter}(t+u, u+1) = \mathrm{true}$*, then* $\mathrm{maskparity}(t + s + m \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicmaskparityzeroagreeofblackfirstsuperdiag}(T, P,$
□

Theorem 9.1175. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+p$, then if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(m \cdot P, z)$*, then if* $\mathrm{settledby}(z + 1, z + (p + 1))$*, then if* $\mathrm{settledby}(z + 1 - m \cdot P, z + (p + 1))$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if for every natural number u, if $u \leq S + 1$, then* $\mathrm{rhcenter}(z + p + m \cdot P + u, u + 1) = \mathrm{rhcenter}(z + p + u, u + 1)$*, then if for every natural number u, if $u \leq S + 1$, then if* $\mathrm{rhcenter}(z + p + u, u + 1) = \mathrm{false}$*, then* $\mathrm{rhcenter}(z + p + m \cdot P + u, u + 2) = \mathrm{rhcenter}(z + p + u, u + 2)$*, then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from $\mathrm{notcenterperiodicofcorridorgateatzerodiag}(T, P, m, z, p, S)$.
□

Theorem 9.1176. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+p$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z + 1, z + (p + 1))$, *then if* settledby$(z + 1 - m \cdot P, z + (p + 1))$, *then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if for every natural number u, if $u \leq S + 1$, then* rhcenter$(z + p + m \cdot P + u, u + 1) =$ true, *then if for every natural number u, if $u \leq S + 1$, then* rhcenter$(z + p + u, u + 1) =$ true, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofblackfirstsuperdiagatzerodiag(T, P, m, z, p, S). □

Theorem 9.1177. *For every natural number td and natural number u, if* center$(td) =$ false, *then if* center$(td + 1) =$ true, *then* rdrv$(td, 0, u + 1, u) =$ true.

Proof. The goal follows from rdrvsubdiagoneeqtrueoffalsetrue(td, u). □

Theorem 9.1178. *For every natural number td and natural number d and natural number k and natural number p, if for every natural number j, if $j < d$, then* center$(td+j) =$ false, *then if $k < p+d$, then* rdrv$(td, 0, k, p) =$ false.

Proof. The goal follows from rdrvwhiteseedeqfalseofwhitedriveprefix(td, d, k, p). □

Theorem 9.1179. *For every natural number td and natural number d and natural number k and natural number p, if for every natural number j, if $j < d$, then* center$(td+j) =$ false, *then* rdrv$(td, 0, d+k, p) =$ rdrv$(td+d, 0, k, p)$.

Proof. The goal follows from rdrvwhiteseedshiftofwhitedriveprefix(td, d, k, p). □

Theorem 9.1180. *For every natural number td and natural number d and natural number u, if for every natural number j, if $j < d$, then* center$(td + j) =$ false, *then* rdrv$(td, 0, u + d, u) =$ center$(td + d)$.

Proof. The goal follows from rdrvwhiteseedsubdiageqcenterofwhitedriveprefix(td, d, u). □

Theorem 9.1181. *For every natural number td and natural number $tx2$ and natural number tx and natural number d and natural number k and natural number p, if for every natural number u, if $1 \leq u$, then if $u + d \leq k$, then* rdrv$(td, tx2, u + d, u) =$ rdrv$(td, tx, u + d, u)$, *then if for every natural*

number u, if $1 \leq u$, then if $u + d \leq k$, then if $\mathrm{rdrv}(td, tx, u + d, u) = \mathrm{false}$*, then* $\mathrm{rdrv}(td, tx2, u + d, u + 1) = \mathrm{rdrv}(td, tx, u + d, u + 1)$*, then if $p + d \leq k$, then* $\mathrm{rdrv}(td, tx2, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from $\mathrm{rdrvsubconeagreeofsubdiaggate}(td, tx2, tx, d, k, p)$. □

Theorem 9.1182. *For every natural number td and natural number $tx2$ and natural number tx and natural number d and natural number k and natural number p, if for every natural number u, if $1 \leq u$, then if $u + d \leq k$, then* $\mathrm{rdrv}(td, tx2, u + d, u) = \mathrm{true}$*, then if for every natural number u, if $1 \leq u$, then if $u + d \leq k$, then* $\mathrm{rdrv}(td, tx, u + d, u) = \mathrm{true}$*, then if $p + d \leq k$, then* $\mathrm{rdrv}(td, tx2, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from $\mathrm{rdrvsubconeagreeofblacksubdiag}(td, tx2, tx, d, k, p)$. □

Theorem 9.1183. *For every natural number td and natural number tx and natural number d and natural number k and natural number p, if for every natural number j, if $j < d$, then* $\mathrm{center}(td+j) = \mathrm{false}$*, then if* $\mathrm{center}(td+d) = \mathrm{true}$*, then if for every natural number u, if $1 \leq u$, then if $u + d \leq k$, then* $\mathrm{rdrv}(td, tx, u + d, u) = \mathrm{true}$*, then if $p + d \leq k$, then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{rdrv}(td, 0, k, p)$.

Proof. The goal follows from $\mathrm{rdrvsubconeeqwhiteseedofwhiteprefixandblacksubdiag}(td, tx, d, k, p)$. □

Theorem 9.1184. *For every natural number td and natural number tx and natural number k and natural number p, if* $\mathrm{center}(td) = \mathrm{false}$*, then if* $\mathrm{center}(td + 1) = \mathrm{true}$*, then if for every natural number u, if $1 \leq u$, then if $u + 1 \leq k$, then* $\mathrm{rdrv}(td, tx, u + 1, u) = \mathrm{true}$*, then if $p + 1 \leq k$, then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{rdrv}(td, 0, k, p)$.

Proof. The goal follows from $\mathrm{rdrvsubconeoneeqwhiteseedoffalsetrueandblacksubdiagone}(td, tx, k, p)$. □

Theorem 9.1185. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number d and natural number s, if $T \leq t$, then if $1 + d \leq s$, then if for every natural number u, if $1 \leq u$, then if $u+d \leq s$, then* $\mathrm{rdrv}(t, t+m \cdot P, u+d, u) = \mathrm{rdrv}(t, t, u+d, u)$*, then if for every natural number u, if $1 \leq u$, then if $u + d \leq s$, then if* $\mathrm{rdrv}(t, t, u + d, u) = \mathrm{false}$*, then* $\mathrm{rdrv}(t, t + m \cdot P, u + d, u + 1) = \mathrm{rdrv}(t, t, u + d, u + 1)$*, then* $\mathrm{maskparity}(t + s + m \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofsubdiaggate(T, P, hP, hp
□

Theorem 9.1186. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number d and natural number s, if $T \leq t$, then if $1 + d \leq s$, then if for every natural number u, if $1 \leq u$, then if $u + d \leq s$, then* $\mathrm{rdrv}(t, t + m \cdot P, u + d, u) = \mathrm{true}$*, then if for every natural number u, if $1 \leq u$, then if $u + d \leq s$, then* $\mathrm{rdrv}(t, t, u + d, u) = \mathrm{true}$*, then* $\mathrm{maskparity}(t + s + m \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofblacksubdiag(T, P, hP, h
□

Theorem 9.1187. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number t and natural number d and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t+(d+1) = z+(p+1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if $p+3 \cdot (m \cdot P) \leq S+1$, then if for every natural number u, if $1 \leq u$, then if $u+d \leq S+(d+1)$, then* $\mathrm{rdrv}(t, t+m \cdot P, u+d, u) = \mathrm{rdrv}(t, t, u + d, u)$*, then if for every natural number u, if $1 \leq u$, then if $u + d \leq S + (d + 1)$, then if* $\mathrm{rdrv}(t, t, u + d, u) = \mathrm{false}$*, then* $\mathrm{rdrv}(t, t + m \cdot P, u + d, u + 1) = \mathrm{rdrv}(t, t, u + d, u + 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsubdiaggateatzerodiag(T, P, m, z, p, t, d, S).
□

Theorem 9.1188. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number t and natural number d and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t+(d+1) = z+(p+1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if $p+3 \cdot (m \cdot P) \leq S+1$, then if for every natural number u, if $1 \leq u$, then if $u+d \leq S+(d+1)$, then* $\mathrm{rdrv}(t, t+m \cdot P, u+d, u) = \mathrm{true}$*, then if for every natural number u, if $1 \leq u$, then if $u+d \leq S+(d+1)$, then* $\mathrm{rdrv}(t, t, u + d, u) = \mathrm{true}$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofblacksubdiagatzerodiag(T, P, m, z, p, t, d, S).
□

Theorem 9.1189. *For every natural number td and natural number tx and natural number d and natural number p, if for every natural number j, if $j < d$, then* $\mathrm{center}(td+j) = \mathrm{center}(tx+j)$, *then if* $\mathrm{center}(td+d) = \mathrm{not}(\mathrm{center}(tx+d))$, *then if* $\mathrm{rhcenter}(tx+(p+d),p) = \mathrm{false}$, *then* $\mathrm{rdrv}(td,tx,p+d,p) = \mathrm{true}$.

Proof. The goal follows from $\mathrm{rdrvdrivewavefronteqtrueoffirstdiffandwhiterhcenter}(td,tx,d,p)$. □

Theorem 9.1190. *For every natural number td and natural number tx and natural number d and natural number p, if for every natural number j, if $j \leq d$, then* $\mathrm{center}(td+j) = \mathrm{center}(tx+j)$, *then* $\mathrm{rdrv}(td,tx,p+d,p) = \mathrm{rhcenter}(tx+(p+d),p)$.

Proof. The goal follows from $\mathrm{rdrvdrivewavefronteqrhcenterofcenterprefixeq}(td,tx,d,p)$. □

Theorem 9.1191. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T,P)$, *then for every natural number m and natural number t and natural number d and natural number p, if $T \leq t$, then* $\mathrm{rdrv}(t,t+m\cdot P,p+d,p) = \mathrm{rhcenter}(t+m\cdot P+(p+d),p)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicrdrvsubdiageqrhcenter}(T,P,hP,hper)$. □

Theorem 9.1192. *For every natural number td and natural number tx and natural number Q and natural number B and natural number k and natural number p, if* $\mathrm{rhbandagree}(tx,Q,1,B)$, *then if for every natural number u, if $u \leq k$, then* $\mathrm{rhcenter}(tx+Q+u,u+(B+1)) = \mathrm{rhcenter}(tx+u,u+(B+1))$, *then if for every natural number u, if $u \leq k$, then if* $\mathrm{rhcenter}(tx+u,u+(B+1)) = \mathrm{false}$, *then* $\mathrm{rhcenter}(tx+Q+u,u+(B+2)) = \mathrm{rhcenter}(tx+u,u+(B+2))$, *then if $p \leq k+(B+1)$, then* $\mathrm{rdrv}(td,tx+Q,k,p) = \mathrm{rdrv}(td,tx,k,p)$.

Proof. The goal follows from $\mathrm{rdrvtrapezoidagreeofsuperdiaggate}(td,tx,Q,B,k,p)$. □

Theorem 9.1193. *For every natural number td and natural number tx and natural number Q and natural number B and natural number k and natural number p, if* $\mathrm{rhbandagree}(tx,Q,1,B)$, *then if for every natural number u, if $u \leq k$, then* $\mathrm{rhcenter}(tx+Q+u,u+(B+1)) = \mathrm{true}$, *then if for every natural number u, if $u \leq k$, then* $\mathrm{rhcenter}(tx+u,u+(B+1)) = \mathrm{true}$, *then if $p \leq k+(B+1)$, then* $\mathrm{rdrv}(td,tx+Q,k,p) = \mathrm{rdrv}(td,tx,k,p)$.

Proof. The goal follows from $\mathrm{rdrvtrapezoidagreeofblacksuperdiag}(td,tx,Q,B,k,p)$. □

Theorem 9.1194. *For every natural number td and natural number tx and natural number B and natural number k and natural number p, if for every natural number i, if $i \leq B$, then* $\mathrm{rhcenter}(tx, 1+i) = \mathrm{false}$*, then if for every natural number u, if $u \leq k$, then* $\mathrm{rhcenter}(tx+u, u+(B+1)) = \mathrm{false}$ *and* $\mathrm{rhcenter}(tx+u, u+(B+2)) = \mathrm{false}$*, then if $p \leq k+(B+1)$, then* $\mathrm{rdrv}(td, tx, k, p) = \mathrm{rdrv}(td, 0, k, p)$.

Proof. The goal follows from $\mathrm{rdrvtrapezoideqwhiteseedofwhitebandandcorridor}(td, tx, B, k, p)$. □

Theorem 9.1195. *For every natural number t and natural number B and natural number k and natural number p, if for every natural number i, if $i \leq B$, then* $\mathrm{rhcenter}(t, 1+i) = \mathrm{false}$*, then if for every natural number u, if $u \leq k$, then* $\mathrm{rhcenter}(t+u, u+(B+1)) = \mathrm{false}$ *and* $\mathrm{rhcenter}(t+u, u+(B+2)) = \mathrm{false}$*, then if $p \leq k+(B+1)$, then* $\mathrm{rhcenter}(t+k, p) = \mathrm{rdrv}(t, 0, k, p)$.

Proof. The goal follows from $\mathrm{rhcentertrapezoideqwhiteseedofwhitebandandcorridor}(t, B, k, p)$. □

Theorem 9.1196. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number B and natural number s, if $T \leq t$, then if* $\mathrm{rhbandagree}(t, m \cdot P, 1, B)$*, then if for every natural number u, if $u \leq s$, then* $\mathrm{rhcenter}(t+m \cdot P+u, u+(B+1)) = \mathrm{rhcenter}(t+u, u+(B+1))$*, then if for every natural number u, if $u \leq s$, then if* $\mathrm{rhcenter}(t+u, u+(B+1)) = \mathrm{false}$*, then* $\mathrm{rhcenter}(t+m \cdot P+u, u+(B+2)) = \mathrm{rhcenter}(t+u, u+(B+2))$*, then* $\mathrm{maskparity}(t+s+m \cdot P, 0) = \mathrm{maskparity}(t+s, 0)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicmaskparityzeroagreeofsuperdiaggate}(T, P, hP, h$ □

Theorem 9.1197. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$*, then for every natural number m and natural number t and natural number B and natural number s, if $T \leq t$, then if* $\mathrm{rhbandagree}(t, m \cdot P, 1, B)$*, then if for every natural number u, if $u \leq s$, then* $\mathrm{rhcenter}(t+m \cdot P+u, u+(B+1)) = \mathrm{true}$*, then if for every natural number u, if $u \leq s$, then* $\mathrm{rhcenter}(t+u, u+(B+1)) = \mathrm{true}$*, then* $\mathrm{maskparity}(t+s+m \cdot P, 0) = \mathrm{maskparity}(t+s, 0)$.

Proof. Assume hP, $hper$. The goal follows from $\mathrm{periodicmaskparityzeroagreeofblacksuperdiag}(T, P, hP,$ □

Theorem 9.1198. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number B and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z+1-m \cdot P, z+(p+1))$, *then if* $p+3 \cdot (m \cdot P) \leq S+1$, *then if* rhbandagree$(z+(p+1), m \cdot P, 1, B)$, *then if for every natural number u, if $u \leq S$, then* rhcenter$(z+(p+1)+m \cdot P+u, u+(B+1)) =$ rhcenter$(z+(p+1)+u, u+(B+1))$, *then if for every natural number u, if $u \leq S$, then if* rhcenter$(z+(p+1)+u, u+(B+1)) =$ false, *then* rhcenter$(z+(p+1)+m \cdot P+u, u+(B+2)) =$ rhcenter$(z+(p+1)+u, u+(B+2))$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsuperdiaggateatzerodiag(T, P, m, z, p, B, S). □

Theorem 9.1199. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number B and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z+1-m \cdot P, z+(p+1))$, *then if* $p+3 \cdot (m \cdot P) \leq S+1$, *then if* rhbandagree$(z+(p+1), m \cdot P, 1, B)$, *then if for every natural number u, if $u \leq S$, then* rhcenter$(z+(p+1)+m \cdot P+u, u+(B+1)) =$ true, *then if for every natural number u, if $u \leq S$, then* rhcenter$(z+(p+1)+u, u+(B+1)) =$ true, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofblacksuperdiagatzerodiag(T, P, m, z, p, B, S). □

Theorem 9.1200. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number s, if $T \leq t$, then if for every natural number u, if $u \leq s$, then* rhcenter$(t+m \cdot P+u, u+1) =$ rhcenter$(t+u, u+1)$, *then if for every natural number u, if $u \leq s$, then if* rhcenter$(t+u, u+1) =$ false, *then* rhcenter$(t+m \cdot P+u, u+2) =$ rhcenter$(t+u, u+2)$, *then* maskparity$(t+s+m \cdot P, 0) =$ maskparity$(t+s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofcorridorgateall$(T, P, hP,$ □

Theorem 9.1201. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number s, if $T \leq t$, then if for every natural*

number u, if $u \leq s$, then $\mathrm{rhcenter}(t + m \cdot P + u, u + 1) = \mathrm{true}$*, then if for every natural number u, if $u \leq s$, then* $\mathrm{rhcenter}(t + u, u + 1) = \mathrm{true}$*, then* $\mathrm{maskparity}(t + s + m \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofblackfirstsuperdiagall(T, □

Theorem 9.1202. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(m \cdot P, z)$*, then if* $\mathrm{settledby}(z+1, z+(p+1))$*, then if* $\mathrm{settledby}(z + 1 - m \cdot P, z + (p + 1))$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if for every natural number u, if $u \leq S$, then* $\mathrm{rhcenter}(z + (p + 1) + m \cdot P + u, u + 1) = \mathrm{rhcenter}(z + (p + 1) + u, u + 1)$*, then if for every natural number u, if $u \leq S$, then if* $\mathrm{rhcenter}(z + (p + 1) + u, u + 1) = \mathrm{false}$*, then* $\mathrm{rhcenter}(z + (p + 1) + m \cdot P + u, u + 2) = \mathrm{rhcenter}(z + (p + 1) + u, u + 2)$*, then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofcorridorgateatzerodiagdirect(T, P, m, z, p, S). □

Theorem 9.1203. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* $\mathrm{zerodiagonal}(z)$*, then if* $\mathrm{doublesat}(m \cdot P, z)$*, then if* $\mathrm{settledby}(z+1, z+(p+1))$*, then if* $\mathrm{settledby}(z + 1 - m \cdot P, z + (p + 1))$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if for every natural number u, if $u \leq S$, then* $\mathrm{rhcenter}(z + (p + 1) + m \cdot P + u, u + 1) = \mathrm{true}$*, then if for every natural number u, if $u \leq S$, then* $\mathrm{rhcenter}(z + (p + 1) + u, u + 1) = \mathrm{true}$*, then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofblackfirstsuperdiagatzerodiagdirect(T, P, m, z, p, S). □

Theorem 9.1204. *For every natural number T and natural number P and natural number m and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $m \cdot P \leq \mathrm{twopow}(k)$, then if $T + m \cdot P \leq \mathrm{twopow}(k)$, then if* $\mathrm{twopow}(k) \leq S + 1$*, then if for every natural number u, if $u \leq S$, then* $\mathrm{rhcenter}(\mathrm{twopow}(k) - m \cdot P + m \cdot P + u, u + 1) = \mathrm{rhcenter}(\mathrm{twopow}(k) - m \cdot P + u, u + 1)$*, then if for every natural number u, if $u \leq S$, then if* $\mathrm{rhcenter}(\mathrm{twopow}(k) - m \cdot P + u, u + 1) = \mathrm{false}$*, then* $\mathrm{rhcenter}(\mathrm{twopow}(k) - m \cdot P + m \cdot P + u, u + 2) = \mathrm{rhcenter}(\mathrm{twopow}(k) - m \cdot P + u, u + 2)$*, then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofcorridorgatesubtwopow(T, P, m, k, S). □

Theorem 9.1205. *For every natural number T and natural number P and natural number m and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $m \cdot P \leq$ twopow(k), then if $T + m \cdot P \leq$ twopow(k), then if* twopow$(k) \leq S+1$*, then if for every natural number u, if $u \leq S$, then* rhcenter(twopow$(k)-m\cdot P+m\cdot P+u, u+1) =$ true*, then if for every natural number u, if $u \leq S$, then* rhcenter(twopow$(k) - m \cdot P + u, u + 1) =$ true*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofblackfirstsuperdiagsubtwopow(T, P, m, k, S). □

Theorem 9.1206. *For every natural number T and natural number P and natural number m and natural number k and natural number B and natural number S, if $0 < P$, then if $0 < m$, then if $m \cdot P \leq$ twopow(k), then if $T + m \cdot P \leq$ twopow(k), then if* twopow$(k) \leq S + 1$*, then if* rhbandagree(twopow$(k)-m\cdot P, m\cdot P, 1, B)$*, then if for every natural number u, if $u \leq S$, then* rhcenter(twopow$(k) - m \cdot P + m \cdot P + u, u + (B + 1)) =$ rhcenter(twopow$(k)-m\cdot P+u, u+(B+1))$*, then if for every natural number u, if $u \leq S$, then if* rhcenter(twopow$(k)-m\cdot P+u, u+(B+1)) =$ false*, then* rhcenter(twopow$(k)-m\cdot P+m\cdot P+u, u+(B+2)) =$ rhcenter(twopow$(k)-m \cdot P + u, u + (B + 2))$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsuperdiaggatesubtwopow(T, P, m, k, B, S). □

Theorem 9.1207. *For every natural number td and natural number tx and natural number k and natural number p, if* rdrv$(td, tx, k, p) =$ false*, then if* rdrv$(td, tx, k, p + 1) =$ true*, then* rdrv$(td, tx, k + 1, p + 1) =$ true.

Proof. The goal follows from rdrvblackstaysstep(td, tx, k, p). □

Theorem 9.1208. *For every natural number td and natural number tx and natural number a and natural number p and natural number n and natural number k, if for every natural number s, if $s < n$, then* rdrv$(td, tx, a+s, p) =$ false*, then if* rdrv$(td, tx, a, p + 1) =$ true*, then if $k \leq n$, then* rdrv$(td, tx, a + k, p + 1) =$ true.

Proof. The goal follows from rdrvblackstays(td, tx, a, p, n, k). □

Theorem 9.1209. *For every natural number td and natural number tx and natural number a and natural number r and natural number k, if for every natural number s, if $s < r$, then* $\operatorname{center}(td + (a + s)) = \text{false}$*, then if* $\operatorname{rdrv}(td, tx, a, 1) = \text{true}$*, then if $k \leq r$, then* $\operatorname{rdrv}(td, tx, a + k, 1) = \text{true}$*.*

Proof. The goal follows from $\operatorname{rdrvcoloneblackstaysofwhitedrive}(td, tx, a, r, k)$. □

Theorem 9.1210. *For every natural number td and natural number tx and natural number a and natural number r, if for every natural number s, if $s < r$, then* $\operatorname{center}(td + (a + s)) = \text{false}$*, then if* $\operatorname{center}(td + (a + r)) = \text{true}$*, then if* $\operatorname{rdrv}(td, tx, a, 1) = \text{true}$*, then* $\operatorname{rdrv}(td, tx, a + (r + 1), 1) = \text{false}$*.*

Proof. The goal follows from $\operatorname{rdrvcoloneflipafterwhitedrive}(td, tx, a, r)$. □

Theorem 9.1211. *For every natural number td and natural number tx and natural number a and natural number p and natural number n, if for every natural number s, if $s < n$, then* $\operatorname{rdrv}(td, tx, a + s, p) = \text{false}$*, then if* $\operatorname{rdrv}(td, tx, a, p + (n + 1)) = \text{true}$*, then* $\operatorname{rdrv}(td, tx, a + n, p + 1) = \text{true}$*.*

Proof. The goal follows from $\operatorname{rdrvblackreachesboundary}(td, tx, a, p, n)$. □

Theorem 9.1212. *For every natural number td and natural number tx and natural number a and natural number n and natural number r and natural number k, if for every natural number s, if $s < r$, then* $\operatorname{center}(td + (a + s)) = \text{false}$*, then if* $\operatorname{rdrv}(td, tx, a, n + 1) = \text{true}$*, then if $n \leq k$, then if $k \leq r$, then* $\operatorname{rdrv}(td, tx, a + k, 1) = \text{true}$*.*

Proof. The goal follows from $\operatorname{rdrvcoloneblackofwhitedriveandblackcell}(td, tx, a, n, r, k)$. □

Theorem 9.1213. *For every natural number td and natural number tx and natural number n and natural number r and natural number k, if for every natural number s, if $s < r$, then* $\operatorname{center}(td + s) = \text{false}$*, then if* $\operatorname{rhcenter}(tx, n + 1) = \text{true}$*, then if $n \leq k$, then if $k \leq r$, then* $\operatorname{rdrv}(td, tx, k, 1) = \text{true}$*.*

Proof. The goal follows from $\operatorname{rdrvcoloneblackfrominitofwhitedrive}(td, tx, n, r, k)$. □

Theorem 9.1214. *For every natural number td and natural number tx and natural number r and natural number k, if $1 \leq tx$, then if for every natural number s, if $s < r$, then* $\operatorname{center}(td + s) = \text{false}$*, then if $tx \leq k + 1$, then if $k \leq r$, then* $\operatorname{rdrv}(td, tx, k, 1) = \text{true}$*.*

Proof. The goal follows from rdrvcoloneblackfromrightedgeofwhitedrive(td, tx, r, k). □

Theorem 9.1215. *For every natural number td and natural number $tx2$ and natural number tx and natural number a and natural number $n2$ and natural number $n1$ and natural number r and natural number k, if for every natural number s, if $s < r$, then* center$(td + (a + s)) =$ false*, then if* rdrv$(td, tx2, a, n2 + 1) =$ true*, then if* rdrv$(td, tx, a, n1 + 1) =$ true*, then if $n2 \leq k$, then if $n1 \leq k$, then if $k \leq r$, then* rdrv$(td, tx2, a + k, 1) =$ rdrv$(td, tx, a + k, 1)$.

Proof. The goal follows from rdrvcoloneagreeofwhitedriveandblackcells($td, tx2, tx, a, n2, n1, r, k$). □

Theorem 9.1216. *For every natural number td and natural number $tx2$ and natural number tx and natural number a and natural number $n2$ and natural number $n1$ and natural number r, if for every natural number s, if $s < r$, then* center$(td+(a+s)) =$ false*, then if* rdrv$(td, tx2, a, n2+1) =$ true*, then if* rdrv$(td, tx, a, n1 + 1) =$ true*, then if $n2 \leq r$, then if $n1 \leq r$, then* rdrv$(td, tx2, a + (r + 1), 1) =$ rdrv$(td, tx, a + (r + 1), 1)$.

Proof. The goal follows from rdrvcoloneagreeafterwhitedriveandblackcells($td, tx2, tx, a, n2, n1, r$). □

Theorem 9.1217. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number a and natural number $n2$ and natural number $n1$ and natural number r and natural number k, if $T \leq t$, then if for every natural number s, if $s < r$, then* center$(t + (a + s)) =$ false*, then if* rdrv$(t, t + m \cdot P, a, n2 + 1) =$ true*, then if* rdrv$(t, t, a, n1 + 1) =$ true*, then if $n2 \leq k$, then if $n1 \leq k$, then if $k \leq r$, then* maskparity$(t+(a+k)+m \cdot P, 0) =$ maskparity$(t + (a + k), 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofwhitedriveandblackcells(□

Theorem 9.1218. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number $n2$ and natural number $n1$ and natural number r and natural number k, if $T \leq t$, then if for every natural number s, if $s < r$, then* center$(t+s) =$ false*, then if* rhcenter$(t+m \cdot P, n2+1) =$ true*, then if* rhcenter$(t, n1 + 1) =$ true*, then if $n2 \leq k$, then if $n1 \leq k$, then if $k \leq r$, then* maskparity$(t + k + m \cdot P, 0) =$ maskparity$(t + k, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreefrominitofwhitedrive($T, P,$ □

Theorem 9.1219. *For every natural number td and natural number tx and natural number a and natural number r and natural number q, if for every natural number s, if $s < r$, then* $\mathrm{center}(td + (a + s)) = \mathrm{false}$*, then if* $\mathrm{rdrv}(td, tx, a + (q - 1), 1) = \mathrm{false}$*, then if* $1 \leq q$*, then if* $q \leq r + 1$*, then* $\mathrm{rdrv}(td, tx, a, q) = \mathrm{false}$.

Proof. The goal follows from rdrvwhiteprefixforcedofcolonewhite(td, tx, a, r, q). □

Theorem 9.1220. *For every natural number t and natural number r and natural number q, if for every natural number s, if $s < r$, then* $\mathrm{center}(t+s) = \mathrm{false}$*, then if* $\mathrm{rhcenter}(t + (q - 1), 1) = \mathrm{false}$*, then if* $1 \leq q$*, then if* $q \leq r + 1$*, then* $\mathrm{rhcenter}(t, q) = \mathrm{false}$.

Proof. The goal follows from rhcenterwhiteprefixforcedofcolonewhite(t, r, q). □

Theorem 9.1221. *For every natural number t and natural number r and natural number q, if for every natural number s, if $s < r$, then* $\mathrm{center}(t+s) = \mathrm{false}$*, then if for every natural number s, if $s \leq r$, then* $\mathrm{rhcenter}(t + s, 1) = \mathrm{false}$*, then if* $1 \leq q$*, then if* $q \leq r + 1$*, then* $\mathrm{rhcenter}(t, q) = \mathrm{false}$.

Proof. The goal follows from rhcenterwhitebandforcedofcolonewhiterun(t, r, q). □

Theorem 9.1222. *For every natural number td and natural number tx and natural number c and natural number u,* $\mathrm{rdrv}(td, tx, u + 1 + (c + 2), u + 1) = \mathrm{bxor}(\mathrm{rdrv}(td, tx, u+(c+2), u), \mathrm{rdrv}(td, tx, u+1+(c+1), u+1)$ or $\mathrm{rdrv}(td, tx, u+2+c, u+2))$.

Proof. The goal follows from rdrvsubdiagsucctwostep(td, tx, c, u). □

Theorem 9.1223. *For every natural number td and natural number tx2 and natural number tx and natural number c and natural number k, if for every natural number u, if $u < k$, then* $(\mathrm{rdrv}(td, tx2, u + 1 + (c + 1), u + 1)$ or $\mathrm{rdrv}(td, tx2, u+2+c, u+2)) = (\mathrm{rdrv}(td, tx, u+1+(c+1), u+1)$ or $\mathrm{rdrv}(td, tx, u+2+c, u+2))$*, then* $\mathrm{rdrv}(td, tx2, k + (c + 2), k) = \mathrm{rdrv}(td, tx, k + (c + 2), k)$.

Proof. The goal follows from rdrvsubdiagagreeofprevoragree($td, tx2, tx, c, k$). □

Theorem 9.1224. *For every natural number td and natural number $tx2$ and natural number tx and natural number c and natural number k, if for every natural number u, if $u \leq k$, then* $\mathrm{rdrv}(td, tx2, u+(c+1), u) = \mathrm{rdrv}(td, tx, u+(c+1), u)$*, then if for every natural number u, if $u \leq k+1$, then* $\mathrm{rdrv}(td, tx2, u+c, u) = \mathrm{rdrv}(td, tx, u+c, u)$*, then* $\mathrm{rdrv}(td, tx2, k+(c+2), k) = \mathrm{rdrv}(td, tx, k+(c+2), k)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagagreeoftwoprevsubdiags}(td, tx2, tx, c, k)$. □

Theorem 9.1225. *For every natural number td and natural number $tx2$ and natural number tx and natural number c and natural number k and natural number p, if for every natural number u, if $u \leq k$, then* $\mathrm{rdrv}(td, tx2, u+(c+1), u) = \mathrm{rdrv}(td, tx, u+(c+1), u)$*, then if for every natural number u, if $u \leq k$, then* $\mathrm{rdrv}(td, tx2, u+1+c, u+1) = \mathrm{rdrv}(td, tx, u+1+c, u+1)$*, then if $p+(c+1) \leq k$, then* $\mathrm{rdrv}(td, tx2, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from $\mathrm{rdrvsubconeagreeoftwosubdiags}(td, tx2, tx, c, k, p)$. □

Theorem 9.1226. *For every natural number td and natural number tx and natural number c and natural number k, if for every natural number u, if $u < k$, then* $(\mathrm{rdrv}(td, tx, u+1+(c+1), u+1) \text{ or } \mathrm{rdrv}(td, tx, u+2+c, u+2)) = (\mathrm{rdrv}(td, 0, u+1+(c+1), u+1) \text{ or } \mathrm{rdrv}(td, 0, u+2+c, u+2))$*, then* $\mathrm{rdrv}(td, tx, k+(c+2), k) = \mathrm{rdrv}(td, 0, k+(c+2), k)$.

Proof. The goal follows from $\mathrm{rdrvsubdiageqwhiteseedofprevoreq}(td, tx, c, k)$. □

Theorem 9.1227. *For every natural number td and natural number tx and natural number c and natural number a and natural number H and natural number k, if for every natural number u, if $u < k$, then* $(\mathrm{rdrv}(td, tx, a+H+(u+1)+(c+1), a+H+(u+1)) \text{ or } \mathrm{rdrv}(td, tx, a+H+(u+2)+c, a+H+(u+2))) = (\mathrm{rdrv}(td, tx, a+(u+1)+(c+1), a+(u+1)) \text{ or } \mathrm{rdrv}(td, tx, a+(u+2)+c, a+(u+2)))$*, then* $\mathrm{bxor}(\mathrm{rdrv}(td, tx, a+H+k+(c+2), a+H+k), \mathrm{rdrv}(td, tx, a+k+(c+2), a+k)) = \mathrm{bxor}(\mathrm{rdrv}(td, tx, a+H+(c+2), a+H), \mathrm{rdrv}(td, tx, a+(c+2), a))$.

Proof. The goal follows from $\mathrm{rdrvsubdiagaccumshiftxor}(td, tx, c, a, H, k)$. □

Theorem 9.1228. *For every natural number td and natural number tx and natural number c and natural number a and natural number H and natural number k, if for every natural number u, if $u < H+k$, then* $(\mathrm{rdrv}(td, tx, a+$

$H+(u+1)+(c+1), a+H+(u+1))$ or $\mathrm{rdrv}(td, tx, a+H+(u+2)+c, a+H+(u+2)))$ = $(\mathrm{rdrv}(td, tx, a+(u+1)+(c+1), a+(u+1))$ or $\mathrm{rdrv}(td, tx, a+(u+2)+c, a+(u+2)))$, *then* $\mathrm{rdrv}(td, tx, a+2 \cdot H+k+(c+2), a+2 \cdot H+k) = \mathrm{rdrv}(td, tx, a+k+(c+2), a+k)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagperioddoubleofprevorperiod}(td, tx, c, a, H, k)$. □

Theorem 9.1229. *For every natural number td and natural number tx and natural number c and natural number a and natural number H and natural number k, if for every natural number u, if $u \leq H+k$, then* $\mathrm{rdrv}(td, tx, a+H+u+(c+1), a+H+u) = \mathrm{rdrv}(td, tx, a+u+(c+1), a+u)$, *then if for every natural number u, if $u \leq H+k+1$, then* $\mathrm{rdrv}(td, tx, a+H+u+c, a+H+u) = \mathrm{rdrv}(td, tx, a+u+c, a+u)$, *then* $\mathrm{rdrv}(td, tx, a+2 \cdot H+k+(c+2), a+2 \cdot H+k) = \mathrm{rdrv}(td, tx, a+k+(c+2), a+k)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagperioddoubleoftwoprevperiods}(td, tx, c, a, H, k)$. □

Theorem 9.1230. *For every natural number td and natural number tx and natural number c and natural number u, if* $\mathrm{rdrv}(td, tx, u+1+(c+1), u+1) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx, u+1+(c+2), u+1) = \mathrm{not}(\mathrm{rdrv}(td, tx, u+(c+2), u))$.

Proof. The goal follows from $\mathrm{rdrvsubdiagsucctwostepofblackprev}(td, tx, c, u)$. □

Theorem 9.1231. *For every natural number td and natural number tx and natural number u, if* $\mathrm{rdrv}(td, tx, u+1, u+1) = \mathrm{true}$, *then if* $\mathrm{rdrv}(td, tx, u+2, u+2) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx, u+2+1, u+2) = \mathrm{rdrv}(td, tx, u+1, u)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagoneperiodtwoofblackmaindiag}(td, tx, u)$. □

Theorem 9.1232. *For every natural number td and natural number $tx2$ and natural number tx and natural number k, if for every natural number u, if $u < k$, then* $\mathrm{rdrv}(td, tx2, u+1, u+1) = \mathrm{true}$ *and* $\mathrm{rdrv}(td, tx, u+1, u+1) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx2, k+1, k) = \mathrm{rdrv}(td, tx, k+1, k)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagoneagreeofblackmaindiag}(td, tx2, tx, k)$. □

Theorem 9.1233. *For every natural number td and natural number tx and natural number u, if* $\mathrm{rdrv}(td, tx, u+2, u+2) = \mathrm{true}$, *then* $\mathrm{rdrv}(td, tx, u+1+2, u+1) = \mathrm{not}(\mathrm{rdrv}(td, tx, u+2, u))$.

Proof. The goal follows from rdrvsubdiagtwostepofblackmaindiag(td, tx, u). □

Theorem 9.1234. *For every natural number td and natural number tx and natural number u, if* rdrv$(td, tx, u + 2, u + 2) =$ true*, then if* rdrv$(td, tx, u + 3, u + 3) =$ true*, then* rdrv$(td, tx, u + 2 + 2, u + 2) =$ rdrv$(td, tx, u + 2, u)$.

Proof. The goal follows from rdrvsubdiagtwoperiodtwoofblackmaindiag(td, tx, u). □

Theorem 9.1235. *For every natural number td and natural number $tx2$ and natural number tx and natural number k, if for every natural number u, if $u < k$, then* rdrv$(td, tx2, u + 2, u + 2) =$ true *and* rdrv$(td, tx, u + 2, u + 2) =$ true*, then* rdrv$(td, tx2, k + 2, k) =$ rdrv$(td, tx, k + 2, k)$.

Proof. The goal follows from rdrvsubdiagtwoagreeofblackmaindiag$(td, tx2, tx, k)$. □

Theorem 9.1236. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number p, if for every natural number u, if $u \leq k$, then* rdrv$(td, tx2, u + 1, u + 1) =$ true *and* rdrv$(td, tx, u + 1, u + 1) =$ true*, then if $p + 2 \leq k$, then* rdrv$(td, tx2, k, p) =$ rdrv(td, tx, k, p).

Proof. The goal follows from rdrvsubconetwoagreeofblackmaindiag$(td, tx2, tx, k, p)$. □

Theorem 9.1237. *For every natural number t and natural number c and natural number u, if $c \leq t$, then* rhcenter$(t + u, u + c) =$ rdiag$(t - c, t + u)$.

Proof. The goal follows from rhcentersuperdiageqrdiag(t, c, u). □

Theorem 9.1238. *For every natural number td and natural number tx and natural number c and natural number u, if $1 \leq c$, then if $c \leq tx$, then* rdrv$(td, tx, u, u + c) =$ rdiag$(tx - c, tx + u)$.

Proof. The goal follows from rdrvsuperdiageqrdiag(td, tx, c, u). □

Theorem 9.1239. *For every natural number t and natural number c and natural number k and natural number u and natural number v, if $t - c \leq k$, then* rhcenter$(t + (u + v \cdot$ twopow$(k)), u + v \cdot$ twopow$(k) + c) =$ rhcenter$(t + u, u + c)$.

Proof. The goal follows from rhcentersuperdiagperiodtwopow(t, c, k, u, v). □

Theorem 9.1240. *For every natural number t and natural number c and natural number k and natural number J and natural number u and natural number v, if* $\mathrm{rightthreshold}(k, J)$*, then if* $t - c \leq J$*, then* $\mathrm{rhcenter}(t + (u + v \cdot \mathrm{twopow}(k)), u + v \cdot \mathrm{twopow}(k) + c) = \mathrm{rhcenter}(t + u, u + c)$.

Proof. The goal follows from $\mathrm{rhcentersuperdiagperiodtwopowofthreshold}(t, c, k, J, u, v)$. □

Theorem 9.1241. *For every natural number td and natural number tx and natural number c and natural number k and natural number u and natural number v, if $1 \leq c$, then if $tx - c \leq k$, then* $\mathrm{rdrv}(td, tx, u + v \cdot \mathrm{twopow}(k), u + v \cdot \mathrm{twopow}(k) + c) = \mathrm{rdrv}(td, tx, u, u + c)$.

Proof. The goal follows from $\mathrm{rdrvsuperdiagperiodtwopow}(td, tx, c, k, u, v)$. □

Theorem 9.1242. *For every natural number td and natural number tx and natural number c and natural number k and natural number J and natural number u and natural number v, if $1 \leq c$, then if* $\mathrm{rightthreshold}(k, J)$*, then if $tx - c \leq J$, then* $\mathrm{rdrv}(td, tx, u + v \cdot \mathrm{twopow}(k), u + v \cdot \mathrm{twopow}(k) + c) = \mathrm{rdrv}(td, tx, u, u + c)$.

Proof. The goal follows from $\mathrm{rdrvsuperdiagperiodtwopowofthreshold}(td, tx, c, k, J, u, v)$. □

Theorem 9.1243. *For every natural number t and natural number Q and natural number c and natural number k and natural number u, if $t+Q-c \leq k$, then if for every natural number r, if $r <$* $\mathrm{twopow}(k)$*, then* $\mathrm{rhcenter}(t + Q + r, r + c) = \mathrm{rhcenter}(t + r, r + c)$*, then* $\mathrm{rhcenter}(t + Q + u, u + c) = \mathrm{rhcenter}(t + u, u + c)$.

Proof. The goal follows from $\mathrm{rhcentersuperdiagagreeallofltwopow}(t, Q, c, k, u)$. □

Theorem 9.1244. *For every natural number t and natural number Q and natural number c and natural number k and natural number J and natural number u, if* $\mathrm{rightthreshold}(k, J)$*, then if $t + Q - c \leq J$, then if for every natural number r, if $r <$* $\mathrm{twopow}(k)$*, then* $\mathrm{rhcenter}(t + Q + r, r + c) = \mathrm{rhcenter}(t + r, r + c)$*, then* $\mathrm{rhcenter}(t + Q + u, u + c) = \mathrm{rhcenter}(t + u, u + c)$.

Proof. The goal follows from $\mathrm{rhcentersuperdiagagreeallofthreshold}(t, Q, c, k, J, u)$. □

Theorem 9.1245. *For every natural number t and natural number Q and natural number c and natural number k and natural number u, if $t+Q-c \leq k$, then if for every natural number r, if $r <$ twopow(k), then if* rhcenter$(t + r, r+c) =$ false*, then* rhcenter$(t+Q+r, r+(c+1)) =$ rhcenter$(t+r, r+(c+1))$*, then if* rhcenter$(t + u, u + c) =$ false*, then* rhcenter$(t + Q + u, u + (c + 1)) =$ rhcenter$(t + u, u + (c + 1))$.

Proof. The goal follows from rhcentersuperdiaggateallofltttwopow(t, Q, c, k, u). □

Theorem 9.1246. *For every natural number t and natural number Q and natural number c and natural number k and natural number J and natural number u, if* rightthreshold(k, J)*, then if $t + Q - c \leq J$, then if for every natural number r, if $r <$ twopow(k), then if* rhcenter$(t + r, r + c) =$ false*, then* rhcenter$(t + Q + r, r + (c + 1)) =$ rhcenter$(t + r, r + (c + 1))$*, then if* rhcenter$(t + u, u + c) =$ false*, then* rhcenter$(t + Q + u, u + (c + 1)) =$ rhcenter$(t + u, u + (c + 1))$.

Proof. The goal follows from rhcentersuperdiaggateallofthreshold(t, Q, c, k, J, u). □

Theorem 9.1247. *For every natural number td and natural number tx and natural number Q and natural number B and natural number k and natural number s and natural number p, if $tx + Q - (B + 1) \leq k$, then if* rhbandagree$(tx, Q, 1, B)$*, then if for every natural number r, if $r <$ twopow(k), then* rhcenter$(tx+Q+r, r+(B+1)) =$ rhcenter$(tx+r, r+(B+1))$*, then if for every natural number r, if $r <$ twopow(k), then if* rhcenter$(tx + r, r + (B + 1)) =$ false*, then* rhcenter$(tx+Q+r, r+(B+2)) =$ rhcenter$(tx+r, r+(B+2))$*, then if $p \leq s + (B + 1)$, then* rdrv$(td, tx + Q, s, p) =$ rdrv(td, tx, s, p).

Proof. The goal follows from rdrvtrapezoidagreeofsuperdiaggatetwopow(td, tx, Q, B, k, s, p). □

Theorem 9.1248. *For every natural number td and natural number tx and natural number Q and natural number B and natural number k and natural number J and natural number s and natural number p, if* rightthreshold(k, J)*, then if $tx + Q - (B + 1) \leq J$, then if* rhbandagree$(tx, Q, 1, B)$*, then if for every natural number r, if $r <$ twopow(k), then* rhcenter$(tx + Q + r, r + (B + 1)) =$ rhcenter$(tx + r, r + (B + 1))$*, then if for every natural number r, if $r <$ twopow(k), then if* rhcenter$(tx + r, r + (B + 1)) =$ false*, then* rhcenter$(tx + Q + r, r + (B + 2)) =$ rhcenter$(tx + r, r + (B + 2))$*, then if $p \leq s + (B + 1)$, then* rdrv$(td, tx + Q, s, p) =$ rdrv(td, tx, s, p).

Proof. The goal follows from rdrvtrapezoidagreeofsuperdiaggatethreshold$(td, tx, Q, B, k, J, s, p)$. □

Theorem 9.1249. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number k and natural number s, if $T \leq t$, then if $t + m \cdot P - (B + 1) \leq k$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if for every natural number r, if $r <$* twopow(k)*, then* rhcenter$(t + m \cdot P + r, r + (B + 1)) =$ rhcenter$(t + r, r + (B + 1))$*, then if for every natural number r, if $r <$* twopow(k)*, then if* rhcenter$(t + r, r + (B + 1)) =$ false*, then* rhcenter$(t + m \cdot P + r, r + (B + 2)) =$ rhcenter$(t + r, r + (B + 2))$*, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofsuperdiaggatetwopow(T, □

Theorem 9.1250. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number B and natural number k, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z + 1, z + (p + 1))$*, then if* settledby$(z + 1 - m \cdot P, z + (p + 1))$*, then if $z + (p + 1) + m \cdot P - (B + 1) \leq k$, then if* rhbandagree$(z + (p + 1), m \cdot P, 1, B)$*, then if for every natural number r, if $r <$* twopow(k)*, then* rhcenter$(z + (p + 1) + m \cdot P + r, r + (B + 1)) =$ rhcenter$(z + (p + 1) + r, r + (B + 1))$*, then if for every natural number r, if $r <$* twopow(k)*, then if* rhcenter$(z + (p + 1) + r, r + (B + 1)) =$ false*, then* rhcenter$(z + (p + 1) + m \cdot P + r, r + (B + 2)) =$ rhcenter$(z + (p + 1) + r, r + (B + 2))$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsuperdiaggatetwopowatzerodiag(T, P, m, z, p, B, k). □

Theorem 9.1251. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number B and natural number k and natural number J, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z + 1, z + (p + 1))$*, then if* settledby$(z + 1 - m \cdot P, z + (p + 1))$*, then if* rightthreshold(k, J)*, then if $z + (p + 1) + m \cdot P - (B + 1) \leq J$, then if* rhbandagree$(z + (p + 1), m \cdot P, 1, B)$*, then if for every natural number r, if $r <$* twopow(k)*, then* rhcenter$(z + (p + 1) + m \cdot P + r, r + (B + 1)) =$ rhcenter$(z + (p + 1) + r, r + (B + 1))$*, then if for every natural number r, if $r <$* twopow(k)*, then if* rhcenter$(z + (p + 1) + r, r + (B + 1)) =$ false*, then*

rhcenter$(z+(p+1)+m\cdot P+r, r+(B+2))$ = rhcenter$(z+(p+1)+r, r+(B+2))$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsuperdiaggatethresholdatzerodiag(T, P, m, z, p, B, k, J). □

Theorem 9.1252. *For every natural number td and natural number $tx2$ and natural number tx and natural number a and natural number B and natural number k and natural number p, if for every natural number q, if $q \leq B + 1$, then* rdrv$(td, tx2, a, q)$ = rdrv(td, tx, a, q), *then if $p + k \leq B + 1$, then* rdrv$(td, tx2, a + k, p)$ = rdrv$(td, tx, a + k, p)$.

Proof. The goal follows from rdrvinwardconeagreefromstep$(td, tx2, tx, a, B, k, p)$. □

Theorem 9.1253. *For every natural number td and natural number $tx2$ and natural number tx and natural number B and natural number k and natural number p, if for every natural number q, if $q \leq B+1$, then* rdrv$(td, tx2, 0, q)$ = rdrv$(td, tx, 0, q)$, *then if $p+k \leq B+1$, then* rdrv$(td, tx2, k, p)$ = rdrv(td, tx, k, p).

Proof. The goal follows from rdrvinwardconeagreeofinitagree$(td, tx2, tx, B, k, p)$. □

Theorem 9.1254. *For every natural number td and natural number tx and natural number Q and natural number B and natural number k and natural number p, if* rhbandagree$(tx, Q, 1, B)$, *then if $p+k \leq B+1$, then* rdrv$(td, tx+Q, k, p)$ = rdrv(td, tx, k, p).

Proof. The goal follows from rdrvinwardconeagreeofrhbandagree(td, tx, Q, B, k, p). □

Theorem 9.1255. *For every natural number td and natural number tx and natural number Q and natural number B and natural number k, if* rhbandagree$(tx, Q, 1, B)$, *then if $k \leq B$, then* rdrv$(td, tx+Q, k, 1)$ = rdrv$(td, tx, k, 1)$.

Proof. The goal follows from rdrvcoloneagreeofrhbandagree(td, tx, Q, B, k). □

Theorem 9.1256. *For every natural number td and natural number tx and natural number B and natural number k and natural number p, if for every natural number i, if $i \leq B$, then* rhcenter$(tx, 1 + i)$ = false, *then if $p + k \leq B + 1$, then* rdrv(td, tx, k, p) = rdrv$(td, 0, k, p)$.

Proof. The goal follows from rdrvinwardconeeqwhiteseedofinitialwhiteband(td, tx, B, k, p). □

Theorem 9.1257. *For every natural number t and natural number B and natural number k and natural number p, if for every natural number i, if $i \leq B$, then* rhcenter$(t, 1+i) =$ false*, then if $p+k \leq B+1$, then* rhcenter$(t+k, p) =$ rdrv$(t, 0, k, p)$.

Proof. The goal follows from rhcenterinwardconeeqwhiteseedofinitialwhiteband(t, B, k, p). □

Theorem 9.1258. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number s and natural number p, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if $p+s \leq B+1$, then* rhcenter$(t+s+m \cdot P, p) =$ rhcenter$(t+s, p)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhcenterinwardconeagreeofrhbandagree$(T, P, hI$ □

Theorem 9.1259. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if $s \leq B$, then* rhbandagree$(t+s, m \cdot P, 1, B-s)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhbandagreestepofrhbandagree$(T, P, hP, hper)$. □

Theorem 9.1260. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if $s \leq B$, then* rhcenter$(t+s+m \cdot P, 1) =$ rhcenter$(t+s, 1)$.

Proof. Assume hP, $hper$. The goal follows from periodicrhcenteroneagreeofrhbandagree$(T, P, hP, hper)$. □

Theorem 9.1261. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if $s \leq B$, then* maskparity$(t+s+m \cdot P, 0) =$ maskparity$(t+s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofrhbandagree(T, P, hP, hp
□

Theorem 9.1262. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number B, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z), *then if* doublesat($m \cdot P, z$), *then if* settledby($z+1, z+(p+1)$), *then if* settledby($z + 1 - m \cdot P, z + (p + 1)$), *then if $p + 3 \cdot (m \cdot P) \leq B + 1$, then if* rhbandagree($z + (p + 1), m \cdot P, 1, B$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreeatzerodiagdoubles(T, P, m, z, p, B).
□

Theorem 9.1263. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number B, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$* twopow(k), *then if* twopow(k) $\leq B + 1$, *then if* rhbandagree($t, m \cdot P, 1, B$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreetwopow(T, P, m, t, k, B).
□

Theorem 9.1264. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number B, if $0 < P$, then if* zerodiagonal(z), *then if $2 \leq z + m \cdot P$, then if* zerodiagonal($z+m \cdot P-2$), *then if $T \leq z+$*twopow(k), *then if* settledby($z, z+$ twopow(k)), *then if* settledby($z + m \cdot P, z + m \cdot P +$ twopow(k)), *then if* twopow(k) $\leq B + 1$, *then if* rhbandagree($z +$ twopow(k), $m \cdot P, 1, B$), *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreeatzerodiagpair(T, P, m, z, k, B).
□

Theorem 9.1265. *For every natural number td and natural number tx and natural number Q and natural number B and natural number a and natural number k and natural number p, if* rhbandagree($tx, Q, 1, B$), *then if for every natural number u, if $u < a$, then if* rdrv($td, tx, u, B + 1$) = false, *then* rdrv($td, tx + Q, u, B + 2$) = rdrv($td, tx, u, B + 2$), *then if $p + k \leq B + 1$, then* rdrv($td, tx + Q, a + k, p$) = rdrv($td, tx, a + k, p$).

Proof. The goal follows from rdrvinwardconeagreeofshortboundarygate(td, tx, Q, B, a, k, p).
□

Theorem 9.1266. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number B and natural number a and natural number k, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, B)$*, then if for every natural number u, if $u < a$, then if* rdrv$(t, t, u, B+1) =$ false*, then* rdrv$(t, t+m \cdot P, u, B+2) =$ rdrv$(t, t, u, B+2)$*, then if $k \leq B$, then* maskparity$(t + (a + k) + m \cdot P, 0) =$ maskparity$(t + (a + k), 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofshortboundarygate$(T, P,$ □

Theorem 9.1267. *For every natural number td and natural number tx and natural number a and natural number p and natural number n and natural number r and natural number k, if for every natural number s, if $s < r$, then* rdrv$(td, tx, a + s, p) =$ false*, then if* rdrv$(td, tx, a, p + (n + 1)) =$ true*, then if $n \leq k$, then if $k \leq r$, then* rdrv$(td, tx, a + k, p + 1) =$ true.

Proof. The goal follows from rdrvblackreachesandstaysboundary(td, tx, a, p, n, r, k). □

Theorem 9.1268. *For every natural number td and natural number tx and natural number a and natural number p and natural number n and natural number r, if for every natural number s, if $s < r$, then* rdrv$(td, tx, a+s, p) =$ false*, then if* rdrv$(td, tx, a+r, p) =$ true*, then if* rdrv$(td, tx, a, p+(n+1)) =$ true*, then if $n \leq r$, then* rdrv$(td, tx, a + (r + 1), p + 1) =$ false.

Proof. The goal follows from rdrvcolsuccflipafterwhitecolandblackcell(td, tx, a, p, n, r). □

Theorem 9.1269. *For every natural number td and natural number $tx2$ and natural number tx and natural number a and natural number p and natural number $n2$ and natural number $n1$ and natural number r and natural number k, if for every natural number s, if $s < r$, then* rdrv$(td, tx2, a + s, p) =$ false *and* rdrv$(td, tx, a + s, p) =$ false*, then if* rdrv$(td, tx2, a, p + (n2 + 1)) =$ true*, then if* rdrv$(td, tx, a, p + (n1 + 1)) =$ true*, then if $n2 \leq k$, then if $n1 \leq k$, then if $k \leq r$, then* rdrv$(td, tx2, a + k, p + 1) =$ rdrv$(td, tx, a + k, p + 1)$.

Proof. The goal follows from rdrvcolsuccagreeofwhitecolandblackcells$(td, tx2, tx, a, p, n2, n1, r, k)$. □

Theorem 9.1270. *For every natural number td and natural number $tx2$ and natural number tx and natural number a and natural number p and natural number $n2$ and natural number $n1$ and natural number r, if for every natural*

number s, *if* $s < r$, *then* $\mathrm{rdrv}(td, tx2, a + s, p) = \mathrm{false}$ *and* $\mathrm{rdrv}(td, tx, a + s, p) = \mathrm{false}$, *then if* $\mathrm{rdrv}(td, tx2, a + r, p) = \mathrm{rdrv}(td, tx, a + r, p)$, *then if* $\mathrm{rdrv}(td, tx2, a, p+(n2+1)) = \mathrm{true}$, *then if* $\mathrm{rdrv}(td, tx, a, p+(n1+1)) = \mathrm{true}$, *then if* $n2 \leq r$, *then if* $n1 \leq r$, *then* $\mathrm{rdrv}(td, tx2, a + (r + 1), p + 1) = \mathrm{rdrv}(td, tx, a + (r + 1), p + 1)$.

Proof. The goal follows from rdrvcolsuccagreeafterwhitecolandblackcells$(td, tx2, tx, a, p, n2, n1, r)$. □

Theorem 9.1271. *For every natural number* td *and natural number* tx *and natural number* a *and natural number* p *and natural number* r *and natural number* q, *if for every natural number* s, *if* $s < r$, *then* $\mathrm{rdrv}(td, tx, a+s, p) = \mathrm{false}$, *then if* $\mathrm{rdrv}(td, tx, a + (q - 1), p + 1) = \mathrm{false}$, *then if* $1 \leq q$, *then if* $q \leq r + 1$, *then* $\mathrm{rdrv}(td, tx, a, p + q) = \mathrm{false}$.

Proof. The goal follows from rdrvwhiteoffsetforcedofwhitecol(td, tx, a, p, r, q). □

Theorem 9.1272. *For every natural number* td *and natural number* tx *and natural number* a *and natural number* p *and natural number* r *and natural number* u *and natural number* q, *if for every natural number* s, *if* $s < r$, *then* $\mathrm{rdrv}(td, tx, a + s, p) = \mathrm{false}$, *then if for every natural number* s, *if* $s \leq r$, *then* $\mathrm{rdrv}(td, tx, a + s, p + 1) = \mathrm{false}$, *then if* $1 \leq q$, *then if* $u + q \leq r + 1$, *then* $\mathrm{rdrv}(td, tx, a + u, p + q) = \mathrm{false}$.

Proof. The goal follows from rdrvwhitetriangleforcedoftwowhitecols(td, tx, a, p, r, u, q). □

Theorem 9.1273. *For every natural number* td *and natural number* tx *and natural number* a *and natural number* r *and natural number* u *and natural number* q, *if for every natural number* s, *if* $s < r$, *then* $\mathrm{center}(td+(a+s)) = \mathrm{false}$, *then if for every natural number* s, *if* $s \leq r$, *then* $\mathrm{rdrv}(td, tx, a+s, 1) = \mathrm{false}$, *then if* $1 \leq q$, *then if* $u + q \leq r + 1$, *then* $\mathrm{rdrv}(td, tx, a + u, q) = \mathrm{false}$.

Proof. The goal follows from rdrvwhitetriangleforcedofwhitedriveandcolone(td, tx, a, r, u, q). □

Theorem 9.1274. *For every natural number* t *and natural number* r *and natural number* u *and natural number* q, *if for every natural number* s, *if* $s < r$, *then* $\mathrm{center}(t+s) = \mathrm{false}$, *then if for every natural number* s, *if* $s \leq r$, *then* $\mathrm{rhcenter}(t + s, 1) = \mathrm{false}$, *then if* $1 \leq q$, *then if* $u + q \leq r + 1$, *then* $\mathrm{rhcenter}(t + u, q) = \mathrm{false}$.

Proof. The goal follows from rhcenterwhitetriangleforcedofwhitedriveandcolone(t, r, u, q). □

Theorem 9.1275. *For every natural number td and natural number tx and natural number k and natural number p, if* rdrv$(td, tx, k, p) =$ false*, then if* rdrv$(td, tx, k, p + 1) =$ false*, then if* rdrv$(td, tx, k, p + 2) =$ false*, then* rdrv$(td, tx, k + 1, p + 1) =$ false*.*

Proof. The goal follows from rdrvwhitestepofthreewhite(td, tx, k, p). □

Theorem 9.1276. *For every natural number td and natural number tx and natural number a and natural number p and natural number r and natural number u and natural number q, if for every natural number s, if $s < r$, then* rdrv$(td, tx, a + s, p) =$ false*, then if for every natural number j, if $1 \leq j$, then if $j \leq r + 1$, then* rdrv$(td, tx, a, p + j) =$ false*, then if $1 \leq q$, then if $u + q \leq r + 1$, then* rdrv$(td, tx, a + u, p + q) =$ false*.*

Proof. The goal follows from rdrvwhitetriangleofwhitecolandspatialinterval(td, tx, a, p, r, u, q). □

Theorem 9.1277. *For every natural number td and natural number tx and natural number a and natural number p and natural number L and natural number u and natural number q, if for every natural number j, if $1 \leq j$, then if $j \leq L$, then* rdrv$(td, tx, a, p + j) =$ false*, then if $u + 1 \leq q$, then if $u + q \leq L$, then* rdrv$(td, tx, a + u, p + q) =$ false*.*

Proof. The goal follows from rdrvwhitediamondofspatialwhiteinterval(td, tx, a, p, L, u, q). □

Theorem 9.1278. *For every natural number td and natural number tx and natural number a and natural number p and natural number L and natural number u, if for every natural number j, if $1 \leq j$, then if $j \leq L$, then* rdrv$(td, tx, a, p + j) =$ false*, then if $2 \cdot u + 2 \leq L$, then* rdrv$(td, tx, a + u, p + (u + 1)) =$ false *and* rdrv$(td, tx, a + u, p + (u + 2)) =$ false*.*

Proof. The goal follows from rdrvsuperdiagwhitecorridorofspatialwhiteinterval(td, tx, a, p, L, u). □

Theorem 9.1279. *For every natural number t and natural number p and natural number L and natural number u and natural number q, if for every natural number j, if $1 \leq j$, then if $j \leq L$, then* rhcenter$(t, p + j) =$ false*, then if $u + 1 \leq q$, then if $u + q \leq L$, then* rhcenter$(t + u, p + q) =$ false*.*

Proof. The goal follows from rhcenterwhitediamondofspatialwhiteinterval(t, p, L, u, q). □

Theorem 9.1280. *For every natural number t and natural number B and natural number L and natural number u, if for every natural number j, if $1 \leq j$, then if $j \leq L$, then* rhcenter$(t, B+j) =$ false*, then if* $2 \cdot u+2 \leq L$*, then* rhcenter$(t+u, u+(B+1)) =$ false *and* rhcenter$(t+u, u+(B+2)) =$ false.

Proof. The goal follows from rhcentersuperdiagwhitecorridorofspatialwhiteinterval(t, B, L, u). □

Theorem 9.1281. *For every natural number td and natural number tx and natural number B and natural number k and natural number p, if for every natural number i, if $i \leq B + (2 \cdot k + 2)$, then* rhcenter$(tx, 1+i) =$ false*, then if* $p \leq k + (B + 1)$*, then* rdrv$(td, tx, k, p) =$ rdrv$(td, 0, k, p)$.

Proof. The goal follows from rdrvtrapezoideqwhiteseedofwidewhiteband(td, tx, B, k, p). □

Theorem 9.1282. *For every natural number td and natural number tx and natural number u,* rdrv$(td, tx, u+1, u+1) =$ bxor(rdrv(td, tx, u, u), rhcenter$(tx+u, u+1)$ or rhcenter$(tx+u, u+2)$).

Proof. The goal follows from rdrvmaindiagstepeqrhcenter(td, tx, u). □

Theorem 9.1283. *For every natural number td and natural number tx and natural number a and natural number H and natural number n, if for every natural number u, if $u < n$, then* (rhcenter$(tx + (a + H + u), a + H + u + 1)$ or rhcenter$(tx + (a + H + u), a + H + u + 2)$) = (rhcenter$(tx + (a + u), a + u + 1)$ or rhcenter$(tx + (a + u), a + u + 2)$)*, then* bxor(rdrv$(td, tx, a + H + n, a + H + n)$, rdrv$(td, tx, a + n, a + n)$) = bxor(rdrv$(td, tx, a + H, a + H)$, rdrv$(td, tx, a, a)$).

Proof. The goal follows from rdrvmaindiagaccumshiftxor(td, tx, a, H, n). □

Theorem 9.1284. *For every natural number td and natural number tx and natural number a and natural number H and natural number n, if for every natural number u, if $u < H + n$, then* (rhcenter$(tx + (a + H + u), a + H + u + 1)$ or rhcenter$(tx + (a + H + u), a + H + u + 2)$) = (rhcenter$(tx + (a + u), a + u + 1)$ or rhcenter$(tx + (a + u), a + u + 2)$)*, then* rdrv$(td, tx, a + 2 \cdot H + n, a + 2 \cdot H + n) =$ rdrv$(td, tx, a + n, a + n)$.

Proof. The goal follows from rdrvmaindiagperioddoubleofsuperdiagorperiod(td, tx, a, H, n). □

Theorem 9.1285. *For every natural number td and natural number tx and natural number k and natural number u, if $tx - 1 \leq k$, then* $\mathrm{rdrv}(td, tx, u + \mathrm{twopow}(k+1), u + \mathrm{twopow}(k+1)) = \mathrm{rdrv}(td, tx, u, u)$.

Proof. The goal follows from $\mathrm{rdrvmaindiagperiodtwopowsucc}(td, tx, k, u)$. □

Theorem 9.1286. *For every natural number td and natural number tx and natural number k and natural number J and natural number u, if* $\mathrm{rightthreshold}(k, J)$, *then if $tx - 1 \leq J$, then* $\mathrm{rdrv}(td, tx, u + \mathrm{twopow}(k+1), u + \mathrm{twopow}(k+1)) = \mathrm{rdrv}(td, tx, u, u)$.

Proof. The goal follows from $\mathrm{rdrvmaindiagperiodtwopowsuccofthreshold}(td, tx, k, J, u)$. □

Theorem 9.1287. *For every natural number td and natural number tx and natural number k and natural number u and natural number v, if $tx - 1 \leq k$, then* $\mathrm{rdrv}(td, tx, u + v \cdot \mathrm{twopow}(k+1), u + v \cdot \mathrm{twopow}(k+1)) = \mathrm{rdrv}(td, tx, u, u)$.

Proof. The goal follows from $\mathrm{rdrvmaindiagperiodtwopowsuccmul}(td, tx, k, u, v)$. □

Theorem 9.1288. *For every natural number td and natural number tx and natural number Q and natural number k and natural number u, if $tx + Q - 1 \leq k$, then if for every natural number r, if $r < \mathrm{twopow}(k+1)$, then* $\mathrm{rdrv}(td, tx + Q, r, r) = \mathrm{rdrv}(td, tx, r, r)$, *then* $\mathrm{rdrv}(td, tx + Q, u, u) = \mathrm{rdrv}(td, tx, u, u)$.

Proof. The goal follows from $\mathrm{rdrvmaindiagagreeallofltttwopowsucc}(td, tx, Q, k, u)$. □

Theorem 9.1289. *For every natural number td and natural number tx and natural number u,* $\mathrm{rdrv}(td, tx, u + 1 + 1, u + 1) = \mathrm{bxor}(\mathrm{rdrv}(td, tx, u + 1, u), \mathrm{rdrv}(td, tx, u + 1, u + 1)$ or $\mathrm{rhcenter}(tx + (u + 1), u + 1 + 1))$.

Proof. The goal follows from $\mathrm{rdrvsubdiagonestepeqrhcenter}(td, tx, u)$. □

Theorem 9.1290. *For every natural number td and natural number tx and natural number a and natural number H and natural number n, if for every natural number u, if $u < n$, then* $(\mathrm{rdrv}(td, tx, a + H + (u+1), a + H + (u+1))$ or $\mathrm{rhcenter}(tx + (a + H + (u+1)), a + H + (u+1) + 1)) = (\mathrm{rdrv}(td, tx, a + (u+1), a + (u+1))$ or $\mathrm{rhcenter}(tx + (a + (u+1)), a + (u+1) + 1))$, *then* $\mathrm{bxor}(\mathrm{rdrv}(td, tx, a + H + n + 1, a + H + n), \mathrm{rdrv}(td, tx, a + n + 1, a + n)) = \mathrm{bxor}(\mathrm{rdrv}(td, tx, a + H + 1, a + H), \mathrm{rdrv}(td, tx, a + 1, a))$.

Proof. The goal follows from rdrvsubdiagoneaccumshiftxor(td, tx, a, H, n). □

Theorem 9.1291. *For every natural number td and natural number tx and natural number a and natural number H and natural number n, if for every natural number u, if* $u < H + n$, *then* $(\mathrm{rdrv}(td, tx, a + H + (u + 1), a + H + (u + 1))$ or $\mathrm{rhcenter}(tx + (a + H + (u + 1)), a + H + (u + 1) + 1)) = (\mathrm{rdrv}(td, tx, a+(u+1), a+(u+1))$ or $\mathrm{rhcenter}(tx+(a+(u+1)), a+(u+1)+1))$, *then* $\mathrm{rdrv}(td, tx, a+2 \cdot H+n+1, a+2 \cdot H+n) = \mathrm{rdrv}(td, tx, a+n+1, a+n)$.

Proof. The goal follows from rdrvsubdiagoneperioddoubleoforperiod(td, tx, a, H, n). □

Theorem 9.1292. *For every natural number td and natural number tx and natural number k and natural number u, if* $tx - 1 \leq k$, *then* $\mathrm{rdrv}(td, tx, u + \mathrm{twopow}(k + 2) + 1, u + \mathrm{twopow}(k + 2)) = \mathrm{rdrv}(td, tx, u + 1, u)$.

Proof. The goal follows from rdrvsubdiagoneperiodtwopowsucctwo(td, tx, k, u). □

Theorem 9.1293. *For every natural number td and natural number tx and natural number k and natural number J and natural number u, if* $\mathrm{rightthreshold}(k, J)$, *then if* $tx-1 \leq J$, *then* $\mathrm{rdrv}(td, tx, u+\mathrm{twopow}(k+2)+1, u+\mathrm{twopow}(k+2)) = \mathrm{rdrv}(td, tx, u + 1, u)$.

Proof. The goal follows from rdrvsubdiagoneperiodtwopowsucctwoofthreshold(td, tx, k, J, u). □

Theorem 9.1294. *For every natural number td and natural number tx and natural number k and natural number c and natural number u, if* $tx - 1 \leq k$, *then* $\mathrm{rdrv}(td, tx, u + \mathrm{twopow}(k + 1 + c) + c, u + \mathrm{twopow}(k + 1 + c)) = \mathrm{rdrv}(td, tx, u + c, u)$.

Proof. The goal follows from rdrvsubdiagperiodtwopowall(td, tx, k, c, u). □

Theorem 9.1295. *For every natural number td and natural number tx and natural number k and natural number c and natural number u and natural number v, if* $tx - 1 \leq k$, *then* $\mathrm{rdrv}(td, tx, u + v \cdot \mathrm{twopow}(k + 1 + c) + c, u + v \cdot \mathrm{twopow}(k + 1 + c)) = \mathrm{rdrv}(td, tx, u + c, u)$.

Proof. The goal follows from rdrvsubdiagperiodtwopowallmul(td, tx, k, c, u, v). □

Theorem 9.1296. *For every natural number td and natural number tx and natural number k and natural number J and natural number c and natural number u, if* $\mathrm{rightthreshold}(k, J)$*, then if* $tx - 1 \leq J$*, then* $\mathrm{rdrv}(td, tx, u + \mathrm{twopow}(k + 1 + c) + c, u + \mathrm{twopow}(k + 1 + c)) = \mathrm{rdrv}(td, tx, u + c, u)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagperiodtwopowallofthreshold}(td, tx, k, J, c, u)$. □

Theorem 9.1297. *For every natural number td and natural number tx and natural number k and natural number v, if $tx - 1 \leq k$, then* $\mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(k + 1), v \cdot \mathrm{twopow}(k + 1)) = \mathrm{center}(td)$.

Proof. The goal follows from $\mathrm{rdrvmaindiagattwopowsucceqcenter}(td, tx, k, v)$. □

Theorem 9.1298. *For every natural number td and natural number tx and natural number k and natural number c and natural number v, if $tx - 1 \leq k$, then* $\mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(k+1+c)+c, v \cdot \mathrm{twopow}(k+1+c)) = \mathrm{center}(td+c)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagattwopowalleqcenter}(td, tx, k, c, v)$. □

Theorem 9.1299. *For every natural number td and natural number tx and natural number k and natural number J and natural number c, if* $\mathrm{rightthreshold}(k, J)$*, then if $tx-1 \leq J$, then* $\mathrm{rdrv}(td, tx, \mathrm{twopow}(k+1+c)+c, \mathrm{twopow}(k+1+c)) = \mathrm{center}(td + c)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagattwopowalleqcenterofthreshold}(td, tx, k, J, c)$. □

Theorem 9.1300. *For every natural number t and natural number k and natural number c and natural number v, if $t - 1 \leq k$, then* $\mathrm{rhcenter}(t + (v \cdot \mathrm{twopow}(k + 1 + c) + c), v \cdot \mathrm{twopow}(k + 1 + c)) = \mathrm{center}(t + c)$.

Proof. The goal follows from $\mathrm{rhcentersubdiagattwopowalleqcenter}(t, k, c, v)$. □

Theorem 9.1301. *For every natural number t and natural number k and natural number J and natural number c, if* $\mathrm{rightthreshold}(k, J)$*, then if $t - 1 \leq J$, then* $\mathrm{rhcenter}(t + (\mathrm{twopow}(k + 1 + c) + c), \mathrm{twopow}(k + 1 + c)) = \mathrm{center}(t + c)$.

Proof. The goal follows from $\mathrm{rhcentersubdiagattwopowalleqcenterofthreshold}(t, k, J, c)$. □

Theorem 9.1302. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number c and natural number v, if $tx2 - 1 \leq k$, then if $tx - 1 \leq k$, then* $\mathrm{rdrv}(td, tx2, v \cdot \mathrm{twopow}(k + 1 + c) + c, v \cdot \mathrm{twopow}(k + 1 + c)) = \mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(k + 1 + c) + c, v \cdot \mathrm{twopow}(k + 1 + c))$.

Proof. The goal follows from $\mathrm{rdrvsubdiagattwopowallagree}(td, tx2, tx, k, c, v)$. □

Theorem 9.1303. *For every natural number td and natural number tx and natural number k and natural number c and natural number v, if $tx - 1 \leq k$, then* $\mathrm{rdrv}(td, tx, 1 + c, 1) = \mathrm{rdrv}(td, tx, 1 + v \cdot \mathrm{twopow}(k + 1 + c) + c, 1 + v \cdot \mathrm{twopow}(k + 1 + c))$.

Proof. The goal follows from $\mathrm{rdrvcoloneeqsubdiagattwopowall}(td, tx, k, c, v)$. □

Theorem 9.1304. *For every natural number td and natural number tx and natural number k and natural number J and natural number c, if* $\mathrm{rightthreshold}(k, J)$*, then if $tx - 1 \leq J$, then* $\mathrm{rdrv}(td, tx, 1 + c, 1) = \mathrm{rdrv}(td, tx, 1 + \mathrm{twopow}(k + 1 + c) + c, 1 + \mathrm{twopow}(k + 1 + c))$.

Proof. The goal follows from $\mathrm{rdrvcoloneeqsubdiagattwopowallofthreshold}(td, tx, k, J, c)$. □

Theorem 9.1305. *For every natural number t and natural number k and natural number c and natural number v, if $t - 1 \leq k$, then* $\mathrm{rhcenter}(t + (1 + c), 1) = \mathrm{rhcenter}(t + (1 + v \cdot \mathrm{twopow}(k + 1 + c) + c), 1 + v \cdot \mathrm{twopow}(k + 1 + c))$.

Proof. The goal follows from $\mathrm{rhcentercoloneeqsubdiagattwopowall}(t, k, c, v)$. □

Theorem 9.1306. *For every natural number t and natural number k and natural number J and natural number c, if* $\mathrm{rightthreshold}(k, J)$*, then if $t - 1 \leq J$, then* $\mathrm{rhcenter}(t + (1 + c), 1) = \mathrm{rhcenter}(t + (1 + \mathrm{twopow}(k + 1 + c) + c), 1 + \mathrm{twopow}(k + 1 + c))$.

Proof. The goal follows from $\mathrm{rhcentercoloneeqsubdiagattwopowallofthreshold}(t, k, J, c)$. □

Theorem 9.1307. *For every natural number td and natural number $tx2$ and natural number tx and natural number k and natural number c and natural number v, if $tx2 - 1 \leq k$, then if $tx - 1 \leq k$, then if* $\mathrm{rdrv}(td, tx2, 1 + v \cdot$

$\mathrm{twopow}(k+1+c)+c, 1+v\cdot\mathrm{twopow}(k+1+c)) = \mathrm{rdrv}(td, tx, 1+v\cdot\mathrm{twopow}(k+1+c)+c, 1+v\cdot\mathrm{twopow}(k+1+c))$, *then* $\mathrm{rdrv}(td, tx2, 1+c, 1) = \mathrm{rdrv}(td, tx, 1+c, 1)$.

Proof. The goal follows from rdrvcoloneagreeofsubdiagtwopowagree$(td, tx2, tx, k, c, v)$. □

Theorem 9.1308. *For every natural number td and natural number tx and natural number Q and natural number k and natural number c and natural number u, if $tx + Q - 1 \leq k$, then if for every natural number r, if $r < \mathrm{twopow}(k+1+c)$, then $\mathrm{rdrv}(td, tx+Q, r+c, r) = \mathrm{rdrv}(td, tx, r+c, r)$, then $\mathrm{rdrv}(td, tx+Q, u+c, u) = \mathrm{rdrv}(td, tx, u+c, u)$.*

Proof. The goal follows from rdrvsubdiagagreeallofltttwopowall(td, tx, Q, k, c, u). □

Theorem 9.1309. *For every natural number td and natural number tx and natural number k and natural number J and natural number c and natural number u and natural number v, if $\mathrm{rightthreshold}(k, J)$, then if $tx - 1 \leq J$, then $\mathrm{rdrv}(td, tx, u+v\cdot\mathrm{twopow}(k+1+c)+c, u+v\cdot\mathrm{twopow}(k+1+c)) = \mathrm{rdrv}(td, tx, u+c, u)$.*

Proof. The goal follows from rdrvsubdiagperiodtwopowallmulofthreshold(td, tx, k, J, c, u, v). □

Theorem 9.1310. *For every natural number td and natural number tx and natural number Q and natural number k and natural number J and natural number c and natural number u, if $\mathrm{rightthreshold}(k, J)$, then if $tx + Q - 1 \leq J$, then if for every natural number r, if $r < \mathrm{twopow}(k+1+c)$, then $\mathrm{rdrv}(td, tx+Q, r+c, r) = \mathrm{rdrv}(td, tx, r+c, r)$, then $\mathrm{rdrv}(td, tx+Q, u+c, u) = \mathrm{rdrv}(td, tx, u+c, u)$.*

Proof. The goal follows from rdrvsubdiagagreeallofthreshold(td, tx, Q, k, J, c, u). □

Theorem 9.1311. *For every natural number T and natural number P, if $0 < P$, then if $\mathrm{centerperiodic}(T, P)$, then for every natural number m and natural number t and natural number k and natural number c and natural number v, if $T \leq t$, then if $t + m\cdot P - 1 \leq k$, then if $\mathrm{rdrv}(t, t+m\cdot P, 1+v\cdot\mathrm{twopow}(k+1+c)+c, 1+v\cdot\mathrm{twopow}(k+1+c)) = \mathrm{rdrv}(t, t, 1+v\cdot\mathrm{twopow}(k+1+c)+c, 1+v\cdot\mathrm{twopow}(k+1+c))$, then $\mathrm{maskparity}(t+(1+c)+m\cdot P, 0) = \mathrm{maskparity}(t+(1+c), 0)$.*

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofsubdiagtwopowagree(T, l □

Theorem 9.1312. *For every natural number a and natural number b,* twopow($a+b$) = twopow(b) · twopow(a).

Proof. The goal follows from twopowaddeqmul(a, b). □

Theorem 9.1313. *For every natural number k and natural number c and natural number C and natural number v, if $c \leq C$, then* $v \cdot$ twopow($k + 1 + C$) = $v \cdot$ twopow($C - c$) · twopow($k + 1 + c$).

Proof. The goal follows from twopowsubdiagscalesplit(k, c, C, v). □

Theorem 9.1314. *For every natural number td and natural number tx and natural number k and natural number C and natural number v and natural number j and natural number q, if $tx - 1 \leq k$, then if $j \leq q + C$, then* rdrv($td, tx, v \cdot$ twopow($k + 1 + C$) $+ j, v \cdot$ twopow($k + 1 + C$) $+ q$) = rdrv(td, tx, j, q).

Proof. The goal follows from rdrvdyadicquadranteq(td, tx, k, C, v, j, q). □

Theorem 9.1315. *For every natural number td and natural number tx and natural number k and natural number J and natural number C and natural number v and natural number j and natural number q, if* rightthreshold(k, J), *then if $tx - 1 \leq J$, then if $j \leq q + C$, then* rdrv($td, tx, v \cdot$ twopow($k + 1 + C$) $+ j, v \cdot$ twopow($k + 1 + C$) $+ q$) = rdrv(td, tx, j, q).

Proof. The goal follows from rdrvdyadicquadranteqofthreshold(td, tx, k, J, C, v, j, q). □

Theorem 9.1316. *For every natural number td and natural number tx and natural number k and natural number C and natural number v and natural number j and natural number q, if $tx - 1 \leq k$, then if $j \leq C$, then* rdrv($td, tx, v \cdot$ twopow($k + 1 + C$) $+ j, v \cdot$ twopow($k + 1 + C$) $+ q$) = rdrv(td, tx, j, q).

Proof. The goal follows from rdrvdyadicslabeq(td, tx, k, C, v, j, q). □

Theorem 9.1317. *For every natural number td and natural number tx and natural number k and natural number v and natural number q, if $tx - 1 \leq k$, then* rdrv($td, tx, v \cdot$ twopow($k + 1$), $v \cdot$ twopow($k + 1$) $+ q$) = rdrv($td, tx, 0, q$).

Proof. The goal follows from rdrvdyadicroweqinit(td, tx, k, v, q). □

Theorem 9.1318. *For every natural number t and natural number k and natural number C and natural number v and natural number j and natural number q, if $t - 1 \leq k$, then if $j \leq q + C$, then* $\mathrm{rhcenter}(t + (v \cdot \mathrm{twopow}(k + 1 + C) + j), v \cdot \mathrm{twopow}(k + 1 + C) + q) = \mathrm{rhcenter}(t + j, q)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicquadranteq}(t, k, C, v, j, q)$. □

Theorem 9.1319. *For every natural number t and natural number k and natural number J and natural number C and natural number v and natural number j and natural number q, if* $\mathrm{rightthreshold}(k, J)$, *then if $t - 1 \leq J$, then if $j \leq q + C$, then* $\mathrm{rhcenter}(t + (v \cdot \mathrm{twopow}(k + 1 + C) + j), v \cdot \mathrm{twopow}(k + 1 + C) + q) = \mathrm{rhcenter}(t + j, q)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicquadranteqofthreshold}(t, k, J, C, v, j, q)$. □

Theorem 9.1320. *For every natural number t and natural number k and natural number C and natural number v and natural number j and natural number q, if $t - 1 \leq k$, then if $j \leq C$, then* $\mathrm{rhcenter}(t + (v \cdot \mathrm{twopow}(k + 1 + C) + j), v \cdot \mathrm{twopow}(k + 1 + C) + q) = \mathrm{rhcenter}(t + j, q)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicslabeq}(t, k, C, v, j, q)$. □

Theorem 9.1321. *For every natural number t and natural number k and natural number v and natural number q, if $t - 1 \leq k$, then* $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(k + 1), v \cdot \mathrm{twopow}(k + 1) + q) = \mathrm{rhcenter}(t, q)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicroweq}(t, k, v, q)$. □

Theorem 9.1322. *For every natural number t and natural number k and natural number J and natural number v and natural number q, if* $\mathrm{rightthreshold}(k, J)$, *then if $t - 1 \leq J$, then* $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(k + 1), v \cdot \mathrm{twopow}(k + 1) + q) = \mathrm{rhcenter}(t, q)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicroweqofthreshold}(t, k, J, v, q)$. □

Theorem 9.1323. *For every natural number t and natural number Q and natural number k and natural number v and natural number a and natural number B, if $t + Q - 1 \leq k$, then if* $\mathrm{rhbandagree}(t + v \cdot \mathrm{twopow}(k + 1), Q, v \cdot \mathrm{twopow}(k + 1) + a, B)$, *then* $\mathrm{rhbandagree}(t, Q, a, B)$.

Proof. The goal follows from $\mathrm{rhbandagreeofdyadicrowshiftagree}(t, Q, k, v, a, B)$. □

Theorem 9.1324. *For every natural number t and natural number Q and natural number k and natural number v and natural number a and natural number B, if $t + Q - 1 \leq k$, then if* rhbandagree(t, Q, a, B), *then* rhbandagree$(t + v \cdot \text{twopow}(k+1), Q, v \cdot \text{twopow}(k+1) + a, B)$.

Proof. The goal follows from rhbandagreedyadicrowshiftofrhbandagree(t, Q, k, v, a, B). □

Theorem 9.1325. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number k and natural number v and natural number B and natural number s, if $T \leq t$, then if $t + m \cdot P - 1 \leq k$, then if* rhbandagree$(t + v \cdot \text{twopow}(k+1), m \cdot P, v \cdot \text{twopow}(k+1) + 1, B)$, *then if $s \leq B$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeofdyadicrowshiftagree$(T, P$ □

Theorem 9.1326. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number k and natural number v and natural number B, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z+1-m \cdot P, z+(p+1))$, *then if $z + (p+1) + m \cdot P - 1 \leq k$, then if $p + 3 \cdot (m \cdot P) \leq B$, then if* rhbandagree$(z + (p+1) + v \cdot \text{twopow}(k+1), m \cdot P, v \cdot \text{twopow}(k+1) + 1, B)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofdyadicrowshiftagreeatzerodiagdoubles$(T, P, m, z, p, k, v$ □

Theorem 9.1327. *For every natural number td and natural number u,* rdrv$(td, 0, u+1+1, u+1) =$ bxor(rdrv$(td, 0, u+1, u)$, center(td)).

Proof. The goal follows from rdrvwhiteseedsubdiagonestepeqcenter(td, u). □

Theorem 9.1328. *For every natural number td and natural number u,* rdrv$(td, 0, u+2+1, u+2) =$ rdrv$(td, 0, u+1, u)$.

Proof. The goal follows from rdrvwhiteseedsubdiagoneperiodtwoall(td, u). □

Theorem 9.1329. *For every natural number td and natural number c and natural number u,* $\mathrm{rdrv}(td, 0, u+\mathrm{twopow}(c)+c, u+\mathrm{twopow}(c)) = \mathrm{rdrv}(td, 0, u+c, u)$.

Proof. The goal follows from $\mathrm{rdrvwhiteseedsubdiagperiodtwopowall}(td, c, u)$. □

Theorem 9.1330. *For every natural number td and natural number c and natural number u and natural number v,* $\mathrm{rdrv}(td, 0, u + v \cdot \mathrm{twopow}(c) + c, u + v \cdot \mathrm{twopow}(c)) = \mathrm{rdrv}(td, 0, u + c, u)$.

Proof. The goal follows from $\mathrm{rdrvwhiteseedsubdiagperiodtwopowallmul}(td, c, u, v)$. □

Theorem 9.1331. *For every natural number td and natural number c and natural number v,* $\mathrm{rdrv}(td, 0, v \cdot \mathrm{twopow}(c)+c, v \cdot \mathrm{twopow}(c)) = \mathrm{center}(td+c)$.

Proof. The goal follows from $\mathrm{rdrvwhiteseedsubdiagattwopoweqcenter}(td, c, v)$. □

Theorem 9.1332. *For every natural number td and natural number c and natural number v,* $\mathrm{rdrv}(td, 0, 1 + c, 1) = \mathrm{rdrv}(td, 0, 1 + v \cdot \mathrm{twopow}(c) + c, 1 + v \cdot \mathrm{twopow}(c))$.

Proof. The goal follows from $\mathrm{rdrvwhiteseedcoloneeqsubdiagattwopow}(td, c, v)$. □

Theorem 9.1333. *For every natural number td and natural number C and natural number v and natural number j and natural number q, if $j \leq q + C$, then* $\mathrm{rdrv}(td, 0, v \cdot \mathrm{twopow}(C) + j, v \cdot \mathrm{twopow}(C) + q) = \mathrm{rdrv}(td, 0, j, q)$.

Proof. The goal follows from $\mathrm{rdrvwhiteseeddyadicquadranteq}(td, C, v, j, q)$. □

Theorem 9.1334. *For every natural number td and natural number C and natural number v and natural number j and natural number q, if $j \leq C$, then* $\mathrm{rdrv}(td, 0, v \cdot \mathrm{twopow}(C) + j, v \cdot \mathrm{twopow}(C) + q) = \mathrm{rdrv}(td, 0, j, q)$.

Proof. The goal follows from $\mathrm{rdrvwhiteseeddyadicslabeq}(td, C, v, j, q)$. □

Theorem 9.1335. *For every natural number c and natural number u,* $\mathrm{rhcenter}(u+\mathrm{twopow}(c) + c, u + \mathrm{twopow}(c)) = \mathrm{rhcenter}(u + c, u)$.

Proof. The goal follows from $\mathrm{rhcentersubdiagperiodtwopowzerodelay}(c, u)$. □

Theorem 9.1336. *For every natural number c and natural number u and natural number v,* $\mathrm{rhcenter}(u + v \cdot \mathrm{twopow}(c) + c, u + v \cdot \mathrm{twopow}(c)) = \mathrm{rhcenter}(u + c, u)$.

Proof. The goal follows from $\mathrm{rhcentersubdiagperiodtwopowzerodelaymul}(c, u, v)$. □

Theorem 9.1337. *For every natural number c and natural number v,* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(c) + c, v \cdot \mathrm{twopow}(c)) = \mathrm{center}(c)$.

Proof. The goal follows from $\mathrm{rhcentersubdiagattwopowzerodelayeqcenter}(c, v)$. □

Theorem 9.1338. *For every natural number c and natural number v,* $\mathrm{rhcenter}(1 + c, 1) = \mathrm{rhcenter}(1 + v \cdot \mathrm{twopow}(c) + c, 1 + v \cdot \mathrm{twopow}(c))$.

Proof. The goal follows from $\mathrm{rhcentercoloneeqsubdiagattwopowzerodelay}(c, v)$. □

Theorem 9.1339. *For every natural number C and natural number v and natural number j and natural number q, if $j \leq q + C$, then* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(C) + j, v \cdot \mathrm{twopow}(C) + q) = \mathrm{rhcenter}(j, q)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicquadrantzerodelayeq}(C, v, j, q)$. □

Theorem 9.1340. *For every natural number C and natural number v and natural number j and natural number q, if $j \leq C$, then* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(C) + j, v \cdot \mathrm{twopow}(C) + q) = \mathrm{rhcenter}(j, q)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicslabzerodelayeq}(C, v, j, q)$. □

Theorem 9.1341. *For every natural number C and natural number v and natural number j and natural number q, if $j < q$, then* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(C) + j, v \cdot \mathrm{twopow}(C) + q) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{rhcenterdyadictrianglewhitecorridor}(C, v, j, q)$. □

Theorem 9.1342. *For every natural number td and natural number tx and natural number c and natural number u,* $\mathrm{rdrv}(td, tx, u + \mathrm{twopow}(tx + c) + c, u + \mathrm{twopow}(tx + c)) = \mathrm{rdrv}(td, tx, u + c, u)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagperiodtwopowtxadd}(td, tx, c, u)$. □

Theorem 9.1343. *For every natural number td and natural number tx and natural number c and natural number u and natural number v,* $\mathrm{rdrv}(td, tx, u + v \cdot \mathrm{twopow}(tx + c) + c, u + v \cdot \mathrm{twopow}(tx + c)) = \mathrm{rdrv}(td, tx, u + c, u)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagperiodtwopowtxaddmul}(td, tx, c, u, v)$. □

Theorem 9.1344. *For every natural number td and natural number tx and natural number u and natural number v,* $\mathrm{rdrv}(td, tx, u + v \cdot \mathrm{twopow}(tx), u + v \cdot \mathrm{twopow}(tx)) = \mathrm{rdrv}(td, tx, u, u)$.

Proof. The goal follows from $\mathrm{rdrvmaindiagperiodtwopowtx}(td, tx, u, v)$. □

Theorem 9.1345. *For every natural number td and natural number tx and natural number v,* $\mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(tx), v \cdot \mathrm{twopow}(tx)) = \mathrm{center}(td)$.

Proof. The goal follows from $\mathrm{rdrvmaindiagattwopowtxeqcenter}(td, tx, v)$. □

Theorem 9.1346. *For every natural number td and natural number tx and natural number c and natural number v,* $\mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(tx + c) + c, v \cdot \mathrm{twopow}(tx + c)) = \mathrm{center}(td + c)$.

Proof. The goal follows from $\mathrm{rdrvsubdiagattwopowtxaddeqcenter}(td, tx, c, v)$. □

Theorem 9.1347. *For every natural number td and natural number tx and natural number c and natural number v,* $\mathrm{rdrv}(td, tx, 1+c, 1) = \mathrm{rdrv}(td, tx, 1 + v \cdot \mathrm{twopow}(tx + c) + c, 1 + v \cdot \mathrm{twopow}(tx + c))$.

Proof. The goal follows from $\mathrm{rdrvcoloneeqsubdiagattwopowtxadd}(td, tx, c, v)$. □

Theorem 9.1348. *For every natural number td and natural number tx and natural number C and natural number v and natural number j and natural number q, if $j \leq q+C$, then* $\mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(tx+C)+j, v \cdot \mathrm{twopow}(tx + C) + q) = \mathrm{rdrv}(td, tx, j, q)$.

Proof. The goal follows from $\mathrm{rdrvdyadicquadranttxaddeq}(td, tx, C, v, j, q)$. □

Theorem 9.1349. *For every natural number td and natural number tx and natural number C and natural number v and natural number j and natural number q, if $j \leq C$, then* $\mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(tx + C) + j, v \cdot \mathrm{twopow}(tx + C) + q) = \mathrm{rdrv}(td, tx, j, q)$.

Proof. The goal follows from rdrvdyadicslabtxaddeq(td, tx, C, v, j, q). □

Theorem 9.1350. *For every natural number td and natural number tx and natural number v and natural number q,* rdrv$(td, tx, v \cdot \text{twopow}(tx), v \cdot \text{twopow}(tx) + q) = \text{rdrv}(td, tx, 0, q)$.

Proof. The goal follows from rdrvdyadicrowtxeqinit(td, tx, v, q). □

Theorem 9.1351. *For every natural number t and natural number C and natural number v and natural number j and natural number q, if $j \leq q + C$, then* rhcenter$(t+(v\cdot\text{twopow}(t+C)+j), v\cdot\text{twopow}(t+C)+q) = \text{rhcenter}(t+j, q)$.

Proof. The goal follows from rhcenterdyadicquadranttaddeq(t, C, v, j, q). □

Theorem 9.1352. *For every natural number t and natural number C and natural number v and natural number j and natural number q, if $j \leq C$, then* rhcenter$(t+(v\cdot\text{twopow}(t+C)+j), v\cdot\text{twopow}(t+C)+q) = \text{rhcenter}(t+j, q)$.

Proof. The goal follows from rhcenterdyadicslabtaddeq(t, C, v, j, q). □

Theorem 9.1353. *For every natural number t and natural number v and natural number q,* rhcenter$(t+v\cdot\text{twopow}(t), v\cdot\text{twopow}(t)+q) = \text{rhcenter}(t, q)$.

Proof. The goal follows from rhcenterdyadicrowteq(t, v, q). □

Theorem 9.1354. *For every natural number t and natural number K and natural number v and natural number q, if $t \leq K$, then* rhcenter$(t + v \cdot \text{twopow}(K), v \cdot \text{twopow}(K) + q) = \text{rhcenter}(t, q)$.

Proof. The goal follows from rhcenterdyadicrowscaleeq(t, K, v, q). □

Theorem 9.1355. *For every natural number t and natural number Q and natural number K and natural number v and natural number a and natural number B, if $t + Q \leq K$, then if* rhbandagree$(t + v \cdot \text{twopow}(K), Q, v \cdot \text{twopow}(K) + a, B)$, *then* rhbandagree$(t, Q, a, B)$.

Proof. The goal follows from rhbandagreeofdyadicrowscaleagree(t, Q, K, v, a, B). □

Theorem 9.1356. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number K and natural number v and natural number B and natural number s, if $T \leq t$, then if $t + m \cdot P \leq K$, then if* rhbandagree$(t + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) + 1, B)$, *then if $s \leq B$, then* maskparity$(t + s + m \cdot P, 0) = \text{maskparity}(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeofdyadicrowscaleagree(T, P □

Theorem 9.1357. *For every natural number td and natural number tx and natural number q, if* $tx < q$, *then* $\mathrm{rdrv}(td, tx, 0, q) = \mathrm{false}$.

Proof. The goal follows from rdrvinitgteqfalse(td, tx, q). □

Theorem 9.1358. *For every natural number td and natural number tx and natural number C and natural number v and natural number j and natural number q, if* $j \leq C$, *then if* $tx + j < q$, *then* $\mathrm{rdrv}(td, tx, v \cdot \mathrm{twopow}(tx + C) + j, v \cdot \mathrm{twopow}(tx + C) + q) = \mathrm{false}$.

Proof. The goal follows from rdrvdyadicslabtxgteqfalse(td, tx, C, v, j, q). □

Theorem 9.1359. *For every natural number td and natural number tx2 and natural number tx and natural number q, if* $tx2 < q$, *then if* $tx < q$, *then* $\mathrm{rdrv}(td, tx2, 0, q) = \mathrm{rdrv}(td, tx, 0, q)$.

Proof. The goal follows from rdrvinitagreeofgt($td, tx2, tx, q$). □

Theorem 9.1360. *For every natural number td and natural number tx2 and natural number tx and natural number k and natural number p, if* $tx2 + k < p$, *then if* $tx + k < p$, *then* $\mathrm{rdrv}(td, tx2, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from rdrvagreeofgt($td, tx2, tx, k, p$). □

Theorem 9.1361. *For every natural number tx and natural number Q and natural number B, if* rhbandagree($tx, Q, 1, tx+Q$), *then* rhbandagree($tx, Q, 1, B$).

Proof. The goal follows from rhbandagreeallofsupportagree(tx, Q, B). □

Theorem 9.1362. *For every natural number td and natural number tx and natural number Q and natural number k and natural number p, if* rhbandagree($tx, Q, 1, tx + Q$), *then* $\mathrm{rdrv}(td, tx + Q, k, p) = \mathrm{rdrv}(td, tx, k, p)$.

Proof. The goal follows from rdrvagreeallofsupportrhbandagree(td, tx, Q, k, p). □

Theorem 9.1363. *For every natural number T and natural number P, if* $0 < P$, *then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number s and natural number p, if* $T \leq t$, *then if* rhbandagree($t, m \cdot P, 1, t + m \cdot P$), *then* $\mathrm{rhcenter}(t + s + m \cdot P, p) = \mathrm{rhcenter}(t + s, p)$.

Proof. Assume hP, *hper*. The goal follows from periodicrhcenteragreeallofsupportrhbandagree(T, P, hP,
□

Theorem 9.1364. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number s, if $T \leq t$, then if* rhbandagree$(t, m \cdot P, 1, t + m \cdot P)$*, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, *hper*. The goal follows from periodicmaskparityzeroagreeallofsupportrhbandagree(T
□

Theorem 9.1365. *For every natural number T and natural number P and natural number m and natural number z and natural number p, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if* rhbandagree$(z+(p+1), m \cdot P, 1, z+(p+1)+m \cdot P)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsupportrhbandagreeatzerodiagdoubles(T, P, m, z, p).
□

Theorem 9.1366. *For every natural number T and natural number P and natural number m and natural number t and natural number k, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$* twopow(k)*, then if* rhbandagree$(t, m \cdot P, 1, t + m \cdot P)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsupportrhbandagreetwopow(T, P, m, t, k).
□

Theorem 9.1367. *For every natural number T and natural number P and natural number m and natural number z and natural number k, if $0 < P$, then if* zerodiagonal(z)*, then if $2 \leq z + m \cdot P$, then if* zerodiagonal$(z + m \cdot P - 2)$*, then if $T \leq z +$* twopow(k)*, then if* settledby$(z, z +$ twopow$(k))$*, then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$*, then if* rhbandagree$(z +$ twopow$(k), m \cdot P, 1, z +$ twopow$(k) + m \cdot P)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofsupportrhbandagreeatzerodiagpair(T, P, m, z, k).
□

Theorem 9.1368. *For every natural number td and natural number tx and natural number v and natural number q, if $tx < q$, then* rdrv$(td, tx, v \cdot$ twopow$(tx), v \cdot$ twopow$(tx) + q) =$ false.

Proof. The goal follows from rdrvdyadicrowtxgteqfalse(td, tx, v, q). □

Theorem 9.1369. *For every natural number t and natural number v and natural number q, if $t < q$, then* rhcenter$(t+v\cdot\text{twopow}(t), v\cdot\text{twopow}(t)+q) =$ false.

Proof. The goal follows from rhcenterdyadicrowtgteqfalse(t, v, q). □

Theorem 9.1370. *For every natural number t and natural number Q and natural number K and natural number v and natural number B, if $t+Q \leq K$, then if* rhbandagree$(t + v \cdot \text{twopow}(K), Q, v \cdot \text{twopow}(K) + 1, t + Q)$*, then* rhbandagree$(t, Q, 1, B)$.

Proof. The goal follows from rhbandagreeallofdyadicrowsupportagree(t, Q, K, v, B). □

Theorem 9.1371. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number K and natural number v and natural number s, if $T \leq t$, then if $t + m \cdot P \leq K$, then if* rhbandagree$(t + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) + 1, t + m \cdot P)$*, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$. The goal follows from periodicmaskparityzeroagreeallofdyadicrowsupportagre □

Theorem 9.1372. *For every natural number td and natural number tx and natural number k and natural number p,* rdrv$(td, tx, k, p) =$ bxor(rdrv$(td, tx, k+1, p+1)$, rdrv$(td, tx, k, p+1)$ or rdrv$(td, tx, k, p+2)$).

Proof. The goal follows from rdrvleftstepinv(td, tx, k, p). □

Theorem 9.1373. *For every natural number td and natural number tx and natural number k and natural number U and natural number d, if $d+1 \leq U$, then* rdrv$(td, tx, k, U-(d+1)) =$ bxor(rdrv$(td, tx, k+1, U-d)$, rdrv$(td, tx, k, U-d)$ or rdrv$(td, tx, k, U-d+1)$).

Proof. The goal follows from rdrvleftstepinvsub(td, tx, k, U, d). □

Theorem 9.1374. *For every natural number t and natural number p,* rhcenter$(t, p) =$ bxor(rhcenter$(t+1, p+1)$, rhcenter$(t, p+1)$ or rhcenter$(t, p+2)$).

Proof. The goal follows from rhcenterleftstepinv(t, p). □

Theorem 9.1375. *For every natural number t and natural number U and natural number d, if $d+1 \leq U$, then* rhcenter$(t, U-(d+1)) =$ bxor(rhcenter$(t+1, U-d)$, rhcenter$(t, U-d)$ or rhcenter$(t, U-d+1)$).

Proof. The goal follows from rhcenterleftstepinvsub(t, U, d). □

Theorem 9.1376. *For every natural number td and natural number tx and natural number C and natural number v and natural number j, if $1 \leq v$, then if $j+1 \leq C$, then* rdrv$(td, tx, v \cdot \text{twopow}(tx+C)+j, v \cdot \text{twopow}(tx+C)-1) =$ bxor(center$(td + (j + 1))$, center$(td + j)$ or rdrv$(td, tx, j, 1)$).

Proof. The goal follows from rdrvdyadicpredcoleqcenterxor(td, tx, C, v, j). □

Theorem 9.1377. *For every natural number t and natural number C and natural number v and natural number j, if $1 \leq v$, then if $j + 1 \leq C$, then* rhcenter$(t+(v \cdot \text{twopow}(t+C)+j), v \cdot \text{twopow}(t+C)-1) = \text{lhcenter}(t+j, 1)$.

Proof. The goal follows from rhcenterdyadicpredcoltaddeqlhcenterone(t, C, v, j). □

Theorem 9.1378. *For every natural number C and natural number v and natural number j, if $1 \leq v$, then if $j+1 \leq C$, then* rhcenter$(v \cdot \text{twopow}(C) + j, v \cdot \text{twopow}(C) - 1) = \text{lhcenter}(j, 1)$.

Proof. The goal follows from rhcenterdyadicpredcolzerodelayeqlhcenterone(C, v, j). □

Theorem 9.1379. *For every natural number t and natural number $T0$ and natural number U and natural number C and natural number d and natural number j, if $d \leq U$, then if $j + d \leq C$, then if for every natural number s, if $s \leq C$, then* rhcenter$(T0 + s, U) = \text{center}(t + s)$, *then if for every natural number s, if $s + 1 \leq C$, then* rhcenter$(T0 + s, U - 1) = \text{lhcenter}(t + s, 1)$, *then* rhcenter$(T0 + j, U - d) = \text{lhcenter}(t + j, d)$.

Proof. The goal follows from rhcentersubeqlhcenterofcoleqcenterandone$(t, T0, U, C, d, j)$. □

Theorem 9.1380. *For every natural number t and natural number C and natural number v and natural number d and natural number j, if $1 \leq v$, then if $d \leq v \cdot \text{twopow}(t+C)$, then if $j+d \leq C$, then* rhcenter$(t+(v \cdot \text{twopow}(t+C)+j), v \cdot \text{twopow}(t+C)-d) = \text{lhcenter}(t+j, d)$.

Proof. The goal follows from rhcenterdyadicsubeqlhcentertadd(t, C, v, d, j). □

Theorem 9.1381. *For every natural number C and natural number v and natural number d and natural number j, if $1 \leq v$, then if $d \leq v \cdot \mathrm{twopow}(C)$, then if $j + d \leq C$, then* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(C) + j, v \cdot \mathrm{twopow}(C) - d) = \mathrm{lhcenter}(j, d)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicsubeqlhcenterzerodelay}(C, v, d, j)$. □

Theorem 9.1382. *For every natural number t and natural number C and natural number v and natural number d and natural number j, if $1 \leq v$, then if $d \leq v \cdot \mathrm{twopow}(t + C)$, then if $j + d \leq C$, then if $t + j < d$, then* $\mathrm{rhcenter}(t + (v \cdot \mathrm{twopow}(t + C) + j), v \cdot \mathrm{twopow}(t + C) - d) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{rhcenterdyadicleftwhitecorridortadd}(t, C, v, d, j)$. □

Theorem 9.1383. *For every natural number C and natural number v and natural number d and natural number j, if $1 \leq v$, then if $d \leq v \cdot \mathrm{twopow}(C)$, then if $j + d \leq C$, then if $j < d$, then* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(C) + j, v \cdot \mathrm{twopow}(C) - d) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{rhcenterdyadicleftwhitecorridorzerodelay}(C, v, d, j)$. □

Theorem 9.1384. *For every natural number t and natural number C and natural number v and natural number d, if $1 \leq v$, then if $d \leq C$, then if $t < d$, then* $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(t + C), v \cdot \mathrm{twopow}(t + C) - d) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{rhcenterdyadicrowleftwhitecorridortadd}(t, C, v, d)$. □

Theorem 9.1385. *For every natural number C and natural number v and natural number d, if $1 \leq v$, then if $1 \leq d$, then if $d \leq C$, then* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(C), v \cdot \mathrm{twopow}(C) - d) = \mathrm{false}$.

Proof. The goal follows from $\mathrm{rhcenterdyadicrowleftwhitecorridorzerodelay}(C, v, d)$. □

Theorem 9.1386. *For every natural number t and natural number C and natural number v and natural number d and natural number j, if $1 \leq v$, then if $d \leq v \cdot \mathrm{twopow}(t + C)$, then if $j + d \leq C$, then if $d \leq t + j$, then* $\mathrm{rhcenter}(t+(v \cdot \mathrm{twopow}(t+C)+j), v \cdot \mathrm{twopow}(t+C)-d) = \mathrm{ldiag}(t+j-d, t+j)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicsubeqldiagtadd}(t, C, v, d, j)$. □

Theorem 9.1387. *For every natural number K and natural number v and natural number d and natural number j, if $1 \leq v$, then if $j + d \leq K$, then* $\mathrm{rhcenter}(v \cdot \mathrm{twopow}(K) + j, v \cdot \mathrm{twopow}(K) - d) = \mathrm{lhcenter}(j, d)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicsubeqlhcenterofle}(K, v, d, j)$. □

Theorem 9.1388. *For every natural number t and natural number K and natural number v and natural number d and natural number s, if $1 \leq v$, then if $t + s + d \leq K$, then* $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K) + s, v \cdot \mathrm{twopow}(K) - d) = \mathrm{lhcenter}(t + s, d)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicsubshifteqlhcenter}(t, K, v, d, s)$. □

Theorem 9.1389. *For every natural number t and natural number Q and natural number K and natural number v and natural number d and natural number s, if $1 \leq v$, then if $t+s+Q+d \leq K$, then* $\mathrm{rhcenter}(t+v \cdot \mathrm{twopow}(K)+ s + Q, v \cdot \mathrm{twopow}(K) - d) = \mathrm{lhcenter}(t + s + Q, d)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicsubperiodshifteqlhcenter}(t, Q, K, v, d, s)$. □

Theorem 9.1390. *For every natural number t and natural number K and natural number v and natural number s, if $1 \leq v$, then if $t + s + 1 \leq K$, then* $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K) + s, v \cdot \mathrm{twopow}(K) - 1) = \mathrm{lhcenter}(t + s, 1)$.

Proof. The goal follows from $\mathrm{rhcenterdyadicpredcolshifteqlhcenterone}(t, K, v, s)$. □

Theorem 9.1391. *For every natural number t and natural number Q and natural number K and natural number v and natural number d and natural number s, if $1 \leq v$, then if $t + s + Q + d \leq K$, then if* $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K) + s + Q, v \cdot \mathrm{twopow}(K) - d) = \mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K) + s, v \cdot \mathrm{twopow}(K) - d)$, *then* $\mathrm{lhcenter}(t + s + Q, d) = \mathrm{lhcenter}(t + s, d)$.

Proof. The goal follows from $\mathrm{lhcentershiftagreeofrhcenterdyadicsubagree}(t, Q, K, v, d, s)$. □

Theorem 9.1392. *For every natural number t and natural number Q and natural number K and natural number v and natural number d and natural number S, if $1 \leq v$, then if $t + S + Q + d \leq K$, then if* $\mathrm{rhcolagree}(t + v \cdot \mathrm{twopow}(K), Q, v \cdot \mathrm{twopow}(K) - d, S)$, *then* $\mathrm{lhcolagree}(t, Q, d, S)$.

Proof. The goal follows from lhcolagreeofrhcolagreedyadicsub(t, Q, K, v, d, S).
□

Theorem 9.1393. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t + S + Q + 1 \leq K$, then if* rhcolagree$(t + v \cdot \text{twopow}(K), Q, v \cdot \text{twopow}(K) - 1, S)$, *then* lhcolagree$(t, Q, 1, S)$.

Proof. The goal follows from lhcolagreeoneofrhcolagreedyadicpredcol(t, Q, K, v, S).
□

Theorem 9.1394. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number K and natural number v and natural number S and natural number s, if $T \leq t$, then if $1 \leq v$, then if $t + S + m \cdot P + 1 \leq K$, then if* rhcolagree$(t + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - 1, S)$, *then if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$, m, t, K, v, S, s. The goal follows from periodicmaskparityzeroagreeofrhcolagre
□

Theorem 9.1395. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number K and natural number v and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z+1-m \cdot P, z+(p+1))$, *then if $p+3 \cdot (m \cdot P) \leq S+1$, then if $1 \leq v$, then if $z+(p+1)+S+m \cdot P+1 \leq K$, then if* rhcolagree$(z + (p + 1) + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - 1, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhcolagreedyadicpredcolatzerodiagdoubles$(T, P, m, z, p$
□

Theorem 9.1396. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number K and natural number v and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$ twopow(k), then if* twopow$(k) \leq S + 1$, *then if $1 \leq v$, then if $t + S + m \cdot P + 1 \leq K$, then if* rhcolagree$(t + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - 1, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhcolagreedyadicpredcoltwopow(T, P, m, t, k, K, v, S).
□

Theorem 9.1397. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number K and natural number v and natural number S, if $0 < P$, then if* zerodiagonal(z)*, then if $2 \leq z + m \cdot P$, then if* zerodiagonal$(z + m \cdot P - 2)$*, then if $T \leq z +$* twopow(k)*, then if* settledby$(z, z +$ twopow$(k))$*, then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$*, then if* twopow$(k) \leq S + 1$*, then if $1 \leq v$, then if $z +$* twopow$(k) + S + m \cdot P + 1 \leq K$*, then if* rhcolagree$(z +$ twopow$(k) + v \cdot$ twopow$(K), m \cdot P, v \cdot$ twopow$(K) - 1, S)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhcolagreedyadicpredcolatzerodiagpair$(T, P, m, z, k, K,$ □

Theorem 9.1398. *For every natural number K and natural number v and natural number d and natural number j, if $1 \leq v$, then if $j + d \leq K$, then if $d \leq j$, then* rhcenter$(v \cdot$ twopow$(K) + j, v \cdot$ twopow$(K) - d) =$ ldiag$(j - d, j)$.

Proof. The goal follows from rhcenterdyadicsubeqldiagofle(K, v, d, j). □

Theorem 9.1399. *For every natural number K and natural number v and natural number z and natural number d, if $1 \leq v$, then if* zerodiagonal(z)*, then if* settledby$(z, z+d)$*, then if $z+d+d \leq K$, then* rhcenter$(v \cdot$twopow$(K)+(z + d), v \cdot$ twopow$(K) - d) =$ false.

Proof. The goal follows from rhcenterdyadicsubzerodiageqfalse(K, v, z, d). □

Theorem 9.1400. *For every natural number K and natural number v and natural number z and natural number d, if $1 \leq v$, then if* zerodiagonal(z)*, then if* settledby$(z+2, z+2+d)$*, then if $z+2+d+d \leq K$, then* rhcenter$(v \cdot$ twopow$(K) + (z + 2 + d), v \cdot$ twopow$(K) - d) =$ true.

Proof. The goal follows from rhcenterdyadicsubafterzerodiageqtrue(K, v, z, d). □

Theorem 9.1401. *For every natural number K and natural number v and natural number Q and natural number z and natural number d, if $1 \leq v$, then if* zerodiagonal(z)*, then if* doublesat(Q, z)*, then if* settledby$(z+1, z+1+d)$*, then if $z + 1 + d + Q + (d + Q) \leq K$, then* rhcenter$(v \cdot$ twopow$(K) + (z + 1 + d + Q), v \cdot$ twopow$(K) - (d + Q)) =$ not(rhcenter$(v \cdot$ twopow$(K) + (z + 1 + d), v \cdot$ twopow$(K) - d))$.

Proof. The goal follows from rhcenterdyadicsubdoublingdiagantiperiodic(K, v, Q, z, d). □

Theorem 9.1402. *For every natural number t and natural number p,* rhcenter$(t+1, p+1) =$ bxor(rhcenter(t, p), rhcenter$(t, p+1)$ or rhcenter$(t, p+2)$).

Proof. The goal follows from rhcenterstepsuccinterior(t, p). □

Theorem 9.1403. *For every natural number $t2$ and natural number $t1$ and natural number a and natural number L and natural number i, if for every natural number q, if $q \leq L+2$, then* rhcenter$(t2, a+q) =$ rhcenter$(t1, a+q)$, *then if $i \leq L$, then* rhcenter$(t2+1, a+1+i) =$ rhcenter$(t1+1, a+1+i)$.

Proof. The goal follows from rhcenterbilateralconeagreestep$(t2, t1, a, L, i)$. □

Theorem 9.1404. *For every natural number $t2$ and natural number $t1$ and natural number a and natural number L and natural number s and natural number i, if for every natural number q, if $q \leq L+2 \cdot s$, then* rhcenter$(t2, a+q) =$ rhcenter$(t1, a+q)$, *then if $i \leq L$, then* rhcenter$(t2+s, a+s+i) =$ rhcenter$(t1+s, a+s+i)$.

Proof. The goal follows from rhcenterbilateralconeagree$(t2, t1, a, L, s, i)$. □

Theorem 9.1405. *For every natural number $t2$ and natural number $t1$ and natural number $P0$ and natural number s, if $s \leq P0$, then if for every natural number q, if $q \leq 2 \cdot s$, then* rhcenter$(t2, P0-s+q) =$ rhcenter$(t1, P0-s+q)$, *then* rhcenter$(t2+s, P0) =$ rhcenter$(t1+s, P0)$.

Proof. The goal follows from rhcenterbilateralconepointagree$(t2, t1, P0, s)$. □

Theorem 9.1406. *For every natural number $T0$ and natural number Q and natural number $P0$ and natural number S, if $S \leq P0$, then if* rhbandagree$(T0, Q, P0-S, 2 \cdot S)$, *then* rhcolagree$(T0, Q, P0, S)$.

Proof. The goal follows from rhcolagreeofrhbandagreebilateralcone$(T0, Q, P0, S)$. □

Theorem 9.1407. *For every natural number t and natural number Q and natural number K and natural number v and natural number d and natural number S, if $1 \leq v$, then if $t+S+Q+d \leq K$, then if* rhbandagree$(t+v \cdot$ twopow$(K), Q, v \cdot$ twopow$(K)-d-S, 2 \cdot S)$, *then* lhcolagree(t, Q, d, S).

Proof. The goal follows from lhcolagreeofrhbandagreedyadicsubcone(t, Q, K, v, d, S). □

Theorem 9.1408. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t + S + Q + 1 \leq K$, then if* rhbandagree$(t + v \cdot \text{twopow}(K), Q, v \cdot \text{twopow}(K) - 1 - S, 2 \cdot S)$*, then* lhcolagree$(t, Q, 1, S)$.

Proof. The goal follows from lhcolagreeoneofrhbandagreedyadicpredcolcone(t, Q, K, v, S). □

Theorem 9.1409. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number K and natural number v and natural number S and natural number s, if $T \leq t$, then if $1 \leq v$, then if $t+S+m \cdot P+1 \leq K$, then if* rhbandagree$(t+v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K)-1-S, 2 \cdot S)$*, then if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$, m, t, K, v, S, s. The goal follows from periodicmaskparityzeroagreeofrhbandag □

Theorem 9.1410. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number K and natural number v and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if $p+3 \cdot (m \cdot P) \leq S+1$, then if $1 \leq v$, then if $z+(p+1)+S+m \cdot P+1 \leq K$, then if* rhbandagree$(z + (p + 1) + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - 1 - S, 2 \cdot S)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadicpredcolconeatzerodiagdoubles$(T, P,$ □

Theorem 9.1411. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number K and natural number v and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$ twopow(k), then if* twopow$(k) \leq S+1$*, then if $1 \leq v$, then if $t+S+m \cdot P+1 \leq K$, then if* rhbandagree$(t+v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K)-1-S, 2 \cdot S)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadicpredcolconetwopow$(T, P, m, t, k, K,$ □

Theorem 9.1412. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number K and natural number v and natural number S, if $0 < P$, then if*

zerodiagonal(z), *then if* $2 \leq z + m \cdot P$, *then if* zerodiagonal$(z + m \cdot P - 2)$, *then if* $T \leq z +$ twopow(k), *then if* settledby$(z, z +$ twopow$(k))$, *then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$, *then if* twopow$(k) \leq S + 1$, *then if* $1 \leq v$, *then if* $z +$ twopow$(k) + S + m \cdot P + 1 \leq K$, *then if* rhbandagree$(z +$ twopow$(k) + v \cdot$ twopow$(K), m \cdot P, v \cdot$ twopow$(K) - 1 - S, 2 \cdot S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadicpredcolconeatzerodiagpair$(T, P, m, z$ □

Theorem 9.1413. *For every natural number t and natural number K and natural number v and natural number d, if $1 \leq v$, then if $t + d \leq K$, then if $t < d$, then* rhcenter$(t + v \cdot$ twopow$(K), v \cdot$ twopow$(K) - d) =$ false.

Proof. The goal follows from rhcenterdyadicrowleftwhitecorridorofle(t, K, v, d). □

Theorem 9.1414. *For every natural number t and natural number Q and natural number K and natural number v and natural number d, if $1 \leq v$, then if $t + Q + d \leq K$, then if $t + Q < d$, then* rhcenter$(t + v \cdot$ twopow$(K) + Q, v \cdot$ twopow$(K) - d) =$ rhcenter$(t + v \cdot$ twopow$(K), v \cdot$ twopow$(K) - d)$.

Proof. The goal follows from rhcenterdyadicrowleftwhiteagree(t, Q, K, v, d). □

Theorem 9.1415. *For every natural number t and natural number K and natural number v and natural number r, if $t \leq K$, then if $t < r$, then* rhcenter$(t + v \cdot$ twopow$(K), v \cdot$ twopow$(K) + r) =$ false.

Proof. The goal follows from rhcenterdyadicrowrightwhiteofle(t, K, v, r). □

Theorem 9.1416. *For every natural number t and natural number Q and natural number K and natural number v and natural number r, if $t + Q \leq K$, then if $t + Q < r$, then* rhcenter$(t + v \cdot$ twopow$(K) + Q, v \cdot$ twopow$(K) + r) =$ rhcenter$(t + v \cdot$ twopow$(K), v \cdot$ twopow$(K) + r)$.

Proof. The goal follows from rhcenterdyadicrowrightwhiteagree(t, Q, K, v, r). □

Theorem 9.1417. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t + S + Q + (t + Q) + 1 \leq K$, then if* rhbandagree$(t + v \cdot$ twopow$(K), Q, v \cdot$ twopow$(K) - (t + Q), 2 \cdot (t + Q))$, *then* rhbandagree$(t + v \cdot$ twopow$(K), Q, v \cdot$ twopow$(K) - 1 - S, 2 \cdot S)$.

Proof. The goal follows from rhbandagreedyadiccoreexpandtocone(t, Q, K, v, S).
□

Theorem 9.1418. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t+S+Q+(t+Q)+1 \leq K$, then if* rhbandagree$(t+v \cdot \text{twopow}(K), Q, v \cdot \text{twopow}(K) - (t+Q), 2 \cdot (t+Q))$, *then* lhcolagree$(t, Q, 1, S)$.

Proof. The goal follows from lhcolagreeoneofrhbandagreedyadiccore(t, Q, K, v, S).
□

Theorem 9.1419. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number K and natural number v and natural number S and natural number s, if $T \leq t$, then if $1 \leq v$, then if $t + S + m \cdot P + (t + m \cdot P) + 1 \leq K$, then if* rhbandagree$(t + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - (t + m \cdot P), 2 \cdot (t + m \cdot P))$, *then if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) = \text{maskparity}(t + s, 0)$.

Proof. Assume hP, $hper$, m, t, K, v, S, s. The goal follows from periodicmaskparityzeroagreeofrhbandag
□

Theorem 9.1420. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number K and natural number v, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z + 1, z + (p + 1))$, *then if* settledby$(z + 1 - m \cdot P, z + (p + 1))$, *then if $1 \leq v$, then if $z + (p+1) + (p + 3 \cdot (m \cdot P)) + m \cdot P + (z + (p+1) + m \cdot P) + 1 \leq K$, then if* rhbandagree$(z + (p+1) + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - (z + (p+1) + m \cdot P), 2 \cdot (z + (p+1) + m \cdot P))$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadiccoreatzerodiagdoubles$(T, P, m, z, p,$
□

Theorem 9.1421. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number K and natural number v, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P = \text{twopow}(k)$, then if $1 \leq v$, then if $t + \text{twopow}(k) + m \cdot P + (t + m \cdot P) + 1 \leq K$, then if* rhbandagree$(t + v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - (t + m \cdot P), 2 \cdot (t + m \cdot P))$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadiccoretwopow(T, P, m, t, k, K, v).
□

Theorem 9.1422. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number K and natural number v, if $0 < P$, then if* zerodiagonal(z)*, then if* $2 \leq z + m \cdot P$*, then if* zerodiagonal$(z + m \cdot P - 2)$*, then if* $T \leq z +$ twopow(k)*, then if* settledby$(z, z +$ twopow$(k))$*, then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$*, then if* $1 \leq v$*, then if* $z +$ twopow$(k) +$ twopow$(k) + m \cdot P + (z +$ twopow$(k) + m \cdot P) + 1 \leq K$*, then if* rhbandagree$(z +$ twopow$(k) + v \cdot$ twopow$(K), m \cdot P, v \cdot$twopow$(K) - (z +$twopow$(k) + m \cdot P), 2 \cdot (z +$twopow$(k) + m \cdot P))$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadiccoreatzerodiagpair$(T, P, m, z, k, K, v$
□

Theorem 9.1423. *For every natural number t and natural number Q and natural number K and natural number v, if $1 \leq v$, then if $2 \cdot (t + Q) \leq K$, then if* lhbandagree$(t, Q, 0, t + Q)$*, then* rhbandagree$(t + v \cdot$ twopow$(K), Q, v \cdot$ twopow$(K) - (t + Q), t + Q)$.

Proof. The goal follows from rhbandagreedyadiclefthalfoflhbandagree(t, Q, K, v).
□

Theorem 9.1424. *For every natural number t and natural number Q and natural number K and natural number v, if $t + Q \leq K$, then if* rhbandagree$(t, Q, 0, t + Q)$*, then* rhbandagree$(t + v \cdot$ twopow$(K), Q, v \cdot$ twopow$(K), t + Q)$.

Proof. The goal follows from rhbandagreedyadicrighthalfofrhbandagree(t, Q, K, v).
□

Theorem 9.1425. *For every natural number t and natural number Q and natural number K and natural number v, if $1 \leq v$, then if $2 \cdot (t + Q) \leq K$, then if* lhbandagree$(t, Q, 0, t + Q)$*, then if* rhbandagree$(t, Q, 0, t + Q)$*, then* rhbandagree$(t + v \cdot$ twopow$(K), Q, v \cdot$ twopow$(K) - (t + Q), 2 \cdot (t + Q))$.

Proof. The goal follows from rhbandagreedyadiccoreoflhbandagreeandrhbandagree(t, Q, K, v).
□

Theorem 9.1426. *For every natural number t and natural number Q and natural number S, if* lhbandagree$(t, Q, 0, t + Q)$*, then if* rhbandagree$(t, Q, 0, t + Q)$*, then* lhcolagree$(t, Q, 1, S)$.

Proof. The goal follows from lhcolagreeoneoffullrowbandagree(t, Q, S). □

Theorem 9.1427. *For every natural number t,* lhcenter$(t, 0) =$ rhcenter$(t, 0)$.

Proof. The goal follows from lhcenterzeroeqrhcenterzero(t). □

Theorem 9.1428. *For every natural number t and natural number Q and natural number B, if* lhbandagree($t, Q, 1, B$)*, then if* rhbandagree($t, Q, 0, B$)*, then* lhbandagree($t, Q, 0, B$).

Proof. The goal follows from lhbandagreezerooflhbandagreeoneandrhbandagreezero(t, Q, B). □

Theorem 9.1429. *For every natural number t and natural number Q and natural number S, if* lhbandagree($t, Q, 1, t+Q$)*, then if* rhbandagree($t, Q, 0, t+Q$)*, then* lhcolagree($t, Q, 1, S$).

Proof. The goal follows from lhcolagreeoneoflhbandagreeoneandrhbandagreezero(t, Q, S). □

Theorem 9.1430. *For every natural number T and natural number P and natural number m and natural number z and natural number p, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat($m \cdot P, z$)*, then if* settledby($z+1, z+(p+1)$)*, then if* settledby($z+1-m \cdot P, z+(p+1)$)*, then if* lhbandagree($z+(p+1), m \cdot P, 1, z+(p+1)+m \cdot P$)*, then if* rhbandagree($z + (p + 1), m \cdot P, 0, z + (p + 1) + m \cdot P$)*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhbandagreeoneandrhbandagreezeroatzerodiagdoubles(□

Theorem 9.1431. *For every natural number T and natural number P and natural number m and natural number t and natural number k, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$* twopow(k)*, then if* lhbandagree($t, m \cdot P, 1, t + m \cdot P$)*, then if* rhbandagree($t, m \cdot P, 0, t + m \cdot P$)*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhbandagreeoneandrhbandagreezerotwopow($T, P, m, t,$ □

Theorem 9.1432. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t + S + Q + 1 \leq K$, then if* lhbandagree($t, Q, 1, S$)*, then* rhbandagree($t + v \cdot$ twopow(K)$, Q, v \cdot$ twopow(K)$ - 1 - S, S$).

Proof. The goal follows from rhbandagreedyadicleftconeoflhbandagree(t, Q, K, v, S). □

Theorem 9.1433. *For every natural number t and natural number Q and natural number K and natural number v and natural number B, if $t+Q \leq K$, then if* rhbandagree$(t, Q, 0, B)$, *then* rhbandagree$(t + v \cdot \text{twopow}(K), Q, v \cdot \text{twopow}(K), B)$.

Proof. The goal follows from rhbandagreedyadicrightconeofrhbandagree(t, Q, K, v, B). □

Theorem 9.1434. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t+S+Q+1 \leq K$, then if* lhbandagree$(t, Q, 1, S)$, *then if* rhbandagree$(t, Q, 0, S)$, *then* rhbandagree$(t + v \cdot \text{twopow}(K), Q, v \cdot \text{twopow}(K) - 1 - S, 2 \cdot S)$.

Proof. The goal follows from rhbandagreedyadicpredcolconeoflhbandagreeandrhbandagree(t, Q, K, v, S) □

Theorem 9.1435. *For every natural number t and natural number Q and natural number S, if* lhbandagree$(t, Q, 1, S)$, *then if* rhbandagree$(t, Q, 0, S)$, *then* lhcolagree$(t, Q, 1, S)$.

Proof. The goal follows from lhcolagreeoneoflhbandagreeoneandrhbandagreezerocone(t, Q, S). □

Theorem 9.1436. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number m and natural number t and natural number S and natural number s, if $T \leq t$, then if* lhbandagree$(t, m \cdot P, 1, S)$, *then if* rhbandagree$(t, m \cdot P, 0, S)$, *then if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$, m, t, S, s. The goal follows from periodicmaskparityzeroagreeoflhbandagreeon □

Theorem 9.1437. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z+1-m \cdot P, z+(p+1))$, *then if $p+3 \cdot (m \cdot P) \leq S+1$, then if* lhbandagree$(z+(p+1), m \cdot P, 1, S)$, *then if* rhbandagree$(z+(p+1), m \cdot P, 0, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhbandagreeoneandrhbandagreezeroconeatzerodiagdoul □

Theorem 9.1438. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$ twopow(k), then if twopow$(k) \leq S + 1$, then if* lhbandagree$(t, m \cdot P, 1, S)$, *then if* rhbandagree$(t, m \cdot P, 0, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhbandagreeoneandrhbandagreezeroconetwopow$(T, P, r$
□

Theorem 9.1439. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if $2 \leq z + m \cdot P$, then if* zerodiagonal$(z+m\cdot P-2)$, *then if $T \leq z+$*twopow(k), *then if* settledby$(z, z+$ twopow$(k))$, *then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$, *then if* twopow$(k) \leq S + 1$, *then if* lhbandagree$(z +$ twopow$(k), m \cdot P, 1, S)$, *then if* rhbandagree$(z +$ twopow$(k), m \cdot P, 0, S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhbandagreeoneandrhbandagreezeroconeatzerodiagpair
□

Theorem 9.1440. *For every natural number t and natural number K and natural number v and natural number w and natural number d, if $1 \leq v$, then if $t+d \leq K$, then* rhcenter$(t+v\cdot$twopow$(K)+w\cdot$twopow$(K), v\cdot$twopow$(K)+$ $w \cdot$ twopow$(K) - d) =$ rhcenter$(t + v \cdot$ twopow$(K), v \cdot$ twopow$(K) - d)$.

Proof. The goal follows from rhcenterdyadicdiagshiftlefteq(t, K, v, w, d). □

Theorem 9.1441. *For every natural number t and natural number K and natural number v and natural number w and natural number q, if $t \leq K$, then* rhcenter$(t+v\cdot$twopow$(K)+w\cdot$twopow$(K), v\cdot$twopow$(K)+w\cdot$twopow$(K)+$ $q) =$ rhcenter$(t + v \cdot$ twopow$(K), v \cdot$ twopow$(K) + q)$.

Proof. The goal follows from rhcenterdyadicdiagshiftrighteq(t, K, v, w, q). □

Theorem 9.1442. *For every natural number t and natural number K and natural number v and natural number w and natural number W and natural number i, if $1 \leq v$, then if $t + W \leq K$, then* rhcenter$(t + v \cdot$ twopow$(K) +$ $w \cdot$ twopow$(K), v \cdot$ twopow$(K) + w \cdot$ twopow$(K) - W + i) =$ rhcenter$(t + v \cdot$ twopow$(K), v \cdot$ twopow$(K) - W + i)$.

Proof. The goal follows from rhcenterdyadicdiagshiftbandeq(t, K, v, w, W, i).
□

Theorem 9.1443. *For every natural number t and natural number K and natural number v and natural number w and natural number W and natural number B, if $1 \leq v$, then if $t + W \leq K$, then if for every natural number i, if $i \leq B$, then* $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K) + w \cdot \mathrm{twopow}(K), v \cdot \mathrm{twopow}(K) - W + i) = \mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K) + w \cdot \mathrm{twopow}(K), v \cdot \mathrm{twopow}(K) + w \cdot \mathrm{twopow}(K) - W + i)$, *then* $\mathrm{rhbandagree}(t + v \cdot \mathrm{twopow}(K), w \cdot \mathrm{twopow}(K), v \cdot \mathrm{twopow}(K) - W, B)$.

Proof. The goal follows from $\mathrm{rhbandagreedyadicofspatialperiodshiftagree}(t, K, v, w, W, B)$. □

Theorem 9.1444. *For every natural number t and natural number K and natural number L and natural number u and natural number w and natural number d, if $1 \leq u$, then if $K \leq L$, then if $t + d \leq K$, then* $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K) - d) = \mathrm{lhcenter}(t, d)$.

Proof. The goal follows from $\mathrm{rhcentertwoscalerowlefteqlhcenter}(t, K, L, u, w, d)$. □

Theorem 9.1445. *For every natural number t and natural number K and natural number L and natural number u and natural number w and natural number q, if $K \leq L$, then if $t \leq K$, then* $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K) + q) = \mathrm{rhcenter}(t, q)$.

Proof. The goal follows from $\mathrm{rhcentertwoscalerowrighteqrhcenter}(t, K, L, u, w, q)$. □

Theorem 9.1446. *For every natural number t and natural number K and natural number L and natural number u and natural number w and natural number S, if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + w \cdot \mathrm{twopow}(K) + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq 2 \cdot S$, then* $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) - 1 - S + i) = \mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K) - 1 - S + i)$, *then* $\mathrm{lhcolagree}(t, w \cdot \mathrm{twopow}(K), 1, S)$.

Proof. The goal follows from $\mathrm{lhcolagreeoneoftwoscalesinglerowspatialagree}(t, K, L, u, w, S)$. □

Theorem 9.1447. *For every natural number t and natural number P and natural number K and natural number L and natural number u and natural number c and natural number S, if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural*

number i, if $i \leq 2 \cdot S$, then rhcenter$(t + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) - 1 - S + i) =$ rhcenter$(t + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P - 1 - S + i)$, *then* lhcolagree$(t, c \cdot \text{twopow}(K) \cdot P, 1, S)$.

Proof. The goal follows from lhcolagreeoneoftwoscalesinglerowmulperiodagree(t, P, K, L, u, c, S). □

Theorem 9.1448. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P), *then for every natural number t and natural number K and natural number L and natural number u and natural number c and natural number S and natural number s, if $T \leq t$, then if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot \text{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq 2 \cdot S$, then* rhcenter$(t + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) - 1 - S + i) =$ rhcenter$(t + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P - 1 - S + i)$, *then if $s \leq S$, then* maskparity$(t + s + c \cdot \text{twopow}(K) \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$, t, K, L, u, c, S, s. The goal follows from periodicmaskparityzeroagreeoftwoscal □

Theorem 9.1449. *For every natural number T and natural number P and natural number c and natural number z and natural number p and natural number K and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < c$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z), *then if* doublesat$(c \cdot \text{twopow}(K) \cdot P, z)$, *then if* settledby$(z + 1, z + (p + 1))$, *then if* settledby$(z + 1 - c \cdot \text{twopow}(K) \cdot P, z + (p + 1))$, *then if $p + 3 \cdot (c \cdot \text{twopow}(K) \cdot P) \leq S + 1$, then if $1 \leq u$, then if $z + (p + 1) + S + 1 \leq K$, then if $z + (p + 1) + S + c \cdot \text{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq 2 \cdot S$, then* rhcenter$(z + (p + 1) + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) - 1 - S + i) =$ rhcenter$(z + (p + 1) + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P - 1 - S + i)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoftwoscalesinglerowatzerodiagdoubles$(T, P, c, z, p, K, L, u$ □

Theorem 9.1450. *For every natural number T and natural number P and natural number c and natural number t and natural number k and natural number K and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < c$, then if $T \leq t$, then if $t + c \cdot \text{twopow}(K) \cdot P =$ twopow(k), then if* twopow$(k) \leq S + 1$, *then if $1 \leq u$, then if $t + S + 1 \leq K$,*

then if $t + S + c \cdot \text{twopow}(K) \cdot P + 1 \leq L$, *then if* $K \leq L$, *then if for every natural number* i, *if* $i \leq 2 \cdot S$, *then* $\text{rhcenter}(t + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) - 1 - S + i) = \text{rhcenter}(t + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P - 1 - S + i)$, *then not* $\text{centerperiodic}(T, P)$.

Proof. The goal follows from $\text{notcenterperiodicoftwoscalesinglerowtwopow}(T, P, c, t, k, K, L, u, S)$. □

Theorem 9.1451. *For every natural number* T *and natural number* P *and natural number* c *and natural number* z *and natural number* k *and natural number* K *and natural number* L *and natural number* u *and natural number* S, *if* $0 < P$, *then if* $\text{zerodiagonal}(z)$, *then if* $2 \leq z + c \cdot \text{twopow}(K) \cdot P$, *then if* $\text{zerodiagonal}(z + c \cdot \text{twopow}(K) \cdot P - 2)$, *then if* $T \leq z + \text{twopow}(k)$, *then if* $\text{settledby}(z, z + \text{twopow}(k))$, *then if* $\text{settledby}(z + c \cdot \text{twopow}(K) \cdot P, z + c \cdot \text{twopow}(K) \cdot P + \text{twopow}(k))$, *then if* $\text{twopow}(k) \leq S + 1$, *then if* $1 \leq u$, *then if* $z + \text{twopow}(k) + S + 1 \leq K$, *then if* $z + \text{twopow}(k) + S + c \cdot \text{twopow}(K) \cdot P + 1 \leq L$, *then if* $K \leq L$, *then if for every natural number* i, *if* $i \leq 2 \cdot S$, *then* $\text{rhcenter}(z + \text{twopow}(k) + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) - 1 - S + i) = \text{rhcenter}(z + \text{twopow}(k) + u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P, u \cdot \text{twopow}(L) + c \cdot \text{twopow}(K) \cdot P - 1 - S + i)$, *then not* $\text{centerperiodic}(T, P)$.

Proof. The goal follows from $\text{notcenterperiodicoftwoscalesinglerowatzerodiagpair}(T, P, c, z, k, K, L, u, S)$ □

Theorem 9.1452. *For every natural number* L *and natural number* u *and natural number* d, *if* $1 \leq u$, *then if* $d \leq L$, *then if* $0 < d$, *then* $\text{rhcenter}(u \cdot \text{twopow}(L), u \cdot \text{twopow}(L) - d) = \text{false}$.

Proof. The goal follows from $\text{rhcenterdyadicexactrowlefteqfalse}(L, u, d)$. □

Theorem 9.1453. *For every natural number* L *and natural number* u, $\text{rhcenter}(u \cdot \text{twopow}(L), u \cdot \text{twopow}(L)) = \text{true}$.

Proof. The goal follows from $\text{rhcenterdyadicexactrowselfeqtrue}(L, u)$. □

Theorem 9.1454. *For every natural number* L *and natural number* u *and natural number* r, *if* $0 < r$, *then* $\text{rhcenter}(u \cdot \text{twopow}(L), u \cdot \text{twopow}(L) + r) = \text{false}$.

Proof. The goal follows from $\text{rhcenterdyadicexactrowrighteqfalse}(L, u, r)$. □

Theorem 9.1455. *For every natural number* L *and natural number* u *and natural number* v *and natural number* W *and natural number* i, *if* $1 \leq u$, *then if* $1 \leq v$, *then if* $W \leq L$, *then* $\text{rhcenter}(v \cdot \text{twopow}(L), v \cdot \text{twopow}(L) - W + i) = \text{rhcenter}(u \cdot \text{twopow}(L), u \cdot \text{twopow}(L) - W + i)$.

Proof. The goal follows from rhcenterdyadicexactrowtranslateagree(L, u, v, W, i). □

Theorem 9.1456. *For every natural number L and natural number u and natural number v and natural number W and natural number s and natural number i, if $1 \leq u$, then if $1 \leq v$, then if $s + W \leq L$, then* rhcenter$(v \cdot \text{twopow}(L) + s, v \cdot \text{twopow}(L) - W + i) =$ rhcenter$(u \cdot \text{twopow}(L) + s, u \cdot \text{twopow}(L) - W + i)$.

Proof. The goal follows from rhcenterdyadicexactrowevolvetranslateagree(L, u, v, W, s, i). □

Theorem 9.1457. *For every natural number td and natural number L and natural number u and natural number d, if $1 \leq u$, then if $d \leq L$, then if $0 < d$, then* rdrv$(td, u \cdot \text{twopow}(L), 0, u \cdot \text{twopow}(L) - d) =$ false.

Proof. The goal follows from rdrvdyadicexactrowlefteqfalse(td, L, u, d). □

Theorem 9.1458. *For every natural number td and natural number L and natural number u, if $1 \leq u$, then* rdrv$(td, u \cdot \text{twopow}(L), 0, u \cdot \text{twopow}(L)) =$ true.

Proof. The goal follows from rdrvdyadicexactrowselfeqtrue(td, L, u). □

Theorem 9.1459. *For every natural number td and natural number L and natural number u and natural number r, if $0 < r$, then* rdrv$(td, u \cdot \text{twopow}(L), 0, u \cdot \text{twopow}(L) + r) =$ false.

Proof. The goal follows from rdrvdyadicexactrowrighteqfalse(td, L, u, r). □

Theorem 9.1460. *For every natural number td and natural number L and natural number u and natural number d and natural number s, if $1 \leq u$, then if $s+d \leq L$, then* rdrv$(td, u \cdot \text{twopow}(L), s, u \cdot \text{twopow}(L) - d) =$ lhcenter(s, d).

Proof. The goal follows from rdrvdyadicexactrowshiftlefteq(td, L, u, d, s). □

Theorem 9.1461. *For every natural number td and natural number L and natural number u and natural number q and natural number s, if $1 \leq u$, then if $s \leq L$, then* rdrv$(td, u \cdot \text{twopow}(L), s, u \cdot \text{twopow}(L) + q) =$ rhcenter(s, q).

Proof. The goal follows from rdrvdyadicexactrowshiftrighteq(td, L, u, q, s). □

Theorem 9.1462. *For every natural number td and natural number t and natural number L and natural number u and natural number d and natural number s, if $1 \leq u$, then if $t + s + d \leq L$, then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + s, u \cdot \mathrm{twopow}(L) - d) = \mathrm{lhcenter}(t + s, d)$.

Proof. The goal follows from $\mathrm{rdrvdyadicexactrowtimeshiftlefteq}(td, t, L, u, d, s)$. □

Theorem 9.1463. *For every natural number td and natural number t and natural number L and natural number u and natural number q and natural number s, if $1 \leq u$, then if $t + s \leq L$, then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + s, u \cdot \mathrm{twopow}(L) + q) = \mathrm{rhcenter}(t + s, q)$.

Proof. The goal follows from $\mathrm{rdrvdyadicexactrowtimeshiftrighteq}(td, t, L, u, q, s)$. □

Theorem 9.1464. *For every natural number td and natural number t and natural number Q and natural number L and natural number u and natural number d and natural number S, if $1 \leq u$, then if $t + S + Q + d \leq L$, then if for every natural number s, if $s \leq S$, then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + s + Q, u \cdot \mathrm{twopow}(L) - d) = \mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + s, u \cdot \mathrm{twopow}(L) - d)$, *then* $\mathrm{lhcolagree}(t, Q, d, S)$.

Proof. The goal follows from $\mathrm{lhcolagreeofrdrvdyadicexactrowstepagree}(td, t, Q, L, u, d, S)$. □

Theorem 9.1465. *For every natural number td and natural number t and natural number Q and natural number L and natural number u and natural number S, if $1 \leq u$, then if $t + S + Q + 1 \leq L$, then if for every natural number s, if $s \leq S$, then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + s + Q, u \cdot \mathrm{twopow}(L) - 1) = \mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + s, u \cdot \mathrm{twopow}(L) - 1)$, *then* $\mathrm{lhcolagree}(t, Q, 1, S)$.

Proof. The goal follows from $\mathrm{lhcolagreeoneofrdrvdyadicexactrowstepagree}(td, t, Q, L, u, S)$. □

Theorem 9.1466. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number td and natural number m and natural number t and natural number L and natural number u and natural number S and natural number s, if $T \leq t$, then if $1 \leq u$, then if $t + S + m \cdot P + 1 \leq L$, then if for every natural number j, if $j \leq S$, then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + j + m \cdot P, u \cdot \mathrm{twopow}(L) - 1) = \mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + j, u \cdot \mathrm{twopow}(L) - 1)$, *then if $s \leq S$, then* $\mathrm{maskparity}(t + s + m \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, $hper$, td, m, t, L, u, S, s. The goal follows from periodicmaskparityzeroagreeofrdrvdyadicexactrowstepagree$(T, P, hP, hper, td, m, t, L, u, S, s)$. □

Theorem 9.1467. *For every natural number T and natural number P and natural number td and natural number m and natural number z and natural number p and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z+1-m \cdot P, z+(p+1))$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if $1 \leq u$, then if $z + (p+1) + S + m \cdot P + 1 \leq L$, then if for every natural number s, if $s \leq S$, then* rdrv$(td, u \cdot$ twopow$(L), z + (p+1) + s + m \cdot P, u \cdot$ twopow$(L) - 1) =$ rdrv$(td, u \cdot$ twopow$(L), z + (p+1) + s, u \cdot$ twopow$(L) - 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrdrvdyadicexactrowstepagreeatzerodiagdoubles$(T, P, td$ □

Theorem 9.1468. *For every natural number T and natural number P and natural number td and natural number m and natural number t and natural number k and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P =$* twopow(k)*, then if* twopow$(k) \leq S+1$*, then if $1 \leq u$, then if $t + S + m \cdot P + 1 \leq L$, then if for every natural number s, if $s \leq S$, then* rdrv$(td, u \cdot$ twopow$(L), t + s + m \cdot P, u \cdot$ twopow$(L) - 1) =$ rdrv$(td, u \cdot$ twopow$(L), t + s, u \cdot$ twopow$(L) - 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrdrvdyadicexactrowstepagreetwopow$(T, P, td, m, t, k, L$ □

Theorem 9.1469. *For every natural number T and natural number P and natural number td and natural number m and natural number z and natural number k and natural number L and natural number u and natural number S, if $0 < P$, then if* zerodiagonal(z)*, then if $2 \leq z + m \cdot P$, then if* zerodiagonal$(z + m \cdot P - 2)$*, then if $T \leq z +$* twopow(k)*, then if* settledby$(z, z +$ twopow$(k))$*, then if* settledby$(z + m \cdot P, z + m \cdot P +$ twopow$(k))$*, then if* twopow$(k) \leq S+1$*, then if $1 \leq u$, then if $z+$*twopow$(k)+ S + m \cdot P + 1 \leq L$*, then if for every natural number s, if $s \leq S$, then* rdrv$(td, u \cdot$ twopow$(L), z +$ twopow$(k) + s + m \cdot P, u \cdot$ twopow$(L) - 1) =$ rdrv$(td, u \cdot$ twopow$(L), z +$ twopow$(k) + s, u \cdot$ twopow$(L) - 1)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrdrvdyadicexactrowstepagreeatzerodiagpair(T, P, td, m
□

Theorem 9.1470. *For every natural number td and natural number t and natural number L and natural number u and natural number d and natural number s, if $1 \leq u$, then if $t+s+d \leq L$, then* $\mathrm{rdrv}(td, t+u \cdot \mathrm{twopow}(L), s, u \cdot \mathrm{twopow}(L) - d) = \mathrm{lhcenter}(t+s, d)$.

Proof. The goal follows from rdrvdyadicshiftedinitlefteqlhcenter(td, t, L, u, d, s).
□

Theorem 9.1471. *For every natural number td and natural number t and natural number L and natural number u and natural number q and natural number s, if $1 \leq u$, then if $t + s \leq L$, then* $\mathrm{rdrv}(td, t + u \cdot \mathrm{twopow}(L), s, u \cdot \mathrm{twopow}(L) + q) = \mathrm{rhcenter}(t + s, q)$.

Proof. The goal follows from rdrvdyadicshiftedinitrighteqrhcenter(td, t, L, u, q, s).
□

Theorem 9.1472. *For every natural number td and natural number t and natural number Q and natural number L and natural number u and natural number d and natural number S, if $1 \leq u$, then if $t + S + Q + d \leq L$, then if for every natural number s, if $s \leq S$, then* $\mathrm{rdrv}(td, t + u \cdot \mathrm{twopow}(L) + Q, s, u \cdot \mathrm{twopow}(L) - d) = \mathrm{rdrv}(td, t + u \cdot \mathrm{twopow}(L), s, u \cdot \mathrm{twopow}(L) - d)$, *then* $\mathrm{lhcolagree}(t, Q, d, S)$.

Proof. The goal follows from lhcolagreeofrdrvdyadicshiftedinitagree(td, t, Q, L, u, d, S).
□

Theorem 9.1473. *For every natural number td and natural number t and natural number Q and natural number L and natural number u and natural number S, if $1 \leq u$, then if $t + S + Q + 1 \leq L$, then if for every natural number s, if $s \leq S$, then* $\mathrm{rdrv}(td, t+u \cdot \mathrm{twopow}(L)+Q, s, u \cdot \mathrm{twopow}(L)-1) = \mathrm{rdrv}(td, t + u \cdot \mathrm{twopow}(L), s, u \cdot \mathrm{twopow}(L) - 1)$, *then* $\mathrm{lhcolagree}(t, Q, 1, S)$.

Proof. The goal follows from lhcolagreeoneofrdrvdyadicshiftedinitagree(td, t, Q, L, u, S).
□

Theorem 9.1474. *For every natural number T and natural number P and natural number td and natural number m and natural number z and natural number p and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(m \cdot P, z)$, *then if* $\mathrm{settledby}(z+1, z+(p+1))$, *then if* $\mathrm{settledby}(z+$

$1 - m \cdot P, z + (p + 1))$, *then if* $p + 3 \cdot (m \cdot P) \leq S + 1$, *then if* $1 \leq u$, *then if* $z + (p + 1) + S + m \cdot P + 1 \leq L$, *then if for every natural number* s, *if* $s \leq S$, *then* $\mathrm{rdrv}(td, z + (p + 1) + u \cdot \mathrm{twopow}(L) + m \cdot P, s, u \cdot \mathrm{twopow}(L) - 1) = \mathrm{rdrv}(td, z + (p + 1) + u \cdot \mathrm{twopow}(L), s, u \cdot \mathrm{twopow}(L) - 1)$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofrdrvdyadicshiftedinitagreeatzerodiagdoubles(T, P, td, m □

Theorem 9.1475. *For every natural number* T *and natural number* P *and natural number* td *and natural number* m *and natural number* t *and natural number* k *and natural number* L *and natural number* u *and natural number* S, *if* $0 < P$, *then if* $0 < m$, *then if* $T \leq t$, *then if* $t + m \cdot P = \mathrm{twopow}(k)$, *then if* $\mathrm{twopow}(k) \leq S + 1$, *then if* $1 \leq u$, *then if* $t + S + m \cdot P + 1 \leq L$, *then if for every natural number* s, *if* $s \leq S$, *then* $\mathrm{rdrv}(td, t + u \cdot \mathrm{twopow}(L) + m \cdot P, s, u \cdot \mathrm{twopow}(L) - 1) = \mathrm{rdrv}(td, t + u \cdot \mathrm{twopow}(L), s, u \cdot \mathrm{twopow}(L) - 1)$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofrdrvdyadicshiftedinitagreetwopow($T, P, td, m, t, k, L, u,$ □

Theorem 9.1476. *For every natural number* T *and natural number* P *and natural number* td *and natural number* m *and natural number* z *and natural number* k *and natural number* L *and natural number* u *and natural number* S, *if* $0 < P$, *then if* $\mathrm{zerodiagonal}(z)$, *then if* $2 \leq z + m \cdot P$, *then if* $\mathrm{zerodiagonal}(z + m \cdot P - 2)$, *then if* $T \leq z + \mathrm{twopow}(k)$, *then if* $\mathrm{settledby}(z, z + \mathrm{twopow}(k))$, *then if* $\mathrm{settledby}(z + m \cdot P, z + m \cdot P + \mathrm{twopow}(k))$, *then if* $\mathrm{twopow}(k) \leq S + 1$, *then if* $1 \leq u$, *then if* $z + \mathrm{twopow}(k) + S + m \cdot P + 1 \leq L$, *then if for every natural number* s, *if* $s \leq S$, *then* $\mathrm{rdrv}(td, z + \mathrm{twopow}(k) + u \cdot \mathrm{twopow}(L) + m \cdot P, s, u \cdot \mathrm{twopow}(L) - 1) = \mathrm{rdrv}(td, z + \mathrm{twopow}(k) + u \cdot \mathrm{twopow}(L), s, u \cdot \mathrm{twopow}(L) - 1)$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofrdrvdyadicshiftedinitagreeatzerodiagpair($T, P, td, m, z,$ □

Theorem 9.1477. *For every natural number* T *and natural number* P, *if* $0 < P$, *then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number* m *and natural number* t *and natural number* L, *if* $T \leq t$, *then if* $\mathrm{lhbandagree}(t, m \cdot P, 1, L + 1)$, *then* $\mathrm{lhbandagree}(t + 1, m \cdot P, 1, L)$.

Proof. Assume hP, $hper$, m, t, L. The goal follows from periodiclhbandagreeonestep($T, P, hP, hper, m, t$ □

Theorem 9.1478. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number L and natural number s, if $T \leq t$, then if* lhbandagree$(t, m \cdot P, 1, L + s)$*, then* lhbandagree$(t + s, m \cdot P, 1, L)$.

Proof. Assume hP, $hper$, m, t, L, s. The goal follows from periodiclhbandagreeonesteps$(T, P, hP, hper, m$
□

Theorem 9.1479. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S and natural number s, if $T \leq t$, then if* lhbandagree$(t, m \cdot P, 1, S)$*, then if $s \leq S$, then* lhcenter$(t + s + m \cdot P, 1) =$ lhcenter$(t + s, 1)$.

Proof. Assume hP, $hper$, m, t, S, s. The goal follows from periodiclhcenteroneagreeoflhbandagreeone$(T,$
□

Theorem 9.1480. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S, if $T \leq t$, then if* lhbandagree$(t, m \cdot P, 1, S)$*, then* lhcolagree$(t, m \cdot P, 1, S)$.

Proof. Assume hP, $hper$, m, t, S. The goal follows from periodiclhcolagreeoneoflhbandagreeone$(T, P, hP$
□

Theorem 9.1481. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number m and natural number t and natural number S and natural number s, if $T \leq t$, then if* lhbandagree$(t, m \cdot P, 1, S)$*, then if $s \leq S$, then* maskparity$(t + s + m \cdot P, 0) =$ maskparity$(t + s, 0)$.

Proof. Assume hP, $hper$, m, t, S, s. The goal follows from periodicmaskparityzeroagreeoflhbandagreeon
□

Theorem 9.1482. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(m \cdot P, z)$*, then if* settledby$(z+1, z+(p+1))$*, then if* settledby$(z + 1 - m \cdot P, z + (p + 1))$*, then if $p + 3 \cdot (m \cdot P) \leq S + 1$, then if* lhbandagree$(z + (p + 1), m \cdot P, 1, S)$*, then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicoflhbandagreeoneatzerodiagdoubles(T, P, m, z, p, S).
□

Theorem 9.1483. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t + m \cdot P = \mathrm{twopow}(k)$, then if $\mathrm{twopow}(k) \leq S + 1$, then if $\mathrm{lhbandagree}(t, m \cdot P, 1, S)$, then not $\mathrm{centerperiodic}(T, P)$.*

Proof. The goal follows from $\mathrm{notcenterperiodicoflhbandagreeonetwopow}(T, P, m, t, k, S)$. □

Theorem 9.1484. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number S, if $0 < P$, then if $\mathrm{zerodiagonal}(z)$, then if $2 \leq z + m \cdot P$, then if $\mathrm{zerodiagonal}(z + m \cdot P - 2)$, then if $T \leq z + \mathrm{twopow}(k)$, then if $\mathrm{settledby}(z, z + \mathrm{twopow}(k))$, then if $\mathrm{settledby}(z + m \cdot P, z + m \cdot P + \mathrm{twopow}(k))$, then if $\mathrm{twopow}(k) \leq S + 1$, then if $\mathrm{lhbandagree}(z + \mathrm{twopow}(k), m \cdot P, 1, S)$, then not $\mathrm{centerperiodic}(T, P)$.*

Proof. The goal follows from $\mathrm{notcenterperiodicoflhbandagreeoneatzerodiagpair}(T, P, m, z, k, S)$. □

Theorem 9.1485. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t + Q + (S + 1) \leq K$, then if for every natural number i, if $i \leq S$, then $\mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K) + Q, v \cdot \mathrm{twopow}(K) - (1 + i)) = \mathrm{rhcenter}(t + v \cdot \mathrm{twopow}(K), v \cdot \mathrm{twopow}(K) - (1 + i))$, then $\mathrm{lhbandagree}(t, Q, 1, S)$.*

Proof. The goal follows from $\mathrm{lhbandagreeoneofrhcenterdyadicleftcorridoragree}(t, Q, K, v, S)$. □

Theorem 9.1486. *For every natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t + Q + (S + 1) \leq K$, then if $\mathrm{rhbandagree}(t + v \cdot \mathrm{twopow}(K), Q, v \cdot \mathrm{twopow}(K) - (S + 1), S)$, then $\mathrm{lhbandagree}(t, Q, 1, S)$.*

Proof. The goal follows from $\mathrm{lhbandagreeoneofrhbandagreedyadicleftcorridor}(t, Q, K, v, S)$. □

Theorem 9.1487. *For every natural number T and natural number P, if $0 < P$, then if $\mathrm{centerperiodic}(T, P)$, then for every natural number m and natural number t and natural number K and natural number v and natural number S, if $T \leq t$, then if $1 \leq v$, then if $t + m \cdot P + (S + 1) \leq K$, then if $\mathrm{rhbandagree}(t + v \cdot \mathrm{twopow}(K), m \cdot P, v \cdot \mathrm{twopow}(K) - (S + 1), S)$, then $\mathrm{lhcolagree}(t, m \cdot P, 1, S)$.*

Proof. Assume hP, $hper$, m, t, K, v, S. The goal follows from periodiclhcolagreeoneofrhbandagreedyadic
□

Theorem 9.1488. *For every natural number T and natural number P and natural number m and natural number z and natural number p and natural number K and natural number v and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq z+(p+1)$, then if* zerodiagonal(z), *then if* doublesat$(m \cdot P, z)$, *then if* settledby$(z+1, z+(p+1))$, *then if* settledby$(z+1-m \cdot P, z+(p+1))$, *then if $p+3 \cdot (m \cdot P) \leq S+1$, then if $1 \leq v$, then if $z+(p+1)+m \cdot P+(S+1) \leq K$, then if* rhbandagree$(z+(p+1)+v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K) - (S+1), S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadicleftcorridoratzerodiagdoubles$(T, P, r$
□

Theorem 9.1489. *For every natural number T and natural number P and natural number m and natural number t and natural number k and natural number K and natural number v and natural number S, if $0 < P$, then if $0 < m$, then if $T \leq t$, then if $t+m \cdot P = \text{twopow}(k)$, then if* twopow$(k) \leq S+1$, *then if $1 \leq v$, then if $t+m \cdot P+(S+1) \leq K$, then if* rhbandagree$(t+v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K)-(S+1), S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadicleftcorridortwopow$(T, P, m, t, k, K, v$
□

Theorem 9.1490. *For every natural number T and natural number P and natural number m and natural number z and natural number k and natural number K and natural number v and natural number S, if $0 < P$, then if* zerodiagonal(z), *then if $2 \leq z+m \cdot P$, then if* zerodiagonal$(z+m \cdot P-2)$, *then if $T \leq z+\text{twopow}(k)$, then if* settledby$(z, z+\text{twopow}(k))$, *then if* settledby$(z+m \cdot P, z+m \cdot P+\text{twopow}(k))$, *then if* twopow$(k) \leq S+1$, *then if $1 \leq v$, then if $z+\text{twopow}(k)+m \cdot P+(S+1) \leq K$, then if* rhbandagree$(z+\text{twopow}(k)+v \cdot \text{twopow}(K), m \cdot P, v \cdot \text{twopow}(K)-(S+1), S)$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofrhbandagreedyadicleftcorridoratzerodiagpair$(T, P, m, z$
□

Theorem 9.1491. *For every natural number td and natural number t and natural number Q and natural number K and natural number v and natural number S, if $1 \leq v$, then if $t+Q+(S+1) \leq K$, then if for every natural number i, if $i \leq S$, then* rdrv$(td, v \cdot \text{twopow}(K), t+Q, v \cdot \text{twopow}(K)-(1+i)) =$ rdrv$(td, v \cdot \text{twopow}(K), t, v \cdot \text{twopow}(K)-(1+i))$, *then* lhbandagree$(t, Q, 1, S)$.

Proof. The goal follows from lhbandagreeoneofrdrvdyadicexactrowleftcorridoragree(td, t, Q, K, v, S). □

Theorem 9.1492. *For every natural number t and natural number K and natural number L and natural number u and natural number w and natural number S, if $1 \leq u$, then if $t+S+1 \leq K$, then if $t+S+w \cdot \mathrm{twopow}(K)+1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) - (1+i)) = \mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K) - (1+i))$, *then* $\mathrm{lhbandagree}(t, w \cdot \mathrm{twopow}(K), 1, S)$.

Proof. The goal follows from lhbandagreeoneoftwoscalesinglerowleftcorridorspatialagree(t, K, L, u, w, S) □

Theorem 9.1493. *For every natural number t and natural number P and natural number c and natural number K and natural number L and natural number u and natural number S, if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1+i)) = \mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1+i))$, *then* $\mathrm{lhbandagree}(t, c \cdot \mathrm{twopow}(K) \cdot P, 1, S)$.

Proof. The goal follows from lhbandagreeoneofdyadicmulperiodsinglerowleftcorridorspatialagree$(t, P, c,$ □

Theorem 9.1494. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number c and natural number t and natural number K and natural number L and natural number u and natural number S, if $T \leq t$, then if $1 \leq u$, then if $t+S+1 \leq K$, then if $t + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1+i)) = \mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1+i))$, *then* $\mathrm{lhcolagree}(t, c \cdot \mathrm{twopow}(K) \cdot P, 1, S)$.

Proof. Assume hP, $hper$, c, t, K, L, u, S. The goal follows from periodiclhcolagreeoneofdyadicmulperiod □

Theorem 9.1495. *For every natural number T and natural number P, if $0 < P$, then if* $\mathrm{centerperiodic}(T, P)$, *then for every natural number c and natural number t and natural number K and natural number L and natural*

number u and natural number S and natural number s, if $T \leq t$, then if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1 + i)) = \mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1 + i))$, *then if $s \leq S$, then* $\mathrm{maskparity}(t + s + c \cdot \mathrm{twopow}(K) \cdot P, 0) = \mathrm{maskparity}(t + s, 0)$.

Proof. Assume hP, $hper$, c, t, K, L, u, S, s. The goal follows from periodicmaskparityzeroagreeofdyadicr
□

Theorem 9.1496. *For every natural number T and natural number P and natural number c and natural number z and natural number p and natural number K and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < c$, then if $T \leq z + (p + 1)$, then if* $\mathrm{zerodiagonal}(z)$, *then if* $\mathrm{doublesat}(c \cdot \mathrm{twopow}(K) \cdot P, z)$, *then if* $\mathrm{settledby}(z + 1, z + (p + 1))$, *then if* $\mathrm{settledby}(z + 1 - c \cdot \mathrm{twopow}(K) \cdot P, z + (p + 1))$, *then if $p + 3 \cdot (c \cdot \mathrm{twopow}(K) \cdot P) \leq S + 1$, then if $1 \leq u$, then if $z + (p + 1) + S + 1 \leq K$, then if $z + (p + 1) + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* $\mathrm{rhcenter}(z + (p + 1) + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1 + i)) = \mathrm{rhcenter}(z + (p + 1) + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1 + i))$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofdyadicmulperiodsinglerowleftcorridorspatialagreeatzero
□

Theorem 9.1497. *For every natural number T and natural number P and natural number c and natural number t and natural number k and natural number K and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < c$, then if $T \leq t$, then if $t + c \cdot \mathrm{twopow}(K) \cdot P = \mathrm{twopow}(k)$, then if* $\mathrm{twopow}(k) \leq S + 1$, *then if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* $\mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1 + i)) = \mathrm{rhcenter}(t + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1 + i))$, *then not* $\mathrm{centerperiodic}(T, P)$.

Proof. The goal follows from notcenterperiodicofdyadicmulperiodsinglerowleftcorridorspatialagreetwopo
□

Theorem 9.1498. *For every natural number T and natural number P and natural number c and natural number z and natural number k and natural number K and natural number L and natural number u and natural number*

S, *if* $0 < P$, *then if* zerodiagonal(z), *then if* $2 \leq z + c \cdot \mathrm{twopow}(K) \cdot P$, *then if* zerodiagonal$(z + c \cdot \mathrm{twopow}(K) \cdot P - 2)$, *then if* $T \leq z + \mathrm{twopow}(k)$, *then if* settledby$(z, z + \mathrm{twopow}(k))$, *then if* settledby$(z + c \cdot \mathrm{twopow}(K) \cdot P, z + c \cdot \mathrm{twopow}(K) \cdot P + \mathrm{twopow}(k))$, *then if* $\mathrm{twopow}(k) \leq S + 1$, *then if* $1 \leq u$, *then if* $z + \mathrm{twopow}(k) + S + 1 \leq K$, *then if* $z + \mathrm{twopow}(k) + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, *then if* $K \leq L$, *then if for every natural number* i, *if* $i \leq S$, *then* $\mathrm{rhcenter}(z + \mathrm{twopow}(k) + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1 + i)) = \mathrm{rhcenter}(z + \mathrm{twopow}(k) + u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1 + i))$, *then not* centerperiodic(T, P).

Proof. The goal follows from notcenterperiodicofdyadicmulperiodsinglerowleftcorridorspatialagreeatzero
□

Theorem 9.1499. *For every natural number* td *and natural number* t *and natural number* K *and natural number* L *and natural number* u *and natural number* w *and natural number* d, *if* $1 \leq u$, *then if* $K \leq L$, *then if* $t + d \leq K$, *then if* $t + w \cdot \mathrm{twopow}(K) + d \leq L$, *then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K) - d) = \mathrm{lhcenter}(t, d)$.

Proof. The goal follows from rdrvtwoscalerowlefteqlhcenter(td, t, K, L, u, w, d).
□

Theorem 9.1500. *For every natural number* td *and natural number* t *and natural number* K *and natural number* L *and natural number* u *and natural number* w *and natural number* q, *if* $1 \leq u$, *then if* $K \leq L$, *then if* $t \leq K$, *then if* $t + w \cdot \mathrm{twopow}(K) \leq L$, *then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K) + q) = \mathrm{rhcenter}(t, q)$.

Proof. The goal follows from rdrvtwoscalerowrighteqrhcenter(td, t, K, L, u, w, q).
□

Theorem 9.1501. *For every natural number* td *and natural number* t *and natural number* K *and natural number* L *and natural number* u *and natural number* w *and natural number* S, *if* $1 \leq u$, *then if* $t + S + 1 \leq K$, *then if* $t + S + w \cdot \mathrm{twopow}(K) + 1 \leq L$, *then if* $K \leq L$, *then if for every natural number* i, *if* $i \leq S$, *then* $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) - (1 + i)) = \mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + w \cdot \mathrm{twopow}(K), u \cdot \mathrm{twopow}(L) + w \cdot \mathrm{twopow}(K) - (1 + i))$, *then* $\mathrm{lhbandagree}(t, w \cdot \mathrm{twopow}(K), 1, S)$.

Proof. The goal follows from lhbandagreeoneofrdrvtwoscalesinglerowleftcorridorspatialagree$(td, t, K, L,$
□

Theorem 9.1502. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number td and natural number c and natural number t and natural number K and natural number L and natural number u and natural number S, if $T \leq t$, then if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot$ twopow$(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* rdrv$(td, u \cdot$ twopow$(L), t + c \cdot$ twopow$(K) \cdot P, u \cdot$ twopow$(L) - (1 + i)) =$ rdrv$(td, u \cdot$ twopow$(L), t + c \cdot$ twopow$(K) \cdot P, u \cdot$ twopow$(L) + c \cdot$ twopow$(K) \cdot P - (1 + i))$*, then* lhcolagree$(t, c \cdot$ twopow$(K) \cdot P, 1, S)$*.*

Proof. Assume hP, $hper$, td, c, t, K, L, u, S. The goal follows from periodiclhcolagreeoneofrdrvdyadicmulperiodsinglerowleftcorridorspatialagree$(T, P, hP, hper, td, c, t, K,$ □

Theorem 9.1503. *For every natural number T and natural number P, if $0 < P$, then if* centerperiodic(T, P)*, then for every natural number td and natural number c and natural number t and natural number K and natural number L and natural number u and natural number S and natural number s, if $T \leq t$, then if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot$ twopow$(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* rdrv$(td, u \cdot$ twopow$(L), t + c \cdot$ twopow$(K) \cdot P, u \cdot$ twopow$(L) - (1 + i)) =$ rdrv$(td, u \cdot$ twopow$(L), t + c \cdot$ twopow$(K) \cdot P, u \cdot$ twopow$(L) + c \cdot$ twopow$(K) \cdot P - (1 + i))$*, then if $s \leq S$, then* maskparity$(t + s + c \cdot$ twopow$(K) \cdot P, 0) =$ maskparity$(t + s, 0)$*.*

Proof. Assume hP, $hper$, td, c, t, K, L, u, S, s. The goal follows from periodicmaskparityzeroagreeofrdrvdyadicmulperiodsinglerowleftcorridorspatialagree$(T, P, hP, hper, td,$ □

Theorem 9.1504. *For every natural number T and natural number P and natural number td and natural number c and natural number z and natural number p and natural number K and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < c$, then if $T \leq z + (p + 1)$, then if* zerodiagonal(z)*, then if* doublesat$(c \cdot$ twopow$(K) \cdot P, z)$*, then if* settledby$(z + 1, z + (p + 1))$*, then if* settledby$(z + 1 - c \cdot$ twopow$(K) \cdot P, z + (p + 1))$*, then if $p + 3 \cdot (c \cdot$ twopow$(K) \cdot P) \leq S + 1$, then if $1 \leq u$, then if $z + (p + 1) + S + 1 \leq K$, then if $z + (p + 1) + S + c \cdot$ twopow$(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then* rdrv$(td, u \cdot$ twopow$(L), z + (p + 1) + c \cdot$ twopow$(K) \cdot P, u \cdot$ twopow$(L) - (1 + i)) =$ rdrv$(td, u \cdot$ twopow$(L), z + (p + 1) + c \cdot$ twopow$(K) \cdot P, u \cdot$ twopow$(L) + c \cdot$ twopow$(K) \cdot P - (1 + i))$*, then not* centerperiodic(T, P)*.*

Proof. The goal follows from notcenterperiodicofrdrvdyadicmulperiodsinglerowleftcorridorspatialagreea
□

Theorem 9.1505. *For every natural number T and natural number P and natural number td and natural number c and natural number t and natural number k and natural number K and natural number L and natural number u and natural number S, if $0 < P$, then if $0 < c$, then if $T \leq t$, then if $t + c \cdot \mathrm{twopow}(K) \cdot P = \mathrm{twopow}(k)$, then if $\mathrm{twopow}(k) \leq S + 1$, then if $1 \leq u$, then if $t + S + 1 \leq K$, then if $t + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1 + i)) = \mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), t + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1 + i))$, then not $\mathrm{centerperiodic}(T, P)$.*

Proof. The goal follows from notcenterperiodicofrdrvdyadicmulperiodsinglerowleftcorridorspatialagreetw
□

Theorem 9.1506. *For every natural number T and natural number P and natural number td and natural number c and natural number z and natural number k and natural number K and natural number L and natural number u and natural number S, if $0 < P$, then if $\mathrm{zerodiagonal}(z)$, then if $2 \leq z + c \cdot \mathrm{twopow}(K) \cdot P$, then if $\mathrm{zerodiagonal}(z + c \cdot \mathrm{twopow}(K) \cdot P - 2)$, then if $T \leq z + \mathrm{twopow}(k)$, then if $\mathrm{settledby}(z, z + \mathrm{twopow}(k))$, then if $\mathrm{settledby}(z + c \cdot \mathrm{twopow}(K) \cdot P, z + c \cdot \mathrm{twopow}(K) \cdot P + \mathrm{twopow}(k))$, then if $\mathrm{twopow}(k) \leq S + 1$, then if $1 \leq u$, then if $z + \mathrm{twopow}(k) + S + 1 \leq K$, then if $z + \mathrm{twopow}(k) + S + c \cdot \mathrm{twopow}(K) \cdot P + 1 \leq L$, then if $K \leq L$, then if for every natural number i, if $i \leq S$, then $\mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), z + \mathrm{twopow}(k) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) - (1 + i)) = \mathrm{rdrv}(td, u \cdot \mathrm{twopow}(L), z + \mathrm{twopow}(k) + c \cdot \mathrm{twopow}(K) \cdot P, u \cdot \mathrm{twopow}(L) + c \cdot \mathrm{twopow}(K) \cdot P - (1 + i))$, then not $\mathrm{centerperiodic}(T, P)$.*

Proof. The goal follows from notcenterperiodicofrdrvdyadicmulperiodsinglerowleftcorridorspatialagreea
□